Inequality and Misallocation under Production Networks

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Abstract

This paper examines the equity–efficiency tradeoff by showing that changes in the joint distributions of factor income and household expenditure influence aggregate productivity (TFP). I develop a general equilibrium aggregation and incidence theory for a distorted production network economy with heterogeneous households, deriving a sufficient-statistic decomposition of TFP. I define and decompose the positional terms of trade (PTT) — a household-specific efficiency wedge that maps shocks into real-consumption incidence. The model ranks factors, households, and firms by distortion, expenditure, and revenue centralities, showing that changes in who earns and who spends are generically non-neutral: allocative efficiency rises when factor income shifts from high to low distortion centrality factors and when demand reallocates toward high expenditure centrality households or high revenue centrality firms, thereby alleviating bottlenecks. Benchmarking against a constrained social planner allocation that internalizes reallocation externalities, I show that decentralization lacks the planner's balancing property and generates first-order externalities for aggregate efficiency and welfare. Estimating the model on U.S. data (1997–2021), I find that changes in the income distribution explain approximately 20% of TFP variation. Finally, budget-neutral, progressive lump-sum transfers targeted by expenditure centrality raise aggregate TFP and improve the PTTs of lower- and middle-income households, aligning efficiency gains with favorable equity outcomes.

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1 Introduction

> *"While we often must focus on aggregates for macroeconomic policy, it is impossible to think coherently about national well-being while ignoring inequality and poverty, neither of which is visible in aggregate data. Indeed, and except in exceptional cases, macroeconomic aggregates themselves depend on distribution."*
>
> — Deaton (2016)

Modern economies are a complex web of market interactions shaped by the decisions of interconnected firms and households. These interactions take place in multilayered input-output and payment networks through which goods, services, intermediates, and payments are routed. In these economies, small, idiosyncratic distortions, such as markups, amplify along input-output chains, reallocating inputs across firms and affecting TFP. Less explored is the complementary implication of Deaton's point: do changes in inequality — who earns income, who receives profits, and how spending is routed — move TFP via reallocation? This connection matters for distributional macroeconomics and aggregation: it speaks to whether the classic "*big tradeoff*" (Okun, 1975) between efficiency and equity is inevitable, or whether *targeted* redistribution can also raise aggregate efficiency in second-best economies.

This paper studies inequality and misallocation in distorted production network economies with heterogeneous households. Firms are linked by input-output linkages and face idiosyncratic price-cost wedges (e.g., markups); households differ in preferences, factor incomes, and equity ownership. Unlike a representative agent economy, this environment makes distribution a state variable: shifts in who receives factor income and profits, and how expenditure is allocated across goods, feed back into equilibrium prices, allocations, and aggregate productivity. To track these channels, I introduce a layered system of centralities that keeps factor-firm-household linkages separate and yields sufficient statistics for aggregate propagation and household-level incidence. This structure decomposes aggregate TFP variation into components driven by technology, distortions, income and spending reallocations, and shifts in demand composition. In short, it clarifies when and how household heterogeneity in income and spending interacts with firm-level distortions to shape allocative efficiency in a networked economy. Developing such a theory is challenging: it requires tracking how shocks and redistribution propagate through input-output linkages and how incidence varies with households' and firms' network positions.

This paper aims to explain not only how inequality affects aggregate efficiency, but also to measure incidence, i.e., who gains and who loses in response to microeconomic shocks in a networked economy. Standard aggregate TFP decompositions are silent on incidence. Even when output barely changes, real consumption can vary markedly across households because they differ in their consumption baskets and in where their factor income and profits are earned. To address this gap, I introduce the *positional terms of trade* (PTT): a household-level incidence wedge that maps shocks into real consumption outcomes. PTT is the household analogue of TFP. Formally, $Y = TFP\, F$ measures how efficiently primary factors produce aggregate output, whereas $C_h = PTT_h\, f_h$ gauges how efficiently the same factors translate into real consumption for household h (with F and f_h representing factor indexes). Treating PTT as an efficiency wedge links micro incidence to macro outcomes: aggregate TFP growth is an expenditure-weighted average of household PTT growth,

$$d \log TFP = \sum_{h \in \mathscr{H}} \chi_h \, d \log PTT_h, \text{ where } \chi_h \text{ is household } h\text{'s final expenditure share.}$$

A central contribution is new decompositions of TFP and PTT that isolate the role of inequality and deliver analytic sufficient statistics for first-order variations. These statistics decompose aggregate productivity and household-level incidence into three forces: (i) technology (holding allocations fixed), (ii) distortions (holding demand composition fixed), and (iii) demand pattern and distributional shifts. A key implication is that, in second-best economies, who earns and who spends shapes allocative efficiency. In a first-best economy, an aggregate distributional envelope result holds: redistribution is locally neutral for output because the allocation already lies on the efficient frontier. By contrast, with distortions, redistribution can worsen or mitigate misallocation even when technology and resources are unchanged. The paper introduces three network centralities — *distortion*, *expenditure*, and *revenue* — that characterize the first-order impact of changes in distributions and demand composition on aggregate efficiency and incidence. Intuitively, a primary factor's *distortion centrality* measures the extent to which its compensation is exposed to heavily distorted supply chains composed of high-markup firms that act as bottlenecks to aggregate output. A household's *expenditure centrality* is the distortion centrality exposure of the factors embedded in its consumption bundle. A firm's *revenue centrality* is the distortion centrality exposure of the factors used to produce its output. High expenditure or revenue centrality indicates that a large share of spending is routed toward high distortion centrality factors. In short, the sufficient statistics imply that TFP rises by alleviating bottlenecks when (i) factor income shifts from high to low *distortion centrality* factors, (ii) expenditure shifts towards higher *expenditure centrality* households, or (iii) demand shifts toward higher *revenue centrality* firms. These are inherently second-best effects: with microeconomic distortions, distributional changes have tangible aggregate consequences by moving the economy along the production possibility frontier. This reframes the equity-efficiency debate in general equilibrium: when production networks feature wedges, inequality is not only an outcome but also a driver of aggregate efficiency.

These second-best mechanisms motivate a diagnostic benchmark that isolates distortions from decentralization: what changes when a *constrained* social planner — respecting feasibility, technology, and existing wedges — centralizes household consumption and factor supply decisions and internalizes the general equilibrium reallocation externalities? The key result is that centralization balances heterogeneity precisely along the margins that generate inequality-driven misallocation. The planner chooses the joint allocation of expenditure and factor supplies across households so that centrality-weighted exposures to distortions are equalized across households along two dimensions. First, the planner neutralizes factor income reallocation by choosing factor supplies that equalize distortion centralities across all factors. Second, it neutralizes the dependence of the factor income distribution on the expenditure distribution via budget-neutral lump-sum taxes and transfers, thereby breaking the first-order dependence of TFP on expenditure composition. By contrast, in the decentralized economy, households do not internalize the reallocation externalities their consumption and factor supply decisions impose on aggregate welfare. As a result, allocations lack the planner's balancing property, generating economy-wide reallocation externality and making distributions state variables for aggregate efficiency. Hence, the centralized-versus-decentralized benchmark is not a matter of tractability but a foundational modeling assumption with first-order implications for measuring TFP and interpreting both aggregate and incidence effects of distributional policy.

Beyond the theoretical framework, the paper's empirical contribution is to bring the model to U.S. data. I assemble a U.S. mapping for 1997-2021 that links input-output structure to the distribution of labor and capital incomes and to household consumption patterns across sectors and household groups. In essence, I map the U.S. production network onto the joint distribution of factor incomes and household spending. The mapping leverages standard sources: BEA Industry Accounts for input-output structure; BLS Quarterly Census of Employment and Wages (QCEW) and Occupational and Employment Wage Statistics (OEWS) for the geographic- and occupation-level distribution of labor income; and the Consumer Expenditure Survey (CEX) and Personal Consumption Expenditure (PCE) for heterogeneity in consumption bundles. Estimating the model on 1997-2021 allows me to quantify how much of observed TFP variation is attributable to changes in sector-level productivity and distortions, and changes in the income distribution and demand composition.

The empirical findings show that distributional variation is a relevant driver of U.S. productivity dynamics. In particular, changes in the income distribution explain about 20% of the variance in U.S. TFP (1997-2021). Quantitatively, this implies that a meaningful share of aggregate productivity fluctuations can be traced to shifts in who earns income across labor, capital, and profits. Crucially, the cross-sectional mapping from earnings to expenditure centrality implies a time-varying *distributional multiplier*: prior to 2021, the data support a lower-income multiplier — strengthening purchasing power among lower-earning occupations raises productivity — whereas by 2021 the relationship becomes U-shaped and the strongest lever shifts to the middle of the earnings distribution, yielding a middle-income multiplier. More broadly, the U.S. evidence substantiates the model's prediction that distribution matters: ignoring income distribution would omit a sizable portion of business-cycle variation in aggregate productivity.

The analysis has policy implications for redistribution in second-best economies. A conventional view holds that there is an equity-efficiency tradeoff: redistributive policies, while desirable for equity, inevitably reduce efficiency. The model implies a more nuanced picture in distorted economies. In such second-best environments, targeted redistribution can raise efficiency and equity, holding technology and wedges fixed. Thus, the tradeoff can be relaxed under identifiable conditions. Using the model's sufficient statistics, I study budget-neutral lump-sum transfers across household groups. Targeting transfers toward households with higher *expenditure centrality* raises aggregate TFP and improves the distribution of real incomes in the model. Imposing progressivity constraints on these transfers improves lower-income households' PTTs while preserving efficiency gains. A simple *sufficient-statistics targeting* rule based on the M-hierarchies (ranking households by expenditure centrality) jointly advances efficiency and equity, raising TFP and improving households' PTTs at the bottom of the distribution. This provides a constructive example of how to exploit the inequality-misallocation interaction to achieve dual goals. Properly accounting for heterogeneity is critical for policy design: ignoring distribution can misstate an economy's efficiency and the impact of shocks. More concretely, targeted transfers can increase aggregate output while reducing inequality; a win-win outcome that overturns the presumption that equity necessarily comes at the expense of efficiency.

Related Literature

This paper connects to macroeconomic aggregation, production networks, national accounts, and distributional macroeconomics. Aggregation in multisector input-output economies builds on Hulten (1978), whose envelope result implies that, in a perfectly competitive representative-household economy, first-order aggregate effects of productivity shocks depend only on Domar sales weights, rendering network structure irrelevant to a first order. A subsequent literature shows how idiosyncratic disturbances propagate along input–output linkages to shape aggregate fluctuations: early multi-sector business-cycle analyses include Horvath (1998, 2000) and Dupor (1999); Foerster, Sarte, & Watson (2011) document industrial comovement with aggregate shocks; and Acemoglu, Carvalho, Ozdaglar, & Tahbaz-Salehi (2012) formalize how network topology can scale micro shocks into macro volatility. Recent evidence further underscores supply-chain transmission (Carvalho et al., 2021). Beyond pure technology shocks, work on distorted environments studies how wedges propagate through networks: early contributions such as Basu (1995), Ciccone (2002), and Jones (2011) characterize propagation in specific structures; later analyses deliver general results — Jones (2013) provide multiplier formulas, while Liu (2019) and Baqaee & Farhi (2020) derive nonparametric aggregation for distorted economies with a representative household. Relatedly, Bigio & La'O (2020) quantify how sectoral wedges reallocate resources and affect the labor wedge and TFP relative to the first best. More recent open-economy work integrates household heterogeneity into networked general equilibrium, showing cross-country transmission through input–output linkages (Devereux et al., 2023; Baqaee & Farhi, 2024). Relative to this literature, I develop nonparametric first-order sufficient statistics in a production-network model with heterogeneous households that map shifts in the distributions of income and expenditure, and demand composition into aggregate TFP; the representation is layered across factors, firms, and households and yields novel bilateral centralities (distortion, expenditure, revenue) to rank households and firms by their efficiency implications. The equity-efficiency tradeoff also requires measuring the distributional effect. For this reason, to the best of my knowledge, this paper, using the positional terms of trade (PTT), is the first to deliver household-level incidence metrics in a production network model. On the empirical side, I operationalize the layered representation in U.S. data to estimate centralities, TFP variations, and PTTs, and show that there is space for progressive redistribution through budget-neutral lump-sum transfers and taxes driven by sufficient-statistics targeting.

In a related setting, Dávila & Schaab (2025) develop an axiomatic framework for decomposing changes in social welfare. This framework is agnostic to any notion of equilibrium and is built exclusively on economic primitives — preferences, production technologies, and resource constraints — so allocations need not be generated by optimizing agents and markets need not clear. Conceptually, they separate welfare changes into efficiency and redistribution components; the latter is normative and reflects heterogeneity in households' marginal utilities. By contrast, my equilibrium-based approach targets aggregate efficiency (TFP) rather than welfare: redistribution and household heterogeneity feed back into TFP by creating second-best mismatches between where resources should go and where they do go under decentralized demand. Additionally, Baqaee & Burstein (2025) define a heterogeneous-household analogue to Lucas (1987): aggregate efficiency equals the maximal uniform contraction of the consumption-possibility set that keeps every agent at least as well off as the status quo. They show that, with identical, homothetic preferences and common relative prices, there exists a Hicksian representative agent that delivers a single welfare scalar for efficiency — the *miraculous consensus*. Evaluated around a *compensated equilibrium* — an allocation in which transfers return each household to its baseline utility — their first-order decomposition characterizes changes in aggregate efficiency. By contrast, my TFP and PTT decompositions, and their sufficient statistics, operate in decentralized, distorted economies with preference heterogeneity, where the income and expenditure distributions are state variables and are not neutralized by compensation.

This paper contributes to the misallocation literature (Solow, 1957; Hulten, 1978; Jorgenson et al., 1987; Hall & Diamond, 1990; Basu & Fernald, 2002; Petrin & Levinsohn, 2012). More recently, Baqaee & Farhi (2020) formalize a general aggregation theory for inefficient multi-sector economies, showing that both technology shocks and reallocation terms generate first-order changes in TFP. This paper identifies and quantifies a *distribution-driven* channel for TFP variation: shifts in the income and consumption distributions translate into changes in allocative efficiency. This mechanism connects to the misallocation literature of Restuccia & Rogerson (2008) and Hsieh & Klenow (2009), which quantify TFP losses from resource misallocation. The innovation here is to show that redistributive changes are generically non-neutral for efficiency and can amplify or mitigate misallocation.

Finally, a parallel literature documents differences in household consumption behavior across the income distribution. High-income households have lower MPCs (Jappelli & Pistaferri, 2014), purchase higher-priced varieties (Jaimovich et al., 2020), are less price sensitive (Auer, Burstein, Lein, & Vogel, 2024), pay higher markups for identical products and purchase baskets with higher average markups (Sangani, 2024). Because MPCs differ across households, distributional shocks interact with nominal rigidities — downward nominal wage rigidities as in Auclert & Rognlie (2020), or a zero-lower-bound as in Mian, Straub, & Sufi (2021) — to affect aggregate demand. Transfers from high-MPC to low-MPC households reduce demand and can generate Keynesian unemployment. By contrast, the distributional effects on aggregate output in this paper do not operate through capacity constraints or nominal rigidities; they operate via demand recomposition that alters aggregate supply via TFP.

Layout

The paper proceeds as follows. Section 2 sets up a multisector input–output economy with heterogeneous households and generic distortions, defines prices and allocations, and introduces the layered centralities. Section 3 delivers nonparametric, first-order sufficient statistics that map shifts in productivities, distortions, the joint income–expenditure distribution, and demand composition into changes in aggregate TFP and PTTs, and benchmarks these results against a constrained social-planner allocation. Section 4 embeds the accounting in a nested-CES general-equilibrium environment to discipline key elasticities, quantify nonlinear interactions, and characterize conditions for allocative neutrality and distributional incidence. Section 5 details the empirical implementation and documents heterogeneity in the sufficient statistics across households. Section 6 shows that redistribution targeted by expenditure centrality can improve aggregate efficiency while, on average, delivering positive incidence for low-income households. Section 7 concludes.

2 General Framework

In this section, I develop a static general equilibrium, production network model with heterogeneous firms and households. Firms are connected by input-output linkages and face firm-specific distortions (exogenous markups) that create wedges between prices and marginal costs. Households differ in preferences, endowments, and asset ownership, which jointly determine the distribution of income and expenditure. The model is nonparametric, characterizing equilibrium allocations without functional form restrictions. I derive centrality measures that are sufficient statistics for first-order changes in TFP and household real consumption. Let firms be indexed by $i \in \mathcal{N} := \{1, \cdots, N\}$ and household types by $h \in \mathcal{H} := \{1, \cdots, H\}$.

2.1 Production

Firm $i \in \mathcal{N}$ produces output y_i according to

$$y_i = A_i \, Q_i(L_i, X_i), \qquad L_i = Q_i^\ell(\{\ell_{ih}\}_{h \in \mathcal{H}}), \qquad X_i = Q_i^x(\{x_{ij}\}_{j \in \mathcal{N}}). \tag{1}$$

Here A_i is Hicks-neutral productivity; L_i is the primary-factor composite; X_i is the intermediate-input composite; ℓ_{ih} is the quantity of the primary-factor supplied by household type h employed by firm i; and x_{ij} is the quantity of input from firm j used by firm i.[1]

Firms take prices as given and choose input bundles to minimize total costs

$$\bar{c}_i = p_i^\ell \, L_i + p_i^x \, X_i = \sum_{h \in \mathcal{H}} w_h \, \ell_{ih} + \sum_{j \in \mathcal{N}} p_j \, x_{ij}, \tag{2}$$

where w_h and p_j are the prices of primary factors and intermediate inputs, respectively; p_i^ℓ and p_i^x are the unit expenditure indices dual to Q_i^ℓ and Q_i^x. A central feature of the model is the presence of firm-level distortions: firm i's marginal cost satisfies $mc_i = \mu_i \, p_i$ with $0 < \mu_i \leq 1$. The inverse markup μ_i measures the fraction of revenue paid to suppliers of primary factors and intermediates. When $\mu_i < 1$, price exceeds marginal cost and the unit rent is $(1 - \mu_i) \, p_i$.

[1] The technologies $Q_i : \mathbb{R}_+^2 \to \mathbb{R}_+$, $Q_i^\ell : \mathbb{R}_+^H \to \mathbb{R}_+$, and $Q_i^x : \mathbb{R}_+^N \to \mathbb{R}_+$ are neoclassical and satisfy the following regularity conditions: they are positive, twice continuously differentiable, and strictly concave; for inputs actually used, marginal products satisfy the Inada conditions.

This wedge shifts income shares without altering the production possibility set. Distortions do not mechanically shrink the production possibility frontier; they alter pricing and the allocation of resources in general equilibrium. Online Appendix Section 1 microfounds μ_i via a firm-level Dixit & Stiglitz (1977) structure with input-specific productivities.

2.2 Households

Each household type h chooses (C_h, L_h) to maximize $U_h(C_h, L_h)$, where C_h is a consumption index and L_h is primary factor supply. Consumption satisfies $C_h = Q_h^c\left(\{C_{hi}\}_{i\in\mathcal{N}}\right)$ where C_{hi} is the quantity purchased from firm i. Let p_h^c denote the unit expenditure index dual to Q_h^c.[2] Households take prices and wages as given and face the budget constraint

$$E_h \equiv p_h^c C_h = \sum_{i\in\mathcal{N}} p_i C_{hi} \leq J_h + \Pi_h + T_h, \tag{3}$$

where $J_h = w_h L_h$, $\Pi_h = \sum_{i\in\mathcal{N}} \kappa_{ih}\,\pi_i$, and T_h denote factor income, dividends, and lump-sum transfers, respectively. Ownership shares satisfy $\kappa_{ih} \in [0,1]$ and $\sum_{h\in\mathcal{H}} \kappa_{ih} = 1$ for each firm i.

Households differ in preferences, primary factor supply, and equity holdings. Preference heterogeneity is only one device: the same dispersion in equilibrium expenditure shares can arise from non-homothetic aggregators, idiosyncratic taste shifters or amenity terms, household composition or home-production differences, and quality choice; all can be represented as reduced-form shifts in Q_h^c holding the common price vector fixed. Financial markets are incomplete — there are no state-contingent claims spanning idiosyncratic household or firm shocks. As a result, households cannot fully insure idiosyncratic factor income or firm-specific dividend risk, so income responses to shocks differ systematically across household types.

2.3 Market Clearing

The model closes with standard market-clearing conditions. Goods market clearing requires:

$$y_i = \sum_{h\in\mathcal{H}} C_{hi} + \sum_{j\in\mathcal{N}} x_{ji} \quad \forall i \in \mathcal{N}. \tag{4}$$

Primary factor markets clear for each household type:

$$L_h = \sum_{i\in\mathcal{N}} \ell_{ih} \quad \forall h \in \mathcal{H}. \tag{5}$$

The fiscal authority maintains budget neutrality: $\sum_{h\in\mathcal{H}} T_h = 0$. Zero-sum lump-sum transfers enable redistribution or targeted compensation without altering aggregate resources available to the private sector.

[2] The utility $U_h : \mathbb{R}_+^2 \to \mathbb{R}_+$ is increasing in C_h, nonincreasing in L_h, twice continuously differentiable, strictly concave, and the Inada conditions hold; $Q_h^c : \mathbb{R}_+^N \to \mathbb{R}_+$ is positive, homogeneous of degree one, and for the goods actually consumed, marginal utilities and aggregator derivatives satisfy the Inada conditions.

2.4 Equilibrium Characterization

The equilibrium characterizes how firm-level technologies and distortions, and household-level heterogeneity jointly determine resource allocation and the distribution of income across the production network. It maps exogenous primitives — technologies, distortions, preferences, endowments, and equity holdings — into equilibrium prices and allocations.

Let the aggregate state be $e \equiv (A, \mu, \kappa)$, and let $\mathscr{E}$ denote the set of all possible realizations of this state. The vectors $A \equiv (A_1, \cdots, A_N)'$ and $\mu \equiv (\mu_1, \cdots, \mu_N)'$ collect firm-specific productivities and inverse-markups, respectively. The asset-ownership matrix $\kappa \equiv (\kappa_1, \cdots, \kappa_N)'$ is the $N \times H$ matrix with rows $\kappa_i \equiv (\kappa_{i1}, \cdots, \kappa_{iH})'$, each recording the ownership distribution of firm i across household types. Each row satisfies the normalization $\kappa_i' \mathbb{1}_H = 1$, where $\mathbb{1}_H$ denotes the H-dimensional vector of ones. Fix a state e, and for notational simplicity, suppress dependence on e, and drop functional arguments (e.g., $w_h(e) \to w_h$). Under the maintained regularity conditions, a competitive equilibrium exists, so the mapping $e \mapsto$ equilibrium prices and allocations is well defined.

Definition 1. For any realization of the aggregate state $e \in \mathscr{E}$, a competitive equilibrium is a pair consisting of an allocation $(\{y_i\}, \{x_{ij}\}, \{\ell_{ih}\}, \{C_{hi}\}, \{L_h\})$ and a price system $(\{p_i\}, \{w_h\})$ such that:

(i) given prices $\{p_i\}_{i \in \mathscr{N}}$ and $\{w_h\}_{h \in \mathscr{H}}$, each firm i chooses primary factors $\{\ell_{ih}\}_{h \in \mathscr{H}}$ and intermediate inputs $\{x_{ij}\}_{j \in \mathscr{N}}$ to minimize total cost of producing y_i, subject to its technology, and the pricing condition $mc_i = \mu_i \, p_i$ holds;

(ii) given prices $\{p_i\}_{i \in \mathscr{N}}$ and $\{w_h\}_{h \in \mathscr{H}}$, each household h chooses $(L_h, \{C_{hi}\}_{i \in \mathscr{N}})$ to maximize $U_h(C_h, L_h)$ subject to the definition of the consumption aggregator and the budget constraint;

(iii) goods markets and primary-factor markets clear, and the fiscal authority maintains budget neutrality, $\sum_{h \in \mathscr{H}} T_h = 0$.

The role of the distortions is to reallocate revenue via rents across the network without shrinking the production possibility set. By altering the distribution of income and the sources of final demand, they distort relative prices and thereby affect real allocations. Household heterogeneity matters because it shapes how revenue shifts and the propagation of shocks. Changes in wages, dividends, and transfers shift labor supply and aggregate consumption patterns, which feed back through the production network via input-output linkages — potentially amplifying or dampening misallocation. Proposition 4 in the Online Appendix characterizes the necessary conditions for equilibrium allocations and prices. Existence requires a mild connectivity condition: there exists at least one essential factor — used directly or indirectly in the production of every good. This assumption ensures that all firms are exposed to a common system of relative prices, precluding disconnected production clusters and ensuring that aggregate measures are well defined.

2.5 Measures of Centrality

In the network literature, "centrality" denotes a node's systemwide importance or influence — for example, eigenvector or degree centrality. Here, node-level centralities — such as firms' Domar weights (sales-to-GDP ratios) — retain this interpretation as measure of systemwide exposure. By contrast, bilateral centralities measure node i's directed influence on node j through a single edge type or specific class of transactions. This subsection first introduces direct bilateral centralities, which isolate one node's marginal contribution to another along a single economic interaction. Bilateral centralities are generally asymmetric: the influence of i on j need not equal the influence of j on i. These direct centralities serve as building blocks for network-adjusted centralities that aggregate direct and indirect paths through the entire input-output structure, characterizing how shocks propagate through the economy to affect aggregate and distributional outcomes.

In what follows, downstream (or cost) centrality refers to how input-cost shocks transmit from suppliers to buyers through supply chains (from primary factors and intermediate inputs to final goods). Conversely, upstream (revenue) centrality captures the propagation of expenditure from final demand back to suppliers, i.e., the flow of payments generated by demand for goods and factors through the network of transactions. Each of these centralities has an equilibrium interpretation via the envelope theorem — specifically, Shephard's lemma — linking the measures to cost and expenditure shares evaluated at the reference equilibrium. Table 1 summarizes the direct centralities; Table 2 reports the corresponding network-adjusted measures.

2.5.1 Direct Centralities

The vectors $\omega_\ell \equiv \left(\omega_1^\ell, \cdots, \omega_N^\ell\right)'$ and $\omega_x \equiv (\omega_1^x, \cdots, \omega_N^x)'$ collect firms' direct cost centralities with respect to the primary-factor and intermediate-input composites. Each element measures the elasticity of firm i's total costs $\bar{c}_i$ with respect to the primary-factor and intermediate-input composite prices p_i^ℓ and p_i^x: $\omega_i^\ell \equiv \frac{\partial \log \bar{c}_i}{\partial \log p_i^\ell} = \frac{p_i^\ell L_i}{\bar{c}_i}$ and $\omega_i^x \equiv \frac{\partial \log \bar{c}_i}{\partial \log p_i^x} = \frac{p_i^x X_i}{\bar{c}_i}$. Under cost minimization (Shephard's lemma) and linear homogeneity of the cost function in (p_i^ℓ, p_i^x), these elasticities equal the corresponding cost shares and therefore satisfy $\omega_i^\ell + \omega_i^x = 1$ for each i.

The matrices $\widetilde{\Omega}_\ell$ and $\widetilde{\Omega}_x$ capture firms' direct exposure to factor and intermediate-goods price changes, respectively. Each element measures the elasticity of firm i's total cost $\bar{c}_i$ with respect to w_h or p_j: $\widetilde{\Omega}_{ih}^\ell \equiv \frac{\partial \log \bar{c}_i}{\partial \log w_h} = \frac{w_h \ell_{ih}}{\bar{c}_i}$ and $\widetilde{\Omega}_{ij}^x \equiv \frac{\partial \log \bar{c}_i}{\partial \log p_j} = \frac{p_j x_{ij}}{\bar{c}_i}$. Under cost minimization, these elasticities equal the cost shares of factor h and inputs from firm j. The identity $\sum_{h\in\mathscr{H}} \widetilde{\Omega}_{ih}^\ell + \sum_{j\in\mathscr{N}} \widetilde{\Omega}_{ij}^x = 1$ follows from linear homogeneity of the cost function in $(\{w_h\}, \{p_j\})$.

Using these definitions, I construct the factor network $\alpha \equiv diag(\omega_\ell)^{-1}\,\widetilde{\Omega}_\ell$ and the input-output network $\mathscr{W} \equiv diag(\omega_x)^{-1}\,\widetilde{\Omega}_x$, where $diag(.)$ denotes the diagonal operator. The elements of these matrices measure the elasticity of the composite costs with respect to individual input prices: $\alpha_{ih} \equiv \frac{\partial \log p_i^\ell L_i}{\partial \log w_h} = \frac{w_h \ell_{ih}}{p_i^\ell L_i}$ and $\omega_{ij} \equiv \frac{\partial \log p_i^x X_i}{\partial \log p_j} = \frac{p_j x_{ij}}{p_i^x X_i}$. In equilibrium, these elasticities equal the corresponding expenditure shares, on factor h within the primary-factor composite and on input j within the intermediate-input composite. Accordingly, each row of these matrices sums

Table 1: Direct Centralities

Object	*Definition*	*Equilibrium*	*Properties*
Cost Centralities			
ω_ℓ	$\omega_i^\ell \equiv \frac{\partial \log \bar{c}_i}{\partial \log p_i^\ell}$	$\omega_i^\ell = \frac{p_i^\ell L_i}{\bar{c}_i}$	$\omega_i^\ell + \omega_i^x = 1$
ω_x	$\omega_i^x \equiv \frac{\partial \log \bar{c}_i}{\partial \log p_i^x}$	$\omega_i^x = \frac{p_i^x X_i}{\bar{c}_i}$	
$\widetilde{\Omega}_\ell$	$\widetilde{\Omega}_{ih}^\ell \equiv \frac{\partial \log \bar{c}_i}{\partial \log w_h}$	$\widetilde{\Omega}_{ih}^\ell = \frac{w_h \ell_{ih}}{\bar{c}_i}$	$\sum_{h\in\mathscr{H}} \widetilde{\Omega}_{ih}^\ell + \sum_{j\in\mathscr{N}} \widetilde{\Omega}_{ij}^x = 1$
$\widetilde{\Omega}_x$	$\widetilde{\Omega}_{ij}^x \equiv \frac{\partial \log \bar{c}_i}{\partial \log p_j}$	$\widetilde{\Omega}_{ij}^x = \frac{p_j x_{ij}}{\bar{c}_i}$	
$diag(\omega_\ell)\, \alpha = \widetilde{\Omega}_\ell$	$\alpha_{ih} \equiv \frac{\partial \log p_i^\ell L_i}{\partial \log w_h}$	$\alpha_{ih} = \frac{w_h \ell_{ih}}{p_i^\ell L_i}$	$\sum_{h\in\mathscr{H}} \alpha_{ih} = 1$
$diag(\omega_x)\, \mathscr{W} = \widetilde{\Omega}_x$	$\omega_{ij} \equiv \frac{\partial \log p_i^x X_i}{\partial \log p_j}$	$\omega_{ij} = \frac{p_j x_{ij}}{p_i^x X_i}$	$\sum_{j\in\mathscr{N}} \omega_{ij} = 1$
Revenue Centralities			
$\Omega_\ell \equiv diag(\mu)\, \widetilde{\Omega}_\ell$	$\Omega_{ih}^\ell \equiv \frac{\partial \log S_i}{\partial \log w_h}$	$\Omega_{ih}^\ell = \frac{w_h \ell_{ih}}{S_i}$	$\sum_{h\in\mathscr{H}} \left(\Omega_{ih}^\ell + \Omega_{ih}^\pi\right) + \sum_{j\in\mathscr{N}} \Omega_{ij}^x = 1$
$\Omega_x \equiv diag(\mu)\, \widetilde{\Omega}_x$	$\Omega_{ij}^x \equiv \frac{\partial \log S_i}{\partial \log p_j}$	$\Omega_{ij}^x = \frac{p_j x_{ij}}{S_i}$	
$\Omega_\pi \equiv diag(\mathbb{1}_N - \mu)\, \kappa$	$\Omega_{ih}^\pi = \frac{\kappa_{ih} \pi_i}{S_i}$		
Household Exposure			
β	$\beta_{hi} \equiv \frac{\partial \log E_h}{\partial \log p_i}$	$\beta_{hi} = \frac{p_i C_{hi}}{E_h}$	$\sum_{i\in\mathscr{N}} \beta_{hi} = 1$
κ	$\kappa_{ih} \equiv \frac{d\,\Pi_h}{d\,\pi_i}$		$\sum_{h\in\mathscr{H}} \kappa_{ih} = 1$
τ	$\tau_h \equiv \frac{T_h}{GDP}$		$\sum_{h\in\mathscr{H}} \tau_h = 0$

to one: $\sum_{h\in\mathscr{H}} \alpha_{ih} = 1$ and $\sum_{j\in\mathscr{N}} \omega_{ij} = 1$, that is, α and $\mathscr{W}$ are row-stochastic.

Define the revenue-based (sales) upstream centrality matrices as $\Omega_\ell \equiv diag(\mu)\, \widetilde{\Omega}_\ell$ and $\Omega_x \equiv diag(\mu)\, \widetilde{\Omega}_x$. Since $\mu_i \in (0, 1]$, revenue-based centralities are elementwise no larger than their cost-based counterparts: $\widetilde{\Omega}_\ell \succcurlyeq \Omega_\ell$ and $\widetilde{\Omega}_x \succcurlyeq \Omega_x$ (with equality iff $\mu_i = 1$), where $\succcurlyeq$ denotes elementwise inequality. This reflects the fact that distortions ($\mu_i < 1$) reduce the fraction of revenue passed on to upstream suppliers, with the residual $(1 - \mu_i)$ paid as profits. The elements $\Omega_{ih}^\ell \equiv \frac{\partial \log S_i}{\partial \log w_h} = \frac{w_h \ell_{ih}}{S_i}$ and $\Omega_{ij}^x \equiv \frac{\partial \log S_i}{\partial \log p_j} = \frac{p_j x_{ij}}{S_i}$ represent the elasticities of firm i's sales with respect to w_h and p_j (where $S_i = p_i y_i$). In equilibrium, these elasticities equal the revenue shares paid to household type h and to supplier j, respectively. Additionally, define $\Omega_\pi \equiv diag(\mathbb{1}_N - \mu)\, \kappa$, so $\Omega_{ih}^\pi \equiv \frac{\kappa_{ih} \pi_i}{S_i}$ captures the profits (dividend) share of sales accruing to type h. The accounting identity $\sum_{h\in\mathscr{H}} \Omega_{ih}^\ell + \sum_{j\in\mathscr{N}} \Omega_{ij}^x + \sum_{b\in\mathscr{H}} \Omega_{ib}^\pi = 1$ ensures that all revenue is allocated to payment for factors, intermediate inputs, or profits.

For households, the consumption network is represented by the matrix $\beta = (\beta_1, \cdots, \beta_H)'$; the row vector $\beta_h \equiv (\beta_{h1}, \cdots, \beta_{hN})'$ describes the exposure of household type h to final good prices. Each element $\beta_{hi} \equiv \frac{\partial \log E_h}{\partial \log p_i} = \frac{p_i C_{hi}}{E_h}$ measures the elasticity of household h's total expenditure E_h with respect to the price of good i, and by Shephard's lemma, it equals the expenditure share on good i. Accordingly, each row of β is row-stochastic and sums to one: $\sum_{i\in\mathcal{N}}\beta_{hi}=1$. The variable $\tau_h\equiv T_h/GDP$ denotes the lump-sum transfer share for household h. Since transfers are budget-neutral, $\sum_{h\in\mathcal{H}}\tau_h=0$ holds by construction.

2.5.2 Network-Adjusted Centralities

The firm-to-firm downstream centrality matrix, or cost-based Leontief inverse, is defined as $\widetilde{\Psi}_x\equiv\left(I-\widetilde{\Omega}_x\right)^{-1}=\sum_{q=0}^{\infty}\widetilde{\Omega}_x^q$, where I is the $N\times N$ identity. Each element $\widetilde{\psi}_{ij}^x$ captures the cumulative cost influence through all direct and indirect input-output paths from supplier j to buyer i. Similarly, the firm-to-firm upstream centrality matrix, or revenue-based Leontief inverse, is defined as $\Psi_x\equiv(I-\Omega_x)^{-1}=\sum_{q=0}^{\infty}\Omega_x^q$. Each element ψ_{ij}^x gives the total revenue accruing to supplier j per unit of firm i's sales (a revenue multiplier), aggregating all upstream chains of transactions.

The firm-to-consumer downstream centrality matrix is defined as $\widetilde{\mathscr{B}}\equiv\beta\widetilde{\Psi}_x$. Each element $\widetilde{\mathscr{B}}_{hi}=\sum_{j\in\mathcal{N}}\beta_{hj}\widetilde{\psi}_{ji}^x$ aggregates all direct and indirect input-output paths through which firm i's price affects household h's expenditure. The cost-based sales Domar centrality, $\widetilde{\lambda}_i\equiv\sum_{h\in\mathcal{H}}\chi_h\widetilde{\mathscr{B}}_{hi}$, measures firm i's average household-exposure centrality, where $\chi_h=E_h/GDP$ is household h's expenditure share. In general, $\widetilde{\lambda}_i$ is an elasticity-based centrality that tracks the share of aggregated value-added routed downstream through firm i. Similarly, the consumer-to-firm upstream centrality matrix is defined as $\mathscr{B}\equiv\beta\Psi_x$. Each element $\mathscr{B}_{hi}=\sum_{j\in\mathcal{N}}\beta_{hj}\psi_{ji}^x$ represents the share of household type h's expenditure that ultimately accrues to firm i (aggregating all upstream chains). The revenue-based Domar weight $\lambda_i=\sum_{h\in\mathcal{H}}\chi_h\mathscr{B}_{hi}=S_i/GDP$ measures firm i's average consumer-to-firm centrality and equals its sales-to-GDP ratio. When $\widetilde{\lambda}_i>\lambda_i$, the gap reflects value added captured as downstream profits along the paths from i to households. These definitions extend the supplier centrality vector of Baqaee (2018) and the influence vector of Acemoglu et al. (2012) to a setting with heterogeneous households and firm-level distortions.

The factor-to-firm downstream centrality matrix is defined as $\widetilde{\Psi}_\ell\equiv\widetilde{\Psi}_x\widetilde{\Omega}_\ell$. Each element $\widetilde{\psi}_{ih}^\ell$ is the value-added share of factor h in firm i's unit cost, accounting for all direct and indirect input-output paths through the production network. Since $\sum_{h\in\mathcal{H}}\widetilde{\psi}_{ih}^\ell=1$, firm i's entire production cost is attributed to factor payments once intermediate layers are resolved. Thus, $\widetilde{\psi}_{ih}^\ell$ can be interpreted as factor h's value-added share in firm i's production. The firm-to-factor upstream centrality matrix is defined as $\Psi_\ell\equiv\Psi_x\Omega_\ell$, and ψ_{ih}^ℓ represents the share of firm i's revenue that ultimately accrues to factor h.

The factor-to-consumer downstream centrality matrix is defined as $\widetilde{\mathscr{C}}\equiv\beta\widetilde{\Psi}_\ell$. Since $\sum_{b\in\mathcal{H}}\widetilde{\mathscr{C}}_{hb}=1$, each element $\widetilde{\mathscr{C}}_{hb}$ represents the share of value-added in the consumption of household type h attributable to factor b. The cost-based factor weight, $\widetilde{\Lambda}_h=\sum_{b\in\mathcal{H}}\chi_b\widetilde{\mathscr{C}}_{bh}$, measures the average factor-to-consumer centrality of factor h. Consequently, $\widetilde{\Lambda}_h$ equals the share of aggregate value-added generated by that factor. All costs in this economy ultimately originate from factor payments, so $\sum_{h\in\mathcal{H}}\widetilde{\Lambda}_h=1$. Similarly, the consumer-to-factor upstream centrality matrix is defined as $\mathscr{C}\equiv\beta\Psi_\ell$, where each element $\mathscr{C}_{hb}$ measures the share of household type h's consumption expenditure that ultimately accrues to factor b. The revenue-based

Table 2: Network-Adjusted Centralities

Matrix	*Equilibrium expression & interpretation*	*Properties*
	Downstream / Cost-Based Centralities	
$\widetilde{\Psi}_x = \left(I - \widetilde{\Omega}_x\right)^{-1}$	$\widetilde{\psi}^x_{ij}$ *firm-to-firm* Centrality of j in the costs of i	
$\widetilde{\mathscr{B}} = \beta\, \widetilde{\Psi}_x$	$\widetilde{\mathscr{B}}_{hi}$ *firm-to-consumer* Centrality of i in the costs of h	
$\widetilde{\Psi}_\ell = \widetilde{\Psi}_x\, \widetilde{\Omega}_\ell$	$\widetilde{\psi}^\ell_{ih}$ *factor-to-firm* Factor $h's$ value-added share in i's costs	$\sum_{h \in \mathscr{H}} \widetilde{\psi}^\ell_{ih} = 1$
$\widetilde{\mathscr{C}} = \beta\, \widetilde{\Psi}_\ell$	$\widetilde{\mathscr{C}}_{hb}$ *factor-to-consumer* Factor $b's$ value-added share in$h$'s costs	$\sum_{b \in \mathscr{H}} \widetilde{\mathscr{C}}_{hb} = 1$
$\widetilde{\lambda} = \widetilde{\mathscr{B}}'\, \chi$	$\widetilde{\lambda}_i$ *cost-based Domar weight* Share of aggregate value-added routed through i	$\sum_{i \in \mathscr{N}} \omega^\ell_i\, \widetilde{\lambda}_i = 1$
$\widetilde{\Lambda} = \widetilde{\mathscr{C}}'\, \chi$	$\widetilde{\Lambda}_h$ *value-added factor share* Share of aggregate value-added generated by h	$\sum_{h \in \mathscr{H}} \widetilde{\Lambda}_h = 1$
	Upstream / Revenue-Based Centralities	
$\Psi_x = (I - \Omega_x)^{-1}$	ψ^x_{ij} *firm-to-firm* Share of S_i that reaches S_j	
$\mathscr{B} = \beta\, \Psi x$	$\mathscr{B}_{hi}$ *consumer-to-firm* Share of E_h that reaches S_i	
$\Psi_\ell = \Psi_x\, \Omega_\ell$	ψ^ℓ_{ih} *firm-to-factor* Share of S_i that reaches J_h	$\psi^\ell_i = \sum_{h \in \mathscr{H}} \psi^\ell_{ih}$
$\mathscr{C} = \beta\, \Psi_\ell$	$\mathscr{C}_{hb}$ *consumer-to-factor* Share of E_h that reaches J_h	$\mathscr{C}_h = \sum_{b \in \mathscr{H}} \mathscr{C}_{hb}$
$\lambda = \mathscr{B}'\, \chi$	λ_i *revenue-based Domar weight* Aggregate sales share S_i / GDP	$\sum_{i \in \mathscr{N}} \lambda_i \geq 1$
$\Lambda = \mathscr{C}'\, \chi$	Λ_h *revenue-based factor share* Factor income share J_h / GDP	$\Gamma = \sum_{h \in \mathscr{H}} \Lambda_h \leq 1$
$\chi = (\Omega_\ell + \Omega_\pi)'\ \lambda + \tau$	χ_h *expenditure share* Consumption expenditure share E_h / GDP	$\sum_{h \in \mathscr{H}} \chi_h = 1$
	Other Definitions	
$\delta = diag(\Lambda)^{-1}\ \Lambda$	δ_h *distortion centrality* Measure of undercompensation of factor L_h	$\delta_h = \widetilde{\Lambda}_h / \Lambda_h \geq 1$
$M = \mathscr{C}\, \delta$	M_h *expenditure centrality* Average δ faced by E_h	$M_h = \sum_{b \in \mathscr{H}} \mathscr{C}_{hb}\, \delta_b$
$F = \Psi_\ell\, \delta$	F_i *revenue centrality* Average δ faced by S_i	$F_i = \sum_{h \in \mathscr{H}} \psi^\ell_{ih}\, \delta_h$

factor weight $\Lambda_h = \sum_{b \in \mathscr{H}} \chi_b\, \mathscr{C}_{bh} = J_h / GDP$ measures the average consumer-to-factor centrality of factor h and equals its income share.

To further clarify the previous centralities, I unpack the intuition behind $\widetilde{\mathscr{C}}$ and $\mathscr{C}$. The factor-to-consumer downstream centrality matrix is defined as $\widetilde{\mathscr{C}} = \beta\, \widetilde{\Psi}_x\, \widetilde{\Omega}_\ell$ and captures how factor costs propagate into household consumption expenditure: $\widetilde{\Omega}_\ell$ captures firms' direct exposure to factor costs, $\widetilde{\Psi}_x$ traces how these costs flow through the input-output network, and β allocates firm-level costs onto household-level expenditure. Conversely, the consumer-to-factor upstream centrality matrix is defined as $\mathscr{C} = \beta\, \Psi_x\, \Omega_\ell$ and captures how consumption expenditure propagates upstream into factor income: β allocates household expenditure across firms, Ψ_x traces how this revenue flows through the network, and Ω_ℓ captures how firm revenue is allocated to factor compensation.

Cost-based centralities are greater than or equal to revenue-based centralities, i.e., $\widetilde{\Psi}_x \succcurlyeq \Psi_x$, $\widetilde{\mathscr{B}} \succcurlyeq \mathscr{B}$, $\widetilde{\Psi}_\ell \succcurlyeq \Psi_\ell$, $\widetilde{\mathscr{C}} \succcurlyeq \mathscr{C}$, $\widetilde{\lambda} \succcurlyeq \lambda$, and $\widetilde{\Lambda} \succcurlyeq \Lambda$. Accordingly, for factor h, define the *distortion centrality* ratio $\delta_h \equiv \widetilde{\Lambda}_h / \Lambda_h \geq 1$, which captures how undercompensated relative to its value-added contribution a factor is in equilibrium. The larger δ_h, the greater the wedge between the factor's value-added share ($\widetilde{\Lambda}_h$) and its income share (Λ_h). When households supply factors to firms embedded in highly distorted supply chains (low μ along downstream paths), δ_h is high: a larger fraction of value added created by h is captured and paid out as downstream profits rather than factor income.

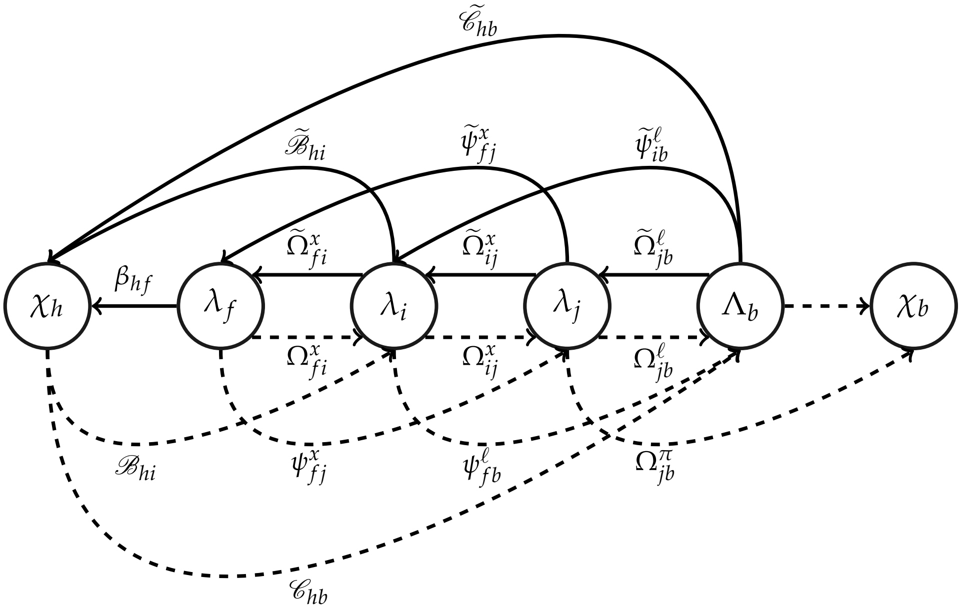
Define household h's expenditure-centrality as $M_h = \sum_{b\in\mathscr{H}} \mathscr{C}_{hb}\,\delta_b$ and firm i's revenue-centrality as $F_i = \sum_{h\in\mathscr{H}} \psi^\ell_{ih}\,\delta_h$. These indices capture the exposure-weighted distortion centrality embedded in the consumption expenditure of household h and the revenue stream of firm i. A household has high expenditure centrality when the dot product of its consumer-to-factor centrality vector $\mathscr{C}_{\uparrow h} = (\mathscr{C}_{h1}, \ldots, \mathscr{C}_{hH})'$ and the distortion centrality vector δ is large. Similarly, a firm has high revenue centrality when its firm-to-factor centrality vector $\psi^\ell_{\uparrow i} = \left(\psi^\ell_{i1}, \ldots, \psi^\ell_{iH}\right)'$ is closely aligned with δ. In both cases, higher centrality reflects greater reliance on undervalued factors along paths with limited profit leakage — i.e., where downstream μ are relatively high so that a larger share of payments reaches factors. Thus, households and firms with high M_h and F_i tend to source from relatively competitive (high-μ) supply chains that rely heavily on factors with high δ, often because those factors are essential inputs into more distorted chains. These metrics are referred to as the *expenditure centrality* of household h and the *revenue centrality* of firm i.

Finally, in equilibrium, household expenditure shares satisfy the accounting identity $\chi_h = \Lambda_h + \sum_{i\in\mathscr{N}} \Omega^\pi_{ih}\,\lambda_i + \tau_h$. Under budget neutrality $\sum_{h\in\mathscr{H}} \chi_h = 1$.

2.6 A Diagrammatic Recap

Figure 1 illustrates the centrality relationships among firms $f, i, j \in \mathscr{N}$ and households $h, b \in \mathscr{H}$. Household b supplies labor to firm j, firm j supplies intermediate inputs to firm i, firm i supplies intermediate inputs to firm f, and firm f supplies final goods to household h. Firm f does not purchase directly from firm j, but it is exposed through firm i's transactions; the indirect linkages between f and j are measured by $\widetilde{\psi}^x_{fj}$ and ψ^x_{fj}. Firm i does not hire labor from b, but it is exposed through firm j's labor demand and input provision; the indirect linkages between i and b are captured by $\widetilde{\psi}^\ell_{ib}$ and ψ^ℓ_{ib}. Household h does not purchase from firm i directly, but it is exposed to i's costs through firm f's upstream dependencies; the indirect linkages between h and i are captured by $\widetilde{\mathscr{B}}_{hi}$ and $\mathscr{B}_{hi}$. Household h is also exposed to b's labor costs via the production chain linking f, i, and j; the indirect linkages between h and b are captured by $\widetilde{\mathscr{C}}_{hb}$ and $\mathscr{C}_{hb}$. Household b receives income from wages and distributed profits from firm j, captured by Ω^ℓ_{jb} and Ω^π_{jb}. This list is illustrative rather than exhaustive and analogous paths are omitted for brevity.

Figure 1: Measures of Centrality



Notes: Continuous and dashed arrows represent cost- and revenue-based centrality measures, respectively.

2.7 Remarks

This framework advances the production network literature along several dimensions. First, it introduces heterogeneous households — differentiated by preferences, factor endowments, and asset ownership — within a nonparametric production network framework. This contrasts with Baqaee & Farhi (2020), who allow for rich firm-level heterogeneity in distortions and elasticities while imposing a representative household. It is more closely aligned with models such as Baqaee & Farhi (2024), which incorporate household heterogeneity into a production network setting. Rather than focusing solely on aggregate wedges between social and private returns, this framework extends their logic to a more granular incidence mapping and propagation structure. Specifically, it introduces a modular centrality system that decomposes the economy into layered (factors, firms, households) blocks for production and consumption. By contrast, other models encode the entire economy in a single Markov transition matrix; for example, the downstream generalized Leontief inverse can be written compactly in block form:[3]

$$\widetilde{\Omega} = \begin{pmatrix} \widetilde{\Omega}_x & \widetilde{\Omega}_\ell \\ \beta & 0 \end{pmatrix}, \qquad \left(I - \widetilde{\Omega}\right)^{-1} = \begin{pmatrix} \widetilde{\Psi}_x + \widetilde{\Psi}_\ell \left(I - \widetilde{\mathscr{C}}\right)^{-1} \widetilde{\mathscr{B}} & \widetilde{\Psi}_\ell \left(I - \widetilde{\mathscr{C}}\right)^{-1} \\ \left(I - \widetilde{\mathscr{C}}\right)^{-1} \widetilde{\mathscr{B}} & \left(I - \widetilde{\mathscr{C}}\right)^{-1} \end{pmatrix}.$$

Segmenting the network into distinct layers (factors, firms, households) explicitly highlights distributional propagation channels, allowing precise measurement of the incidence of shocks on identified groups — an exercise less transparent in single-matrix representations. This layered structure resembles earlier models — such as Hulten (1978), Long & Plosser (1983),

[3]The upstream propagation of revenue is captured by $\Omega = \begin{pmatrix} \Omega_x & \Omega_\ell + \Omega_\pi \\ \beta & 0 \end{pmatrix}$.

Acemoglu et al. (2012), Acemoglu et al. (2016), and Bigio & La'O (2020) — but extends them to incorporate heterogeneous household demand and income incidence. By comparison, the generalized Leontief inverse in Baqaee & Farhi (2024) summarizes connectivity in a single (block) operator, whereas the present decomposition keeps factor-firm-household margins separate. This layered, substochastic segmentation is essential for deriving sufficient statistics that link distributional heterogeneity to aggregate productivity and to variation in household-level real consumption.

Second, this model extends the representative household framework of Bigio & La'O (2020) — which features endogenous supply of one factor — by incorporating both household heterogeneity and incidence accounting. This extension enables analysis of distributional incidence and propagation through which distortions affect both the level and composition of income. A key simplification is the one-household/one-factor assumption, effectively segmenting the factor market. This assumption enables parsimony along three important margins. First, it eliminates the need to specify a factor-ownership matrix mapping factor supply shares across household types. Second, it avoids within-household cross-factor supply elasticities that arise when households supply multiple factors. Third, it abstracts from within-factor heterogeneity and cross-household strategic interactions in jointly supplying a given factor. While this assumption restricts the scope for certain distributional spillovers, it enables analytical tractability in characterizing the equilibrium effects of household heterogeneity in a networked economy with endogenous factor supply. This assumption could be relaxed by introducing a factor-ownership matrix and multi-factor household supply as done in Baqaee & Farhi (2024), but doing so would require treating factor supply as exogenous.

3 Aggregate and Distributional Accounting

This section develops a nonparametric, general equilibrium framework to evaluate how productivity, distortions, preferences, and income shocks jointly shape aggregate and distributional outcomes in production networks with heterogeneous households. Its central contribution is a sequence of decomposition results that map firm- and household-specific primitives into sufficient statistics for macroeconomic aggregates and household-level outcomes. Proposition 1 characterizes how cost shocks propagate through the price system via input-output linkages. Theorem 1 shows how variation in household expenditure and the production network structure shape the distribution of factor income. Theorem 2 builds on this by decomposing real GDP into components that isolate the allocative effects of microeconomic shocks, including the channels through which distributions shape aggregate productivity. Theorem 3 examines a constrained social planner benchmark (with endogenous factor supply) that centralizes households' decisions, showing how household heterogeneity alters the measurement of TFP and recasts both the positive and normative role of distributional policy. Finally, Theorem 4 extends the analysis to household-level real consumption by introducing "*positional terms of trade*" (PTT), which quantify how efficiently a household's consumption bundle can be produced from the equilibrium factor allocation. Together, these results constitute an internally consistent and observationally sufficient framework for tracing how income and network heterogeneity amplify or attenuate the aggregate and distributional consequences of shocks in general equilibrium. All terms in these decompositions are, in principle, observable from input-output tables and price data.

Unlike prior literature that fixes nominal GDP as the numéraire, e.g. Baqaee & Farhi (2020, 2024), the decompositions here are invariant to the choice of a numéraire. This is because the objects are accounting identities — share-weighted decompositions and elasticities — that satisfy composition constraints. This distinction matters in heterogeneous-agent settings, where choosing a single aggregate price index can bias distributional comparisons. For example, fixing nominal GDP makes measured real output rise mechanically when the price level falls, conflating nominal revaluation with real efficiency gains and potentially masking the true incidence of gains and losses across households. A numéraire is needed only when computing endogenous responses in a parametric model (see Section 4), where a price level must be pinned down to solve for substitution and income effects; in that setting, when substitution and income effects on factor supply are asymmetric, a nominal-GDP numéraire can deliver different real-output responses than a GDP-deflator numéraire.

3.1 Price Variation

The general equilibrium price system reflects how technology, firm-level distortions, and heterogeneity in factor prices shape the cost structure behind production and consumption. Because prices clear interconnected markets, understanding their responses to shocks is essential for evaluating aggregate and distributional effects. This subsection derives first-order responses of firms' prices, the aggregate price index, and household cost-of-living indices to shocks in productivity, markups, and factor prices.

Proposition 1 characterizes these relationships in terms of network-adjusted centralities. Price responses are mediated by cost-based downstream centralities, which summarize each agent's network exposure elasticity to shocks anywhere in the economy. Accordingly, these elasticities are sufficient statistics for shock pass-through to firms' unit costs and households' real purchasing power.

Proposition 1. Price Variation Across Firms and Households. The first-order variation in firm i's output price p_i, the household type h's consumption price index p_h^c, and the aggregate price index p_Y in response to productivity, markup and factor cost shocks are:

$$
\begin{aligned}
d\,\log\,p_i &= -\sum_{j\in\mathcal{N}} \widetilde{\psi}^x_{ij}\left(d\,\log\,A_j + d\,\log\,\mu_j\right) + \sum_{h\in\mathcal{H}} \widetilde{\psi}^\ell_{ih}\,d\,\log\,w_h,\\
d\log p_h^c &= -\sum_{i\in\mathcal{N}} \widetilde{\mathscr{B}}_{hi}\left(d\,\log\,A_i + d\,\log\,\mu_i\right) + \sum_{b\in\mathcal{H}} \widetilde{\mathscr{C}}_{hb}\,d\,\log\,w_b,\\
d\log p_Y &= -\sum_{i\in\mathcal{N}} \widetilde{\lambda}_i\left(d\,\log\,A_i + d\,\log\,\mu_i\right) + \sum_{h\in\mathcal{H}} \widetilde{\Lambda}_h\,d\,\log\,w_h.
\end{aligned}
$$

These expressions yield six insights. First, cost-based firm-to-firm ($\widetilde{\Psi}_x$) and firm-to-consumer ($\widetilde{\mathscr{B}}$) centralities summarize how upstream productivity and markup shocks transmit to downstream marginal costs and prices. Second, factor-cost shocks propagate in proportion to the cost-based factor-to-firm ($\widetilde{\Psi}_\ell$) and factor-to-consumer ($\widetilde{\mathscr{C}}$) centralities. Third, the GDP deflator's exposure is summarized by the cost-based Domar weights ($\widetilde{\lambda}$) and factor value-added shares ($\widetilde{\Lambda}$). Fourth, productivity and inverse markups are isomorphic cost shifters: a rise in A_i or μ_i lowers prices with the identical cost-based centrality weights. Fifth, household price incidence is heterogenous: differences in β_h tilt $\widetilde{\mathscr{B}}$ and $\widetilde{\mathscr{C}}$, so cost-of-living indices can move differently across households under the same shock. Sixth, heterogeneity in income sources (factor income, equity ownership κ, or transfers τ) do not enter price pass-through directly: they affect incomes, not prices or marginal costs.

3.2 Income Distribution

Theorem 1 decomposes how shifts in expenditure patterns and network structure reallocate the distribution of factor income across households.

Theorem 1. Factor Income Shares. The first-order change in Λ_h in response to shifts in the consumption distribution and consumer-to-factor centralities is

$$d\,\Lambda_h = \overbrace{\sum_{b\in\mathscr{H}} \mathscr{C}_{bh}\, d\,\chi_b}^{\substack{\textit{Distributive} \\ \textit{Income}_h}} + \overbrace{\sum_{b\in\mathscr{H}} \chi_b\; d\,\mathscr{C}_{bh}}^{\substack{\textit{Income} \\ \textit{Centrality}_h}}, \tag{6}$$

$$\substack{\textit{Income} \\ \textit{Centrality}_h} = \overbrace{\sum_{i\in\mathscr{N}} \psi^{\ell}_{ih} \sum_{b\in\mathscr{H}} \chi_b\, d\,\beta_{bi}}^{\substack{\textit{Final-Demand} \\ \textit{Recomposition}_h}} + \overbrace{\sum_{i\in\mathscr{N}} \psi^{\ell}_{ih} \sum_{j\in\mathscr{N}} \mu_j\,\lambda_j\, d\,\widetilde{\Omega}^{x}_{ji}}^{\substack{\textit{Intermediate-Demand} \\ \textit{Recomposition}_h}} + \overbrace{\sum_{i\in\mathscr{N}} \mu_i\,\lambda_i\, d\,\widetilde{\Omega}^{\ell}_{ih}}^{\substack{\textit{Factor-Demand} \\ \textit{Recomposition}_h}} + \overbrace{\sum_{i\in\mathscr{N}} \psi^{\ell}_{ih}\,\lambda_i\, d\log\mu_i}^{\substack{\textit{Competitive} \\ \textit{Income}_h}}. \tag{7}$$

Equation (6) decomposes the first-order change in factor h's income share Λ_h into (i) a distributive-income term from shifts in expenditure shares ($d\,\chi$) and (ii) an income-centrality term from shifts in consumer-to-factor centralities ($d\,\mathscr{C}$). First, the *distributive-income* channel: Λ_h rises when expenditure shifts towards household types with greater upstream exposure to h (larger $\mathscr{C}_{bh}$), holding $\mathscr{C}$ fixed. For example, shifting a dollar of expenditure from type q to type b increases Λ_h if $\mathscr{C}_{bh} > \mathscr{C}_{qh}$. Second, the *income-centrality* channel: Λ_h increases when the expenditure-to-h mapping strengthens — more precisely, when the χ-weighted change in $\mathscr{C}$ toward h is positive, i.e, when $\sum_{b\in\mathscr{H}} \chi_b\; d\,\mathscr{C}_{bh} > 0$.

The income-centrality term aggregates four effects. The *final-demand* and *intermediate-demand* recomposition channels capture how reallocations in household and firm spending, respectively, shift the distribution of income. These channels imply that Λ_h rises when household consumption or firm input demand shifts toward firms with high firm-to-factor centrality for h. For example, reallocating spending or input demand from firm j to firm i raises Λ_h if $\psi^{\ell}_{ih} > \psi^{\ell}_{jh}$. The *factor-demand recomposition* channel captures how changes in factor demand affect Λ_h, with stronger effects in larger and less-distorted firms (higher $\mu_i\,\lambda_i$). The *competitive-income* channel implies that lower markups (higher μ) raise Λ_h in proportion to the firm size (λ_i) and the firm-to-factor centrality toward h (ψ^{ℓ}_{ih}).

Baqaee & Rubbo (2023) derive analogous decompositions for revenue-based Domar weights and factor income shares in a CES production network with a representative household, markups, and labor as the sole factor. Relative to this baseline, Theorem 1 extends the analysis along four key dimensions. First, it introduces distributive channels that link heterogeneity in consumption bundles to reallocation of expenditure across households. Second, it disaggregates cost recomposition into three distinct channels: final-demand, intermediate-demand, and factor-demand recomposition. Third, it generalizes the framework to economies with multiple primary factors. Fourth, the decomposition is nonparametric and thus implementable using standard multi-period input-output data on cost shares and expenditure flows between firms and households. By contrast, Baqaee & Rubbo (2023) rely on structural primitives — such as elasticities of substitution and the distribution of firm-level price changes — that are typically hard to estimate in practice. Moreover, in a representative household economy with labor as the sole factor, Bigio & La'O (2020) show that around the first-best allocation, $d\,log\,\Lambda = \sum_{i\in\mathcal{N}} \lambda_i\, d\, \log\, \mu_i$. Theorem 2 generalizes this result to arbitrary inefficient equilibria with heterogeneous households and multiple primary factors. Finally, Theorem 6 in the Online Appendix characterizes first-order variations in revenue-based Domar weights (λ) as well as cost-based factor and sales Domar weights ($\widetilde{\Lambda}$, $\widetilde{\lambda}$), enriching the characterization of distributional exposure across firms and households.

3.3 Aggregate Accounting in a Heterogeneous Household Economy

This subsection develops a decomposition of aggregate real output for distorted production networks with heterogeneous households. Theorem 2 exploits the equivalence between two representations of real GDP: (i) the expenditure-side representation, which is a function of household consumption and (ii) the production representation, which is given by TFP times a value-added aggregator over primary factors. The theorem decomposes the first-order sources of TFP growth into three components — technology, competitiveness, and distributional reallocation — each tied to distinct structural forces. It also provides four equivalent representations of the distributional reallocation channel, offering complementary perspectives on how changes in the income distribution affect aggregate efficiency. These results generalize and refine the results in Baqaee & Farhi (2020) to settings with heterogeneous households.

Theorem 2. Real GDP and TFP in an Economy with Heterogeneous Households. In equilibrium, real GDP admits expenditure-side and production-side representations:

$$Y = Q_Y\left(\{C_h\}_{h\in\mathscr{H}}\right) = TFP\; F\left(\{L_h\}_{h\in\mathscr{H}}\right), \tag{8}$$

where TFP denotes total factor productivity; Q_Y and F satisfy $d\,\log\,Q_Y/d\,\log\,C_h = \chi_h$ and $d\,\log\,F/d\,\log\,L_h = \widetilde{\Lambda}_h$.

The first order, changes in Y and TFP satisfy

$$d\,\log\,Y = d\,\log\,TFP + \sum_{h\in\mathscr{H}} \widetilde{\Lambda}_h\, d\,\log\,L_h, \tag{9}$$

$$d\,\log\,TFP = \mathit{Technology} + \mathit{Allocative\ Efficiency}, \tag{10}$$

where

$$\textit{Technology} = \sum_{i\in\mathcal{N}} \widetilde{\lambda}_i \, d \log A_i, \qquad \begin{array}{c}\textit{Allocative} \\ \textit{Efficiency}\end{array} = \overbrace{\sum_{i\in\mathcal{N}} \widetilde{\lambda}_i \, d \log \mu_i}^{\textit{Competitiveness}} + \begin{array}{c}\textit{Distributional} \\ \textit{Reallocation}\end{array},$$

and *Distributional Reallocation* has four equivalent representations

$$1. \; -\overbrace{\sum_{h\in\mathcal{H}} \delta_h \, d\,\Lambda_h}^{\textit{Entropic TT}}, \quad 2. \; -\overbrace{\sum_{h\in\mathcal{H}} (\delta_h - 1)\,\Lambda_h \, d \log J_h}^{\textit{Factor Terms of Trade (TT)}} + \overbrace{\sum_{i\in\mathcal{N}} \lambda_i \left((1-\mu_i)\, d \log S_i - d\,\mu_i\right)}^{\textit{Corporate Income}},$$

$$3. \; -\overbrace{\sum_{h\in\mathcal{H}} M_h \, d\,\chi_h}^{\textit{Distributive TT}} - \overbrace{\sum_{h\in\mathcal{H}} \chi_h \sum_{b\in\mathcal{H}} \delta_b \, d\,\mathscr{C}_{hb}}^{\textit{Centrality TT}},$$

$$4. \; -\sum_{h\in\mathcal{H}} M_h \, d\,\chi_h - \overbrace{\sum_{h\in\mathcal{H}} \chi_h \sum_{i\in\mathcal{N}} F_i \, d\,\beta_{hi}}^{\textit{Final-Demand TT}} - \overbrace{\sum_{i\in\mathcal{N}} \mu_i \,\lambda_i \sum_{j\in\mathcal{N}} F_j \, d\,\widetilde{\Omega}^x_{ij}}^{\textit{Intermediate-Demand TT}}$$
$$-\overbrace{\sum_{i\in\mathcal{N}} \mu_i \,\lambda_i \sum_{h\in\mathcal{H}} \delta_h \, d\,\widetilde{\Omega}^{\ell}_{ih}}^{\textit{Factor-Demand TT}} - \overbrace{\sum_{i\in\mathcal{N}} \lambda_i \, F_i \, d \log \mu_i}^{\textit{Competitive TT}},$$

with $\delta_h = \widetilde{\Lambda}_h / \Lambda_h$, $M_h = \sum_{b\in\mathcal{H}} \mathscr{C}_{hb}\, \delta_b$ and $F_i = \sum_{h\in\mathcal{H}} \psi^{\ell}_{ih}\, \delta_h$.

From equation (8), real GDP admits two equivalent representations in equilibrium. First, as a constant returns to scale (CRS) aggregator Q_Y over household consumption, with elasticities given by households' expenditure shares χ. Second, as the product of TFP and a CRS aggregator F over primary factors, with elasticities equal to the value-added shares $\widetilde{\Lambda}$.

Equation (9) decomposes the change in aggregate output into a TFP component and a factor-input component. Equation (10) decomposes the first-order change in TFP into three components. First, technology captures the direct effect of productivity shocks, holding allocations fixed. Second, *competitiveness* captures reallocation effects due to changes in firm-level distortions, holding the factor-income distribution fixed. Together, these two components imply that in the absence of changes to the factor-income distribution, the effect on TFP of a productivity or markup changes in firm i is proportional to firm i's cost-based Domar weight $\widetilde{\lambda}_i$. Third, *distributional reallocation* captures TFP changes due to endogenous shifts in the income distribution that reallocate resources across firms. The latter two terms capture TFP changes arising from input reallocations across firms induced by both exogenous distortions and endogenous shifts in the income distribution. For this reason, Baqaee & Farhi Baqaee & Farhi (2020) interpret *competitiveness* + *distributional reallocation* as *allocative efficiency*. This decomposition can be interpreted as follows: *technology* captures outward shifts of the production possibility frontier, while *allocative efficiency* captures movements along the frontier. Finally, Hulten's theorem Hulten (1978) holds when distortions are absent (i.e., $d \log TFP = \lambda' \, d\log A$), which implies that variations in the factor-income distribution and small distortions around the first-best equilibrium imply *allocative neutrality* (i.e., *competitiveness* + *distributional reallocation* $= 0$).

Theorem 2 presents four equivalent representations of the *distributional reallocation* channel, each offering a distinct lens on how income distribution shapes TFP. All four expressions reflect the idea that TFP rises when inputs shift from firms in less-distorted supply chains with lower marginal product to firms in more-distorted bootleneck supply chains with higher marginal product. Rent-generating distortions attenuate revenue pass-through as revenue flows upstream in the production network. As a result, upstream firms in more-distorted supply chains receive less revenue and demand fewer inputs relative to an undistorted economy, thus operating at a higher marginal product and with larger wedges between value added routed through them and their sales ($\widetilde{\lambda}_i/\lambda_i$). Consequently, factors reliant on income from these firms have larger wedges between their value added and household income received, implying higher distortion centrality (δ) and making them bottleneck inputs for high-marginal-product firms. As demand from these firms declines, high-δ factors reallocate toward less-distorted supply chains where their marginal product is lower, thereby reducing TFP.

In the first representation of Theorem 2, the *entropic terms of trade* quantify changes in the statistical distance between the value-added and the factor-income distributions, thereby capturing how income reallocation across heterogeneous factors affects productivity.[4] This result recovers the main representation from Baqaee & Farhi (2020). The vector of distortion centralities ranks factors by the (negative) marginal impact on TFP of a one-percentage-point increase in their income share. Aggregate productivity rises when income shifts from high-δ to low-δ factors — those that are more essential to high-marginal-product firms. This corresponds to an increase in the statistical distance between the value-added and factor-income distributions. As the statistical distance widens, high-δ factors become more affordable and reallocate toward high-marginal-product firms operating in more-distorted supply chains. This highlights the second-best nature of distributional TFP gains: productivity rises precisely when the income distribution diverges from the first-best benchmark in which factor compensation equals value added. The vector of *distortion centralities*, δ, is a sufficient statistic for the TFP impact of shifts in the factor-income distribution.

The second representation of Theorem 2 decomposes *distributional reallocation* into two nominal components: variation in the *factor terms of trade* and *corporate income*. The *factor terms of trade* capture how, holding nominal dividends constant, TFP rises when compensation falls — particularly for factors with high distortion centrality. As nominal income for high-distortion centrality factors falls, their relative price declines, inducing greater demand from more-distorted firms and thereby raising aggregate productivity. *Corporate income* captures the

[4] A discrete random variable $\mathcal{Q}$ with G mutually exclusive events is distributed according to the probability vector $q = [q_1, \cdots, q_G]'$. The information content of an event g is given by $I(g \mid \mathcal{Q}) = -\log q_g$. This function satisfies two key properties: (i) it is monotonically decreasing ($q_a < q_b$ implies that $I(a \mid \mathcal{Q}) > I(b \mid \mathcal{Q})$), and (ii) it is additive for independent events, i.e., $I(a\,b \mid \mathcal{Q}) = I(a \mid \mathcal{Q}) + I(b \mid \mathcal{Q})$. Shannon's entropy Shannon (1948) captures the average information content of a random draw (the expected "surprise") and is given by $H(q) = \sum_{g=1}^{G} q_g\, I(g \mid \mathcal{Q}) = -\sum_{g=1}^{G} q_g \log q_g$. The excess surprise from using an incorrect distribution $\widetilde{q}$ in place of the true distribution q is given by the Kullback-Leibler (KL) divergence (relative entropy), $\mathcal{K}(q \mid \widetilde{q}) = -\sum_{g=1}^{G} q_g \log\left(\widetilde{q}_g/q_g\right)$. By Gibbs's inequality $\mathcal{K}(q \mid \widetilde{q}) \geq 0$, expressing that using an incorrect probability model $\widetilde{q}$ introduces a positive bias in expected information. This excess surprise measures the statistical distance between the two distributions q and $\widetilde{q}$. The first-order variation in relative entropy as $\widetilde{q}$ changes is given by $d\,\mathcal{K}(q \mid \widetilde{q}) = -\sum_{g=1}^{G} q_g\, d\log \widetilde{q}_g$. When evaluated at $q = \widetilde{q}$, this implies $d\,\mathcal{K}(q \mid \widetilde{q}) = 0$, reflecting that first-order changes in $\widetilde{q}$ around the true distribution add no excess surprise up to a first-order. In other words, the information conveyed by $\widetilde{q}$ satisfies an envelope condition around q.

idea that, holding nominal factor income fixed, TFP rises with increases in aggregate dividends. Aggregate dividends may rise due to higher markups or higher sales volumes. Holding nominal factor income constant, an increase in dividends raises nominal GDP and thereby widens the gap between compensation and value added across factors. This lowers relative factor prices, stimulating firm demand. The demand response is strongest among firms in more distorted supply chains with high marginal products. As a result, input reallocate toward more-distorted supply chains, raising aggregate productivity.

The claim that aggregate productivity rises with an increase in the corporate income share may seem counterintuitive. After all, profits reduce pass-through and thereby contribute to misallocation. How, then, can the very source of inefficiency also promote aggregate productivity? This apparent contradiction is resolved by two key assumptions. First, the result assumes a fixed distribution of nominal factor income. Second, *distributional reallocation* captures only income-distribution effects, while *allocative efficiency* reflects the total impact, including competitive forces. For instance, consider a 1% increase in μ_i, holding both factor-income and sales distributions constant. In this case, *distributional reallocation* falls by $\lambda_i\,\mu_i$ (per 1% change), while *allocative efficiency* increases by $\widetilde{\lambda}_i - \lambda_i\,\mu_i$, which is nonnegative since $\widetilde{\lambda}_i \geq \lambda_i$ and $\mu_i \leq 1$.

Corollary 1. Decomposition of the Factor Terms of Trade. The variation in the factor terms of trade can be segmented into three components

$$\textit{Factor TT} = \overbrace{\sum_{h\in\mathscr{H}} (\delta_h - 1)\,\Lambda_h\, d\,\log\,\mathcal{J}_h}^{\textit{Real Income effect}} + \overbrace{\sum_{h\in\mathscr{H}} (\chi_h - \Lambda_h)\, d\,\log\,p^c_h}^{\textit{CPI effect}} + \overbrace{\sum_{h\in\mathscr{H}} \chi_h\, d\,\log\,\varepsilon_h}^{\textit{RER effect}},$$

where $\mathcal{J}_h = J_h/p^c_h$ denotes real factor income and $d\,\log\,\varepsilon_h = \sum_{b\in\mathscr{H}} \widetilde{\mathscr{C}}_{hb}\, d\,\log\,\varepsilon_{hb}$ is the weighted average change in the bilateral real exchange rate for households of type-h households, with $\varepsilon_{hb} = p^c_b/p^c_h$.

Corollary 1 decomposes the *factor terms of trade* into three components: (i) real income effect, (ii) a consumer price index (CPI) effect, and (iii) a real exchange rate (RER) effect. First, the *real income effect* captures how changes in households' real factor earnings affect TFP, weighted by each type's distortion centrality and income share. For households with unit distortion centrality ($\delta_h = 1$), real income variations are allocatively neutral for TFP. Second, the CPI effect reflects how increases in household consumption prices lower TFP, particularly for households that rely disproportionately on transfers or dividends ($\chi_h \gg \Lambda_h$). Finally, the RER effect captures how a depreciation in household h's average bilateral exchange rate ε_h lowers aggregate productivity, with an impact scaled by its expenditure share.[5]

The final two representations in Theorem 2 build upon the decomposition of factor income shares introduced in Theorem 1. The third representation decomposes *distributional reallocation* into two distinct channels: (i) shifts in the expenditure shares, and (ii) changes in consumer-to-factor centralities. First, the *distributive terms of trade* capture how reallocating expenditure toward households with higher *expenditure centrality* (M_h) reduces aggregate productivity. A

[5]An increase in ε_{hb} denotes a depreciation of the type-h bundle relative to type-b's.

high M_h indicates that household h disproportionately consumes goods produced in less-distorted supply chains that, in turn, rely on inputs essential to more-distorted supply chains. When χ_h increases for households with high M_h, TFP falls as spending shifts toward less-distorted firms that expand their input demand. This reallocation pulls inputs from more-distorted firms with high marginal productivity toward less-distorted firms with low-marginal-product. The vector M of *expenditure centralities* ranks households inversely by their marginal impact on TFP from a one-percentage-point increase in their expenditure share. Hence, M is a sufficient statistic for the TFP impact of redistribution: aggregate productivity rises when expenditure shifts from household h to household b with $M_h > M_b$. In representative-agent economies, the *distributive terms of trade* vanish by construction. Second, the *centrality terms of trade* capture how rising consumer-to-factor centralities depress aggregate productivity, particularly when concentrated on high-distortion centrality factors. This channel isolates the TFP effect of cost recomposition, holding expenditure shares fixed.

Corollary 2. Distributive Neutrality. Variations in the consumption-expenditure distribution have no first-order effect on TFP when expenditure centrality is symmetric across households, i.e., $M_h = M\ \forall h \in \mathscr{H}$. This neutrality condition is satified in several canonical environments: (i) an undistorted economy ($\mu_i = 1\ \forall i \in \mathscr{N}$); (ii) homogeneous consumption bundles across households ($\beta_{hi} = \beta_i\ \forall h \in \mathscr{H}$); (iii) an economy without intermediate inputs and with uniform distortions across firms ($\omega_i^x = 0$ and $\mu_i = \mu\ \forall i \in \mathscr{N}$); and (iv) an economy without intermediate inputs and with firm-specific factor supply ($\omega_i^\ell = \alpha_{ii} = 1\ \forall i \in \mathscr{N}$).

Corollary 2 establishes conditions under which an aggregate production function can abstract from heterogeneity in the consumption-expenditure distribution without inducing first-order bias in TFP measurement. Symmetric expenditure centralities arise in four canonical cases: (i) environments in which the first welfare theorem holds; (ii) economies where all households have identical consumption bundles; (iii) economies without intermediate inputs and with uniform distortions across firms; and (iv) economies without intermediate inputs and with firm-specific factor supply. First, in an efficient economy, all distortion centralities are one, and all household expenditure accrues to factors ($\sum_{b\in\mathscr{H}} \mathscr{C}_{hb} = 1\ \forall h \in \mathscr{H}$). Hence, perturbations in the expenditure distribution may alter expenditure flows and the allocation of resources across firms, but are allocatively neutral. Second, when households share identical consumption bundles, consumer-to-factor centralities and expenditure centralities coincide for all households, regardless of the distortion centrality vector δ. Third, in an economy without intermediate inputs and with uniform distortions, consumers' marginal rates of substitution across goods are unaffected by distortions, thereby eliminating misallocation. As in the efficient benchmark, changes in expenditure may reallocate inputs across firms, but remain allocatively neutral. Finally, misallocation does not arise in economies without intermediate inputs when factor supply is firm-specific. Collectively, these cases demonstrate that while heterogeneity in consumption bundles — and thus aggregate nonhomotheticity induced by shifts in the consumption-expenditure distribution — is necessary, it is not sufficient for such variations to affect aggregate productivity.

The final representation in Theorem 2 decomposes the *centrality terms of trade* into four channels, each corresponding to a distinct type of cost recomposition. The *final-demand* and *intermediate-demand terms of trade* capture how TFP falls when spending shifts toward firms with high revenue centrality F_i. A high F_i implies that firm i operates in a less-distorted supply chain but relies — directly or indirectly — on inputs essential to firms in more-distorted supply chains. The *revenue centrality* vector F inversely ranks firms by their marginal impact on TFP from cost recomposition. Accordingly, F is a sufficient statistic for the aggregate productivity effects of cost recomposition originating from both household and firms' intermediate demand: aggregate TFP rises when household h or firm q reallocates spending from firm i to firm j with $F_i > F_j$. The *factor-demand terms of trade* capture how aggregate productivity responds as firms change their spending across factors with different distortion centralities. The distortion centrality vector δ is a sufficient statistic for the TFP effects of firm-level factor cost recomposition: TFP rises when firm i shifts spending from factor h to b with $\delta_h > \delta_b$. More generally, both the *factor-demand* and *intermediate-demand terms of trade* capture how firm-level substitutions between factors and intermediates shape TFP: aggregate productivity rises when firm i shifts spending from factor h to input j with $\delta_h > F_j$. Finally, the *competitive terms of trade* capture how changes in markups drive distributional reallocation via firm-level demand adjustments. For instance, if μ_i increases by 1%, while holding the expenditure distribution and all other terms of trade fixed, then *distributional reallocation* falls by $\lambda_i F_i$, while total *allocative efficiency* rises by $\widetilde{\lambda}_i - \lambda_i F_i$. In contrast to the *corporate income* channel, *allocative efficiency* may fall. When markups increase for high-F_i firms, inputs are reallocated towards firms in distorted supply chains with higher marginal productivity, thereby increasing TFP.

Revenue centralities, distortion centralities, cost shares, and revenue-based Domar weights are sufficient statistics for the four channels that constitute the *centrality terms of trade*. Corollary 3 characterizes the conditions under which firm- or household-level cost recomposition have no first-order effect on TFP. These cost recompositions may arise exogenously from preference shocks or endogenously in responses to relative price movements.

Corollary 3. Cost Recomposition Neutrality.

1. Cost recompositions have no first-order effect on TFP under the first-best allocation.

2. Household h's expenditure recomposition is neutral on TFP if all firms from which it demands final goods exhibit identical revenue centrality, i.e., $F_i = F \; \forall i \in \mathscr{N}$ with $\beta_{hi} > 0$.

3. Firm i's cost recomposition has no first-order effect on TFP under any of the following conditions. (a) The firm uses no intermediate inputs, and all factors it employs have identical distortion centrality, i.e., $\delta_h = \delta \; \forall h \in \mathscr{H}$ with $\widetilde{\Omega}^{\ell}_{ih} > 0$. (b) The firm uses no primary factors, and all intermediate-input suppliers share the same revenue centrality, i.e., $F_j = F \; \forall j \in \mathscr{N}$ with $\widetilde{\Omega}^{x}_{ij} > 0$. (c) The distortion centralities of all factor suppliers equal the revenue centralities of all intermediate suppliers, i.e., $\delta_h = F_j \; \forall h \in \mathscr{H}$ and $\forall j \in \mathscr{N}$ with $\widetilde{\Omega}^{\ell}_{ih} > 0$ and $\widetilde{\Omega}^{x}_{ij} > 0$.

3.4 Aggregate Accounting in a Constrained Social Planner Economy

I now consider a constrained social planner who centralizes all decisions previously made by individual households. The planner maximizes an aggregate welfare function $W(Y, L)$ subject to the household implementability constraints Equation (3) and the budget-neutrality condition $\sum_{h\in\mathscr{H}} T_h = 0$. Aggregate output and factor supply are defined by $Y = Q_Y(\{C_h\}_{h\in\mathscr{H}})$ and $L = F(\{L_h\}_{h\in\mathscr{H}})$, consistent with Theorem 2.[6] The planner chooses $\{C_h, L_h, T_h, \{C_{hi}\}_{i\in\mathscr{N}}\}_{h\in\mathscr{H}}$ as an allocation that maximizes the welfare function subject to the households' budget constraints and budget-neutrality condition, and the aggregator definitions. Theorem 3 characterizes equilibrium output and TFP in a constrained social planner economy with endogenous factor supply.

Theorem 3. Real GDP and TFP in a Constrained Social Planner Economy. In equilibrium, aggregate output and factor supply satisfy

$$\frac{W_L}{W_Y} + \Gamma\,\frac{Y}{L} = 0 \qquad \text{with} \qquad \Gamma = \sum_{h\in\mathscr{H}} \Lambda_h = \delta_b^{-1} \quad \forall b \in \mathscr{H}. \tag{11}$$

The first-order variations in Y and TFP are given by

$$d\log Y = \overbrace{\sum_{i\in\mathscr{N}} \widetilde{\lambda}_i\, d\log A_i + \sum_{i\in\mathscr{N}} \widetilde{\lambda}_i\, d\log \mu_i - d\log\Gamma}^{d\log TFP} + \sum_{h\in\mathscr{H}} \widetilde{\Lambda}_h\, d\log L_h, \tag{12}$$

and the first-order variations in Λ_h and Γ are, to a first-order,

$$d\Lambda_h = \sum_{i\in\mathscr{N}} \psi^{\ell}_{ih}\, \lambda_i\, d\log\mu_i + \sum_{i\in\mathscr{N}} \mu_i\,\lambda_i\, d\widetilde{\Omega}^{\ell}_{ih} + \sum_{j\in\mathscr{N}} \psi^{\ell}_{jh}\left(d\beta_j + \sum_{i\in\mathscr{N}} \mu_i\,\lambda_i\, d\widetilde{\Omega}^{x}_{ij}\right), \tag{13}$$

$$d\Gamma = \sum_{i\in\mathscr{N}} \psi^{\ell}_{i}\, \lambda_i\, d\log\mu_i + \sum_{i\in\mathscr{N}} \mu_i\,\lambda_i\, d\omega^{\ell}_{i} + \sum_{i\in\mathscr{N}} \psi^{\ell}_{i}\left(d\beta_i + \sum_{j\in\mathscr{N}} \mu_j\,\lambda_j\, d\widetilde{\Omega}^{x}_{ji}\right). \tag{14}$$

where $\beta_i = \sum_{h\in\mathscr{H}} p_i C_{hi}/GDP$ and $\psi^{\ell}_i = \sum_{h\in\mathscr{H}} \psi^{\ell}_{ih}$.

Equation (11) links output, aggregate factor supply, and the wedge Γ. It states that the planner's marginal rate of substitution between output and factor supply equals Γ times the economy's marginal rate of transformation, with Y/L = TFP. In equilibrium, Γ equals both the aggregate factor income share and the inverse of the common distortion centrality (δ) across all factor types. Thus, Γ summarizes how firm-level distortions shape the mapping from factor supply to aggregate output. Because the planner internalizes these distortions, it chooses $\{L_h\}_{h\in\mathscr{H}}$ to equalize distortion-adjusted marginal welfare tradeoffs. This endogenously equalizes distortion centralities across factors ($\delta_h = \delta\ \forall h \in \mathscr{H}$). This symmetry compresses heterogeneity, so no factor is disproportionately misallocated, and the myriad of micro-wedges

[6] The function $W : \mathbb{R}^2_+ \to \mathbb{R}_+$ is strictly concave, twice continuously differentiable, and satisfies $W_Y > 0$, $W_L < 0$, along with the Inada conditions. The aggregator functions $Q_Y : \mathbb{R}^N_+ \to \mathbb{R}_+$ and $F : \mathbb{R}^H_+ \to \mathbb{R}_+$ are neoclassical: homogeneous of degree one, strictly concave, monotonic, twice continuously differentiable, and satisfy the Inada conditions wherever consumption or factor supply is strictly positive.

can be summarized by a single macro-wedge, Γ.

Equation (12) builds on that result by decomposing first-order changes in real GDP into contributions from (i) TFP and (ii) factor supply. As in Theorem 2, TFP decomposes into three channels: *technology*, *competitiveness*, and *distributional reallocation*. In the constrained social planner economy, however, the distributional channel collapses into a single scalar. Because distortion centralities are equalized across factor types, the marginal TFP impact of distributional reallocation is identical across factors. As a result, the multidimensional vector of distributional effects condenses into a single sufficient statistic: the aggregate factor income share Γ. In this setting, Γ governs both the planner's supply choice and the extent to which firm-level distortions pass through to aggregate misallocation via distributional channels. A decline in Γ signals that factors are relatively more affordable, facilitating a broad-based expansion in input demand. This additional demand disproportionately benefits firms in heavily distorted supply chains, raising TFP. This mirrors the *corporate-income* channel: higher profit shares are associated with higher TFP. Both channels underscore how the functional distribution of income — between factor returns and profits — shapes the economy's capacity to reallocate resources toward high-marginal-productivity firms. The key distinction from Theorem 2 is that TFP is invariant to the distribution of income across factor types; only the total split between factor income and profits matters. In a representative household production network economy with markups and a single exogenous factor, Baqaee & Rubbo (2023) show a similar result to that in equation (12). My contribution is to show that an analogous result holds in a heterogeneous household economy when the supply of multiple factors is endogenous and centralized by the constrained social planner.

Equations (13) and (14) provide decompositions of factor income shares and the *distributional reallocation* channel that parallel those in Theorems 1 and 2, but are derived under the constrained social planner's optimality conditions. A key simplification emerges: both the *distributive income* effects on factor shares and the *distributive terms of trade* effects on TFP vanish to a first order. This occurs because the social planner accounts for all income sources and consumption choices, reallocating resources across types via budget-neutral transfers. Unlike in the decentralized setting, the social planner internalizes preferences via $W(Y, L)$, and chooses $\{C_{hi}\}$ to equalize households' marginal rates of substitution, i.e., $p_i/p_Y = (\partial Y/\partial C_{hi})(\partial C_h/\partial C_{hi})$. This compression of heterogeneity implies that variation in household expenditure shares no longer introduces additional wedges between firm-level distortions and aggregate outcomes. In this setting, the sum of firm-to-factor centralities, $\psi_i^\ell = \sum_{h\in\mathscr{H}} \psi_{ih}^\ell$, enters the set of sufficient statistics for both the aggregate factor income share and TFP. This object consolidates distinct interaction effects — such as those arising from firm-level input composition and household-specific expenditure patterns — into a unified network-based metric. It captures how variation in firm-level cost shares, input composition, and household demand jointly determines the economy-wide distribution of income between factor payments and rents.

The contrast between Theorem 2 and Theorem 3 highlights a key point: centralizing heterogeneous households' decisions — whether via a constrained social planner or a representative household — fundamentally changes how TFP is measured. Centralization internalizes the general equilibrium effects of consumption and factor supply on aggregate welfare; decentralization, by contrast, can yield allocations that lack symmetry across the margins of adjustment — distortion centralities δ, and consumption marginal rates of substitution $\partial Y/\partial C_{hi}$ — thereby creating externalities on aggregate social welfare. Consequently, even when a representative household economy with multiple inelastic primary factors — à la Baqaee & Farhi (2020) — and a decentralized heterogeneous household economy share the same functional form for the first-order decomposition for TFP, their measured contributions generally differ because the weights at which the identity is evaluate (e.g., $\widetilde{\lambda}$, $\widetilde{\Lambda}$, λ, λ, δ, χ, M, and F) are allocation-dependent. In the centralized case, consumption marginal rates of substitution are equalized across households, so the equilibrium expenditure and income allocations — and thus the weights — differ from those in the decentralized economy. Hence, coincidence of identities does not imply measurement symmetry: the algebra can match while the magnitude of TFP variation does not, precisely because allocations differ between centralized and decentralized economies.

With endogenous factor supply, the distributional structure that shapes TFP under household heterogeneity collapses into a single macroeconomic wedge: the aggregate factor income share Γ. The implication is that prematurely adopting a centralized entity of decision for households risks mismeasuring the role of inequality in aggregate productivity. In this sense, Theorem 2 and Theorem 3 are not merely distinct accounting identities — they represent fundamentally different views about how heterogeneity and both factor income and expenditure distributions shape aggregate outcomes. Theorem 2 requires bilateral centralities that yield firm-level revenue centralities, household-level expenditure centralities, and factor-specific heterogeneous distortion centralities, whereas Theorem 3 relies only on the aggregate factor income share and the firm-to-factor centrality sum ψ_i^{ℓ}. This contrast offers a diagnostic for empirical work when the assumption of endogenous factor supply is made: if distortion centralities exhibit wide dispersion, heterogeneity matters and must be explicitly modeled; if they are tightly concentrated, the representative household or constrained social planner assumption may suffice, provided consumption marginal rates of substitution are also tightly concentrated. The choice between a centralized and a decentralized economy at the household level is not merely a matter of tractability or measurement — it is a foundational modeling assumption with far-reaching implications for how we interpret TFP growth and the aggregate consequences of distributional policy.

3.5 Distributional Accounting

Theorem 4 characterizes household-level real consumption and its first-order responses to shocks in general equilibrium in a distorted production network. It introduces a new statistic: the *positional terms of trade* (PTT), which measure the efficiency with which a household's consumption bundle can be produced from the economy's equilibrium factor endowment. Unlike the household-specific price index from Proposition 1, which reflects only the expenditure-side composition of consumption, the PTT aggregates both consumption outflows and functional income inflows — including factor earnings, rents, and transfers — thereby capturing the household's full general equilibrium position. This concept parallels TFP in Theorem 2: while TFP gauges the aggregate efficiency of transforming factors into output, the PTT measures how efficiently those same factors generate real consumption for each household, and their first-order changes summarize the incidence of shocks across households.

This distinction is critical for distributional analysis. While Theorem 2 decomposes real GDP into aggregate productivity and factor supply, it does not directly track differences in household-specific real consumption or welfare. In contrast, PTTs show that even when aggregate output remains constant, real household consumption can vary substantially with each household's exposure to productivity shocks, markups, and distributional channels. Moreover, aggregate TFP growth is expressed as a weighted average of household-level PTT growth, $d \log TFP = \sum_{h \in \mathcal{H}} \chi_h \; d \log PTT_h$, where χ_h denotes household h's share of total final expenditure. This identity underscores that aggregate efficiency changes mask significant heterogeneity: aggregate gains are the expenditure-weighted mean of household specific gains.

Theorem 4. Real Consumption and PTTs in a Heterogeneous Household Economy In equilibrium, real consumption for type h households satisfies

$$C_h = Q_h^c \left(\{C_{hi}\}_{i \in \mathcal{N}} \right) = PTT_h \, f_h \left(\{L_b\}_{b \in \mathcal{H}} \right), \tag{15}$$

where PTT_h captures the positional terms of trade, and f_h has elasticities $d \log f_h / d \log L_b = \widetilde{\mathscr{C}}_{hb}$.

The first-order changes in C_h and PTT_h are

$$d \log C_h = d \log PTT_h + \sum_{b \in \mathcal{H}} \widetilde{\mathscr{C}}_{hb} \, d \log L_b, \tag{16}$$

$$d \log PTT_h = \mathit{Technology}_h + \mathit{Allocative\ Efficiency}_h, \tag{17}$$

where

$$\mathit{Technology}_h = \sum_{i \in \mathcal{N}} \widetilde{\mathscr{B}}_{hi} \, d \log A_i, \quad \begin{matrix} \mathit{Allocative} \\ \mathit{Efficiency}_h \end{matrix} = \overbrace{\sum_{i \in \mathcal{N}} \widetilde{\mathscr{B}}_{hi} \, d \log \mu_i}^{\mathit{Competitiveness}_h} + \begin{matrix} \mathit{Distributional} \\ \mathit{Reallocation}_h \end{matrix}.$$

Distributive Reallocation$_h$ has the following four equivalent definitions

$$1. \; d \log \chi_h - \overbrace{\sum_{b \in \mathcal{H}} \delta_{b|h} \, d \Lambda_b}^{\mathit{Entropic\ TT}_h},$$

$$2. \; \overbrace{\frac{\Lambda_h}{\chi_h} d \log J_h - \sum_{b \in \mathcal{H}} \widetilde{\mathscr{C}}_{hb} \, d \log J_b}^{-\mathit{Factor\ TT}_h} + \overbrace{\sum_{i \in \mathcal{N}} \frac{\lambda_i}{\chi_h} \left((1 - \mu_i) \left(d \kappa_{ih} + \kappa_{ih} \, d \log S_i \right) - \kappa_{ih} \, d \mu_i \right)}^{\mathit{Corporate\ Income}_h} + \overbrace{\frac{\tau_h}{\chi_h} d \log T_h}^{\mathit{Transfers}_h},$$

$$3. \; d \log \chi_h - \overbrace{\sum_{b \in \mathcal{H}} M_{b|h} \, d \chi_b}^{\mathit{Distributive\ TT}_h} - \overbrace{\sum_{b \in \mathcal{H}} \chi_b \sum_{q \in \mathcal{H}} \delta_{q|h} \, d \log \mathscr{C}_{bq}}^{\mathit{Centrality\ TT}_h},$$

$$4.\ d\log\chi_h - \sum_{b\in\mathscr{H}} M_{b|h}\, d\chi_b - \overbrace{\sum_{b\in\mathscr{H}}\chi_b \sum_{i\in\mathscr{N}} F_{i|h}\, d\beta_{bi}}^{\textit{Final-Demand } TT_h} - \overbrace{\sum_{i\in\mathscr{N}}\mu_i\,\lambda_i \sum_{j\in\mathscr{N}} F_{j|h}\, d\widetilde{\Omega}^x_{ij}}^{\textit{Intermediate-Demand } TT_h}$$

$$-\overbrace{\sum_{i\in\mathscr{N}}\mu_i\,\lambda_i \sum_{b\in\mathscr{H}} \delta_{b|h}\, d\widetilde{\Omega}^{\ell}_{ib}}^{\textit{Factor-Demand } TT_h} - \overbrace{\sum_{i\in\mathscr{N}}\lambda_i\, F_{i|h}\, d\log\mu_i}^{\textit{Competitive } TT_h},$$

with $\delta_{b|h} = \widetilde{\mathscr{C}}_{hb}/\Lambda_b$, $M_{b|h} = \sum_{q\in\mathscr{H}} \mathscr{C}_{bq}\,\delta_{q|h}$, and $F_{i|h} = \sum_{q\in\mathscr{H}} \psi^{\ell}_{iq}\,\delta_{q|h}$, and

$$d\chi_h = d\Lambda_h + \sum_{i\in\mathscr{N}} \left((1-\mu_i)\,(\lambda_i\, d\kappa_{ih} + \kappa_{ih}\, d\lambda_i) - \kappa_{ih}\,\lambda_i\, d\mu_i\right) + d\tau_h. \tag{18}$$

Equation (15) shows that real consumption for household type h admits two equivalent representations in general equilibrium. First, as a CRS aggregator Q^c_h over final goods consumed by household h. Second, as the product of the household's positional terms of trade PTT_h and a CRS aggregator f_h over primary factors, with elasticities equal to the household-specific factor-to-consumer centrality vector $\widetilde{\mathscr{C}}_{\downarrow h} = (\widetilde{\mathscr{C}}_{h1}, \cdots, \widetilde{\mathscr{C}}_{hH})'$. This dual structure mirrors the representation of aggregate real GDP in equation (8), which can likewise be expressed as either a final-expenditure aggregator or TFP times a value-added aggregator over primary factors.

Equation (16) decomposes the first-order variation in household-level real consumption into two components: a positional terms of trade (PTT) and a factor input component. Equation (17) then decomposes the first-order variation in PTT_h into three structural channels. First, *technology*$_h$ captures the direct effect of changes in firm-level productivity, holding the equilibrium allocation fixed. Second, *competitiveness*$_h$ captures the direct effect of changes in markups, holding the factor income distribution fixed. Absent changes in the income distribution, the elasticity of PTT_h with respect to productivity and markup shocks at firm i is proportional to the firm-to-consumer centrality $\widetilde{\mathscr{B}}_{hi}$. These two channels mirror their aggregate counterparts in Theorem 2, where firm-level productivity and markup shocks propagate downstream via cost-based centralities. Finally, *distributive reallocation*$_h$, captures the household-specific general equilibrium effect of income redistribution on household h's real consumption.

A key observation is that aggregate allocative efficiency is a expenditure-weighted average across households, i.e., *allocative efficiency* $= \sum_{h\in\mathscr{H}} \chi_h$ *allocative efficiency*$_h$. This identity implies that, under the first-best allocation, Hulten's theorem (1978) implies that allocative efficiency sums to zero across households. Thus, the household-level PTT decompositions mirrors the structure of aggregate TFP while revealing how growth channels act asymmetrically and the incidence of shocks across households, depending on their position in the production network.

Theorem 4 also provides four equivalent representations of the household-specific *distributive reallocation* channel. Each representation formalizes how changes in the distribution of income or input allocations benefit household h — either by increasing its expenditure share ($d\log\chi_h > 0$) or by raising the marginal productivity of the factors used in producing its consumption bundle. Equation (18) expresses the first-order variation household h's expenditure share χ_h. It shows that household h's expenditure share rises with factor income, firm rents, and lump-sum transfers. Rents from firm i received by household h rise with the household's equity share κ_{ih}, the firm's Domar weight λ_i, and increases in markups (i.e., $d\mu_i < 0$).

The first household-level representation of the *distributive reallocation* channel is given by the idiosyncratic *entropic terms of trade*, which quantify how changes in the statistical distance between the household's value-added distribution $\widetilde{\mathscr{C}}_{\downarrow h}$ and the aggregate factor income distribution Λ affect how efficiently the economy converts factors into real consumption for household h.[7] PTT_h rises when factor income shifts widen the gap between Λ and $\widetilde{\mathscr{C}}_{\downarrow h}$, that is, when the factors most essential to household h become relatively undercompensated. This mechanism operates through household-specific distortion centralities $\delta_{b|h} = \widetilde{\mathscr{C}}_{hb}/\Lambda_b$, which compare the network-adjusted importance of each factor in h's bundle to its share in aggregate factor income. A factor is undervalued from h's perspective if $\delta_{b|h} > 1$ and overvalued if $\delta_{b|h} < 1$. Unlike the aggregate distortion centrality vector δ, where all its components are weakly above one in distorted economies, household-specific distortion centralities $\delta_{b|h}$ can fall below one — even under the first-best allocation because $\widetilde{\mathscr{C}}_{\downarrow h}$ need not equal Λ. The allocative implication is direct: household h benefits when income shifts from factors it deems undervalued (high $\delta_{b|h}$) toward those it views as overvalued (low $\delta_{b|h}$). For example, if factor income shifts from factor b to q and $\delta_{b|h} > \delta_{q|h}$, then h's positional terms of trade improve.

The second household-level representation of the *distributive reallocation* channel decomposes into three nominal components: *factor terms of trade*, *corporate income*, and *transfers*. The household-specific *factor terms of trade* capture how PTT_h rises when household h's factor income grows relative to the nominal incomes of the factors implicit in its consumption bundle. Income is weighted by the ratio of factor income to total expenditure (Λ_h/χ_h), and the comparison term is weighted by household's factor incidence $\widetilde{\mathscr{C}}_{\downarrow h}$. The *corporate income* channel tracks changes in rent receipts, which increase with the household's equity share κ_{ih}, the firm's revenue S_i, and higher markups $d\,\mu_i < 0$. While equity redistributions affect households' PTTs, their aggregate effect cancels out so long as the underlying factor income and sales distributions remain unchanged. Finally, the *transfer* channel captures the impact of lump-sum transfers, weighted by their share in household h's total expenditure (τ_h/χ_h).

Corollary 4. Decomposition of Household-Specific Factor Terms of Trade. The variation in the factor terms of trade for type h households is given by

$$\text{Factor } TT_h = \overbrace{\sum_{b\in\mathscr{H}} \widetilde{\mathscr{C}}_{hb}\, d\log \mathcal{J}_b - \frac{\Lambda_h}{\chi_h}\, d\log \mathcal{J}_h}^{\text{Real Income effect}_h} + \overbrace{\left(1-\frac{\Lambda_h}{\chi_h}\right) d\log p^c_h}^{\text{CPI effect}_h} + \overbrace{\sum_{b\in\mathscr{H}} \widetilde{\mathscr{C}}_{hb}\, d\log \varepsilon_{hb}}^{\text{RER effect}_h}.$$

Corollary 4 decomposes the household-specific *factor terms of trade* into three distinct channels. First, the *real income effect* captures how PTT_h falls when the real cost of factors embedded in household h's consumption bundle rises relative to household h's real factor earnings. Second, the *CPI effect* reflects how increases in the consumption price index lowers PTT_h, especially for households that depend more heavily on nonfactor income. Third, the *real exchange rate (RER) effect* captures how PTT_h declines when the bilateral price ratio $\varepsilon_{hb} = p^c_b/p^c_h$ depreciates. The magnitude of this effect is weighted by household h's factor-to-consumer centrality $\widetilde{\mathscr{C}}_{hb}$.

[7] The first-order variation in relative entropy is given by $d\,\mathcal{K}(\widetilde{\mathscr{C}}_{\downarrow h} \mid \Lambda) = -\sum_{b\in\mathscr{H}} \widetilde{\mathscr{C}}_{hb}\; d\,\log\,\Lambda_h$, which increases when Λ shifts away from the household's distribution $\widetilde{\mathscr{C}}_{\downarrow h}$.

The third household-specific representation of the *distributive reallocation* channel decomposes the household-specific *entropic terms of trade* into two distinct sources: (i) shifts in the expenditure distribution, and (ii) changes in consumer-to-factor centralities. First, the *distributive terms of trade* for household h measure how PTT_h declines when purchasing power shifts toward households that, from h's perspective, have high *expenditure centrality* (high $M_{b|h}$). The term $M_{b|h}$ denotes the *expenditure centrality* of type-b households relative to h's consumption bundle, capturing how b's spending patterns affect the marginal productivity of the factors used to produce this bundle. When χ_b increases for high-$M_{b|h}$ households, inputs are reallocated toward firms where the factors essential to h's bundle have lower marginal products, thereby reducing PTT_h. The vector $M_{|h} = \left(M_{1|h}, \ldots, M_{H|h}\right)'$ serves as a sufficient statistic for the expenditure redistribution effects on PTT_h: household h gains when expenditure shifts from b to q whenever $M_{b|h} > M_{q|h}$. Second, the *centrality terms of trade* capture how PTT_h deteriorates when the consumer-to-factor centralities $\mathscr{C}_{bq}$ increase (holding χ fixed), especially for factors that h views as disproportionately undercompensated (high $\delta_{q|h}$). This channel isolates the impact of cost-structure recomposition on how efficiently inputs are transformed into real consumption for household h.

The final household-level representation of the *distributive reallocation* channel decomposes the impact of cost recomposition on PTT_h into four mechanisms. First, the *final-demand* and *intermediate-demand* terms of trade capture how PTT_h declines when spending shifts toward firms that, from h's perspective, have high *revenue centrality* (high $F_{i|h}$). The term $F_{i|h}$ denotes the *revenue centrality* of firm i relative to household h's bundle, capturing how spending on i influences the marginal productivity of factors in this bundle. The vector $F_{|h} = \left(F_{1|h}, \ldots, F_{N|h}\right)'$ is a sufficient statistic for the impact on PTT_h of both households' final-demand and firms' intermediate-demand recomposition: PTT_h rises when household b or firm q shifts spending from firm i to firm j with $F_{i|h} > F_{j|h}$. Second, the *factor-demand terms of trade* capture how PTT_h changes when firms reallocate spending across factors with different distortion centralities. The household-specific distortion centrality vector $\delta_{|h} = \left(\delta_{1|h}, \ldots, \delta_{H|h}\right)'$ is a sufficient statistic for the impact on PTT_h from firm-level factor cost recomposition: PTT_h rises when firm i reallocates spending from factor b to m with $\delta_{b|h} > \delta_{m|h}$. More broadly, firm-level substitution between factors and intermediates affects PTT_h: it increases when firm i shifts spending from factor b to the intermediate input from firm j with $\delta_{b|h} > F_{j|h}$. Finally, the *competitive terms of trade* capture the heterogeneous distributional effects across households from markup variations. For example, if μ_i increases by 1% (holding the expenditure distribution and all other terms of trade fixed), PTT_h declines via the *distributional* component by $\lambda_i\, F_{i|h}$, but the net PTT_h change is $\widetilde{\mathscr{B}}_{hi} - \lambda_i\, F_{i|h}$. When markups increase for high-$F_{i|h}$ firms, input reallocation favors firms that use factors more productively from h's perspective ($F_{i|h} > \widetilde{\mathscr{B}}_{hi}/\lambda_i$), thereby improving PTT_h.

4 Parametric General Equilibrium

While Theorems 1 through 4 deliver nonparametric decompositions of real outcomes, their empirical and counterfactual implementation requires a parametric structure to solve for the endogenous general equilibrium responses of prices, expenditures, and cost shares. The following section adopts a normalized CES framework to generate such solutions in closed form.

4.1 Normalized CES Environment

Following Baqaee & Farhi (2019, 2020, 2024), I implement the normalized CES specification of de La Grandville (1989) and Klump & de La Grandville (2000) in an economy with intermediate inputs and heterogeneous households. Overbars denote equilibrium outcomes. Each firm $i \in \mathscr{N}$ produces output using a normalized CES aggregator of primary factors and intermediate inputs

$$\frac{y_i}{\overline{y}_i} = A_i \left(\omega_i^{\ell} \left(\frac{L_i}{\overline{L}_i} \right)^{\frac{\theta_i - 1}{\theta_i}} + \omega_i^x \left(\frac{X_i}{\overline{X}_i} \right)^{\frac{\theta_i - 1}{\theta_i}} \right)^{\frac{\theta_i}{\theta_i - 1}},$$

$$\frac{L_i}{\overline{L}_{z_i}} = \left(\sum_{h \in \mathscr{H}} \alpha_{ih} \left(\frac{\ell_{ih}}{\overline{\ell}_{ih}} \right)^{\frac{\theta_i^{\ell} - 1}{\theta_i^{\ell}}} \right)^{\frac{\theta_i^{\ell}}{\theta_i^{\ell} - 1}}, \qquad \frac{X_i}{\overline{X}_i} = \left(\sum_{j \in \mathscr{N}} \omega_{ij} \left(\frac{x_{ij}}{\overline{x}_{ij}} \right)^{\frac{\theta_i^x - 1}{\theta_i^x}} \right)^{\frac{\theta_i^x}{\theta_i^x - 1}}.$$

Here, θ_i denotes the elasticity of substitution between primary factors and intermediate inputs, θ_i^{ℓ} the elasticity of substitution between primary factors, and θ_i^x the elasticity of substitution between intermediate inputs. Similarly, each household $h \in \mathscr{N}$ consumes a normalized CES aggregate of final goods

$$\frac{C_h}{\overline{C}_h} = \left(\sum_{i \in \mathscr{N}} \beta_{hi} \left(\frac{C_{hi}}{\overline{C}_{hi}} \right)^{\frac{\varrho_h - 1}{\varrho_h}} \right)^{\frac{\varrho_h}{\varrho_h - 1}},$$

where ϱ_h denotes the elasticity of substitution across final goods. A key advantage of the normalized CES specification is that the parameters ω_i^{ℓ}, α_{ih}, ω_i^x, ω_{ij}, and β_{hi} retain their interpretation from Section 2: they serve as cost elasticities and coincide with the cost shares shares in equilibrium.

Each household $h \in \mathscr{H}$, with population size n_h, has preferences represented by the utility function

$$U_h \left(c_h, \widetilde{L}_h \right) = \frac{\left(c_h \left(1 - E_h^{-\gamma_h} \widetilde{L}_h \right)^{\varphi_h} \right)^{1-\sigma} - 1}{1 - \sigma},$$

where $C_h = n_h \, c_h$, $L_h = n_h \, \widetilde{L}_h$, and $\varphi_h > 0$. The variables c_h and $\widetilde{L}_h$ denote per-capita real consumption and factor supply, ensuring that utility is invariant to the household's population size. This function form flexibly separates income and substitution effects in factor supply through the parameters γ_h and φ_h.

Proposition 2. The first-order variation in factor supply from household type h in response to demographic, factor price, and income shocks is given by

$$d\log L_h = \zeta_h^n \, d\log n_h + \zeta_h^w \, d\log w_h - \zeta_h^e \, d\log E_h.$$

The corresponding elasticities are defined as

$$\zeta_h^n = \frac{E_h^{\gamma_h}}{1-\varphi_h\gamma_h}\frac{n_h}{L_h}, \qquad \zeta_h^w = \frac{\varphi_h}{1-\varphi_h\gamma_h}\frac{\chi_h}{\Lambda_h}, \qquad \zeta_h^e = \zeta_h^w - \gamma_h\zeta_h^n.$$

Proposition 2 characterizes the first-order response of household factor supply in terms of three elasticities: (i) the demographic effect ζ_h^n, (ii) the substitution effect ζ_h^w, and (iii) the income effect ζ_h^e. These elasticities depend on both equilibrium objects and the preference parameters γ_h and φ_h, which govern the strength of income and substitution effects. This utility function nests several well-known preference structures. First, when $\gamma_h = 0$ factoral supply reduces to King, Plosser, & Rebelo (1988), which imply symmetric substitution and income effects. In this case, $\zeta_h^n = n_h/L_h$ and $\zeta_h^w = \zeta_h^e = \varphi_h(\chi_h/\Lambda_h)$. Second, imposing the no-income effect condition $\zeta_h^e = 0$ yields the GHH-style restriction on factor supply (Greenwood, Hercowitz, & Huffman, 1988), which holds under the knife-edge relation $\varphi_h = \gamma_h \, E_h^{\gamma_h - 1} \, n_h \, w_h$. Finally, by varying γ_h and φ_h, the specification mimics the factor supply flexibility emphasized by Jaimovich & Rebelo (2009) with asymmetric income and substitution effects.

Theorem 7 in the Online Appendix characterizes the first-order general equilibrium response of consumption expenditure $d\ \log\ E_h$, firm sales $d\ \log\ S_i$, and factor prices $d\ \log\ w_h$ in a CES economy. The result shows that these adjustments propagate through interconnected partial and general equilibrium channels in response to exogenous shocks to: (i) firm-level productivity $d\ \log\ A_i$; (ii) firm-level markups $d\ \log\ \mu_i$; (iii) demographic composition $d\ \log\ n_h$; (iv) household equity participation $d\,\kappa_{ih}$; (v) lump-sum transfers $d\,T_h$; (vi) households' preference parameters $d\,\beta_{hi}$; and (vii) firms' technological parameters $d\,\omega_i^\ell$, $d\,\omega_{ij}$, and $d\,\alpha_{ih}$. This theorem complements the network decomposition framework in Baqaee & Farhi (2024) along several dimensions: (1) it relies on the modular bilateral centralities introduced in this paper, which decompose the propagation channels otherwise summarized by the network covariance operator in Baqaee & Farhi (2019); (2) it explicitly incorporates endogenous and heterogeneous factor supply responses across households, accounting for demographic, substitution, and income elasticities; (3) it separates the effects of each shock into direct impacts and propagation via input output linkages; and (4) it expresses all variations in nominal terms, rather than relative to Domar-weighted aggregates under the nominal GDP numéraire, thereby preserving flexibility over the normalization benchmark.

The expressions in Theorems 1 through 4 involve first-order variations in weights — such as $d\,\beta_{hi}$, $d\,\omega_i^\ell$, $d\,\omega_{ij}$, and $d\,\alpha_{ih}$ — as inputs to decompositions of household consumption, factor income, and aggregate output. In a CES environment, these weights serve as structural elasticities and coincide with equilibrium cost shares at the benchmark allocation. As elasticities, they represent deep parameters that vary only in response to exogenous preference or technology shocks. As cost share, however, they adjust endogenously in response to relative price changes in equilibrium, even when the underlying elasticities remain fixed. The decompositions in Theorems 1-4 remain valid regardless of whether variation arises from parameter shocks or from endogenous share adjustments.

Numéraire Choice and Non-Neutrality. To solve the model, take $2H+N-1$ of the equations in Theorem 7, and normalize the system by taking real output Y as the numéraire, implying $d\,log\,p_Y = 0$. This normalization follows Hulten (1978), Baqaee & Farhi (2019), and Bigio & La'O (2020), who likewise treat Y as the real unit of account. As discussed in Section 3, this is not the only convention: Baqaee & Farhi (2020, 2024) adopt nominal GDP as the numéraire, fixing nominal GDP, which implies $d\log Y = -d\log p_Y$. Proposition 3 characterizes how output variation differs under these two normalization.

Proposition 3. The ratio of first-order variations in real output Y, under a nominal GDP numéraire ($d\log GDP = 0$), relative to a real output numéraire ($d\log p_Y = 0$), is given by:

$$\frac{d\,log\,Y\,|_{\overline{GDP}}}{d\,log\,Y_{|\overline{p}_Y}} = \frac{g\,(d\log GDP = 0)}{g\,(d\log p_Y = 0)} \sum_{h\in\mathscr{H}} \widetilde{\Lambda}_h \frac{1+\zeta_h^e}{1+\zeta_h^w},$$

where $g(\cdot) = \sum_{i\in\mathscr{N}} \widetilde{\lambda}_i\, d\log A_i \mu_i + \sum_{h\in\mathscr{H}} \widetilde{\Lambda}_h \frac{\zeta_h^n\, d\log n_h - d\log\Lambda_h - \zeta_h^e\, d\log\chi_h}{1+\zeta_h^w}$, and is evaluated using the distributional responses implied by each normalization.

The choice of numéraire systematically influences the measurement of real output in general equilibrium. Two mechanisms explain these multiplicative differences across normalizations. First, alternative normalizations induce different endogenous changes in the distributions of factor income Λ and consumption expenditure χ as captured by $g(\cdot)$. Second, differences in factor supply elasticities matter: when income effect dominate substitution effects (i.e., $\zeta_h^e > \zeta_h^w$ for all $h\in\mathscr{H}$), measured real output is larger under a nominal GDP numéraire conditional on the same $g(\cdot)$. This second-channel discrepancy vanishes when income and substitution effects are symmetric (i.e., $\zeta_h^e = \zeta_h^w$ for all $h\in\mathscr{H}$). Unlike the decompositions in Baqaee & Farhi (2020, 2024), the sufficient statistic formulas in Theorems 1-4 are invariant to the choice of numéraire because they are derived using first-order decompositions for accounting identities. For the remainder of this paper, I adopt real output Y as the unit of account (so $d\log P_Y = 0$).

To illustrate, consider a representative household model with labor as the sole factor. in this environment, labor supply responds according to $d\log L = \frac{\zeta^w}{1+\zeta^w} d\log\Lambda + \frac{\zeta_h^w - \zeta_h^e}{1+\zeta_h^w} d\,log\,GDP$. Under a nominal GDP numéraire, an increase in the labor income share Λ (e.g., after a positive productivity shock) raises labor supply by $\frac{\zeta^w}{1+\zeta^w} d\log\Lambda$. Under real output numéraire, the same increase in Λ is accompanied by $d\log GDP = d\,log\,Y > 0$; if $\zeta_h^w > \zeta_h^e$, the GDP term is negative and partially offsets the first term, damping the labor supply response. Hence, for a given $d\log\Lambda$, normalizing p_Y yields a smaller labor supply response than normalizing with nominal GDP.

Nonlinearities. The sufficient statistic decompositions in Theorems 1-4 are linear approximations and are therefore valid only locally. To incorporate nonlinear general equilibrium effects, I employ the *differentiated hat-algebra* method developed by Baqaee & Farhi (2024) and Baqaee & Burstein (2025). The method captures nonlinearities by iteratively updating first-order response from Theorem 7 and recomputing endogenous variables after each round. This provides a tractable alternative to the *exact hat-algebra* approach commonly used in trade models Dekle et al. (2008); Costinot & Rodríguez-Clare (2014), which solves a nonlinear system of equilibrium conditions. Details of implementation are given in Section 3 of the Online Appendix. The main steps are as follows: (i) Initialize by computing the first-order responses $d \log S_i$, $d \log E_h$, and $d \log w_h$ for all $i \in \mathcal{N}$ and $h \in \mathcal{H}$ using Theorem 7. (ii) Use Proposition 1 to compute corresponding variations in equilibrium prices. (iii) Update endogenous cost share matrices β, $\widetilde{\Omega}_\ell$, and $\widetilde{\Omega}_x$ using observed price changes (e.g., $d \log \beta_{hi} = (\varrho - 1)\, d \log p^c_h / p_i$). (iv) Recompute equilibrium allocation using the update structural matrices. (v) Treat the updated structure as a new baseline and take variations in preferences and technologies as a new shocks. Repeat steps (i)-(v) until convergence in $\left(\beta, \widetilde{\Omega}_\ell, \widetilde{\Omega}_x\right)$.

4.2 Allocative Neutrality

Theorem 5 identifies four broad classes of economies in which first-order *allocative neutrality* holds, i.e., cases where the *allocative efficiency* term in Theorem 2 is zero to a first order. These cases identify benchmarks environment with specific combinations of elasticities, distortions, and network structures that neutralize the effect of reallocation on aggregate productivity, even though distributional outcomes may still change.

Theorem 5. For the following economies and classes of shocks, *allocative neutrality* holds:

1. A Cobb-Douglas economy ($\theta_i = \varrho_h = 1$ for all $i \in \mathcal{N}, h \in \mathcal{H}$), in response to productivity or demographic shocks.

2. A Leontief economy ($\theta_i = \varrho_h = 0$ for all $i \in \mathcal{N}, h \in \mathcal{H}$) with inelastic factor supply, in response to markup, preference, or technology shocks. For markup shocks, *allocative neutrality* holds provided either of the following conditions is met: (i) the share of household expenditure reaching factor compensation is constant across households, i.e., $\sum_{b \in \mathcal{H}} \mathscr{C}_{hb} = \tau$ for all $h \in \mathcal{H}$; or (ii) the consumer-to-factor centrality matrix $\mathscr{C}$ is nonsingular and $\mathscr{B}\,\Omega_\pi$ has all eigenvalues strictly inside the unit circle.

3. A horizontal economy with symmetric distortions ($\mu_i = \mu$ for all $i \in \mathcal{N}$), in response to productivity, markup, demographic, equity participation, lump-sum transfer, preference, or technology shocks.

4. In a vertical economy, in response to productivity, markup, demographic, equity participation, lump-sum transfer, preference, or technology shocks.

Cobb-Douglas Economy. In a Cobb-Douglas economy, there are no first-order allocative effects from exogenous productivity or demographic shocks. These shocks affect aggregate TFP only through the direct technology channel, since firm sales shares λ_i, factor income shares Λ_h, and household expenditure shares χ_h are invariant to relative price changes. Consequently, distributional reallocation effects also vanish, and households' PTT satisfies $d \log PTT_h =$

$Technology_h$ for all $h \in \mathscr{H}$. This result extends the neutrality benchmark in Baqaee & Farhi (2020) to settings with heterogeneous households and endogenous factor supply.

Leontief Economy. In a Leontief economy with inelastic factor supply, first-order allocative neutrality for markup shocks requires additional structural and distributional restrictions when households are heterogeneous. In the representative household case, Baqaee & Farhi (2020) show that markup changes leave real allocations unchanged up to a first order. Because both consumption and input bundles are fixed, markups shift prices but not quantities.

Once we consider household heterogeneity, the requirements for allocative neutrality depend on the type of shock. For preference and technology under inelastic factor supply, the exposure matrices, centralities, equilibrium factor income distribution Λ, and expenditure distribution χ are locally fixed. Consequently, there is no substitution margin — full complementarity blocks it — even though prices and wages may change. This logic can break down for markup distortions, which change relative prices and can reweight the factor income and expenditure distributions. As a result, with heterogeneous consumption bundles, shifts in the expenditure distribution tilt aggregate final demand while each household's bundle composition remains fixed. This reweights demand across firms and thereby induces input reallocation across them. Nevertheless, for these economies, the allocative efficiency term can be written as $d\,log\,Y = \sum_{h \in \mathscr{H}} \chi_h \left(1 - \mathscr{C}_h\right) d \log C_h$, where $\mathscr{C}_h = \sum_{b \in \mathscr{H}} \mathscr{C}_{hb}$ captures the share of h's expenditure reaching factor compensation.

This last equation reveals that real GDP can be sensitive to shifts in the distribution of real consumption — especially when a household has a relatively low incidence on factor income compensation (low $\mathscr{C}_h$). Reallocating purchasing power towards low $\mathscr{C}_h$ households can increase output. However, Theorem 5 establishes two sufficient conditions under which gains from reallocating expenditure across households are shut down. First, if the consumer-to-factor centrality matrix $\mathscr{C}$ is nonsingular and $\mathscr{B}\,\Omega_\pi$ has all eigenvalues strictly inside the unit circle, then first-order variations in expenditure across households imply *allocative neutrality*. This spectral condition ensures that dividend-income feedback loops are damped and convergent. For example, $\sum_{i \in \mathscr{N}} \mathscr{B}_{hi} \left(1 - \mu_i\right) \kappa_{ib}$ is the share of type-h consumption expenditure that reaches type-b's corporate income. Hence, this condition limits the strength of inter-household income feedback, preventing explosive mappings from expenditure to rent income. By the Gershgorin circle theorem (Gershgorin, 1931), a sufficient (but not necessary) condition for spectral stability of $\mathscr{B}\,\Omega_\pi$ is that sum of the off-diagonal elements in each row is less than one. For an economy without intermediate inputs, this condition holds because $\sum_{i \in \mathscr{N}} \mathscr{B}_{hi} \leq 1$ for all $h \in \mathscr{H}$.

Second, allocative neutrality holds when the share of expenditure that reaches factor compensation is the same for all households (i.e. $\mathscr{C}_h = \tau$ for all $h \in \mathscr{H}$). This is a stringent condition that clarifies the intuition behind this form of *allocative neutrality*: any redistribution of expenditure is neutralized to a first order in the aggregate when all households have identical expenditure-to-factor incidence. Importantly, this condition can hold even when consumption bundles differ across agents, so long as their total incidence on factor payments is symmetric.

Horizontal Economy. In horizontal economies — where all firms use only primary factors — *allocative neutrality* holds in response to productivity, markup, demographic, equity participation, lump-sum transfers, preference, and technology shocks, provided that markups are uniform across firms. Figure 2 illustrates the real flows in a horizontal economy with two firms and two labor types. Under uniform markups, households' marginal rates of substitution across goods and firms' marginal rates of technical substitution across inputs remain undistorted. As a result, the allocation of factors coincides with the first best, and reallocation effects are neutral for TFP up to a first order. Bigio & La'O (2020) show that in a horizontal representative household economy with a single endogenous labor type and uniform markups, markup shocks leave TFP unchanged to the first order. Theorem 5 generalizes this result to economies with heterogeneous households, multiple primary factors, and a broader set of shocks. This neutrality breaks down once intermediate inputs are introduced, as uniform markups no longer preserve firms' marginal rates of substitution between primary factors and intermediates.

Vertical Economy. In vertical economies — where only the most upstream firm uses primary factors and each downstream firm relies exclusively on intermediate inputs from its immediate upstream supplier — *allocative neutrality* holds in response to productivity, markup, demographic, equity participation, lump-sum transfer, preference, or technology shocks. Figure 3 illustrates the real flows in a vertical economy with two firms and two labor types. This is because both primary factors and intermediate inputs have unique allocations along the chain, leaving no room for misallocation. Bigio & La'O (2020) show that in a vertical representative household economy with a single type of endogenous labor, markup shocks leave TFP unchanged to a first order. Theorem 5 extends this allocative neutrality result to economies with heterogeneous households, multiple primary factors, and additional shocks.

Figure 2: Horizontal Economy

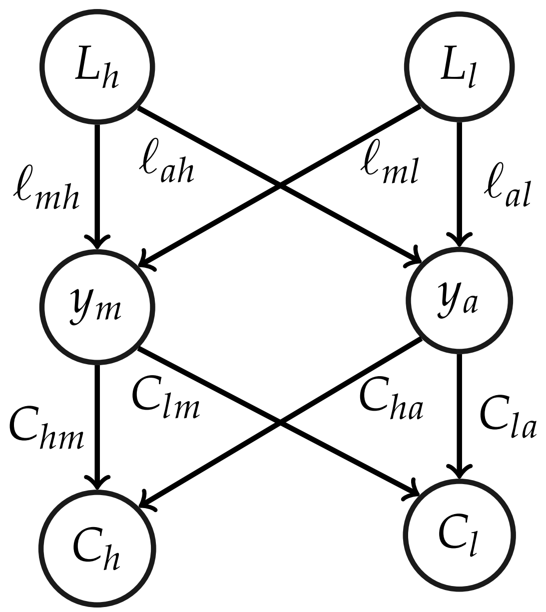
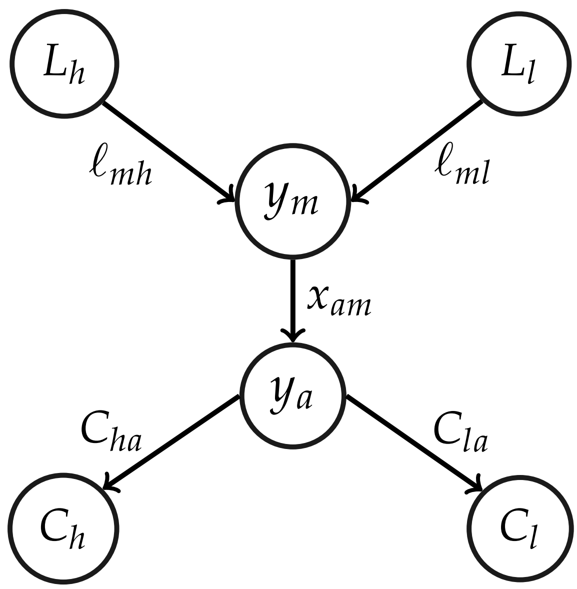


Figure 3: Vertical Economy



Note: Low-skill (l), high-skill (h), manufacturing (m), and agriculture (a).

4.3 Example 1: Redistribution and Aggregate Efficiency

To isolate the pure effect of redistribution on aggregate efficiency, consider a stylized Cobb-Douglas version of the horizontal economy in Figure 2 with two firms, manufacturing (m) and agriculture (a), and two household types, high-skill (h) and low-skill (l). Each firm employs a Cobb-Douglas production function with equal factor shares: $y_i = \sqrt{\ell_{ih}\,\ell_{il}}$ for $i \in \{m, a\}$. Households supply labor inelastically, hold equal equity shares, and consume via Cobb-Douglas preferences: $C_b = C_{bm}^{\beta_b} C_{ba}^{1-\beta_b}$ for $b \in \{h, l\}$. The government imposes a lump-sum transfer τ_h from low-skilled to high-skilled households. Heterogeneity arises from three sources: (i) consumption weights $\beta_h \neq \beta_l$, reflecting differences in preferences; (ii) firm-specific inverse markups $\mu_m \neq \mu_a$, which generate inefficiencies through misallocation; and (iii) redistribution via lump-sum transfers. I examine the introduction of an infinitesimal transfer $d\,\tau_h$, starting from a symmetric baseline with $\tau_h = 0$.

This configuration is analytically tractable and deliberately excludes all channels other than redistribution: no income or substitution effects, no intermediate inputs, no endogenous cost recomposition, and no initial heterogeneity in expenditure shares across households. As a result, any variation in aggregate output arises solely from transfer-induced reallocation across firms with different markups. Lump-sum transfers shift purchasing power across households with distinct preferences, inducing a reallocation of final demand between firms with heterogeneous distortions. This in turn alters the equilibrium allocation of labor across firms and affects aggregate efficiency. This illustrates how redistribution alone — holding productivity, markups, preferences, and technology fixed — can generate first-order effects on TFP.

In the symmetric equilibrium ($\tau_h = 0$), each worker type earns half of total labor income: $\Lambda_h = \Lambda_l = \frac{\mu_a}{2} + \frac{1}{4}(\beta_h + \beta_l)(\mu_m - \mu_a)$. They also generate equal value-added shares ($\widetilde{\Lambda}_h = \widetilde{\Lambda}_l = \frac{1}{2}$) and have the same expenditure shares ($\chi_h = \chi_l = \frac{1}{2}$). The manufacturing firm's sales share is the average of households' spending weights on manufacturing, $\lambda_m = \frac{1}{2}(\beta_h + \beta_l)$, while agriculture's is the complement $\lambda_a = 1 - \lambda_m$. The *expenditure centrality* of household $b \in \{h, l\}$ is $M_b = \frac{\mu_a + (\mu_m - \mu_a)\beta_b}{\mu_a + \frac{1}{2}(\beta_h + \beta_l)(\mu_m - \mu_a)}$. In response to a small positive transfer $d\,\tau_h$ (interpreted as a share of GDP), the endogenous variations are $d\,\chi_h = d\,\tau_h$ (so $d\,\chi_l = -d\,\tau_l$), $d\,\lambda_m = (\beta_h - \beta_l)\,d\,\tau_h$, and $d\,\Lambda_h = d\,\Lambda_l = \frac{1}{2}(\beta_h - \beta_l)(\mu_m - \mu_a)\,d\,\tau_h$. Applying the entropic and distributive terms-of-trade recompositions from Theorem 2, the resulting first-order change in output is

$$d \log Y = -\overbrace{\sum_{b \in \{h,l\}} \left(\widetilde{\Lambda}_b / \Lambda_b\right)\, d\,\Lambda_b}^{\text{Entropic TT}} = -\overbrace{(M_h - M_l)\, d\,\chi_h}^{\text{Distributive TT}} = \frac{(\beta_h - \beta_l)(\mu_a - \mu_m)}{\mu_a + \frac{1}{2}(\beta_h + \beta_l)(\mu_m - \mu_a)}\, d\,\tau_h.$$

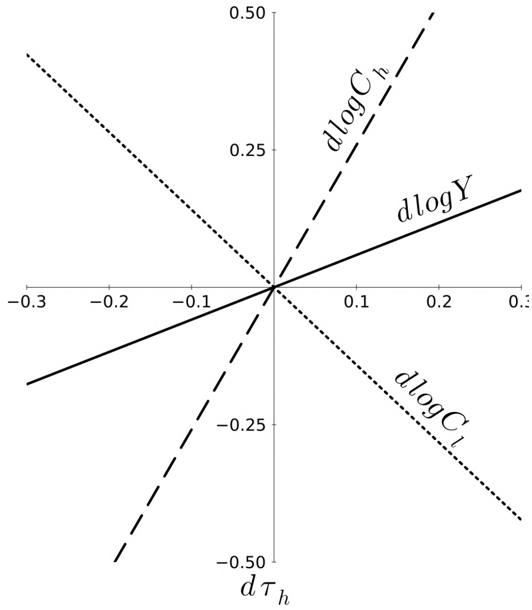
This expression highlights how the interaction between households' preferences and firms' distortions generates aggregate consequences from purely redistributive policies. When the household receiving the transfer has a stronger preference for the more distorted firm — i.e., $\beta_h > \beta_l$ and $\mu_a > \mu_m$, or $\beta_l > \beta_h$ and $\mu_m > \mu_a$ — the reallocation of expenditure increases aggregate efficiency. Demand shifts toward the firm where the marginal product of labor is higher in general equilibrium. Conversely, transferring resources toward households with greater demand for the less distorted firm reallocates labor inefficiently and reduces aggregate output.

From Theorem 4, the heterogeneous effects on real consumption across households are given by

$$d\log C_h = 2\, d\,\tau_h + d\log Y, \qquad d\log C_l = -2\, d\,\tau_h + d\log Y.$$

The second term in each expression reflects the common gain in aggregate efficiency, which accrues equally to both households. However, the increase in aggregate output is accompanied by a lower share of consumption expenditure for the taxed household. For example, if transfers from low- to high-skilled households ($d\,\tau_h > 0$) raise aggregate output, then real consumption increases for the high-skilled and falls for the low-skilled. This example illustrates that redistribution-driven efficiency gains are not necessarily zero-sum. Figure 4 plots the local, first-order responses around the symmetric equilibrium for $(\beta_h, \beta_l, \mu_m, \mu_a) = (0.8, 0.3, 0.2, 0.7)$. A marginal transfer to the high-skilled ($d\tau_h > 0$) raises aggregate output by reallocating expenditure toward the more distorted manufacturing firm. Consequently, the high-skilled gain more and offset the low-skilled's losses.

Figure 4: Output and Consumption



Note: Local (first-order) responses of output and household consumption to a marginal transfer to the high-skilled for $(\beta_h, \beta_l, \mu_m, \mu_a) = (0.8, 0.3, 0.2, 0.7)$. A positive transfer raises aggregate output (solid) and high-skilled consumption (dashed) while lowers low-skilled consumption (dotted); nonlinear and linearized responses are indistinguishable locally.

4.4 Example 2: Redistribution by the Constrained Social Planner

For the same economy as in Example 1, normalize productivities and fix labor supply so that $A_m = A_a = L_h = L_l = 1$. The efficient allocation (without distortions) satisfies $\lambda_m = y_m = \ell_{mh} = \ell_{ml} = \overline{\beta} = \frac{1}{2}(\beta_h + \beta_l)$ and $\lambda_a = y_a = 1 - \overline{\beta}$. Consequently, by Hulten (1978) and Theorem 2, the variation in real output is

$$d\log Y = \overline{\beta}\log A_m + (1 - \overline{\beta})\log A_a.$$

With distortions, the decentralized equilibrium is inefficient, with worker allocation $y_m = \frac{\overline{\beta}\,\mu_m}{\overline{\beta}\,\mu_m + (1-\overline{\beta})\mu_a}$, and $y_a = \frac{(1-\overline{\beta})\,\mu_a}{\overline{\beta}\,\mu_m + (1-\overline{\beta})\mu_a}$, and sales distribution $\lambda_m = \overline{\beta}$, $\lambda_a = 1 - \overline{\beta}$. Consistent with the implications of Theorem 5, worker misallocation arises whenever markups differ, $\mu_m \neq \mu_a$. The aggregate labor share equals the weighted average of inverse markups: $\Gamma = \overline{\beta}\,\mu_m + (1 - \overline{\beta})\mu_a$. With the real GDP aggregator $Y = \sqrt{C_h C_l}$, the ratio of inefficient to efficient output $\frac{Y}{Y_{\mathit{eff}}} = \frac{\mu_m^{\overline{\beta}}\,\mu_a^{1-\overline{\beta}}}{\overline{\beta}\,\mu_m + (1-\overline{\beta})\mu_a} \leq 1$, since the arithmetic mean exceeds the geometric mean. Moreover, by Theorem 2

$$d\,\log\,Y = \overline{\beta}\,\log\,A_m + (1-\overline{\beta})\,\log\,A_a + \frac{(\mu_a - \mu_m)}{\Gamma}\left(\overline{\beta}\,(1-\overline{\beta})\;d\,\log\,\frac{\mu_m}{\mu_a} + (\beta_h - \beta_l)\,d\,\chi_h\right).$$

This decomposition yields three key insights. (i) Higher firm productivity directly raises aggregate output, in proportion to the firm's Domar weight. (ii) Reducing markup dispersion raises aggregate output by easing allocative distortions. (iii) Shifting expenditure toward the household that spends more on the high markup firm increases output.

The constrained social planner in Theorem 3 can modify the equilibrium households' expenditure distribution via lump-sum transfers, because $\chi_h = \frac{1}{2} + \tau_h$ and $\chi_l = \frac{1}{2} - \tau_h$. One option is to restore the efficient allocation of labor, with $y_m = \overline{\beta}$ and $y_a = 1 - \overline{\beta}$, by choosing a transfers from low- to high-skill households equal to

$$\widetilde{\tau}_h = \frac{\overline{\beta}\,(1-\overline{\beta})}{\overline{\beta}\,\mu_a + (1-\overline{\beta})\mu_m}\left(\frac{\mu_a - \mu_m}{\beta_h - \beta_l}\right).$$

For a representative household economy, an intervention that restores the efficient allocation of inputs is sufficient to restore the first-best level of output. However, for an economy with heterogeneous households, where real GDP aggregates over all households' consumption, restoring input efficiency through lump-sum transfers can exacerbate consumption inequality and thereby lower aggregate output. For example, in our economy, $Y = \sqrt{C_h\,C_l}$, so when the social planner's efficient-worker-allocation transfer schedule is followed, real consumption has the potential to reduce Y. Figure 5 summarizes the planner's feasibility set, the required transfer schedule $\widetilde{\tau}_h$, and the output ratio $Y\left(\widetilde{\tau}_h\right)/Y(0)$ relative to the no-transfer equilibrium. For $|\widetilde{\tau}_h| > 1/2$ (gray region), the planner cannot shift enough expenditure from low- to high-skill households to offset markup asymmetry. In Panel A, the left subfigure plots $\widetilde{\tau}_h(\mu_a, \mu_m)$ as a function of inverse markups in manufacturing and agriculture (holding (β_h, β_l) fixed), while the right subfigure plots $\widetilde{\tau}_h(\beta_h, \beta_l)$ as a function of expenditure shares (holding (μ_m, μ_a) fixed). In Panel B, contours report $Y\left(\widetilde{\tau}_h\right)/Y(0)$, measuring the change in real output from implementing $\widetilde{\tau}_h$ relative to the no-transfer baseline. Values > 1 indicate output gains; values < 1 indicate losses; the 1.0 contours are the neutrality loci. Finally, Panel B's non-monotonicity reflects the efficiency-inequality tradeoff: small $|\widetilde{\tau}_h|$ move labor toward the more distorted firm (raising Y), while larger $|\widetilde{\tau}_h|$ increase dispersion in (C_h, C_l), which the aggregator penalizes.

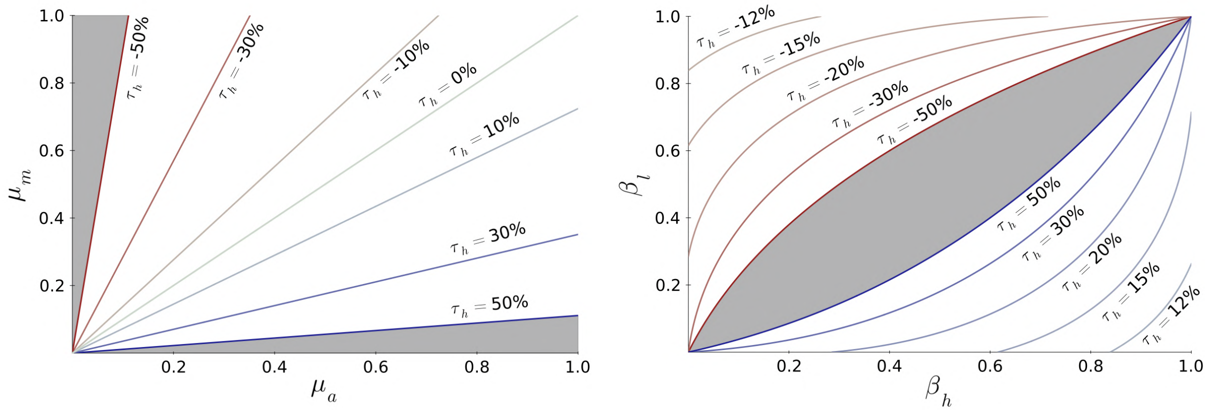
The optimal transfer balances the tradeoff between replicating the first-best allocation of labor and the induced rise in real consumption inequality. The optimal transfer is

$$\tau_h^* = \frac{1}{4}\frac{(\beta_h - \beta_l)\,(\mu_a - \mu_m)}{\overline{\beta}\,\mu_m + (1-\overline{\beta})\mu_a}.$$

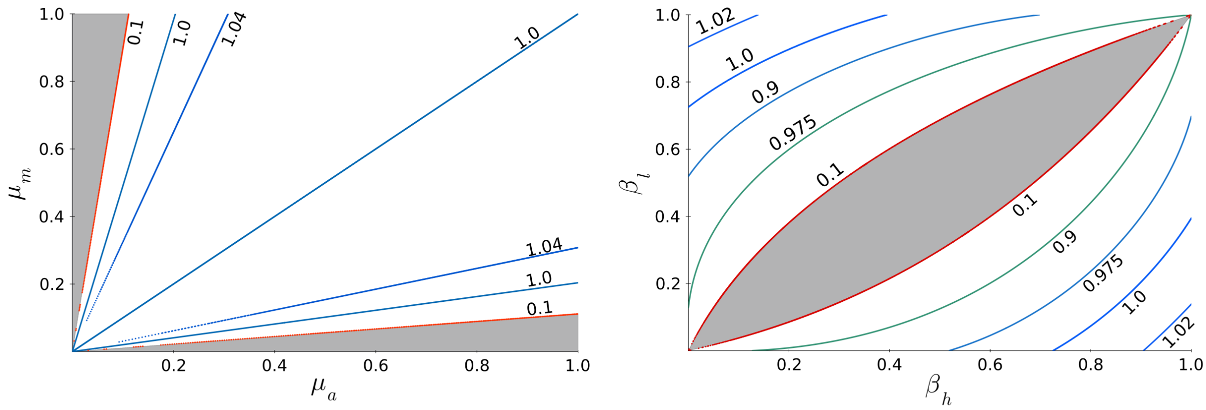
Figure 6 summarizes the transfer schedule τ_h^* and the output ratio $Y\left(\tau_h^*\right)/Y(0)$ relative to the no-transfer equilibrium. In Panel A, the left subfigure plots $\tau_h^*(\mu_a, \mu_m)$ as a function of inverse markups in manufacturing and agriculture (holding (β_h, β_l) fixed), while the right subfigure plots $\tau_h^*(\beta_h, \beta_l)$ as a function of expenditure shares (holding (μ_m, μ_a) fixed). In Panel B, contours

Figure 5: Transfers for Efficient Worker Allocation: Schedule and Output Effects

A. $\widetilde{\tau}_h$ Schedule as a Function of (μ_a, μ_m) and (β_h, β_l)



B. Output Ratio Relative to the No-Transfer Baseline



Note: Panel A (left): $\widetilde{\tau}_h(\mu_a, \mu_m)$ with $(\beta_h, \beta_l) = (0.9, 0.1)$. Panel A (right): $\widetilde{\tau}_h(\beta_h, \beta_l)$ with $(\mu_m, \mu_a) = (0.4, 0.6)$. Panel B reports $Y(\widetilde{\tau}_h)/Y(0)$ under the same parameter values. Shaded gray regions: efficient worker allocation infeasible ($|\widetilde{\tau}_h| > 1/2$). Positive $\widetilde{\tau}_h$ denotes transfers from low- to high-skill households.

report $Y(\tau_h^*)/Y(0)$ and show that the optimal transfer always increases aggregate output.

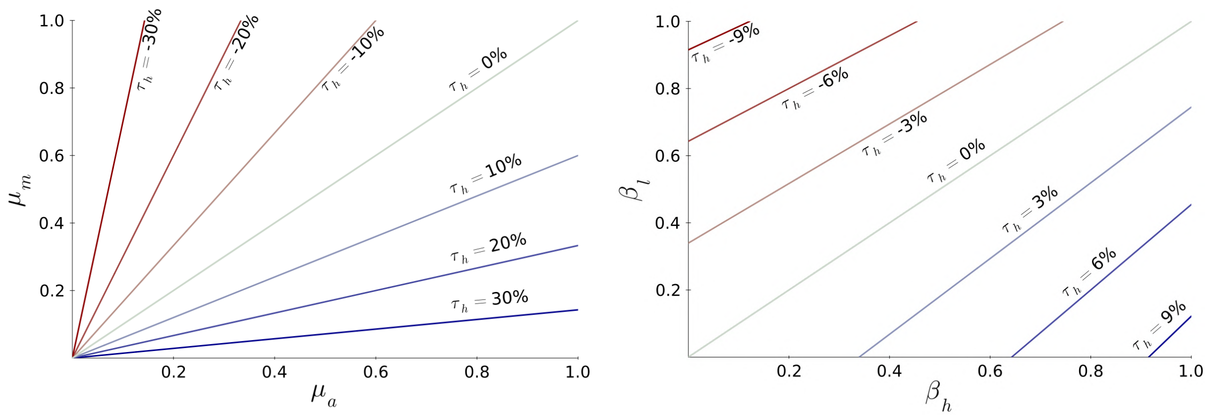
By implementing this transfer, the constrained social planner sets the manufacturing sales share and the aggregate labor share to $\lambda_m = \overline{\beta} + (\beta_h - \beta_l)\,\tau_h^*$ and $\Gamma = \overline{\beta}\,\mu_m + (1-\overline{\beta})\mu_a - (\beta_h - \beta_l)(\mu_a - \mu_m)\,\tau_h^*$. This implies that the higher markup firm expands and the labor share declines relative to the decentralized equilibrium. Accordingly, by Theorem 3, the first-order change in real output is

$$d\,\log\,Y = \overline{\beta}\,\log\,A_m + (1-\overline{\beta})\,\log\,A_a + (\beta_h - \beta_l)\,\tau_h^*\,\log\,\frac{A_m}{A_a} + (\mu_a - \mu_m)\,\frac{\lambda_m(1-\lambda_m)}{\Gamma} d\,\log\,\frac{\mu_m}{\mu_a}.$$

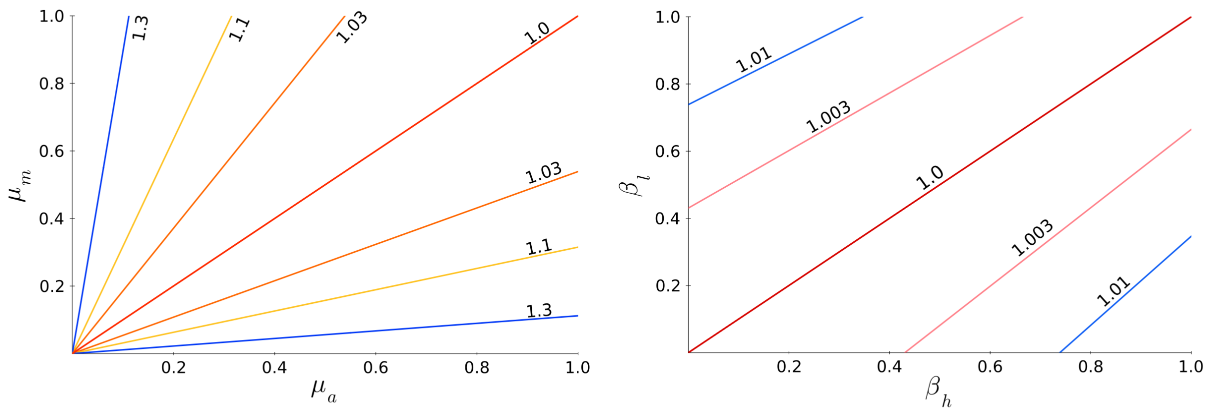
This decomposition yields three additional lessons. (i) The sufficient statistics for $d\,\log\,Y$ differ across the efficient benchmark, the decentralized no-transfer economy, and the optimal transfer economy. (ii) The marginal rates of substitution for consumption are equalized across households; therefore, expenditure reallocation is neutral under the optimal transfer. (iii) Under optimal transfers, movements of the production possibility frontier include an additional wedge linked to relative productivities: as the higher markup firm's relative productivity rises, this

Figure 6: Optimal Transfer: Schedule and Output Effects

A. τ_h^* Schedule as a Function of (μ_a, μ_m) and (β_h, β_l)



B. Output Ratio Relative to the No-Transfer Baseline



Note: Panel A (left): $\tau_h^* (\mu_a, \mu_m)$ with $(\beta_h, \beta_l) = (0.9, 0.1)$. Panel A (right): $\tau_h^* (\beta_h, \beta_l)$ with $(\mu_m, \mu_a) = (0.4, 0.6)$. Panel B reports $Y(\tau_h^*) / Y(0)$ under the same parameter values. Positive τ_h^* denotes transfers from low- to high-skill households.

wedge amplifies measured TFP relative to both the efficient and no-transfer economies. This example underscores the main lesson from Theorem 3: decentralized household decisions create externalities for aggregate output and alter the measurement of TFP. This is not a pecuniary externality, because prices and wages coincide across the baseline (no-transfer) and constrained social-planner economies. This mechanism operates purely through redistribution of purchasing power across households.

5 Quantitative Implementation

This section estimates three versions of the model that escalate household heterogeneity. First, I estimate a representative household benchmark to assess model fit and select the baseline specification. Second, I introduce geographic heterogeneity, where labor income and consumption bundles are state-specific. Third, I introduce occupational heterogeneity, in which earnings reflect each occupation's industry exposure, and consumption bundles vary systematically with income. Section 4 of the Online Appendix details the data sources and code used to estimate the model. For clarity, I introduce each dataset where it enters the model's parameterization.

5.1 Representative Household Model

I combine three BEA/BLS sources. First, the Industry-Level Production Accounts (ILPA) with Extended Capital Detail provide sectoral TFP levels and capital compensation for nine asset classes: communications equipment, computer hardware, R&D, software, entertainment originals, instruments and other office equipment, structures-land-inventories, transportation equipment, and other equipment. I harmonize these series to the Production Accounts industrial classification and build a balanced asset-industry-year panel (1997-2021). Second, BEA USE and MAKE tables (1997-2021) are restricted to the 66 private industries by dropping government and scrap/used categories. The row-normalized version of these tables is used to construct the industry-to-industry input output matrix $\mathscr{W}$. From these accounts, for each industry-year I construct: sales, value added, labor cost, intermediate input costs, capital costs (by nine assets). Inverse markups are μ = Cost/Sales with Cost = Labor + Capital + Intermediate Inputs. Tables 6 and 7 in the Online Appendix report summary statistics for inverse markups by sector and year. For 2004, the unweighted mean equals 0.95, the minimum is 0.71 in arts, sports, and museums, and the maximum is 1.00 in rail transport. Third, sectoral productivity growth is $d \log A_{i,t} = \log (TFP_{i,t}/TFP_{i,t-1})$, mapped from ILPA to the 66 Production Accounts industries. Tables 8 and 9 in the Online Appendix report productivity growth summary statistics by sector and year. For example, educational services has the lowest average productivity growth with -0.76% and computers and electronics has the highest at 4.22%.

The representative household model spans 66 sectors. The household receives income from labor, nine capital assets, and profits. I estimate and compare eight specifications formed by three design choices: (i) labor+capital vs. only labor income; (ii) intermediate production network vs horizontal (no intermediaries); and (iii) β_{IO} from final demand input output shares vs β_{PCE} from PCE-based household bundles mapped to industries. Each specification generates an annual series for $d\, log\, TFP$; I evaluate its fit against the BLS aggregate TFP growth series. The BLS benchmark is not a gold standard. This benchmark is an aggregate Solow residual estimated with Törnqvist chained weights for labor and capital income, i.e., Λ_{labor}/Γ, $\Lambda_{capital}/\Gamma$ with $\Gamma = \Lambda_{labor} + \Lambda_{capital}$. By contrast, the model's $d\, log\, TFP$ is not a residual and is constructed by accounting in Theorem 2 for industry-level variation in productivity, markups, and the observed factor income distribution. Table 3 compares eight model specifications using standard forecast metrics. Beyond traditional error magnitudes (MAE, RMSE, bias), I also report: (i) Nash-Sutcliffe Efficiency (NSE), which measures fit relative to the sample mean; (ii) Index of Agreement (IA), which bounds and normalizes potential errors; (iii) Theil's U (U_1), which provides a unit-free normalized RMSE (lower is better); and (iv) Kling-Gupta Efficiency (KGE), which jointly evaluates correlation, mean bias, and variability. The best-fitting specification across all metrics is column (4) — labor+capital, intermediate-input networks, and β_{IO}. Accordingly, I adopt specification (4) as the baseline for introducing household heterogeneity; in the representative household model, the comparison targets are $IA = 0.923$ and correlation $= 0.875$.

Table 3: **Comprehensive Model Comparison Metrics**

Metric	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
MAE	$6.2e^{-3}$	$1.5e^{-2}$	$6.2e^{-3}$	$4.7e^{-3}$	$7.7e^{-3}$	$1.8e^{-2}$	$7.6e^{-3}$	$7.4e^{-3}$
RMSE	$7.4e^{-3}$	$1.9e^{-2}$	$7.3e^{-3}$	$5.8e^{-3}$	$9.5e^{-3}$	$2.4e^{-2}$	$9.5e^{-3}$	$9.5e^{-3}$
Bias	$5.4e^{-3}$	$-8.9e^{-3}$	$5.3e^{-3}$	$2.1e^{-4}$	$6.9e^{-3}$	$-8.5e^{-3}$	$6.9e^{-3}$	$1.3e^{-3}$
Corr	0.871	0.702	0.873	0.875	0.748	0.597	0.748	0.767
NSE	0.417	-2.99	0.425	0.627	0.033	-5.330	0.029	0.032
IA	0.815	0.646	0.819	0.923	0.760	0.562	0.761	0.839
TU	0.353	0.459	0.349	0.209	0.430	0.53	0.429	0.320
KGE	0.319	-0.679	0.324	0.720	0.191	-1.082	0.190	0.429

Notes: Columns (1)–(8) correspond to: (1) Only labor & no networks + β_{IO}; (2) Only labor & networks + β_{IO}; (3) Labor + capital & no networks + β_{IO}; (4) Labor + capital & networks + β_{IO}; (5) Only labor & no networks + β_{PCE}; (6) Only labor & networks + β_{PCE}; (7) Labor + capital & no networks + β_{PCE}; (8) Labor + capital & networks + β_{PCE}. Lower is better for MAE and RMSE. Closer to 0 is better for Bias. The correlation lies in $[-1, 1]$; higher is better. NSE lies in $[-\infty, 1]$; higher values are better, with 1 indicating a perfect fit and 0 indicating no better than the mean. The IA lies in $[0, 1]$; higher values are better. TU lies in $[0, 1]$, and the lower, the better, with 0 indicating a perfect forecast. KGE lies in $[-\infty, 1]$ and the higher, the better. Section 4.7 of the Online Appendix defines and explains these metrics in more detail.

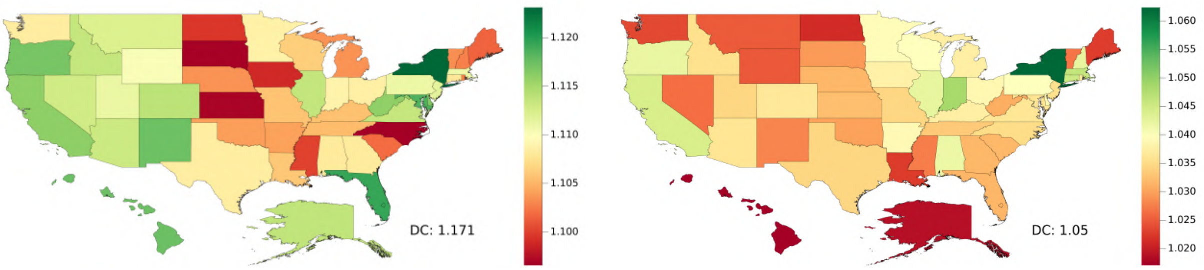
Using this specification, Tables 10 and 11 in the Online Appendix report revenue centralities $F_{i,t}$ by sector and year. Across years, the unweighted mean revenue centrality across sectors equals 0.99. The average across years ranges from 0.77 for arts, sports, and museums to 1.04 for support activities for mining. The lowest observed value is 0.70 (arts, sports, and museums, 2004), and the highest is 1.29 (securities, commodity contracts, and investments, 2008). Securities, commodity contracts, and investments has the most volatile revenue centrality, ranging from 0.83 to 1.29 over the sample. Housing, hospitals, credit intermediation, and chemical products are examples of sectors with high revenue centrality. Arts, sports, and museums, farms, primary metals, construction, air transportation, social assistance, securities, commodity contracts and investments, and ambulatory health care are examples of sectors with low revenue centrality. On average across years, the correlation between revenue centrality and inverse markups μ is 0.94. On average across years, the correlation between revenue centrality and Domar weights λ is 0.01. Year-to-year variation in sectoral revenue centrality is 79% explained by year-to-year changes in its own inverse markup. In other words, sectors with lower markups tend to have higher revenue centrality; variation in revenue centrality is largely orthogonal to size (Domar weights); and most within-sector variation in revenue centrality is explained by its own markup.

5.2 Geographic Heterogeneity

For each year t (1997-2021), I use QCEW to measure state s labor cost shares in industry i as $\alpha_{is,t} = \textit{Labor Income}_{is,t} / \sum_{s'} \textit{Labor Income}_{is',t}$, creating 52 labor types (50 states, District of Columbia, and a residual U.S. labor type) and stacking them with the nine capital-asset cost shares to form the industry-by-factor cost-share matrix α. State s consumption bundles $\beta^{PCE}_{s,t}$ are built from PCE commodity shares mapped to industries; the residual non-labor bundle is $\beta^{NL}_t = (1 - \sum_s \Lambda_{s,t})^{-1} (\beta^{IO}_t - \sum_s \Lambda_{s,t} \beta^{PCE}_{s,t})$ to reconcile aggregate final demand with the Production Accounts. The state-level model spans 66 sectors and 62 household types (52 labor, 9 capital owners, and 1 profit-receiving household). Relative to the representative household model, IA falls from 0.923 to 0.916, and the correlation from 0.875 to 0.867. These near-zero movements ($\Delta\, IA = -7e^{-3}$; Δ correlation $= -8e^{-3}$) indicate that introducing QCEW- and PCE-based state heterogeneity has a negligible effect on aggregate TFP fit.

Figures 7 and 8 display the state-level distortion (δ) and expenditure (M) centralities for 1997 and 2021. From Theorem 2, aggregate productivity rises via the entropic and distributive terms of trade when (i) labor income shifts from high-δ to low-δ workers, and (ii) consumption expenditure shifts from high-M to low-M households. In 1997, the three states with (i) the highest δ were DC, New York, and Florida; (ii) the lowest δ were North Carolina, South Dakota, and Kansas; (iii) the highest M were Arkansas, Wyoming, and DC; and (iv) the lowest M were Hawai'i, Alaska, and California. In 2021, the three states with: (i) the highest δ were New York, DC, and Indiana; (ii) the lowest δ were Hawai'i, Alaska, and North Dakota; (iii) the highest M were DC, Hawai'i, and Maryland; and (iv) the lowest M were Wyoming, Arkansas, and Nebraska. Classifying states by within-year quartiles of δ and M, four groups emerge. **1. High-δ, high-M states:** DC, Maryland (1997); California, DC, New York (2021). **2. High-δ, low-M states:** Alaska, California, Hawai'i, Oregon, West Virginia (1997); Illinois, New Hampshire, Wisconsin (2021). **3. Low-δ, high-M states:** Iowa, Kansas, North Carolina, North Dakota (1997); Hawai'i, New Mexico, Vermont (2021). **4. Low-δ, low-M states:** Maine, New Hampshire, Rhode Island (1997); Montana, Wyoming (2021).

Figure 7: Distortion Centrality by State: 1997 and 2021

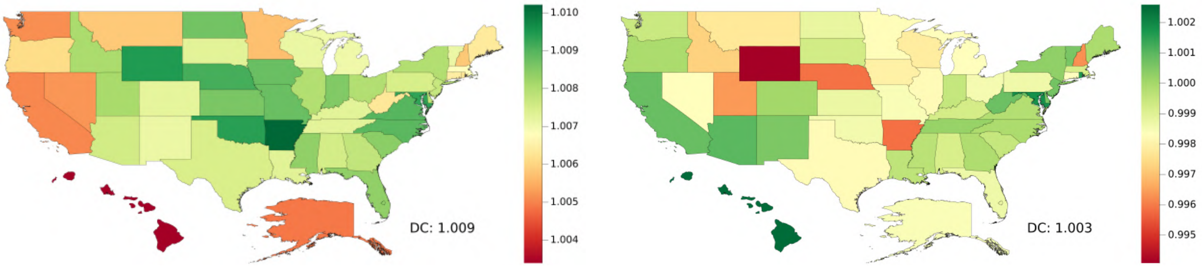


Note: Panels show the distortion centrality index $\delta_{s,t}$ for each state s. Left: 1997; Right: 2021. $\delta_{s,t}$ is computed as the value-added to factor income ratios $\widetilde{\Lambda}_{s,t}/\Lambda_{s,t}$, which are estimated taking the parameterization described above. Darker green (red) denotes high-(low-) expenditure centrality. Label next to "DC" gives its value.

Figures 16 - 21 in the Online Appendix show state-level distortion (δ) and expenditure (M) centralities for 1997–2021. Year-to-year levels and rankings are relatively stable, but comparing 1997 with 2021 reveals several shifts: (i) DC and New York are consistently high-δ, high-M; (ii) Alabama, Kansas, Maine, North and South Dakota, and the Carolinas are consistently low-δ; (iii) Alaska and Hawai'i move from high-δ, low-M to low-δ, high-M; (iv) Arkansas, Nebraska, and Wyoming move from high-M to low-M.

One of the most noticeable differences in the model's implications for the reallocation of expenditure is that, in 1997, TFP could improve through the distributive terms of trade by shifting expenditure from the Interior (Arkansas, Nebraska, Wyoming) to the Pacific states (California, Hawai'i, and Oregon), while in 2021, shifting expenditure from the Pacific states

Figure 8: Expenditure Centrality by State: 1997 and 2021



Note: Panels show the expenditure centrality index $M_{s,t}$ for each state s. Left: 1997; Right: 2021. $M_{s,t} = \sum_{h \in \mathscr{H}} \mathscr{C}_{sh,t}\ \delta_{h,t}$ stands for the weighted summation of distortion centralities, estimated under the parameterization described above. Darker green (red) denotes high-(low-) expenditure centrality. Label next to "DC" gives its value.

to the Interior had the potential to rise TFP. The main mechanism is a compositional shift: the Pacific states, relative to the Interior, increased expenditure in high-F (revenue centrality) sectors such as housing, while reducing expenditure in lower-F sectors such as ambulatory healthcare, and securities, commodity contracts, and investments. On the one hand, housing's share in the Pacific states rose by 1.4 percentage points (pp) (16.6% $\rightarrow$ 18.0%), while the Interior's fell by 0.6 pp (14.1% $\rightarrow$ 13.5%). On the other hand, (i) ambulatory health care in the Interior increased by 1.4 pp (5.3% $\rightarrow$ 6.4%), while the Pacific states rose by 0.7pp (7.6% $\rightarrow$ 8.3%); and (ii) securities, commodity contracts, and investments in the Interior increased by 1.3 pp (1.7% $\rightarrow$ 3.0%), while the Pacific states rose by 0.5 pp (1.0% $\rightarrow$ 1.5%). Hawai'i's M moves from among the lowest (1997) to the highest (2021), driven by its housing share rising from 16.9% to 20.1%.[8]

5.3 Occupational Heterogeneity

For each year t (1997-2021), I use BLS Occupational Employment and Wage Statistics (OEWS) industry-by-occupation employment and wage bills to measure occupation o labor cost shares within industry i as $\alpha_{io,t} = \textit{Labor Income}_{io,t} / \sum_{o'} \textit{Labor Income}_{io',t}$. These 833 occupation types are stacked with the nine capital asset classes to form the factor cost matrix α. To construct consumption bundles, I use the Consumer Expenditure Survey (CEX) to build an income-indexed map from income to a 66-industry share vector: for each household-year, form an income neighborhood of the 100 nearest-income households, aggregate their expenditures, and map commodity expenditure shares to industries. Let $\bar{J}_{o,t}$ denote mean OEWS earnings for occupation o in year t; set $\beta^{CEX}_{o,t}$ to the share vector from the nearest-income neighborhood. Define the residual non-labor bundle so that aggregate final demand matches the Production Accounts: $\beta^{NL}_t = (1 - \sum_s \Lambda_{s,t})^{-1} (\beta^{IO}_t - \sum_s \Lambda_{s,t} \beta^{CEX}_{s,t})$. Relative to the representative household baseline, IA declines from 0.923 to 0.878, and correlation from 0.875 to 0.824. These changes indicate that OEWS- and CEX-based occupation heterogeneity shift the model's implied TFP,

[8] *Housing*: Arkansas (13.9% $\rightarrow$ 12.5%); Nebraska (13.3% $\rightarrow$ 13.8%); Wyoming (15.6% $\rightarrow$ 14.3%); California (17.1% $\rightarrow$ 17.1%); Hawai'i (16.9% $\rightarrow$ 20.1%); Oregon (15.9% $\rightarrow$ 16.9%). *Ambulatory health care services*: Arkansas (6.5% $\rightarrow$ 7.6%); Nebraska (5.1% $\rightarrow$ 7.4%); Wyoming (4.4% $\rightarrow$ 5.2%); California (8.3% $\rightarrow$ 9.1%); Hawai'i (8.1% $\rightarrow$ 8.0%); Oregon (6.5% $\rightarrow$ 8.0%). *Securities, commodity contracts, and investments*: Arkansas (1.2% $\rightarrow$ 2.7%); Nebraska (1.9% $\rightarrow$ 2.7%); Wyoming (1.9% $\rightarrow$ 3.7%); California (1.3% $\rightarrow$ 2.0%); Hawai'i (0.9% $\rightarrow$ 1.4%); Oregon (0.9% $\rightarrow$ 1.3%).

while fit remains strong; lower scores do not necessarily signal model deterioration, since the BLS TFP is a proxy.

Table 4 decomposes the sample variance of $d \log TFP_t$ into the components of Theorem 2 for three models — representative household, geographic, and occupational heterogeneity. Directions and magnitudes are broadly similar across models, with modest shifts under occupational heterogeneity. The nine takeaways are: (1) Technology dominates, explaining 107% to 117% of the variance. (2) Competitiveness (-16% to -18%) and distributive reallocation (17% to 20%) offset one another. (3) Competitiveness is negatively correlated with $d \log TFP_t$, consistent with procyclical markups. (4) Distributional reallocation is positively correlated with $d \log TFP_t$, implying that factor income reallocation amplifies aggregate volatility. (5) The residual term accounts for -9% to -17%. (6) Distributive TT and final-demand TT have small shares; moving from geographic to occupational heterogeneity amplifies the role of distributive TT (0.06% to 0.56%), and final-demand TT (0.05% to -1.17%). (7) Intermediate-demand TT (23% to 30%) and factor-demand TT (-23% to -32%) offset each other. Because both enter with a leading minus sign in Theorem 2, intermediate-demand TT attenuates and factor-demand TT amplifies $d \log TFP_t$ volatility. The procyclicality of intermediate-demand TT is consistent with the procyclical input output multiplier in Ghassibe (2024). (8) Competitive TT is countercyclical (-15% to -17%), again consistent with procyclical markups. (9) The net markup contribution — competitiveness minus competitive TT — is near zero once the indirect channel is netted out (-0.58% to -0.83%). Thus, focusing solely on the direct effects from markups on competitiveness, is misleading. Across models, intermediate- and factor-demand TT rank second to technology, whereas markup effects and distributive/final-demand TT are small — indicating that firm-side demand/cost recomposition is first-order for $d \log TFP_t$, while markups, and household-side recomposition/redistribution are secondary.

These results indicate that the distributive terms of trade have played only a modest aggregate role. A small observed aggregate effect does not rule out meaningful heterogeneity in the sufficient statistics across households or the possibility of aggregate TFP gains from targeted expenditure reallocation. Tables 12, 13, and 14 in the Online Appendix report yearly summary statistics for occupational distortion centralities $\delta_{o,t}$, capital distortion centralities $\delta_{k,t}$, and occupational expenditure centralities $M_{o,t}$. Four patterns stand out. (1) Across all years, mean distortion centrality equals 1.09 for labor and 1.07 for capital, implying that labor (capital) aggregate value added is, on average, 9% (7%) above its compensation. (2) Both series decline over time: in 1997, there are 1.11 (labor) and 1.09 (capital), falling to 1.04 and 1.03 by 2021. This decline tracks an estimated fall in markups — i.e., higher inverse markups, μ — and coincides with more sectors showing operating losses ($\mu > 1$): 2 in 2013 (support activities for mining; rail transportation), 14 during 2014-2019, 24 in 2020, and 27 in 2021.[9] Whether this pattern reflects true profit margin compression, measurement issues at the profit-capital boundary in ILPA (e.g., how to classify capital compensation for structures, land, or office equipment?), or

[9] Before 2013, instances with $\mu > 1$ were rare: data processing, internet publishing, and other information services (1999-2001); securities, commodities, and contracts (2000, 2008, 2011); computer systems design and related activities (2000-2001); and motor vehicles, bodies and trailers, and parts (2009). Section 4 in the Online Appendix reports the industry-year observations with operating losses.

Table 4: d log TFP Variance Decomposition

	Representative Household	Geographic Heterogeneity	Occupational Heterogeneity
Technology	107.44%	108.79%	116.86%
Competitiveness	-15.94%	-18.19%	-17.52%
Distributional Reallocation	17.93%	20.04%	17.48%
Residual	-9.43%	-10.65%	-16.82%
Distributive TT	0.00%	0.06%	0.56%
Final-Demand TT	-0.25%	0.05%	-1.17%
Intermediate-Demand TT	27.09%	29.53%	23.24%
Factor-Demand TT	-29.41%	-32.10%	-23.42%
Competitive TT	-15.37%	-17.59%	-16.69%
Competitiveness − Competitive TT	-0.58%	-0.59 %	-0.83%

Notes: Entries report the share (percent) of the sample variance of d log TFP_t attributable to each component in Theorem 2. Columns correspond to the representative household economy and models with geographic and occupational heterogeneity. d log TFP_t comes from the BLS. *Residual* is given by d log TFP_t - *Technology* - *Competitiveness* - *Distributive Reallocation*. Positive values amplify aggregate d log TFP_t fluctuations; negative values offset them. The line *Competitiveness* - *Competitive TT* reports the net markup contribution after removing the indirect reallocation channel. By construction *Technology* + *Competitiveness* + *Distributive Reallocation* + *Residual* sum to 100%, and *Distributive Reallocation* = - *Distributive TT* - *Final-Demand TT* - *Intermediate-Demand TT* - *Factor-Demand TT* - *Competitive TT*.

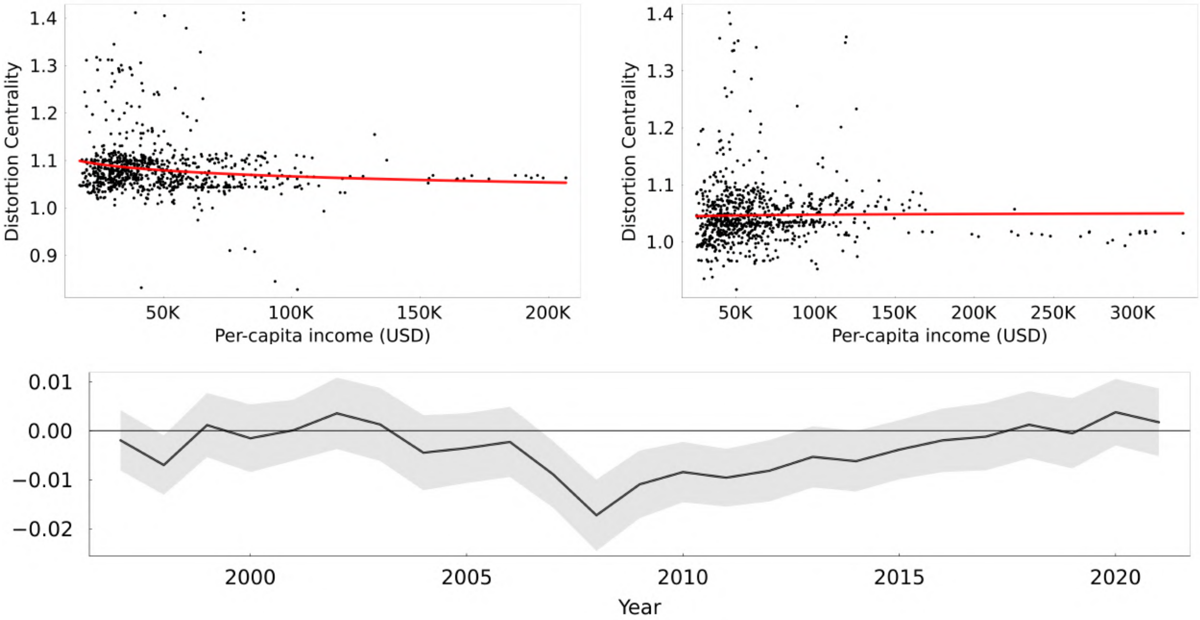
cross-subsidization across industries (e.g., retail platforms such as Amazon financing media endeavors such as Prime Video) is beyond the scope of this paper. (3) Across all years, mean expenditure centrality was 0.99, declining from 1.0 (1997) to 0.98 (2021). (4) Heterogeneity is wider under occupational heterogeneity than geographic heterogeneity. For example, in 2004, the max/min differences in δ and M under geographic heterogeneity were 6.7 pp and 1.4 pp, whereas under occupational heterogeneity they increased to 53 pp and 1.7 pp; standard deviations for δ are an order of magnitude larger than for M.

Figures 9 and 10 plot the cross-occupation relationship between earnings and (i) distortion centralities and (iii) expenditure centralities. For each year t, I estimate

$$\log\, \delta_{o,t} = \beta^{\delta}_{0,t} + \beta^{\delta}_{1,t}\, \log \bar{J}_{o,t} + \epsilon^{\delta}_{o,t} \quad \text{and} \quad \log\, M_{o,t} = \beta^{M}_{0,t} + \beta^{M}_{1,t}\, \log \bar{J}_{o,t} + \epsilon^{M}_{o,t},$$

where $\bar{J}_{o,t}$ is mean OEWS earnings for occupation o in year t. The top panels show the 2008 and 2021 cross-sections with fitted lines; the bottom panels report the yearly slope coefficients $\beta^{\delta}_{1,t}$ and $\beta^{M}_{1,t}$, with 95% confidence intervals. For distortion centralities, $\beta^{\delta}_{1,t} < 0$ in most years and is statistically different from zero in 1998 and 2007-2012. Thus, in those episodes, higher-earning occupations have lower $\delta_{o,t}$, so reallocating labor income toward high earners raises aggregate TFP through the distributional reallocation channel. For expenditure centralities, $\beta^{M}_{1,t}$ is positive and highly significant for all years except 2021; it declines through the 2010s, and flips sign in 2021. When $\beta^{M}_{1,t} > 0$, earnings and $M_{o,t}$ are positively related, so reallocating expenditure from high- to low-earning occupations shifts demand from high-M to low-M consumers and raises TFP. Put differently, in the pre-2021 cross-sections, the model implies a *lower-income multiplier*: strengthening purchasing power in lower-earning occupations raises productivity.

Figure 9: Occupation-Level Distortion Centrality and Earnings: 2008/2021 Cross-Sections and Elasticity Over Time



Note: Top planels plot cross-sectional scatters for occupations in 2008 (left) and 2021 (right); each dot is one of 833 SOC occupations. The red line is the fitted line from the year-specific log-log fit $\log \delta_{o,t} = \beta^{\delta}_{0,t} + \beta^{\delta}_{1,t} \log \bar{J}_{o,t} + \epsilon^{\delta}_{o,t}$, where $\delta_{o,t}$ is the occupation-level distortion centrality and $\bar{J}_{o,t}$ is mean OEWS earnings. The bottom panel reports the estimated elasticity $\beta^{\delta}_{1,t}$ by year (1997-2021); the shaded band shows 95% pointwise confidence intervals. Negative (positive) $\beta_{1,t}$ implies higher-income occupations are associated with lower (higher) distortion centrality in year t.

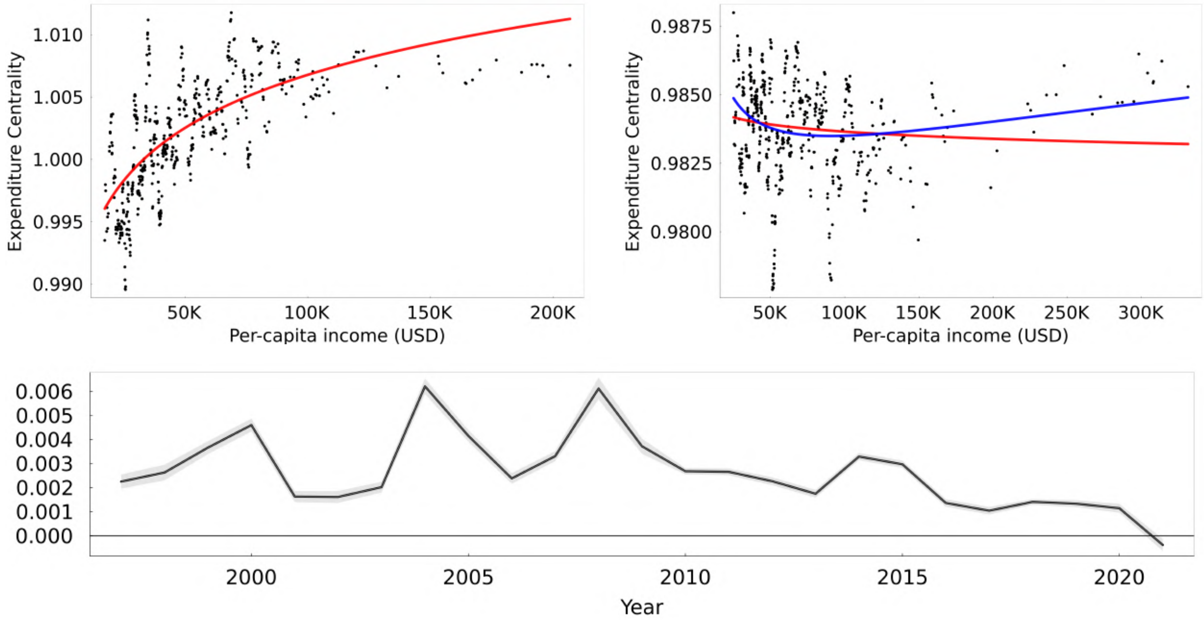
The decline in $\beta^{M}_{1,t}$ over the 2010s foreshadows a change in the shape of this relationship, culminating in 2021. In 2021, the quadratic-in-log fit (blue) is distinctly U-shaped, with a minimum around \$50-\$100k in annual earnings. Consequently, in 2021, the most powerful distributive term-of-trade lever is to shift spending toward the middle-income segment. Put differently, the evidence points to a *middle-income multiplier*: reweighting spending toward middle-income occupations boosts productivity. This interpretation reconciles the negative linear slope in 2021 with the preceding trend: the gradual decline in $\beta^{M}_{1,t}$ through the 2010s signals the transition from a monotone to a U-shaped cross-section.

These M-hierarchies in expenditure centrality reveal the direction and (partial equilibrium) sign of the impact of reallocating spending across households on aggregate productivity. Still, they abstract from general equilibrium (GE) feedbacks driven by input output price adjustments, factor income reallocation, and substitution across inputs. For this reason, I proceed to use CES model to simulate counterfactual expenditure reallocations based on the empirical M-hierarchies.

6 From M to GE: Distributional Engines of Productivity

In this section, I calibrate the normalized nested CES model introduced in Section 4 in the geographic heterogeneity specification to match BLS aggregate TFP (1997-2021). I then use

Figure 10: Occupation-Level Expenditure Centrality and Earnings: 2008/2021 Cross-Sections and Elasticity Over Time



Note: Top planels plot cross-sectional scatters for occupations in 2008 (left) and 2021 (right); each dot is one of 833 SOC occupations. The red line is the year-specific log-log fit $\log M_{o,t} = \beta_{0,t}^M + \beta_{1,t}^M \log \bar{J}_{o,t} + \epsilon_{o,t}^M$, where $M_{o,t}$ is the occupation-level expenditure centrality and $\bar{J}_{o,t}$ is mean OEWS earnings. In the 2021 panel, the blue curve plots the quadratic log-log fit $\log M_o = \beta_0^M + \beta_1^M \log \bar{J}_o + \beta_2^M \left(\log \bar{J}_o\right)^2 + u_o^M$. The bottom panel reports the estimated elasticity $\beta_{1,t}^M$ by year (1997-2021); the shaded band shows 95% pointwise confidence intervals. Negative (positive) $\beta_{1,t}$ implies higher-income occupations are associated with lower (higher) distortion centrality in year t.

this model to quantify general equilibrium effects in two counterfactual designs. First, I study aggregate and distributional effects of a revenue-neutral lump-sum transfer that reallocates expenditure along empirical M-hierarchies using simple, rule-based allocations. Second, I superimpose the same transfers on the historical sequence of productivity and markup shocks in the U.S., assessing their interaction with observed business-cycle forces. The main lesson is that the long-standing equity-efficiency tradeoff articulated by Okun (1975) is apparent: targeted redistribution of expenditure along M-hierarchies yields modest aggregate efficiency gains alongside sizable progressive redistribution. In other words, in these simulations, reductions in consumption expenditure inequality are achievable with at most negligible efficiency costs, and often with slight efficiency gains.

6.1 Calibrating the Economy

I calibrate the normalized nested CES economy by estimating four substitution elasticities — the economy-wide elasticity between primary factors and intermediates (θ), the within primary factor elasticity (θ_ℓ), the within intermediate elasticity (θ_x), and the household consumption elasticity (ϱ). These parameters are collected in $\boldsymbol{\theta} = (\theta, \theta_\ell, \theta_x, \varrho)$, are assumed common across sectors and households, and the search is restricted to $[0,3]^4$. Parameters are estimated by GMM, minimizing $Q(\boldsymbol{\theta}) = g(\boldsymbol{\theta})' W g(\boldsymbol{\theta})$ with $W = I$, where the year-t moment is $g_t(\boldsymbol{\theta}) = d \log TFP_t^{model}(\boldsymbol{\theta}) - d \log TFP_t^{BLS}$ stacked over $t \in \{1997, \cdots, 2020\}$. The model implied $d \log TFP_t^{model}(\boldsymbol{\theta})$ is computed using the same sector-specific estimated productivity and markup shock sequences $d \log A_t$ and $d \log \mu_t$, held fixed across candidate $\boldsymbol{\theta}$. To navigate this potential nonconvex optimization landscape, Stage 1 conducts a global search with adaptive differential evolution; Stage 2 refines the best candidate via gradient-based L-BFGS with automatic differentiation, both under the parameter bounds. Section 4 of the Online Appendix documents the full optimization method.

Table 5: Bootstrap Parameter Estimates and Model Fit Statistics

Parameter	Mean	Std. Dev.	Best	95% CI
θ	0.076	0.058	0.01	[0.005, 0.234]
θ_ℓ	1.482	0.825	0.940	[0.282, 2.855]
θ_x	1.954	0.425	2.398	[1.015, 2.710]
ϱ	0.466	0.346	0.073	[0.015, 1.249]

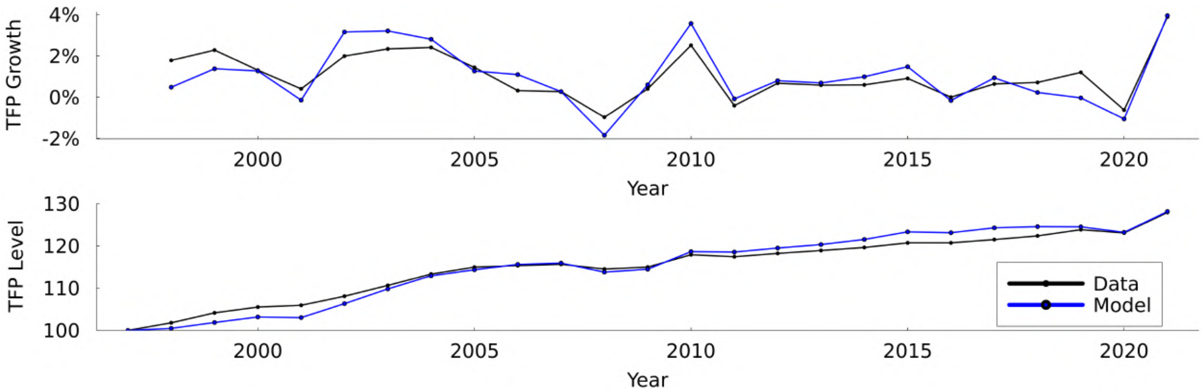
	MAE	RMSE	Bias	Corr	NSE	IA	TU	KGE
Mean	5.32e-03	6.68e-03	3.95e-05	0.892	0.641	0.928	0.204	0.695
Best	5.20e-03	6.55e-03	-7.19e-05	0.891	0.655	0.930	0.201	0.716

Note: First table presents parameter estimates from a bootstrap procedure in which 1000 independent GMM estimations are conducted on the identical full sample (1997—2020, 24 observations), without resampling of time periods. Each iteration employs a two-stage optimization: (i) adaptive differential evolution for global search, and (ii) L-BFGS with automatic differentiation for local refinement. The "Mea'n' column reports the arithmetic average of all 1,000 parameter estimates, while the "Best" column reports the parameters yielding the minimum GMM objective value across all iterations. The 'Std. Dev." column presents the standard deviation of the bootstrap distribution, and the "95% CI" column provides the percentile-based 95% confidence interval (2.5th and 97.5th quantiles). Parameter variation reflects uncertainty rather than temporal or sampling variation. The second table compares eight model fit metrics evaluated on the full sample using the mean and the best bootstrap parameters. The metrics are: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Bias (mean prediction error), Correlation coefficient, Nash-Sutcliffe Efficiency (NSE), Index of Agreement (IA), Theil's U statistic (TU), and Kling-Gupta Efficiency (KGE).

Table 5 reports the distribution of elasticity estimates from 1,000 independent estimations of the GMM problem using the unchanged 1997-2020 sample. Each run uses the same annual moments and the same estimated sectoral productivity and markup shocks, imposes bounds $[0,3]^4$, and follows the same two-stage optimizer. The "Mean" column reports the average across the 1,000 runs; "Best" denotes the parameter vector with the lowest GMM objective. Estimates indicate strong factor-intermediate complementarity ($\theta \approx 0.01$), within primary elasticity slightly below unity ($\theta_\ell \approx 0.94$), high substitutability across intermediates ($\theta_x \approx 2.40$), and a low consumption aggregation elasticity ($\varrho \approx 0.07$) implying high complementarity across household goods. These economy-wide TFP-matching GMM estimates compare to two previous studies. First, Boehm et al. (2014) estimates $\theta \approx 0.03$ and $\theta_x \approx 0.2 - 0.6$ from a firm-level natural experiment based on the 2011 Tōhoku earthquake, finding near Leontief substitution. Second, Atalay (2017) estimates $\theta \approx 0.8 - 0.9$ and $\theta_x \approx 0 - 0.2$ using an industry-level IV price-share regression, instrumenting with military spending demand shocks that propagate through the input-output network. Relative to these papers, the estimates imply stronger factor-intermediate complementarity and substantially higher intermediate substitutability; θ_ℓ and ϱ are not directly identified or targeted in those studies.

Model fit is strong for both the mean and best parameter vectors; for best, the correlation is 0.89 and the index of agreement (IA) is 0.93, exceeding the baseline nonparametric representative household model (0.87 and 0.92, respectively). Figure 11 contrasts annual TFP growth (top panel) with the TFP level indexed to 100 in 1997 (bottom panel). The model reproduces the pre-2008 expansion, the 2008-09 contraction, the 2010 rebound, and the 2021 surge. In levels, the model tracks the BLS TFP index closely and ends within roughly 0.2 index points of the data by 2021. These results support using the model for counterfactual analysis of distributional shocks.

Figure 11: Model Data vs Fit for U.S. TFP 1997-2021



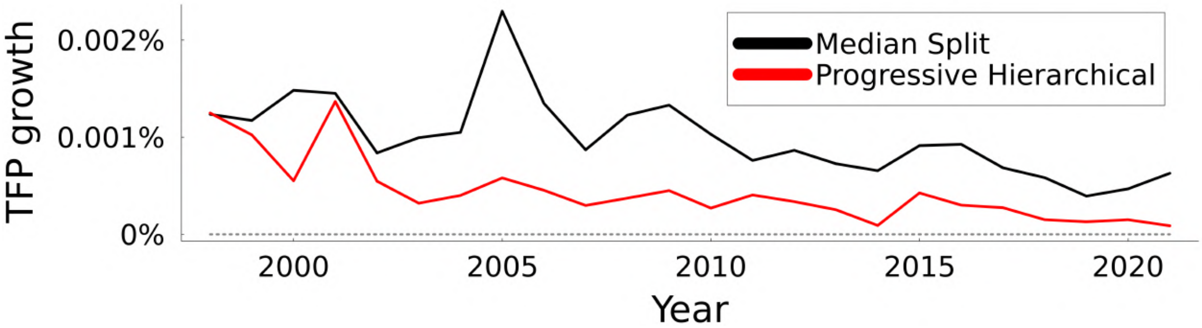
Note: The model series are computed at the bootstrap parameter set that minimizes the GMM objective over 1,000 independent runs on the unchanged 1997—2020 sample. Model growth is generated using the same estimated shock sequences $\{d \log A_t, d \log \mu_t\}$ across runs. The lower panel cumulates growth (index normalized to 100 in 1997). Data are reported BLS TFP variation.

6.2 Pure Redistribution

To evaluate the general equilibrium effects of pure expenditure redistribution, I consider two budget-neutral allocation rules defined over a vector of expenditure centralities (M), where its dimension is the number of workers, and workers in the same occupation share the same M value. First, the *median split rule* taxes workers with M above the median and transfers an equal amount to each worker below the median. Second, the *progressive hierarchical matching rule* permits reallocation only when both the worker's expenditure centrality and the taxed household per capita income exceed those of the recipient. Pairs are formed sequentially along the M ranking, exhausting capacity for high-M donors and low-M recipients first.

Figure 13 plots annual TFP growth from implementing only a \$1,000 transfer each year. To eliminate scale effects, the scale of the transfer implemented each year will be a constant GDP share calibrated to the 2019 share of \$1,000; i.e., $T_t = \tau \, GDP_t$ with $\tau \equiv 1000/GDP_{2019}$. The effect on TFP growth is positive in every year under both allocation rules, with modest magnitudes consistent with the small policy share τ. The median split rule yields larger gains than the progressive hierarchical rule, reflecting differences in the total magnitude of transfers as progressivity leaves some workers unmatched. The response peaks in the early to mid-2000s and declines thereafter. The reported gains arise from reallocation across sectors and input tiers induced by the expenditure reallocation.

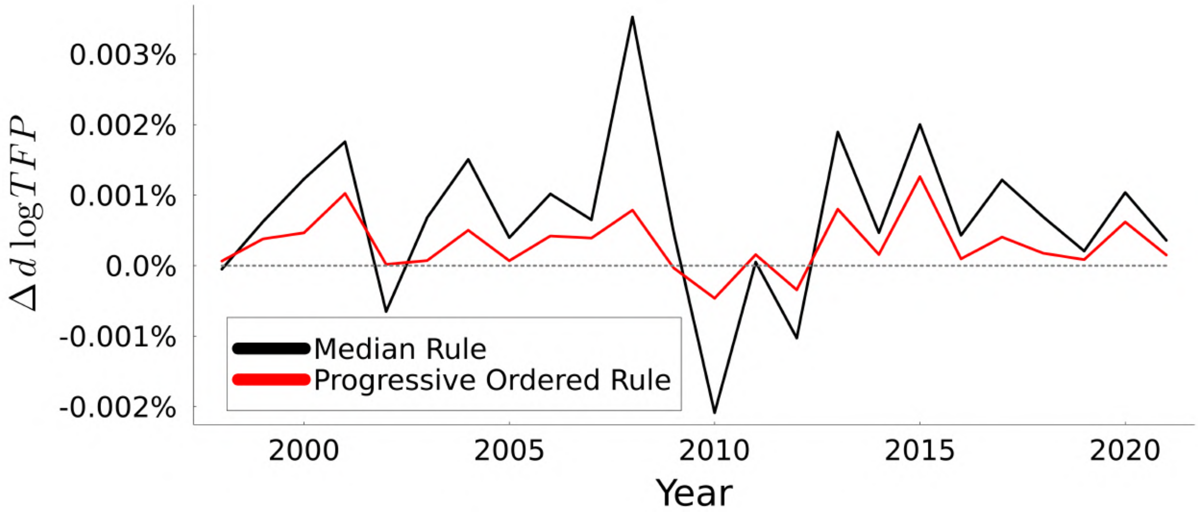
Figure 12: TFP Growth Response to a $1,000 Transfer, by Year



Note: The figure plots the change in annual aggregate TFP growth from a counterfactual transfer implemented each year as a constant GDP share equal to the 2019 share of a $1,000 transfer: $\tau \equiv 1000/GDP_{2019}$ and $T_t = \tau Y_t$, implying $\tau \approx 5.63 \times 10^{-11}$.

6.3 Redistribution with Business Cycle Shocks

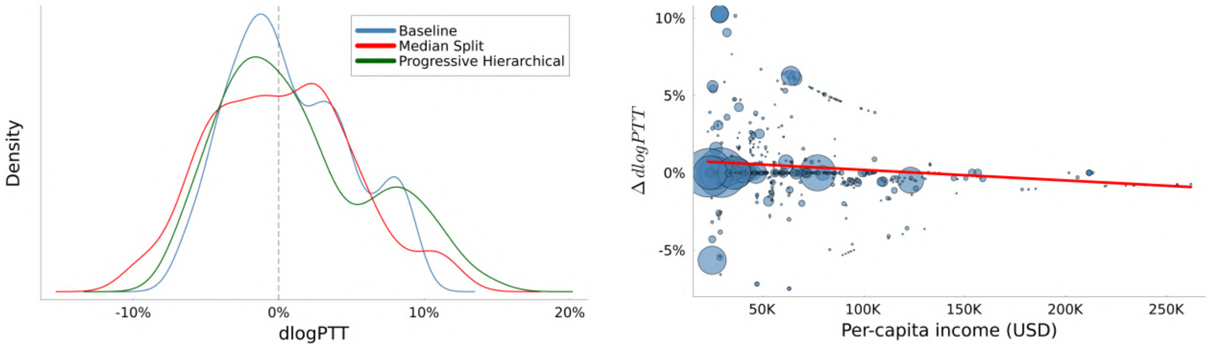
Figure 13: $\Delta\, d\, log\, TFP$ ($1,000 Transfer - Baseline), by Year



Note: The figure plots the annual difference in TFP growth between the transfer counterfactual and the no-transfer baseline: $\Delta\, d \log TFP_t = d \log TFP_t(\tau) - d \log TFP_t(0)$. Each year, the transfer is implemented as a constant GDP share equal to the 2019 share of a $1,000 transfer: $\tau \equiv 1000/GDP_{2019}$ and $T_t = \tau Y_t$, implying $\tau \approx 5.63 \times 10^{-11}$. Sectoral productivity and markup shocks follow their estimated sequences and are identical in the transfer and baseline. The figure isolates the effect of the transfer via reallocation.

Now, I allow for sectoral productivity and markup shocks to follow the same estimated sequences, and on top of those shocks I add expenditure reallocation driven by the median split and progressive hierarchical rules. Figure 13 reports $\Delta\, d\, log\, TFP_t = d \log TFP_t(\tau) - d \log TFP_t(0)$ when each year's transfer is set to a constant GDP share equal to the 2019 share of a $1,000 transfer. The effect is positive in 21 of 24 years under each allocation rule; only three realizations are negative. Hence, even after accounting for contemporaneous technology and markup disturbances, simple consumption allocation rules based on M-hierarchies systematically raise TFP through reallocation. Magnitudes are modest — consistent with the small policy share τ — and the median split rule typically delivers larger gains than the progressive hierarchical rule.

Figure 14: d log PTT Distributions and Progressive Differential by income (2019)



Note: Left figure plots the worker-weighted distribution of occupation level d log PTT (in percent) for a given year under three scenarios: the baseline economy with no trasnfers (blue), the median-split transfer rule (red), and the progressive hierarchical transfer rule based on expenditure centrality (green). Each density is constructed using occupation employment counts as weights and trimmed to exclude the bottom and top 2.5% of cumulative weighted mass to reduce the influence of outliers. The dashed vertical line marks zero, so shifts to the right (left) indicate positive (negative) changes in PTT on average. The right figure plots the occupation-level change in $d \log PTT_{Progressive} - d \log PTT_{Baseline}$ between the progressive hierarchical rule and baseline scenario without transfers against occupation-per-capita labor income for 2019. Each circle represents an occupation, with marker size proportional to the occupation's employment. The red line is the population-weighted least squares linear fit (weighted by employment), summarizing how $\Delta\, d \log PTT$ comoves with income across occupations.

Figure 14 summarizes occupation level heterogeneity in d log PTT for 2019. The left panel shows worker-weighted densities for the baseline (no transfers), the median split rule, and the progressive hierarchical rule. Transfers shift the densities to the right and thicken the upper tail, indicating that modest aggregate TFP effects mask substantial idiosyncratic movements in PTT across occupations. These shifts are more pronounced under the progressive hierarchical rule. The right panel plots the occupation level differential $\Delta\, d \log PTT_i = d \log PTT_i^{Prog} - d \log PTT_i^{Base}$ against occupation per capita labor income; bubble size is employment. The employment weighted OLS fit (red) has a negative slope, significant at the 1% level, implying that lower-income occupations experience on average systematically larger PTT gains under progressive hierarchical relative to baseline.

Taken together, the counterfactuals show that the equity-efficiency trade off is largely apparent in this environment. Simple, budget neutral reallocations of expenditure guided by sufficient-statistics targeting (the M-hierarchies) raise aggregate efficiency in the vast majority of years, even after embedding the transfers in the sequence of estimated productivity and markup shocks. At the same time, the distributional evidence reveals meaningful incidence at the micro level: transfers shift the occupation level distributions of d log PTT to the right, fatten the upper tails, and the progressive hierarchical rule deliver large PTT gains for lower-income occupations. In short, targeting based on expenditure centrality can jointly advance efficiency and equity, raising TFP on average while improving PTTs in a progressive manner, thereby providing a practical policy level that achieves both goals.

7 Conclusion

These findings underscore a broad principle: the distribution of income and expenditures can be a determinant of aggregate productivity. In a distorted production network, who earns and who spends has first-order effects on allocative efficiency. When factor income shifts from high- to low-distortion-centrality occupations or when demand reallocates toward low expenditure centrality households and low revenue centrality firms, the economy alleviates bottlenecks and achieves higher total factor productivity. This reframing of the equity-efficiency tradeoff shows that inequality and misallocation are deeply linked in general equilibrium: changes in the joint distribution of earnings and spending can move the economy to a more efficient allocation along the production possibility frontier. By deriving sufficient statistics for these effects, the theory bridges distributional macroeconomics and misallocation in a networked environment, offering a general toolkit for diagnosing how inequality affects output in any second-best equilibrium.

Empirical evidence underscores the significance of these mechanisms. Estimates from U.S. data indicate that shifts in the income distribution account for roughly 20% of TFP variation from 1997 to 2021. Counterfactual policy experiments further demonstrate that targeted redistribution guided by the model's network centrality measures can raise both efficiency and equity. Simple, budget-neutral lump-sum transfers directed from high- to low-expenditure centrality consumers consistently increase aggregate TFP while improving the real incomes of lower-income households. These results suggest that the conventional equity-efficiency tradeoff need not bind in distorted economies: well-designed redistribution can simultaneously reduce inequality and boost productivity. More broadly, these results point to a new perspective for macroeconomic policy that leverages the structure of production networks and demand to align equitable outcomes with aggregate efficiency gains.

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Online Appendix

1 Proofs for the nonparametric model

1.1 Firms

Firm $i \in \mathscr{N}$ chooses $\left\{\{\ell_{ih}\}_{h\in\mathscr{H}}, \{x_{ij}\}_{j\in\mathscr{N}}\right\}$ to minimize $\bar{c}_i = p_i^\ell L_i + p_i^x X_i = \sum_{h\in\mathscr{H}} w_h \ell_{ih} + \sum_{j\in\mathscr{N}} p_j x_{ij}$ subject to $1 = A_i Q_i(L_i, X_i)$, $L_{z_i} = A_i^\ell Q_i^\ell\left(\left\{A_{ih}^\ell \ell_{ih}\right\}_{h\in\mathscr{H}}\right)$, $X_i = A_i^x Q_i^x\left(\left\{A_{ij}^x x_{ij}\right\}_{j\in\mathscr{N}}\right)$, $mc_i = \mu_i p_i$, and taking prices $\left\{\{w_h\}_{h\in\mathscr{H}}, \{p_j\}_{j\in\mathscr{N}}\right\}$ as given.

The optimality conditions for L_i and X_i are

$$p_i^\ell = \varkappa_i A_i \frac{\partial Q_i(L_i, X_i)}{\partial L_i}, \quad \text{and} \quad p_i^x = \varkappa_i A_i \frac{\partial Q_i(L_i, X_i)}{\partial X_i},$$

where $\varkappa_i$ denotes the lagrange multiplier. Multiplying both conditions by L_i and X_i, summing, and CRS in $Q_i(\cdot)$, we obtain: $\mu_i p_i = p_i^\ell L_i + p_i^x X_i = \varkappa_i A_i Q_i(L_i, X_i) = \varkappa_i$.

The first-order conditions are given by

$$p_i^\ell = \mu_i p_i A_i \frac{\partial Q_i(L_i, X_i)}{\partial L_i}, \quad \text{if} \quad \frac{\partial Q_i(L_i, X_i)}{\partial L_i} > 0, \tag{19}$$

$$p_i^x = \mu_i p_i A_i \frac{\partial Q_i(L_i, X_i)}{\partial X_i}, \quad \text{if} \quad \frac{\partial Q_i(L_i, X_i)}{\partial X_i} > 0, \tag{20}$$

$$w_h = \mu_i p_i A_i \frac{\partial Q_i(L_i, X_i)}{\partial L_i} A_i^\ell \frac{\partial Q_i^\ell\left(\left\{A_{ib}^\ell \ell_{ib}\right\}_{b\in\mathscr{H}}\right)}{\partial \ell_{ih}} \qquad \forall h \in \mathscr{H}: \frac{\partial y_i}{\partial \ell_{ih}} > 0, \tag{21}$$

$$p_j = \mu_i p_i A_i \frac{\partial Q_i(L_i, X_i)}{\partial X_i} A_i^x \frac{\partial Q_i^x\left(\{A_{im}^x x_{im}\}_{m\in\mathscr{N}}\right)}{\partial x_{ij}} \qquad \forall j \in \mathscr{N}: \frac{\partial y_i}{\partial x_{ij}} > 0. \tag{22}$$

Let elasticities be defined as $e(a,b) = (\partial a/\partial b)(b/a)$. The first-order conditions for firm i can equivalently be expressed as $\omega_i^\ell = e(y_i, L_i) = \frac{1}{\mu_i}\frac{p_i^\ell L_i}{p_i y_i}$, $\omega_i^x = e(y_i, X_i) = \frac{1}{\mu_i}\frac{p_i^x X_i}{p_i y_i}$, $\alpha_{ih} = e(L_i, \ell_{ih}) = \frac{w_h \ell_{ih}}{p_i^\ell L_i}$, $\omega_{ij} = e(X_i, x_{ij}) = \frac{p_j x_{ij}}{p_i^x X_i}$, $\widetilde{\Omega}_{ih}^\ell = e(y_i, \ell_{ih})$, and $\widetilde{\Omega}_{ij}^x = e(y_i, x_{ij})$.

Microfoundation for Distortions via Dixit & Stiglitz (1977)

For every firm $i \in \mathscr{N}$, a perfectly competitive aggregator chooses $\left\{y_i, (y_{z_i})_{z_i\in[0,1]}\right\}$ to maximize $\bar{\pi}_i = p_i y_i - \int p_{z_i} y_{z_i}\, dz_i$ by consuming intermediate varieties produced by its own plants subject to $y_i = \left(\int y_{z_i}^{\mu_i}\, dz_i\right)^{\frac{1}{\mu_i}}$ and taking prices $\left\{p_i, (p_{z_i})_{z_i\in[0,1]}\right\}$ as given. The first-order condition yields the standard CES demand function $y_{z_i} = \left(\frac{p_i}{p_{z_i}}\right)^{\frac{1}{1-\mu_i}} y_i \quad \forall z_i \in [0,1]$, which implies that $\frac{\partial p_{z_i}}{\partial y_{z_i}} = -(1-\mu_i)\left(\frac{y_i}{y_{z_i}}\right)^{1-\mu_i}\frac{p_i}{y_{z_i}}$ and $p_i = \left(\int p_{z_i}^{\frac{\mu_i}{\mu_i-1}}\, dz_i\right)^{\frac{\mu_i-1}{\mu_i}}$.

The revenue derivative of plant z_i' with respect to any variable q is given by

$$\frac{\partial p_{z_i} y_{z_i}}{\partial q} = \left(p_{z_i} - (1-\mu_i)\left(\frac{y_{z_i}}{y_i}\right)^{\mu_i-1} p_i\right)\frac{\partial y_{z_i}}{\partial q} = \mu_i p_{z_i}\frac{\partial y_{z_i}}{\partial q}.$$

Moreover, the first-order conditions for plant z_i are identical to those in equations (19)-(22).

1.2 Households' Problem

Household $h \in \mathscr{H}$ chooses $\left\{\{C_{hi}\}_{i\in\mathscr{N}}, L_h\right\}$ to maximize $U_h\left(C_h, L_h\right)$ subject to $C_h = Q_h^c\left(\{C_{hi}\}_{i\in\mathscr{N}}\right)$ and the budget constraint $E_h = p_h^c C_h = \sum_{i\in\mathscr{N}} p_i C_{hi} \leq w_h L_h + \sum_{i\in\mathscr{N}} \kappa_{ih}\, \pi_i + T_h$. The household takes as given $\left\{w_h, T_h, \{p_i, \kappa_{ih}, \pi_i\}_{i\in\mathscr{N}}\right\}$.

The first-order conditions for household $h \in \mathscr{H}$ are given by

$$\frac{w_h}{p_h^c} = -\frac{U_{L_h}}{U_{C_h}}, \tag{23}$$

$$\frac{p_i}{p_h^c} = \frac{\partial C_h}{\partial C_{hi}} \quad \forall i \in \mathscr{N} : \frac{\partial C_h}{\partial C_{hi}} > 0. \tag{24}$$

It follows that the expenditure share of good i in household h's consumption is given by $\beta_{hi} = e\left(C_h, C_{hi}\right) = \frac{p_i C_{hi}}{p_h^c C_h}$. Substitutabilities are captured by $e\left(C_h, L_h\right) = \frac{w_h L_h}{p_h^c C_h}$, and the cross-elasticity condition $e\left(C_{hi}, C_{hm}\right) + \frac{p_m C_{hm}}{p_i C_{hi}} = 0$ holds.

1.3 General Equilibrium Conditions

Proposition 4. The allocation and price system constitute an equilibrium if and only if markets clear, the fiscal budget is balanced, and the following marginal conditions are jointly satisfied:

$$\frac{\frac{\partial C_h}{\partial C_{hj}}}{\frac{\partial C_h}{\partial C_{hi}}} = \mu_i \frac{\partial y_i}{\partial x_{ij}} \quad \forall i, j \in \mathscr{N} : \; C_{hi} > 0, \; C_{hj} > 0, \; x_{ij} > 0, \tag{25}$$

$$-\frac{w_b}{w_h}\frac{U_{L_h}}{U_{C_{hi}}} = \mu_i \frac{\partial y_i}{\partial \ell_{ib}} \quad \forall i \in \mathscr{N}, \; \forall h, b \in \mathscr{H} : \; C_{hi} > 0, \; \ell_{ib} > 0, \; U_{L_h} < 0. \tag{26}$$

1.3.1 Proof of Necessity

First, using equations (22) and (65), I derive the first subset of conditions from Proposition 4

$$\frac{\partial C_h / \partial C_{hj}}{\partial C_h / \partial C_{hi}} = \frac{p_j}{p_i} = \mu_i \frac{\partial y_i}{\partial x_{ij}} \quad \forall i, j \in \mathscr{N}, \; \forall h \in \mathscr{H} \text{ when } C_{hi} > 0, \; C_{hj} > 0, \text{ and } x_{ij} > 0. \tag{27}$$

This condition states that, for any firm i and any household h, the household's marginal rate of substitution between goods i and j must match the cost ratio implied by firm i's optimal sourcing of inputs. Notice that the first equality requires household h to consume from both firms i and j, while the second requires only that firm i demands inputs from firm j.

Second, combining equations (21), and (64) yields

$$-\frac{w_b}{w_h}\frac{U_{L_h}}{U_{C_{hi}}} = \frac{w_b}{p_i} = \mu_i \frac{\partial y_i}{\partial \ell_{ib}} \quad \forall i \in \mathscr{N}, \; \forall h, b \in \mathscr{H} \text{ when } C_{hi} > 0, \; U_{L_h} \neq 0, \text{ and } \ell_{ib} > 0. \tag{28}$$

Note that this second subset of equilibrium conditions links firm i's demand for factor b to household h's marginal rate of substitution between factor supply and consumption of goods from firm i. It is sufficient that firm i demands factor b and that household h consumes from firm i. When $b \neq h$, the wedge w_b / w_h reflects distributional differences in factor prices across household types.

Finally, the following market-clearing and fiscal balance conditions must hold: (i) the goods market clearing condition $y_i = \sum_{h\in\mathscr{H}} C_{hi} + \sum_{j\in\mathscr{N}} x_{ji}$ for all $i \in \mathscr{N}$; (ii) the factor market clearing condition $L_h = \sum_{i\in\mathscr{N}} \ell_{ih}$ for all $h \in \mathscr{H}$; and (iii) the balanced budget condition $\sum_{h\in\mathscr{H}} T_h = 0$.

1.3.2 Proof of Sufficiency

Now, I prove that for any exogenous set of distortions and equity shares $\{\mu_i, \{\kappa_{ih}\}_{h\in\mathscr{H}}\}_{i\in\mathscr{N}}$, there exists a strictly positive price system $\{p_i\}_{i\in\mathscr{N}}, \{w_h\}_{h\in\mathscr{H}}$, that implements a given allocation for firms $\left\{y_i, \{\ell_{ih}\}_{h\in\mathscr{H}}, \{x_{ij}\}_{j\in\mathscr{N}}\right\}_{i\in\mathscr{N}}$, and a household allocation $\{C_h, \{C_{hi}\}_{i\in\mathscr{N}}, L_h\}_{h\in\mathscr{H}}$ as a competitive equilibrium.

Assume, without loss of generality, that prices are normalized such that a CRS function defines the aggregate GDP deflator $\overline{p}_Y = Q^p\left(\{p_i\}_{i\in\mathscr{N}}\right) = 1$. Also assume there exists at least one "essential" factor of production $h_* \in \mathscr{H}$ — meaning all firms are directly or indirectly exposed to its value-added. Using equations (21) and (22), the price for firm i is given by

$$\begin{aligned} p_i &= \frac{w_{h_*}}{\mu_i}\left(\frac{\partial y_i}{\partial \ell_{ih_*}}\right)^{-1} \quad \text{if} \quad \ell_{ih_*} > 0 \quad \text{otherwise} \\ p_i &= \frac{w_{h_*}}{\mu_i}\left(\frac{\partial y_i}{\partial x_{i\underline{j}}}\right)^{-1}\left(\frac{\partial y_{\overline{j}}}{\partial \ell_{\overline{j}h_*}}\right)^{-1}\prod_{j\in\mathscr{N}_{z_i}}\frac{1}{\mu_j}\prod_{j\in\mathscr{N}_i\setminus\{\overline{j}\}}\left(\frac{\partial y_j}{\partial x_{j\,j+1}}\right)^{-1} \end{aligned} \tag{29}$$

where $\mathscr{N}_{z_i} = \left\{\underline{j}, \underline{j}+1, \cdots, \overline{j}-1, \overline{j}\right\}$ denotes a chain of firms connecting i to the supply of factor h_* via intermediate input demand.

From the price normalization, the price for factor h_* is given by

$$\begin{aligned} w_{h_*}^{-1} = Q^p \Bigg(\Bigg\{\frac{1}{\mu_i}\Bigg(&\mathbb{1}\{\ell_{ih_*} > 0\}\left(\frac{\partial y_i}{\partial \ell_{ih_*}}\right)^{-1} \\ +\mathbb{1}\{\ell_{ih_*} = 0\}&\left(\frac{\partial y_i}{\partial x_{i\underline{j}}}\right)^{-1}\left(\frac{\partial y_{\overline{j}}}{\partial \ell_{\overline{j}h_*}}\right)^{-1}\prod_{j\in\mathscr{N}_{z_i}}\frac{1}{\mu_j}\prod_{j\in\mathscr{N}_i\setminus\{\overline{j}\}}\left(\frac{\partial y_j}{\partial x_{j\,j+1}}\right)^{-1}\Bigg)\Bigg\}_{i\in\mathscr{N}}\Bigg). \end{aligned} \tag{30}$$

Hence, $w_{h_*} = 1$, and equation (29) implies that all firm prices are strictly positive, since marginal productivities of both factors and intermediate inputs are strictly positive. It then follows from equation (21) that all factor wages are strictly positive as well.

It remains to show that, under the system of prices defined by equations (29) and (30), the optimality conditions for firms and households hold given the equilibrium conditions in equations (27) and (28), the market clearing conditions, and the balanced budget constraint. In particular, I must verify that the allocation satisfies the first-order conditions for both producers and consumers, thereby establishing that the allocation and prices together constitute a competitive equilibrium.

To derive equations (64) and (65), assume that firm i directly or indirectly demands factor h, and firm j demands factor b. This assumption is made without loss of generality, as it holds for any combination of $i, j \in \mathscr{N}$ and $h, b \in \mathscr{H}$. Substituting the equilibrium relationships from equations (27) and (28) into the pricing expression in equation (29) yields

$$p_i = -\,w_b\,\frac{U_{C_{bi}}}{U_{L_b}}, \quad \text{and} \quad p_j = -w_b\,\frac{U_{C_{bj}}}{U_{L_b}}.$$

From this, I obtain that $\frac{U_{C_{bj}}}{U_{C_{bi}}} = \frac{p_j}{p_i}$. Multiplying both sides by C_{hi} and summing over all firms $j \in \mathscr{N}$ yields

$$\frac{\partial C_h}{\partial C_{hi}}\sum_{j\in\mathscr{N}} p_j C_{hj} = p_i \sum_{j\in\mathscr{N}} C_{hi}\frac{\partial C_h}{\partial C_{hj}} \quad \to \quad U_{C_{bi}}\,p_h^c\,C_h = p_i\,C_h \quad \to \quad \frac{p_i}{p_h^c} = \frac{\partial C_h}{\partial C_{hi}},$$

which is equation (65). Moreover, from equation (28), I recover $\frac{w_h}{p_h^c} = -\frac{U_{L_h}}{U_{C_h}}$, which corresponds to equation (64).

For firms, begin by noting that $\frac{U_{C_{bj}}}{U_{C_{bi}}} = \frac{p_j}{p_i}$. Substituting into equation (27) yields $\frac{p_j}{p_i} = \mu_i \frac{\partial y_i}{\partial x_{ij}}$, which corresponds to equation (22). To derive equation (20), sum (22) across all intermediate inputs

$$\sum_{j\in\mathcal{N}} p_j\, x_{ij} = \mu_i\, p_i \sum_{j\in\mathcal{N}} \frac{\partial y_i}{\partial x_{ij}}\, x_{ij} \quad \rightarrow \quad p_i^x\, X_i = \mu_i\, p_i \frac{\partial y_i}{\partial X_i}\, X_i \quad \rightarrow \quad p_i^x = \mu_i\, p_i \frac{\partial y_i}{\partial X_i}.$$

Second, take $p_i = -w_h \frac{U_{C_{hi}}}{U_{L_h}}$ and use equation (28) to derive $\frac{w_h}{p_i} = \mu_i \frac{\partial y_i}{\partial \ell_{ih}}$, which corresponds to equation (21). To derive equation (19), sum (21) across all households

$$\sum_{h\in\mathcal{H}} w_h\, \ell_{ih} = \mu_i\, p_i \sum_{h\in\mathcal{H}} \frac{\partial y_i}{\partial \ell_{ih}}\, x_{ij} \quad \rightarrow \quad p_i^\ell\, L_i = \mu_i\, p_i \frac{\partial y_i}{\partial L_i}\, i \quad \rightarrow \quad p_i^x = \mu_i\, p_i \frac{\partial y_i}{\partial L_i}.$$

What remains to be verified is that the household budget constraints hold. Summing equation (3) across all households yields

$$\begin{aligned}
\sum_{h\in\mathcal{H}} \sum_{i\in\mathcal{N}} p_i\, C_{hi} &= \sum_{h\in\mathcal{H}} \left(w_h\, L_h + \sum_{i\in\mathcal{N}} \kappa_{ih}\, \pi_i + T_h \right) \\
\sum_{h\in\mathcal{H}} \sum_{i\in\mathcal{N}} p_i\, C_{hi} &= \sum_{h\in\mathcal{H}} w_h\, L_h + \sum_{i\in\mathcal{N}} \pi_i \underbrace{\sum_{h\in\mathcal{H}} \kappa_{ih}}_{=1} + \underbrace{\sum_{h\in\mathcal{H}} T_h}_{=0} \\
\sum_{h\in\mathcal{H}} \sum_{i\in\mathcal{N}} p_i\, C_{hi} &= \sum_{h\in\mathcal{H}} w_h\, L_h + \sum_{i\in\mathcal{N}} p_i\, y_i - \sum_{i\in\mathcal{N}} \sum_{h\in\mathcal{H}} w_h\, \ell_{ih} - \sum_{i\in\mathcal{N}} \sum_{j\in\mathcal{N}} p_j\, x_{ij} \\
0 &= \sum_{i\in\mathcal{N}} p_i \underbrace{\left(y_i - \sum_{h\in\mathcal{H}} C_{hi} - \sum_{j\in\mathcal{N}} x_{ji} \right)}_{=0} + \sum_{h\in} w_h \underbrace{\left(L_h - \sum_{i\in\mathcal{N}} \ell_{ih} \right)}_{=0}.
\end{aligned}$$

1.4 Equilibrium Centralities from Subsection 2.5

Goods market equilibrium conditions. Substituting equations (20), (22), and (65) into the goods market clearing condition (4) for firm $i \in \mathcal{N}$

$$S_i = \sum_{h\in\mathcal{H}} p_i\, C_{hi} + \sum_{j\in\mathcal{N}} p_i\, x_{ji} = \sum_{h\in\mathcal{H}} \beta_{hi}\, E_h + \sum_{j\in\mathcal{N}} \mu_j\, \omega_j^x\, \omega_{ji}\, S_j = \sum_{h\in\mathcal{H}} \beta_{hi}\, E_h + \sum_{j\in\mathcal{N}} \Omega_{ji}^x\, S_j, \tag{31}$$

where $\Omega_{ij}^x \equiv \mu_i\, \omega_i^x\, \omega_{ij}$.

In matrix form, the goods market equilibrium condition can be written as $\left(I_N - \widetilde{\Omega}_x'\, diag(\mu)\right) S = \beta'\, E$ or $S = \mathscr{B}' E$, where $S \equiv [\, S_1, \cdots, S_N]'$, $E \equiv [\, E_1, \cdots, E_H]'$, and $\mu \equiv [\, \mu_1, \cdots, \mu_N]'$. The relevant matrices are defined as

$$\beta \equiv \begin{pmatrix} \beta_{11} & \cdots & \beta_{1N} \\ \vdots & \ddots & \vdots \\ \beta_{H1} & \cdots & \beta_{HN} \end{pmatrix}, \quad \Omega_x \equiv diag(\mu)\, \widetilde{\Omega}_x, \quad \Psi_x \equiv (I_N - \Omega_x)^{-1}, \quad \mathscr{B} \equiv \beta\, \Psi_x.$$

Dividing both sides by nominal GDP yields the following relationship between revenue-based Domar weights and household expenditure shares

$$\lambda = \mathscr{B}' \chi, \tag{32}$$

where $\lambda \equiv [\lambda_1, \cdots, \lambda_N]'$, and $\chi \equiv [\chi_1, \cdots, \chi_H]'$. In equilibrium, λ_i represents the share of aggregate nominal expenditure that accrues to firm i's revenue.

Define the matrix $\widetilde{\mathscr{B}} \equiv \beta \widetilde{\Psi}_x \equiv \beta \Psi_x (I_N - \Omega_x) \widetilde{\Psi}_x \equiv \mathscr{B} (I_N - \Omega_x) \widetilde{\Psi}_x$, where $\widetilde{\Psi}_x \equiv \left(I_N - \widetilde{\Omega}_x\right)^{-1}$. By construction, this implies $\lambda = \Psi_x' \left(I_N - \widetilde{\Omega}_x'\right) \widetilde{\mathscr{B}}' \chi$, which motivates the definition of the cost-based Domar weights

$$\widetilde{\lambda} \equiv \widetilde{\Psi}_x' \left(I_N - \Omega_x'\right) \lambda \equiv \widetilde{\mathscr{B}}' \chi. \tag{33}$$

To understand the cost-based Domar weights, observe that

$$\widetilde{S}_i \equiv \sum_{h \in \mathscr{H}} p_i C_{hi} + \sum_{j \in \mathscr{N}} \widetilde{\Omega}^x_{ji} \widetilde{S}_j = \sum_{h \in \mathscr{H}} \widetilde{\mathscr{B}}_{hi} E_h$$

where $\widetilde{S}_i = \widetilde{\lambda}_i \, GDP$. In equilibrium, $\widetilde{\Omega}^x_{ji}$ captures the cost share in firm j allocated to intermediate goods supplied by firm i. As a result, $\widetilde{S}_i$ measures the total value-added that passes through firm i. For a given consumption expenditure distribution χ, , the cost-based Domar weight $\widetilde{\lambda}_i$ represents the share of aggregate value-added that flows through firm i. Moreover, the identity $\omega_\ell' \widetilde{\lambda} = \mathbb{1}_N' \left(I_N - \widetilde{\Omega}_x'\right) \widetilde{\Psi}_x' \beta' \chi = 1$ implies that $\omega^\ell_i \widetilde{\lambda}_i$ corresponds to the share of aggregate value-added generated by factors employed in firm i.

Finally, I prove that the value-added passing through any firm is weakly greater than its revenue, i.e., that $\widetilde{\lambda}_i \geq \lambda_i$ holds $\forall i \in \mathscr{N}$. Begin with the identity $\widetilde{\Psi}_x - \Psi_x = \widetilde{\Psi}_x - \Psi_x = \sum_{q=1}^{\infty} \left(\widetilde{\Omega}^q_x - \Omega^q_x\right)$. Note that $\widetilde{\Omega}_x - \Omega_x = (I_N - diag(\mu)) \widetilde{\Omega}_x \succcurlyeq 0_N 0_N'$, since $\mu_i \in (0,1]$ and $\widetilde{\Omega}^x \succcurlyeq 0_N 0_N'$. Proceeding by induction, suppose that $\widetilde{\Omega}^{q-1}_x - \Omega^{q-1}_x \succcurlyeq 0_N 0_N'$ for some $q > 1$. Then

$$\begin{aligned}
\widetilde{\Omega}^q_x - \Omega^q_x &= \left(\widetilde{\Omega}^{q-1}_x - \Omega^{q-1}_x diag(\mu)\right) \widetilde{\Omega}_x \\
&= \left(\widetilde{\Omega}^{q-1}_x - \Omega^{q-1}_x + \Omega^{q-1}_x (I_N - diag(\mu))\right) \widetilde{\Omega}_x \succcurlyeq 0_N 0_N'.
\end{aligned}$$

Thus, $\widetilde{\Psi}_x \succcurlyeq \Psi_x$. It follows that $\widetilde{\mathscr{B}} - \mathscr{B} = \beta \left(\widetilde{\Psi}_x - \Psi_x\right) \succcurlyeq 0_H 0_N'$, and therefore $\widetilde{\lambda} - \lambda = \left(\widetilde{\mathscr{B}} - \mathscr{B}\right)' \chi \succcurlyeq 0_N$.

Factor markets equilibrium conditions. Substituting equations (19) and (21) into the factor market clearing condition (5) for each household $h \in \mathscr{H}$ yields

$$J_h = w_h L_h = \sum_{i \in \mathscr{N}} w_h \ell_{ih} = \sum_{i \in \mathscr{N}} \mu_i \omega^\ell_i \alpha_{ih} S_i, \tag{34}$$

where $\widetilde{\Omega}^\ell_{ih} \equiv \omega^\ell_i \alpha_{ih}$.

In matrix form, the factor market equilibrium conditions can be expressed as $J = \widetilde{\Omega}_\ell' diag(\mu) \; S = \Omega_\ell' S$, where the matrices are defined as

$$\Omega_\ell \equiv \begin{pmatrix} \Omega^\ell_{11} & \cdots & \Omega^\ell_{1H} \\ \vdots & \ddots & \vdots \\ \Omega^\ell_{N1} & \cdots & \Omega^\ell_{NH} \end{pmatrix}, \quad \Omega_\ell \equiv diag(\mu) \; \widetilde{\Omega}_\ell,$$

and $J \equiv [J_1, \cdots, J_H]'$.

Dividing by nominal GDP yields a relationship between the factor income shares and the revenue-based Domar weights

$$\Lambda = \Omega_\ell' \lambda, \tag{35}$$

where $\Lambda \equiv [\Lambda_1, \cdots, \Lambda_H]'$.

Similarly, I define the cost-based factor weights as

$$\widetilde{\Lambda} \equiv \widetilde{\Omega}_\ell' \widetilde{\lambda}, \tag{36}$$

where $\mathbb{1}_H' \widetilde{\Lambda} = \mathbb{1}_H' \alpha' \, diag(\omega_\ell) \, \widetilde{\lambda} = \omega_\ell' \widetilde{\lambda} = 1$, ensuring that aggregate value-added is fully accounted for across factors.

Notice that $\widetilde{\Lambda} \succcurlyeq \Lambda$ because $\widetilde{\Lambda} - \Lambda = \widetilde{\Omega}_\ell' \widetilde{\lambda} - \Omega_\ell' \lambda = \underbrace{\widetilde{\Omega}_\ell'}_{\succcurlyeq 0_H 0_N'} \underbrace{\left(\widetilde{\lambda} - \lambda\right)}_{\succcurlyeq 0_N} + \widetilde{\Omega}_\ell' \underbrace{(I_N - diag(\mu))}_{\succcurlyeq 0_N 0_N'} \lambda.$

The firm-to-factor and factor-to-firm centrality matrices are respectively given by $\Psi_\ell = \Psi_x \, \Omega_\ell$ and $\widetilde{\Psi}_\ell = \widetilde{\Psi}_x \, \widetilde{\Omega}_\ell$, where $\widetilde{\Psi}_\ell \, \mathbb{1}_H = \widetilde{\Psi}_x \, \widetilde{\Omega}_\ell \, \mathbb{1}_H = \widetilde{\Psi}_x \, \omega_\ell = \widetilde{\Psi}_x \left(I_N - \widetilde{\Omega}_x\right) \mathbb{1}_N = \mathbb{1}_N$. Additionally $\widetilde{\Psi}_\ell \succcurlyeq \Psi_\ell$ because $\widetilde{\Psi}_\ell - \Psi_\ell = \underbrace{\left(\widetilde{\Psi}_x - \Psi_x\right)}_{\succcurlyeq 0_N 0_N'} \underbrace{\widetilde{\Omega}_\ell}_{\succcurlyeq 0_N 0_H'} + \underbrace{\Psi_x}_{\succcurlyeq 0_N 0_N'} \underbrace{(I_N - diag(\mu))}_{\succcurlyeq 0_N 0_N'} \widetilde{\Omega}_\ell.$

Similarly, the consumer-to-factor and factor-to-consumer centrality matrices are respectively given by $\mathscr{C} = \mathscr{B} \, \Omega_\ell$ and $\widetilde{\mathscr{C}} = \widetilde{\mathscr{B}} \, \widetilde{\Omega}_\ell$, where $\widetilde{\mathscr{C}} \, \mathbb{1}_H = \widetilde{\mathscr{B}} \, \widetilde{\Omega}_\ell \, \mathbb{1}_H = \beta \, \widetilde{\Psi}_x \, \omega_\ell = \beta \, \widetilde{\Psi}_x \left(I_N - \widetilde{\Omega}_x\right) \mathbb{1}_N = \mathbb{1}_H$, $\widetilde{\mathscr{C}}' \chi = \widetilde{\Omega}_\ell' \widetilde{\mathscr{B}}' \chi = \widetilde{\Omega}_\ell' \widetilde{\lambda} = \widetilde{\Lambda}$, $\mathscr{C}' \chi = \Omega_\ell' \mathscr{B}' \chi = \Omega_\ell' \lambda = \Lambda$, and $\widetilde{\mathscr{C}} \succcurlyeq \mathscr{C}$ because $\widetilde{\mathscr{C}} - \mathscr{C} = \underbrace{\left(\widetilde{\mathscr{B}} - \mathscr{B}\right)}_{\succcurlyeq 0_H 0_N'} \underbrace{\widetilde{\Omega}_\ell}_{\succcurlyeq 0_N 0_H'} + \underbrace{\mathscr{B}}_{\succcurlyeq 0_H 0_N'} \underbrace{(I_N - diag(\mu))}_{\succcurlyeq 0_N 0_N'} \widetilde{\Omega}_\ell.$

Household budget equilibrium conditions. Using the fact that $mc_i = \mu_i \, p_i$, firm costs are given $\bar{c}_i = \mu_i \, S_i$, and profits are $\pi_i = (1 - \mu_i) \, S_i$. Applying equation (21), the household budget constraint becomes

$$E_h = \sum_{i \in \mathcal{N}} \left(\mu_i \, \omega_i^\ell \, \alpha_{ih} + \kappa_{ih} \, (1 - \mu_i)\right) p_i \, y_i + T_h = \sum_{i \in \mathcal{N}} \left(\mu_i \, \widetilde{\Omega}_{ih}^\ell + \kappa_{ih} \, (1 - \mu_i)\right) S_i + T_h. \tag{37}$$

In matrix form, these conditions are represented by $E = (\Omega_\ell + \Omega_\pi)' \, S + T$, where $T = [T_1, \cdots, T_H]'$ and $\Omega_\pi = diag(\mathbb{1}_N - \mu) \, \kappa$. hrough by nominal GDP yields a relationship between the expenditure shares and the revenue-based Domar weights

$$\chi = (\Omega_\ell + \Omega_\pi)' \, \lambda + \tau, \tag{38}$$

where $\tau = [\tau_1, \cdots, \tau_H]'$ and $\tau_h = T_h / GDP$. It follows that $\mathbb{1}_H' \, (\Omega_\ell + \Omega_\pi)' \, \lambda = \mathbb{1}_H' \, \Lambda + \mathbb{1}_N' \, diag(\mathbb{1}_N - \mu) \, \lambda = \sum_{h \in \mathcal{H}} \Lambda_h + \sum_{i \in \mathcal{N}} (1 - \mu_i) \, \lambda_i = 1$.

Nominal GDP. Start by aggregating the goods market clearing condition across all firms

$$\sum_{i \in \mathcal{N}} S_i = \sum_{i \in \mathcal{N}} \left(\sum_{h \in \mathcal{H}} p_i C_{hi} + \sum_{j \in \mathcal{N}} p_i \, x_j \right).$$

Then $GDP \equiv \sum_{h\in\mathscr{H}} E_h = \sum_{i\in\mathscr{N}} \left(S_i - \sum_{j\in\mathscr{N}} p_j\, x_{ij}\right)$. Using equations (20) and (22)

$$GDP = \sum_{i\in\mathscr{N}} \left(1 - \mu_i \sum_{j\in\mathscr{N}} \widetilde{\Omega}^x_{ij}\right) S_i = \sum_{i\in\mathscr{N}} \left(1 - \omega^x_i\, \mu_i\right) S_i. \tag{39}$$

1.5 Proof for Propositions in Section 3

1.5.1 Proof for Proposition 1

Prices are given by $p^\ell_i = \frac{\sum_{h\in\mathscr{H}} w_h\, \ell_{ih}}{A^\ell_i\, Q^\ell_i(\{A^\ell_{ih}\, \ell_{ih}\}_{h\in\mathscr{H}})}$, $p^x_i = \frac{\sum_{j\in\mathscr{N}} p_j\, x_{ij}}{A^x_i\, Q^x_i\left(\left\{A^x_{ij}\, x_{ij}\right\}_{j\in\mathscr{N}}\right)}$, $p_i = \frac{(p^\ell_i\, L_i + p^x_i\, X_i)}{\mu_i\, A_i\, Q_i(L_i, X_i)}$, and $p^c_h = \frac{\sum_{i\in\mathscr{N}} p_i\, C_{hi}}{Q^c_h(\{C_{hi}\}_{i\in\mathscr{N}})}$. Taking a first-order approximation of p^ℓ_i

$$\widehat{p}^\ell_i = \frac{A^\ell_i}{p^\ell_i}\overline{\frac{\partial\, p^\ell_i}{\partial\, A^\ell_i}}\widehat{A}^\ell_i + \sum_{h\in\mathscr{H}} \left(\frac{w_h}{p^\ell_i}\overline{\frac{\partial\, p^\ell_i}{\partial\, w_h}}\widehat{w}_h + \frac{A^\ell_{ih}}{p^\ell_i}\overline{\frac{\partial\, p^\ell_i}{\partial\, A^\ell_{ih}}}\widehat{A}^\ell_{ih} + \frac{\ell_{ih}}{p^\ell_{z_i}}\overline{\frac{\partial\, p^\ell_i}{\partial\, \ell_{ih}}}\widehat{\ell}_{ih}\right),$$

where the log deviation of any variable x around its steady-state value is denoted by $\widehat{x} = \log\left(x/\overline{x}\right)$. The elasticities used are $\frac{A^\ell_i}{p^\ell_i}\overline{\frac{\partial\, p^\ell_i}{\partial\, A^\ell_i}} = -1$, $\frac{w_h}{p^\ell_i}\overline{\frac{\partial\, p^\ell_i}{\partial\, w_h}} = \alpha_{ih}$, $\frac{A^\ell_{ih}}{p^\ell_i}\overline{\frac{\partial\, p^\ell_i}{\partial\, A^\ell_{ih}}} = -\alpha_{ih}$, and $\frac{\ell_{ih}}{p^\ell_i}\overline{\frac{\partial\, p^\ell_i}{\partial\, \ell_{ih}}} = \alpha_{ih} - e\left(L_i, \ell_{ih}\right) = 0$.

As a consequence $\widehat{p}^\ell_i = -\widehat{A}^\ell_i + \sum_{h\in\mathscr{H}} \alpha_{ih}\left(\widehat{w}_h - \widehat{A}^\ell_{ih}\right)$, $\widehat{p}^x_i = -\widehat{A}^x_i + \sum_{j\in\mathscr{N}} \omega_{ij}\left(\widehat{p}_j - \widehat{A}^x_{ij}\right)$, $\widehat{p}_i = \omega^\ell_i\, \widehat{p}^\ell_i + \omega^x_i\, \widehat{p}^x_i - \widehat{A}_i - \widehat{\mu}_i$, and $\widehat{p}^c_h = \sum_{i\in\mathscr{N}} \beta_{hi}\, \widehat{p}_i$. In matrix form, these relationships are written as

$$\widehat{p}_\ell = \alpha\, \widehat{w} - \widehat{A}_\ell - \left(\alpha \circ \underline{\widehat{A}}_\ell\right) \mathbb{1}_H, \tag{40}$$

$$\widehat{p}_x = \mathscr{W}\, \widehat{p} - \widehat{A}_x - \left(\mathscr{W} \circ \underline{\widehat{A}}_x\right) \mathbb{1}_N, \tag{41}$$

$$\widehat{p} = diag(\omega_\ell)\, \widehat{p}_\ell + diag(\omega_x)\, \widehat{p}_x - \widehat{A} - \widehat{\mu}, \tag{42}$$

$$\widehat{p}_c = \beta\, \widehat{p}. \tag{43}$$

Substituting equations (40) and (41) into equation (42)

$$\widehat{p} = \widetilde{\Psi}_x \left(\widetilde{\Omega}_\ell\, \widehat{w} - \widehat{\mathcal{A}} - \widehat{\mu}\right). \tag{44}$$

Then, substituting equation (44) into equation (43)

$$\widehat{p}_c = \widetilde{\mathscr{B}} \left(\widetilde{\Omega}_\ell\, \widehat{w} - \widehat{\mathcal{A}} - \widehat{\mu}\right). \tag{45}$$

The matrices previously used are defined by

$$\alpha \equiv \begin{pmatrix} \alpha_{11} & \cdots & \alpha_{1H} \\ \vdots & \ddots & \vdots \\ \alpha_{N1} & \cdots & \alpha_{NH} \end{pmatrix}, \qquad \mathscr{W} \equiv \begin{pmatrix} \omega_{11} & \cdots & \omega_{1N} \\ \vdots & \ddots & \vdots \\ \omega_{N1} & \cdots & \omega_{NN} \end{pmatrix},$$

$$\widetilde{\Psi}_x \equiv \begin{pmatrix} \widetilde{\psi}^x_{11} & \cdots & \widetilde{\psi}^x_{1N} \\ \vdots & \ddots & \vdots \\ \widetilde{\psi}^x_{N1} & \cdots & \widetilde{\psi}^x_{NN} \end{pmatrix}, \qquad \widetilde{\mathscr{B}} \equiv \beta\, \widetilde{\Psi}_x \equiv \begin{pmatrix} \widetilde{\mathscr{B}}_{11} & \cdots & \widetilde{\mathscr{B}}_{1N} \\ \vdots & \ddots & \vdots \\ \widetilde{\mathscr{B}}_{H1} & \cdots & \widetilde{\mathscr{B}}_{HN} \end{pmatrix},$$

$\widehat{\mathcal{A}} \equiv \widehat{A} + diag(\omega_\ell)\, \widehat{A}_\ell + \left(\widetilde{\Omega}_\ell \circ \widehat{\underline{A}}_\ell\right) \mathbb{1}_H + diag(\omega_x)\, \widehat{A}_x + \left(\widetilde{\Omega}_x \circ \widehat{\underline{A}}_x\right) \mathbb{1}_N$, $\widehat{A} \equiv \left[\widehat{A}_1, \cdots, \widehat{A}_N\right]'$, $\widehat{A}_\ell \equiv \left[\widehat{A}^\ell_1, \cdots, \widehat{A}^\ell_N\right]'$, $\widehat{A}_x \equiv \left[\widehat{A}^x_1, \cdots, \widehat{A}^x_N\right]'$, $\widehat{\underline{A}}_\ell = \left[\widehat{\underline{A}}^\ell_1, \cdots, \widehat{\underline{A}}^\ell_N\right]'$, $\widehat{\underline{A}}^\ell_i = \left[\widehat{A}^\ell_{i1}, \cdots, \widehat{A}^\ell_{iH}\right]'$, $\widehat{\underline{A}}_x = \left[\widehat{\underline{A}}^x_1, \cdots, \widehat{\underline{A}}^x_n\right]'$, $\widehat{\underline{A}}^x_i = \left[\widehat{A}^x_{i1}, \cdots, \widehat{A}^x_{iN}\right]'$, $\widehat{p} \equiv [\widehat{p}_1, \cdots, \widehat{p}_N]'$, $\widehat{p}_\ell \equiv \left[\widehat{p}^\ell_1, \cdots, \widehat{p}^\ell_N\right]'$, $\widehat{p}_x \equiv [\widehat{p}^x_1, \cdots, \widehat{p}^x_N]'$, $\widehat{\mu} \equiv [\widehat{\mu}_1, \cdots, \widehat{\mu}_N]'$, and $\widehat{w} \equiv [\widehat{w}_1, \cdots, \widehat{w}_H]'$.

1.5.2 Proof for Theorem 1

Theorem 6. The variations of the sales, factor, and expenditure shares are, to a first-order,

$$d\,\Lambda_h = \overbrace{\sum_{b\in\mathcal{H}} \mathscr{C}_{bh}\, d\,\chi_b}^{\substack{\text{Distributive}\\ \text{Income}_h}} + \overbrace{\sum_{i\in\mathcal{N}} \psi^\ell_{ih}\, \lambda_i\, d\log\mu_i}^{\substack{\text{Competitive}\\ \text{Income}_h}} + \overbrace{\sum_{i\in\mathcal{N}} \mu_i\, \lambda_i\, d\,\widetilde{\Omega}^\ell_{ih}}^{\substack{\text{Factor Demand}\\ \text{Recomposition}\\ \text{on Income}_h}} + \overbrace{\sum_{j\in\mathcal{N}} \psi^\ell_{jh} \big(\sum_{b\in\mathcal{H}} \chi_b\, d\,\beta_{bj} + \sum_{i\in\mathcal{N}} \mu_i\, \lambda_i\, d\,\widetilde{\Omega}^x_{ij} \big)}^{\substack{\text{Final and Intermediate}\\ \text{Demand Recomposition on Income}_h}}, \tag{46}$$

$$d\,\lambda_i = \overbrace{\sum_{h\in\mathcal{H}} \mathscr{B}_{hi}\, d\,\chi_h}^{\substack{\text{Distributive}\\ \text{Revenue}_i}} + \overbrace{\sum_{i\in\mathcal{H}} \psi^x_{ji}\, \lambda_j\, d\log\mu_j}^{\substack{\text{Competitive}\\ \text{Revenue}_i}} + \overbrace{\sum_{j\in\mathcal{N}} \psi^x_{ji} \big(\sum_{h\in\mathcal{H}} \chi_h\, d\,\beta_{hj} + \sum_{m\in\mathcal{N}} \mu_m\, \lambda_m\, d\,\widetilde{\Omega}^x_{mj} \big)}^{\substack{\text{Final and Intermediate}\\ \text{Demand Recomposition on Revenue}_i}}, \tag{47}$$

$$d\,\widetilde{\Lambda}_h = \sum_{b\in\mathcal{H}} \widetilde{\mathscr{C}}_{bh}\, d\,\chi_b + \sum_{i\in\mathcal{N}} \widetilde{\lambda}_i\, d\,\widetilde{\Omega}^\ell_{ih} + \sum_{j\in\mathcal{N}} \widetilde{\psi}^\ell_{jh} \left(\sum_{b\in\mathcal{H}} \chi_b\, d\,\beta_{bj} + \sum_{i\in\mathcal{N}} \widetilde{\lambda}_i\, d\,\widetilde{\Omega}^x_{ij} \right). \tag{48}$$

$$d\,\widetilde{\lambda}_i = \sum_{h\in\mathcal{H}} \widetilde{\mathscr{B}}_{hi}\, d\,\chi_h + \sum_{j\in\mathcal{N}} \widetilde{\psi}^x_{ji} \left(\sum_{b\in\mathcal{H}} \chi_b\, d\,\beta_{bj} + \sum_{m\in\mathcal{N}} \widetilde{\lambda}_m\, d\,\widetilde{\Omega}^x_{mj} \right), \tag{49}$$

Proof.

Revenue-based Λ. From equation (35), $\Lambda_h = \mathbb{1}'_N\, diag\left(\widetilde{\Omega}_\ell\, o_H(h)\right) \left(diag(\mu)^{-1} - \widetilde{\Omega}'_x\right)^{-1} \beta'\,\chi$. Then, the elasticity of Λ with respect to underlying endogenous equilibrium objects

$$\begin{aligned}
\Lambda_h\, \widehat{\Lambda}_h =& \mathbb{1}'_N\, diag(\Omega_\ell\, o_H(h))\, \mathscr{B}' \begin{pmatrix} \chi_1\, \widehat{\chi}_1 \\ \vdots \\ \chi_H\, \widehat{\chi}_H \end{pmatrix} + \mathbb{1}'_N\, diag(\Omega_\ell\, o_H(h))\, \Psi'_x \begin{pmatrix} \sum_{b\in\mathcal{H}} \beta_{b1}\, \chi_b\, \widehat{\beta}_{b1} \\ \vdots \\ \sum_{b\in\mathcal{H}} \beta_{bN}\, \chi_b\, \widehat{\beta}_{bN} \end{pmatrix} \\
&+ \sum_{i\in\mathcal{N}} \Omega^\ell_{ih}\, \lambda_i \left(\widehat{\omega}^\ell_i + \widehat{\alpha}_{ih}\right) + \mathbb{1}'_N\, diag\left(\widetilde{\Omega}_\ell\, o_H(h)\right) \frac{d\left(diag(\mu)^{-1} - \widetilde{\Omega}'_x\right)^{-1}}{d\log\widetilde{\Omega}_x} \beta'\,\chi \\
&+ \mathbb{1}'_N\, diag\left(\widetilde{\Omega}_\ell\, o_H(h)\right) \frac{d\left(diag(\mu)^{-1} - \widetilde{\Omega}'_x\right)^{-1}}{d\log\mu} \beta'\,\chi.
\end{aligned}$$

Using equations (32), (35), and the fact that for any invertible matrix A, $\frac{d\,A^{-1}}{d\,x} = -A^{-1}\frac{d\,A}{d\,x}A^{-1}$, the previous equation becomes

$$\begin{aligned}
d\,\Lambda_h = & \sum_{b\in\mathcal{H}} \mathscr{C}_{bh}\, d\,\chi_b + \sum_{i\in\mathcal{N}} \Omega^\ell_{ih} \sum_{j\in\mathcal{N}} \psi^x_{ji} \sum_{b\in\mathcal{H}} \chi_b\, d\,\beta_{bj} + \sum_{i\in\mathcal{N}} \Omega^\ell_{ih}\, \lambda_i \left(\widehat{\omega}^\ell_i + \widehat{\alpha}_{ih}\right) \\
& - \mathbb{1}'_N diag(\Omega_\ell\, o_H(h))\, \Psi'_x \frac{d\left(diag(\mu)^{-1} - \widetilde{\Omega}'_x\right)}{d\log\widetilde{\Omega}_x} diag(\mu)\;\lambda \\
& - \mathbb{1}'_N\, diag(\Omega_\ell\, o_H(h))\, \Psi'_x \frac{d\left(diag(\mu)^{-1} - \widetilde{\Omega}'_x\right)}{d\log\mu} diag(\mu)\;\lambda.
\end{aligned}$$

$$d\,\Lambda_h = \sum_{b\in\mathscr{H}} \mathscr{C}_{bh}\, d\,\chi_b + \sum_{b\in\mathscr{H}} \chi_b\, d\,\mathscr{C}_{bh} = \sum_{b\in\mathscr{H}} \mathscr{C}_{bh}\, d\,\chi_b + \sum_{i\in\mathscr{N}} \psi^{\ell}_{ih}\, \lambda_i\, d\log\mu_i$$
$$+ \sum_{i\in\mathscr{N}} \mu_i\, \lambda_i\, d\,\widetilde{\Omega}^{\ell}_{ih} + \sum_{j\in\mathscr{N}} \psi^{\ell}_{jh} \left(\sum_{b\in\mathscr{H}} \chi_b\, d\,\beta_{bj} + \sum_{i\in\mathscr{N}} \mu_i\, \lambda_i\, d\,\widetilde{\Omega}^{x}_{ij} \right).$$

Revenue-based λ. From equation (32),

$$\lambda_i = o_N(i)'\, \Psi'_x\, \beta'\, \chi = o_N(i)'\, diag(\mu)^{-1} \left(diag(\mu)^{-1} - \widetilde{\Omega}'_x \right)^{-1} \beta'\, \chi.$$

Hence

$$\lambda_i\, \widehat{\lambda}_i = o_N(i)'\, \mathscr{B}' \begin{pmatrix} \chi_1\, \widehat{\chi}_1 \\ \vdots \\ \chi_H\, \widehat{\chi}_H \end{pmatrix} + o_N(i)'\, \Psi'_x \begin{pmatrix} \sum_{b\in\mathscr{H}} \beta_{b1}\, \chi_b\, \widehat{\beta}_{b1} \\ \vdots \\ \sum_{b\in\mathscr{H}} \beta_{bN}\, \chi_b\, \widehat{\beta}_{bN} \end{pmatrix}$$
$$+ o_N(i)'\, diag(\mu)^{-1} \frac{d\left(diag(\mu)^{-1} - \widetilde{\Omega}'_x \right)^{-1}}{d\log\widetilde{\Omega}_x} \beta'\, \chi + o_N(i)'\, diag(\mu)^{-1} \frac{d\left(diag(\mu)^{-1} - \widetilde{\Omega}'_x \right)^{-1}}{d\log\mu} \beta'\, \chi.$$

$$\lambda_i\, \widehat{\lambda}_i = \sum_{h\in\mathscr{H}} \mathscr{B}_{hi}\, d\,\chi_h + \sum_{j\in\mathscr{N}} \psi^{x}_{ji} \sum_{b\in\mathscr{H}} \chi_b\, d\,\widehat{\beta}_{bj}$$
$$- o_N(i)'\, \Psi'_x \frac{d\left(diag(\mu)^{-1} - \widetilde{\Omega}'_x \right)}{d\log\widetilde{\Omega}_x} diag(\mu)\ \lambda - o_N(i)'\, \Psi'_x \frac{d\left(diag(\mu)^{-1} - \widetilde{\Omega}'_x \right)}{d\log\mu} diag(\mu)\ \lambda.$$

$$d\,\lambda_i = \sum_{h\in\mathscr{H}} \mathscr{B}_{hi}\, d\,\chi_h + \sum_{i\in\mathscr{H}} \psi^{x}_{ji}\, \lambda_j\, d\log\mu_j + \sum_{j\in\mathscr{N}} \psi^{x}_{ji} \left(\sum_{b\in\mathscr{H}} \chi_b\, d\,\beta_{bj} + \sum_{m\in\mathscr{N}} \mu_m\, \lambda_m\, d\,\widetilde{\Omega}^{x}_{mj} \right).$$

Cost-based $\widetilde{\Lambda}$. Similarly, from equation (36), $\widetilde{\Lambda}_h = \mathbb{1}'_N\, diag\left(\widetilde{\Omega}_\ell\, o_H(h)\right) \left(I - \widetilde{\Omega}'_x\right)^{-1} \beta'\, \chi$. Hence

$$d\,\widetilde{\Lambda}_h = \sum_{b\in\mathscr{H}} \widetilde{\mathscr{C}}_{bh}\, d\,\chi_b + \sum_{b\in\mathscr{H}} \chi_b\, d\,\widetilde{\mathscr{C}}_{bh}$$
$$= \sum_{b\in\mathscr{H}} \widetilde{\mathscr{C}}_{bh}\, d\,\chi_b + \sum_{i\in\mathscr{N}} \widetilde{\lambda}_i\, d\,\widetilde{\Omega}^{\ell}_{ih} + \sum_{j\in\mathscr{N}} \widetilde{\psi}^{\ell}_{jh} \left(\sum_{b\in\mathscr{H}} \chi_b\, d\,\beta_{bj} + \sum_{i\in\mathscr{N}} \widetilde{\lambda}_i\, d\,\widetilde{\Omega}^{x}_{ij} \right).$$

Cost-based $\widetilde{\lambda}$. From equation (32), $\widetilde{\lambda}_i = o_N(i)'\, \widetilde{\Psi}'_x\, \beta'\, \chi$. Hence

$$d\,\widetilde{\lambda}_i = \sum_{h\in\mathscr{H}} \widetilde{\mathscr{B}}_{hi}\, d\,\chi_h + \sum_{j\in\mathscr{N}} \widetilde{\psi}^{x}_{ji} \left(\sum_{b\in\mathscr{H}} \chi_b\, d\,\beta_{bj} + \sum_{m\in\mathscr{N}} \widetilde{\lambda}_m\, d\,\widetilde{\Omega}^{x}_{mj} \right).$$

1.5.3 Proof for Theorem 4

The first order approximation to the household budget constraint in equation (3) is

$$E_h\, \widehat{E}_h = \Lambda_h \left(\widehat{w}_h + \widehat{L}_h \right) + \sum_{i\in\mathscr{N}} \kappa_{ih} \left((1-\mu_i) \left(\widehat{\kappa}_{ih} + \widehat{S}_i \right) - \mu_i\, \widehat{\mu}_i \right) S_i + T_h\, \widehat{T}_h. \tag{50}$$

Combining equations (45) and (50) yields

$$\widehat{C}_h = \widehat{E}_h - \widehat{p}^{c}_h$$
$$= \frac{\Lambda_h}{\chi_h} \left(\widehat{w}_h + \widehat{L}_h \right) - \widetilde{\mathscr{C}}'_h\, \widehat{w} + \widetilde{\mathscr{B}}'_h \left(\widehat{\mathcal{A}} + \widehat{\mu} \right) + \sum_{i\in\mathscr{N}} \kappa_{ih} \frac{\lambda_i}{\chi_h} \left((1-\mu_i) \left(\widehat{\kappa}_{ih} + \widehat{S}_i \right) - \mu_i\, \widehat{\mu}_i \right) + \frac{\tau_h}{\chi_h} \widehat{T}_h$$

where $\widetilde{\mathscr{B}}_h = \left[\widetilde{\mathscr{B}}_{h1}, \cdots, \widetilde{\mathscr{B}}_{hN}\right]'$, and $\widetilde{\mathscr{C}}_h = \left[\widetilde{\mathscr{C}}_{h1}, \cdots, \widetilde{\mathscr{C}}_{hH}\right]'$. Therefore

$$\widehat{C}_h = \widetilde{\mathscr{B}}_h' \left(\widehat{\mathcal{A}} + \widehat{\mu}\right) + \frac{\Lambda_h}{\chi_h} \widehat{J}_h - \widetilde{\mathscr{C}}_h' \widehat{J} + \sum_{i \in \mathscr{N}} \kappa_{ih} \frac{\lambda_i}{\chi_h} \left((1-\mu_i)\left(\widehat{\kappa}_{ih} + \widehat{S}_i\right) - \mu_i \widehat{\mu}_i\right) + \frac{\tau_h}{\chi_h} \widehat{T}_h + \widetilde{\mathscr{C}}_h' \widehat{L}. \tag{51}$$

Therefore

$$\frac{C_h}{\overline{C}_h} = \overline{\eta}_h \mathcal{D}_h(\mathcal{A}) \mathcal{D}_h(\mu) \mathcal{D}_h(J) \mathcal{D}_h(\Pi) \mathcal{D}_h(T) f_h\left(\{L_b\}_{b \in \mathscr{H}}\right)$$

where $f_h\left(\{L_b\}_{b \in \mathscr{H}}\right)$ is a CRS function such that $\frac{d \log f_h\left(\{L_b\}_{b \in \mathscr{H}}\right)}{d \log L_b} = \widetilde{\mathscr{C}}_{hb}$, and

$$\mathcal{D}_h(\mathcal{A}) = exp\left\{\widetilde{\mathscr{B}}_h' \widehat{\mathcal{A}}\right\}, \qquad \mathcal{D}_h(\mu) = exp\left\{\widetilde{\mathscr{B}}_h' \widehat{\mu}\right\},$$

$$\mathcal{D}_h(\Pi) = exp\left\{\sum_{i \in \mathscr{N}} \kappa_{ih} \frac{\lambda_i}{\chi_h} \left((1-\mu_i)\left(\widehat{\kappa}_{ih} + \widehat{S}_i\right) - \mu_i \widehat{\mu}_i\right)\right\},$$

$$\mathcal{D}_h(J) = exp\left\{\frac{\Lambda_h}{\chi_h} \widehat{J}_h - \widetilde{\mathscr{C}}_h' \widehat{J}\right\}, \qquad \mathcal{D}_h(J) = exp\left\{\frac{\tau_h}{\chi_h} \widehat{T}_h\right\},$$

and $\overline{\eta}_h$ is a constant specific to household h.

It follows that

$$C_h = \eta_h \mathcal{D}_h(\mathcal{A}) \mathcal{D}_h(\mu) \mathcal{D}_h(J) \mathcal{D}_h(\Pi) \mathcal{D}_h(T) f_h\left(\{L_b\}_{b \in \mathscr{H}}\right) = PTT_h f_h\left(\{L_b\}_{b \in \mathscr{H}}\right) \tag{52}$$

$$PTT_h = \eta_h \mathcal{D}_h(\mathcal{A}) \mathcal{D}_h(\mu) \mathcal{D}_h(\mathcal{J}) \mathcal{D}_h(\Pi) \mathcal{D}_h(T)$$

where $\eta_h = \overline{\eta}_h \overline{C}_h$.

Add and subtract $\widehat{GDP}$ to express equation (52) in terms of Domar weights and factor income shares

$$\begin{aligned}
\widehat{C}_h = {} & \widetilde{\mathscr{B}}_h' \widehat{\mathcal{A}} + \left(\widetilde{\mathscr{B}}_h - \chi_h^{-1} diag(\lambda)\, diag(\mu)\, \kappa\, o_H(h)\right)' \widehat{\mu} + \frac{\Lambda_h}{\chi_h} \widehat{\Lambda}_h - \widetilde{\mathscr{C}}_h' \widehat{\Lambda} \\
& + \chi_h^{-1} o_H(h)' \kappa' diag(\lambda)\, diag(\mathbb{1}_N - \mu) \left(\widehat{\kappa}\, o_H(h) + \widehat{\lambda}\right) + \frac{\tau_h}{\chi_h} \widehat{\tau}_h + \widetilde{\mathscr{C}}_h' \widehat{L} \\
& + \underbrace{\left(\frac{\Lambda_h}{\chi_h} + \sum_{i \in \mathscr{N}} \kappa_{ih} (1-\mu_i) \frac{\lambda_i}{\chi_h} + \frac{\tau_h}{\chi_h} - \sum_{b \in \mathscr{H}} \widetilde{\mathscr{C}}_{hb}\right)}_{=0} \widehat{GDP} \\
= {} & \widetilde{\mathscr{B}}_h' \widehat{\mathcal{A}} + \left(\widetilde{\mathscr{B}}_h - \chi_h^{-1} diag(\lambda)\, diag(\mu)\, \kappa\, o_H(h)\right)' \widehat{\mu} + \Gamma_h \widehat{\Lambda}_h - \widetilde{\mathscr{C}}_h' \widehat{\Lambda} \\
& + \chi_h^{-1} o_H(h)' \kappa' diag(\lambda)\, diag(\mathbb{1}_N - \mu) \left(\widehat{\kappa}\, o_H(h) + \widehat{\lambda}\right) + \frac{\tau_h}{\chi_h} \widehat{\tau}_h + \widetilde{\mathscr{C}}_h' \widehat{L}.
\end{aligned}$$

where the final equality follows from the equilibrium expenditure identity (38).

The $N+2$ vector $\mathcal{R}_h$ summarizes the sources of nominal expenditure for household h

$$\mathcal{R}_h' = \chi_h^{-1} \begin{bmatrix} \Lambda_h & \lambda' diag(\Omega_\pi o_H(h)) & \tau_h \end{bmatrix} = \frac{1}{\chi_h} \begin{bmatrix} \Lambda_h & \kappa_{1h}(1-\mu_1)\lambda_1 & \cdots & \kappa_{Nh}(1-\mu_N)\lambda_N & \tau_h \end{bmatrix}.$$

The first entry corresponds to the share of factor income in household h's expenditure, the next N elements capture the share of profits by each sector in household h's expenditure, and the last element captures the role of lump-sum transfers in household h's expenditure. Since the elements of $\mathcal{R}_h$ sum to one by construction, its first-order approximation satisfies

$$\chi_h \widehat{\chi}_h = \Lambda_h \widehat{\Lambda}_h + \sum_{i \in \mathscr{N}} \kappa_{ih} \lambda_i \left((1-\mu_i) \left(\widehat{\kappa}_{ih} + \widehat{\lambda}_i \right) - \mu_i \widehat{\mu}_i \right) + \tau_h \widehat{\tau}_h. \tag{53}$$

This implies the following first-order approximation for household h's real consumption

$$\widehat{C}_h = \widetilde{\mathscr{B}}_h' \widehat{\mathcal{A}} + \widetilde{\mathscr{B}}_h' \widehat{\mu} + \widehat{\chi}_h - \widetilde{\mathscr{C}}_h' \widehat{\Lambda} + \widetilde{\mathscr{C}}_h' \widehat{L}. \tag{54}$$

Now, using using the definitions $\delta_{b|h} = \widetilde{\mathscr{C}}_{hb} / \widetilde{\Lambda}_b$, $M_{q|h} = \sum_{b \in \mathscr{H}} \mathscr{C}_{qb} \delta_{b|h}$, and $F_{i|h} = \sum_{q \in \mathscr{H}} \psi_{iq}^{\ell} \delta_{q|h}$

$$\begin{aligned}
\widehat{PTT}_h &= \sum_{i \in \mathscr{N}} \widetilde{\mathscr{B}}_{hi} \left(\widehat{\mathcal{A}}_i + \widehat{\mu}_i \right) + \widehat{\chi}_h - \sum_{b \in \mathscr{H}} F_{b|h} \, d\chi_b - \sum_{q \in \mathscr{H}} \chi_q \sum_{b \in \mathscr{H}} \delta_{b|h} \, d\mathscr{C}_{qb} \\
&= \sum_{i \in \mathscr{N}} \widetilde{\mathscr{B}}_{hi} \widehat{\mathcal{A}}_i + \sum_{i \in \mathscr{N}} \widetilde{\mathscr{B}}_{hi} \widehat{\mu}_i + \widehat{\chi}_h - \sum_{b \in \mathscr{H}} M_{b|h} \, d\chi_b - \sum_{i \in \mathscr{N}} \lambda_i F_{i|h} \widehat{\mu}_i \\
&- \sum_{i \in \mathscr{N}} \mu_i \lambda_i \sum_{b \in \mathscr{H}} \delta_{b|h} \, d\widetilde{\Omega}_{ib}^{\ell} - \sum_{b \in \mathscr{H}} \chi_b \sum_{i \in \mathscr{N}} F_{i|h} \, d\beta_{bi} - \sum_{i \in \mathscr{N}} \mu_i \lambda_i \sum_{j \in \mathscr{N}} F_{j|h} \, d\widetilde{\Omega}_{ij}^{x}.
\end{aligned} \tag{55}$$

1.5.4 Proof for Theorem 2

The first-order approximation of nominal GDP is given by $\widehat{GDP} = \sum_{h \in \mathscr{H}} \chi_h \widehat{E}_h = \sum_{h \in \mathscr{H}} \chi_h \left(\widehat{p}_h^c + \widehat{C}_h \right)$. Taking the GDP deflator as the Divisia-weighted average change in household-specific price indices gives

$$\widehat{p}_Y = \sum_{h \in \mathscr{H}} \chi_h \widehat{p}_h^c = \sum_{h \in \mathscr{H}} \widetilde{\Lambda}_h \widehat{w}_h - \sum_{i \in \mathscr{N}} \widetilde{\lambda}_i \left(\widehat{\mathcal{A}}_i + \widehat{\mu}_i \right). \tag{56}$$

Thus, the first-order approximation of real GDP is

$$\widehat{Y} = \widehat{GDP} - \widehat{p}_Y = \sum_{h \in \mathscr{H}} \chi_h \widehat{C}_h. \tag{57}$$

Starting from equation (57) and using (51), the first-order approximation for real GDP is

$$\begin{aligned}
\widehat{Y} = &\sum_{h \in \mathscr{H}} \left(\chi_h \widetilde{\mathscr{B}}_h' \left(\widehat{\mathcal{A}} + \widehat{\mu} \right) + \left(\Lambda_h - \widetilde{\Lambda}_h \right) \widehat{J}_h + \widetilde{\Lambda}_h \widehat{L}_h \right) \\
&+ \sum_{i \in \mathscr{N}} \lambda_i \left((1-\mu_i) \widehat{S}_i - \mu_i \widehat{\mu}_i \right) \underbrace{\sum_{h \in \mathscr{H}} \kappa_{ih}}_{=1} + \sum_{i \in \mathscr{N}} (1-\mu_i) \lambda_i \underbrace{\sum_{h \in \mathscr{H}} \kappa_{ih} \widehat{\kappa}_{ih}}_{=0} + \underbrace{\sum_{h \in \mathscr{H}} \tau_h \widehat{T}_h}_{=0}
\end{aligned}$$

$$\widehat{Y} = \widetilde{\lambda}' \widehat{\mathcal{A}} + \left(\widetilde{\lambda} - diag(\mu) \, \lambda \right)' \widehat{\mu} + \left(\Lambda - \widetilde{\Lambda} \right)' \widehat{J} + \lambda' diag(\mathbb{1}_N - \mu) \, \widehat{S} + \widetilde{\Lambda}' \widehat{L}. \tag{58}$$

To derive the previous equation, I apply the following transformation

$$\begin{aligned}
\sum_{h \in \mathscr{H}} \left(\Lambda_h \widehat{J}_h - \chi_h \widetilde{\mathscr{C}}_h' \widehat{J} \right) &= \sum_{h \in \mathscr{H}} \Lambda_h \widehat{J}_h - \sum_{b \in \mathscr{H}} \widehat{J}_b \sum_{i \in \mathscr{N}} \widetilde{\Omega}_{ib}^{\ell} \sum_{h \in \mathscr{H}} \chi_h \widetilde{\mathscr{B}}_{hi} \\
&= \sum_{h \in \mathscr{H}} \Lambda_h \widehat{J}_h - \sum_{b \in \mathscr{H}} \widehat{J}_b \sum_{i \in \mathscr{N}} \widetilde{\Omega}_{ib}^{\ell} \widetilde{\lambda}_i = \left(\Lambda - \widetilde{\Lambda} \right)' \widehat{J},
\end{aligned} \tag{59}$$

where the second equality follows from the identities in equations (33) and (36).

Equation (58) expressed as deviation from equilibrium is given by

$$Y = Q_Y\left(\{C_h\}_{h\in\mathscr{H}}\right) = \eta_Y\, \mathcal{D}(\mathcal{A})\ \mathcal{D}(\mu)\ \mathcal{D}(J)\ \mathcal{D}(\Pi)\ F\left(\{L_h\}_{h\in\mathscr{H}}\right),$$

where η_Y is a constant, $Q_Y\left(\{C_h\}_{h\in\mathscr{H}}\right)$ is a CRS aggregator with $\frac{d\log Q_Y\left(\{C_h\}_{h\in\mathscr{H}}\right)}{d\log C_h} = \chi_h$. Similarly, $F\left(\{L_h\}_{h\in\mathscr{H}}\right)$ is a CRS function with $\frac{d\log F\left(\{L_h\}_{h\in\mathscr{H}}\right)}{d\log L_h} = \widetilde{\Lambda}_h$. Its components are defined as

$$\mathcal{D}(\mathcal{A}) = exp\left\{\widetilde{\lambda}'\ \widehat{\mathcal{A}}\right\}, \qquad \mathcal{D}(\mu) = exp\left\{\widetilde{\lambda}'\ \widehat{\mu}\right\}, \qquad L = F\left(\{L_h\}_{h\in\mathscr{H}}\right),$$

$$\mathcal{D}(J) = exp\left\{\left(\Lambda - \widetilde{\Lambda}\right)'\widehat{J}\right\}, \qquad \mathcal{D}(\Pi) = exp\left\{\lambda'\, diag(\mathbb{1}_N - \mu)\ \widehat{S} - \lambda'\, diag(\mu)\, \widehat{\mu}\right\}.$$

As a result

$$Y = \eta_Y\, \mathcal{D}(\mathcal{A})\ \mathcal{D}(\mu)\ \mathcal{D}(\Pi)\ \mathcal{D}(J)\ L = TFP\ L, \tag{60}$$

$$TFP = \eta_Y\, \mathcal{D}(\mathcal{A})\ \mathcal{D}(\mu)\ \mathcal{D}(\Pi)\ \mathcal{D}(J).$$

We now re-express equation (58) by adding and subtracting $\widehat{GDP}$, in order to isolate the roles of Domar weights and labor income shares in shaping real GDP changes.

$$\begin{aligned}
\widehat{Y} &= \widetilde{\lambda}'\ \widehat{\mathcal{A}} + \left(\widetilde{\lambda} - diag(\mu)\ \lambda\right)'\widehat{\mu} + \left(\Lambda - \widetilde{\Lambda}\right)'\widehat{\Lambda} + \lambda'\, diag(\mathbb{1}_N - \mu)\ \widehat{\lambda} + \tau'\,\widehat{\tau} + \widetilde{\Lambda}'\ \widehat{L} \\
&\quad + \underbrace{\left(\left(\Lambda - \widetilde{\Lambda}\right)'\mathbb{1}_H + \lambda'\, diag(\mathbb{1}_N - \mu)\ \mathbb{1}_N\right)}_{=\sum_{h\in\mathscr{H}}\Lambda_h + \sum_{i\in\mathscr{N}}(1-\mu_i)\lambda_i - \sum_{h\in\mathscr{H}}\widetilde{\Lambda}_h = 0}\widehat{GDP} - \underbrace{\left(\sum_{h\in\mathscr{H}} T_h\right)}_{=0}\widehat{GDP} \\
&= \widetilde{\lambda}'\ \widehat{\mathcal{A}} + \left(\widetilde{\lambda} - diag(\mu)\ \lambda\right)'\widehat{\mu} + \left(\Lambda - \widetilde{\Lambda}\right)'\ \widehat{\Lambda} + \lambda'\, diag(\mathbb{1}_N - \mu)\ \widehat{\lambda} + \tau'\,\widehat{\tau} + \widetilde{\Lambda}'\ \widehat{L}.
\end{aligned}$$

The final equality follows from the accounting identities in equations (36) and (38), which together imply that the net weight $\widehat{GDP}$ vanishes.

The $3H$-dimensional vector $\mathcal{R}$ captures the decomposition of revenue shares across households

$$\begin{aligned}
\mathcal{R}' &= \begin{bmatrix}\lambda'\,\Omega_\ell & \lambda'\,\Omega_\pi & \tau'\end{bmatrix} \\
&= \begin{bmatrix}\Lambda_1 & \cdots & \Lambda_H & \sum_{i\in\mathscr{N}}\kappa_{i1}(1-\mu_i)\lambda_i & \cdots & \sum_{i\in\mathscr{N}}\kappa_{iH}(1-\mu_i)\lambda_i & \tau_1 & \cdots & \tau_H\end{bmatrix}.
\end{aligned}$$

For each household, the first H entries represent the factor income shares, the next H capture the household's share of distributed profits in total expenditure, and the final H entries correspond to the share of lump-sum transfers. Since the entries of this vector sum to one, its first-oder approximation satisfies

$$
\begin{aligned}
0 &= \sum_{h\in\mathcal{H}} \Lambda_h\,\widehat{\Lambda}_h + \sum_{h\in\mathcal{H}}\sum_{i\in\mathcal{N}} \kappa_{ih}\,\lambda_i\left((1-\mu_i)\left(\widehat{\kappa}_{ih}+\widehat{\lambda}_i\right)-\mu_i\,\widehat{\mu}_i\right)+\underbrace{\sum_{h\in\mathcal{H}} \tau_h\,\widehat{\tau}_h}_{=0} \\
&= \sum_{h\in\mathcal{H}} \Lambda_h\,\widehat{\Lambda}_h + \sum_{i\in\mathcal{N}} \lambda_i\left((1-\mu_i)\left(\underbrace{\sum_{h\in\mathcal{H}}\kappa_{ih}\,\widehat{\kappa}_{ih}}_{=0}+\widehat{\lambda}_i\underbrace{\sum_{h\in\mathcal{H}}\kappa_{ih}}_{=1}\right)-\mu_i\,\widehat{\mu}_i\underbrace{\sum_{h\in\mathcal{H}}\kappa_{ih}}_{=1}\right) \\
&= \sum_{h\in\mathcal{H}} \Lambda_h\,\widehat{\Lambda}_h + \sum_{i\in\mathcal{N}} \lambda_i\left((1-\mu_i)\;\widehat{\lambda}_i-\mu_i\,\widehat{\mu}_i\right).
\end{aligned}
$$

This implies that

$$
\widehat{Y} = \widetilde{\lambda}'\,\widehat{\mathcal{A}} + \widetilde{\lambda}'\,\widehat{\mu} - \widetilde{\Lambda}'\,\widehat{\Lambda} + \widetilde{\Lambda}'\,\widehat{L} \tag{61}
$$

which, under the additional assumption of inelastic factor supply, coincides with Theorem 1 in Baqaee & Farhi (2020).

Now, using equation (46)

$$
\begin{aligned}
\widehat{TFP} &= \sum_{i\in\mathcal{N}} \widetilde{\lambda}_i\left(\widehat{\mathcal{A}}_i+\hat{\mu}_i\right) - \sum_{h\in\mathcal{H}} M_h\,d\,\chi_h - \sum_{h\in\mathcal{H}}\delta_h\sum_{b\in\mathcal{H}}\chi_b\,d\,\mathcal{C}_{bh} \\
&= \sum_{i\in\mathcal{N}} \widetilde{\lambda}_i\widehat{\mathcal{A}}_i + \sum_{i\in\mathcal{N}}\left(\widetilde{\lambda}_i-\lambda_i\,F_i\right)\widehat{\mu}_i - \sum_{h\in\mathcal{H}} M_h\,d\,\chi_h - \sum_{i\in\mathcal{N}}\mu_i\,\lambda_i\sum_{h\in\mathcal{H}}\delta_h\,d\,\widetilde{\Omega}^{\ell}_{ih} \\
&\quad - \sum_{h\in\mathcal{H}}\chi_h\sum_{i\in\mathcal{N}}F_i\,d\,\beta_{hi} - \sum_{i\in\mathcal{N}}\mu_i\,\lambda_i\sum_{j\in\mathcal{N}}F_j\,d\,\widetilde{\Omega}^{x}_{ij}.
\end{aligned} \tag{62}
$$

1.5.5 Proof for Corollary 1

Observe that $\widetilde{\mathcal{B}}_h'\,diag(\omega_\ell)\;\mathbb{1}_N = \beta_h'\,\widetilde{\Psi}_x\,\omega_\ell = \beta_h'\,\widetilde{\Psi}_x\left(I_N-\widetilde{\Omega}_x\right)\mathbb{1}_N = 1$. Therefore, the expression in equation (51) involving variations in factor income can be rewritten as

$$
\begin{aligned}
&(\Lambda_h/\chi_h)\,\widehat{J}_h - \widetilde{\mathcal{C}}_h'\,\widehat{J} \\
&\quad= \widetilde{\mathcal{B}}_h'\,diag(\omega_\ell)\left(\mathbb{1}_N\,(\Lambda_h/\chi_h)\,\widehat{J}_h-\alpha\,\widehat{J}\right) = \widetilde{\mathcal{B}}_h'\,diag(\omega_\ell)\left(\mathbb{1}_N\,(\Lambda_h/\chi_h)\,o_H\,(h)'-\alpha\right)\left(\widehat{J}\pm\widehat{p}_c\right) \\
&\quad= (\Lambda_h/\chi_h)\,\widehat{p}^c_h \pm \widehat{p}^c_h + \widetilde{\mathcal{B}}_h'\,diag(\omega_\ell)\left[\left(\mathbb{1}_N\,(\Lambda_h/\chi_h)\,o_H\,(h)'-\alpha\right)\left(\widehat{J}-\widehat{p}_c\right)-\alpha\,\widehat{p}_c\right] \\
&\quad= -\left(1-(\Lambda_h/\chi_h)\right)\widehat{p}^c_h + \widetilde{\mathcal{B}}_h'\,diag(\omega_\ell)\left(\mathbb{1}_N\,(\Lambda_h/\chi_h)\,,o_H\,(h)'-\alpha\right)\left(\widehat{J}-\widehat{p}_c\right) - \widetilde{\mathcal{C}}_h'\left(\widehat{p}_c-\mathbb{1}_H\,\widehat{p}^c_h\right) \\
&\quad= -\left(1-(\Lambda_h/\chi_h)\right)\widehat{p}^c_h + \widetilde{\mathcal{B}}_h'\,diag(\omega_\ell)\left(\mathbb{1}_N\,(\Lambda_h/\chi_h)\,o_H\,(h)'-\alpha\right)\left(\widehat{J}-\widehat{p}_c\right) - \sum_{b\in\mathcal{H}}\widetilde{\mathcal{C}}_{hb}\,\widehat{\varepsilon}_{hb} \\
&\quad= (\Lambda_h/\chi_h)\left(\widehat{J}_h-\widehat{p}^c_h\right) - \widetilde{\mathcal{C}}_h'\left(\widehat{J}-\widehat{p}_c\right) + \left((\Lambda_h/\chi_h)-1\right)\widehat{p}^c_h - \sum_{b\in\mathcal{H}}\widetilde{\mathcal{C}}_{hb}\,\widehat{\varepsilon}_{hb},
\end{aligned}
$$

where $\varepsilon_{hb} = p^c_b/p^c_h$.

Corollary 1 follows from taking an expenditure-weighted summation of the previous expression,

$$
\begin{aligned}
\sum_{h\in\mathcal{H}}\chi_h\left((\Lambda_h/\chi_h)\,\widehat{J}_h-\widetilde{\mathcal{C}}_h'\,\widehat{J}\right) &= \sum_{h\in\mathcal{H}}\Lambda_h\,\widehat{\mathcal{J}}_h - \sum_{h\in\mathcal{H}}\widetilde{\Lambda}_h\,\widehat{\mathcal{J}}_h + \sum_{h\in\mathcal{H}}(\Lambda_h-\chi_h)\,\widehat{p}^c_h - \sum_{h\in\mathcal{H}}\chi_h\,\widehat{\varepsilon}_h \\
&= \sum_{h\in\mathcal{H}}(1-\delta_h)\,\Lambda_h\,\widehat{\mathcal{J}}_h + \sum_{h\in\mathcal{H}}(\Lambda_h-\chi_h)\,\widehat{p}^c_h - \sum_{h\in\mathcal{H}}\chi_h\,\widehat{\varepsilon}_h.
\end{aligned}
$$

1.5.6 Proof for Corollary 2

If $M_h = M$ for all $h \in \mathscr{H}$, then the distributive terms of trade satisfy $M \sum_{h\in\mathscr{H}} d\,\chi_h = 0$. **(i) Undistorted economy.** $\delta_h = 1$ and $M_h = 1\ \forall h \in \mathscr{H}$. **(ii) Symmetric consumption bundles.** Let $\beta = \mathbb{1}_H \overline{\beta}'$ where $\overline{\beta}$ denotes the common consumption vector. Then, $\mathscr{C} = \beta\,\Psi_x\,\Omega_\ell = \mathbb{1}_H \overline{\beta}'\,\Psi_x\,\Omega_\ell = \mathbb{1}_H \overline{\mathscr{C}}'$. This implies that $M_h = \overline{\mathscr{C}}'\,\delta$ for all $h \in \mathscr{H}$, where δ denotes the vector of distortion centralities. **(iii) No intermediate imputs and symmetric distortions.** In this case, $\mathscr{C} = \beta\,\Omega_\ell = \mu\,\beta\,\alpha$. Then, $\mathscr{C}\,\mathbb{1}_H = \mu\,\beta\,\alpha\,\mathbb{1}_H = \mu\,\beta\,\mathbb{1}_H = \mu\,\mathbb{1}$. Hence $M_h = \sum_{b\in\mathscr{H}} \mathscr{C}_{hb}\delta_b = \mu^{-1}\sum_{b\in\mathscr{H}} \mathscr{C}_{hb} = 1$ for all $h \in \mathscr{H}$, since $\delta_b = \mu^{-1}$ for all $b \in \mathscr{H}$. **(iv) No intermediate imputs and firm specific factor supply.** Let $\widetilde{\Omega}_\ell = I_N$. Then, $\mathscr{C} = \beta\, diag(\mu)$, and $\delta_i = \mu_i^{-1}$. Therefore, $\mathscr{C}\,\delta = \beta\,\mathbb{1}_N = \mathbb{1}_N$.

1.5.7 Proof for Corollary 3

In the absence of distortions, we have $\delta_h = 1$ for all $h \in \mathscr{H}$. Consequently, $M_h = \sum_{b\in\mathscr{H}} \mathscr{C}_{hb} = 1\ \forall h \in \mathscr{H}$, and $F_i = \sum_{h\in\mathscr{H}} \psi^{\ell}_{ih} = 1$ for all $i \in \mathscr{N}$. As a result, the final demand terms of trade simplifies to $\sum_{h\in\mathscr{H}} \chi_h \sum_{i\in\mathscr{N}} d\,\beta_{hi} = 0$. Similarly, the sum of the *factor TT* + *intermediate TT* reduces to $\sum_{i\in\mathscr{N}} \lambda_i \sum_{i\in\mathscr{N}} d\,\beta_{hi} = 0$. Cases 2 and 3 follow directly from the preceding arguments and are omitted for brevity.

1.5.8 Proof for Theorem 3

The constrained social planner chooses $\left\{ Y,\, L,\, \{ C_h,\, L_h,\, T_h,\, \{ C_{hi}\}_{i\in\mathscr{N}}\}_{h\in\mathscr{H}} \right\}$ to maximize $W(Y,\,L)$ subject to $Y = Q_Y\left(\{C_h\}_{h\in\mathscr{H}}\right)$, $C_h = Q^c_h\left(\{C_{hi}\}_{i\in\mathscr{N}}\right)$, $L = F\left(\{L_h\}_{h\in\mathscr{H}}\right)$, $\sum_{h\in\mathscr{H}} T_h = 0$, and the household specific budget constraints (2). First order conditions for T_h imply that the Lagrange multipliers for the idiosyncratic budget constraints are symmetric across households, and as a consequence, the problem is isomorphic to maximizing $W(Y,\,L)$ by choosing $\left\{ Y,\, L,\, \{ C_h,\, L_h,\, \{ C_{hi}\}_{i\in\mathscr{N}}\}_{h\in\mathscr{H}} \right\}$ subject to the consolidated budget constraint

$$GDP = p_Y\,Y = \sum_{h\in\mathscr{H}} p^c_h\,C_h = \sum_{i\in\mathscr{N}} p_i \sum_{h\in\mathscr{H}} C_{hi} \le w\,L + \sum_{i\in\mathscr{N}} \pi_i, \qquad w\,L = \sum_{h\in\mathscr{H}} w_h\,L_h. \tag{63}$$

The planner takes as given $\{ w_h\}_{h\in\mathscr{H}}$ and $\{ p_i,\, \pi_i\}_{i\in\mathscr{N}}$.

The first-order conditions for the household's problem are

$$\frac{w_h}{p_Y} = -\frac{W_L}{W_Y}\,\frac{\partial\,L}{\partial\,L_h}, \tag{64}$$

$$\frac{p_i}{p_Y} = \frac{\partial\,Y}{\partial\,C_h}\,\frac{\partial\,C_h}{\partial\,C_{hi}} \qquad \forall h \in \mathscr{H} \quad \text{and} \quad \forall i \in \mathscr{N} :\ C_{hi} > 0. \tag{65}$$

Let me begin by characterizing the relationship between aggregate output and aggregate factor supply. The optimal condition linking real GDP and aggregate factor supply is

$$-\frac{W_L}{W_Y} = \underbrace{\frac{w\,L}{p_Y\,Y}}_{=\Gamma}\,\frac{Y}{L}. \tag{66}$$

This condition states that the aggregate marginal rate of substitution between real GDP and the aggregate factor supply equals the aggregate marginal rate of transformation scaled by the wedge Γ, which in equilibrium corresponds to the aggregate factor share. Given that $Y = TFP\,L$, the marginal rate of transformation is $TFP = Y/L = -\left(W_L/W_Y\right)\Gamma^{-1}$.

Under the optimal allocation for the planner, the ratio of marginal utilities relates aggregate output to supply of factor h via

$$-\frac{W_L}{W_Y} = \frac{\Lambda_h}{\widetilde{\Lambda}_h}\frac{Y}{L}. \tag{67}$$

This identity follows from the definition $\widetilde{\Lambda}_h = \frac{L_h}{L}\frac{\partial L}{\partial L_h}$.

Combining equations (66) and (67) yields the condition that the planner requires

$$\Gamma = \frac{\Lambda_h}{\widetilde{\Lambda}_h} = \delta_h^{-1} \quad \forall h \in \mathscr{H}. \tag{68}$$

Any deviation from this condition reflects an inefficient composition of aggregate factor supply

Summing over all factors yields $\Gamma \sum_{h\in\mathscr{H}} \widetilde{\Lambda}_h = \sum_{h\in\mathscr{H}} \Lambda_h$ and $\Gamma = \sum_{h\in\mathscr{H}} \Lambda_h$.

Define $\beta_i \equiv \frac{d \log GDP}{d \log p_i}$ as the elasticity of real GDP with respect to the price of good i. By Shephard's lemma, in equilibrium we have $\beta_i = \frac{\sum_{h\in\mathscr{H}} p_i C_{hi}}{GDP} = \sum_{h\in\mathscr{H}} \frac{C_{hi}}{Y}\frac{\partial Y}{\partial C_h}\frac{\partial C_h}{\partial C_{hi}} = \sum_{h\in\mathscr{H}} \chi_h \beta_{hi}$. Hence, equation (32) is given by $\lambda = \Psi_x' \,\overline{\beta}$, where $\overline{\beta} = [\beta_1, \cdots, \beta_N]'$, and define $\Lambda = \Omega_\ell' \,\Psi_x' \,\overline{\beta}$. From equation (35), the share for factor h is given by $\Lambda_h = \mathbb{1}_N' \, diag\left(\widetilde{\Omega}_\ell \, o_H(h)\right) \left(diag(\mu)^{-1} - \widetilde{\Omega}_x'\right)^{-1} \overline{\beta}$. Therefore,

$$\begin{aligned}
\Lambda_h \widehat{\Lambda}_h = {} & \mathbb{1}_N' \, diag(\Omega_\ell \, o_H(h)) \, \Psi_x' \begin{pmatrix} \beta_1 \widehat{\beta}_1 \\ \vdots \\ \beta_N \widehat{\beta}_N \end{pmatrix} + \sum_{i\in\mathscr{N}} \Omega_{ih}^\ell \, \lambda_i \left(\widehat{\omega}_i^\ell + \widehat{\alpha}_{ih}\right) \\
& + \mathbb{1}_N' \, diag\left(\widetilde{\Omega}_\ell \, o_H(h)\right) \left(\frac{d\left(diag(\mu)^{-1} - \widetilde{\Omega}_x'\right)^{-1}}{d\log \widetilde{\Omega}_x} + \frac{d\left(diag(\mu)^{-1} - \widetilde{\Omega}_x'\right)^{-1}}{d\log \mu} \right) \overline{\beta}.
\end{aligned}$$

Using the fact that for any invertible matrix A, $\frac{dA^{-1}}{dx} = -A^{-1}\frac{dA}{dx}A^{-1}$, the previous equation becomes

$$\begin{aligned}
d\,\Lambda_h = {} & \sum_{i\in\mathscr{N}} \Omega_{ih}^\ell \sum_{j\in\mathscr{N}} \psi_{ji}^x \, d\,\beta_j + \sum_{i\in\mathscr{N}} \Omega_{ih}^\ell \, \lambda_i \left(\widehat{\omega}_i^\ell + \widehat{\alpha}_{ih}\right) \\
& - \mathbb{1}_N' diag(\Omega_\ell \, o_H(h)) \, \Psi_x' \left(\frac{d\left(diag(\mu)^{-1} - \widetilde{\Omega}_x'\right)}{d\log \widetilde{\Omega}_x} + \frac{d\left(diag(\mu)^{-1} - \widetilde{\Omega}_x'\right)}{d\log \mu} \right) diag(\mu) \ \lambda.
\end{aligned}$$

$$d\,\Lambda_h = \sum_{i\in\mathscr{N}} \psi_{ih}^\ell \, \lambda_i \, d\log \mu_i + \sum_{i\in\mathscr{N}} \mu_i \, \lambda_i \, d\,\widetilde{\Omega}_{ih}^\ell + \sum_{j\in\mathscr{N}} \psi_{jh}^\ell \left(d\,\beta_j + \sum_{i\in\mathscr{N}} \mu_i \, \lambda_i \, d\,\widetilde{\Omega}_{ij}^x \right).$$

Then

$$d\,\Lambda = \sum_{h\in\mathscr{H}} d\,\Lambda_h = \sum_{i\in\mathscr{H}} \psi_i^\ell \, \lambda_i \, d\log \mu_i + \sum_{i\in\mathscr{N}} \mu_i \, \lambda_i \, d\,\omega_i^\ell + \sum_{j\in\mathscr{N}} \psi_j^\ell \left(d\,\beta_j + \sum_{i\in\mathscr{N}} \mu_i \, \lambda_i \, d\,\widetilde{\Omega}_{ij}^x \right),$$

with $\psi_i^\ell = \sum_{h\in\mathscr{H}} \psi_{ih}^\ell$.

2 Proofs for the normalized nested-CES model

2.1 Environment

Firms. Each firm $i \in \mathscr{N}$ chooses $\left\{y_i, \{\ell_{ih}\}_{h\in\mathscr{H}}, \{x_{ij}\}_{j\in\mathscr{N}}\right\}$ subject to the nested CES production structure defined below.

$$\frac{y_i}{\overline{y}_i} = A_i\left(\omega_i^{\ell}\left(\frac{L_i}{\overline{L}_i}\right)^{\frac{\theta_i-1}{\theta_i}} + \omega_i^{x}\left(\frac{X_i}{\overline{X}_i}\right)^{\frac{\theta_i-1}{\theta_i}}\right)^{\frac{\theta_i}{\theta_i-1}},$$

$$\frac{L_i}{\overline{L}_{z_i}} = \left(\sum_{h\in\mathscr{H}}\alpha_{ih}\left(\frac{\ell_{ih}}{\overline{\ell}_{ih}}\right)^{\frac{\theta_i^{\ell}-1}{\theta_i^{\ell}}}\right)^{\frac{\theta_i^{\ell}}{\theta_i^{\ell}-1}}, \qquad \frac{X_i}{X_i} = \left(\sum_{j\in\mathscr{N}}\omega_{ij}\left(\frac{x_{ij}}{\overline{x}_{ij}}\right)^{\frac{\theta_i^{x}-1}{\theta_i^{x}}}\right)^{\frac{\theta_i^{x}}{\theta_i^{x}-1}}.$$

From here, the first-order conditions are given by

$$p_i^{\ell}L_i = \left(\mu_i\omega_i^{\ell}\right)^{\theta_i}\left(A_i\frac{p_i\overline{y}_i}{p_i^{\ell}\overline{L}_i}\right)^{\theta_i-1} p_iy_i, \tag{69}$$

$$p_i^{x}X_i = \left(\mu_i\omega_i^{x}\right)^{\theta_i}\left(A_i\frac{p_i\overline{y}_i}{p_i^{x}\overline{X}_i}\right)^{\theta_i-1} p_iy_i, \tag{70}$$

$$w_h\ell_{ih} = \left(\mu_i\omega_i^{\ell}\right)^{\theta_i}\alpha_{ih}^{\theta_i^{\ell}}\left(A_i\frac{p_i\overline{y}_i}{p_i^{\ell}\overline{L}_i}\right)^{\theta_i-1}\left(\frac{p_i^{\ell}\overline{L}_i}{w_h\overline{\ell}_{ih}}\right)^{\theta_i^{\ell}-1} p_iy_i, \tag{71}$$

$$p_jx_{ij} = \left(\mu_i\omega_i^{x}\right)^{\theta_i}\omega_{ij}^{\theta_i^{x}}\left(A_i\frac{p_i\overline{y}_i}{p_i^{x}\overline{X}_i}\right)^{\theta_i-1}\left(\frac{p_i^{x}\overline{X}_i}{p_j\overline{x}_{ij}}\right)^{\theta_i^{x}-1} p_iy_i. \tag{72}$$

At the point of normalization, assume $A_i = 1$ for all $i \in \mathscr{N}$. Then, the first-order conditions satisfy in equilibrium: (i) $\overline{p}_i^{\ell}\,\overline{L}_i = \mu_i\,\omega_i^{\ell}\,\overline{p}_i\,\overline{y}_i$, (ii) $\overline{p}_i^{x}\,\overline{X}_i = \mu_i\,\omega_i^{x}\,\overline{p}_i\,\overline{y}_i$, (iii) $\overline{w}_h\,\overline{\ell}_{ih} = \alpha_{ih}\,\overline{p}_i^{\ell}\,\overline{L}_i$, and (iv) $\overline{p}_j\,\overline{x}_{ij} = \omega_{ij}\,\overline{p}_i^{x}\,\overline{X}_i$.

Finally, prices are given by

$$p_i^{\ell} = \frac{1}{\overline{L}_i}\left(\sum_{h\in\mathscr{H}}\alpha_{ih}^{\theta_i^{\ell}}\left(w_h\,\overline{\ell}_{ih}\right)^{1-\theta_i^{\ell}}\right)^{\frac{1}{1-\theta_i^{\ell}}}, \tag{73}$$

$$p_i^{x} = \frac{1}{\overline{X}_i}\left(\sum_{j\in\mathscr{N}}\omega_{ij}^{\theta_i^{x}}\left(p_j\,\overline{x}_{ij}\right)^{1-\theta_i^{x}}\right)^{\frac{1}{1-\theta_i^{x}}}, \tag{74}$$

$$p_i = \frac{1}{A_i\,\mu_i\,\overline{y}_i}\left(\omega_i^{\ell\,\theta_i}\left(p_i^{\ell}\,\overline{L}_i\right)^{1-\theta_i} + \omega_i^{x\,\theta_i}\left(p_i^{x}\,\overline{X}_i\right)^{1-\theta_i}\right)^{\frac{1}{1-\theta_i}}. \tag{75}$$

Households. Each household h chooses $\left\{\{C_{hi}\}_{i\in\mathscr{N}}, L_h\right\}$ to maximize

$$U_h\left(c_h, \widetilde{L}_h\right) = \frac{\left[c_h\left(1 - E_h^{-\gamma_h}\widetilde{L}_h\right)^{\varphi_h}\right]^{1-\sigma} - 1}{1-\sigma},$$

subject to $C_h = n_h c_h$, $L_h = n_h \widetilde{L}_h$,

$$\frac{C_h}{\overline{C}_h} = \left(\sum_{i \in \mathcal{N}} \beta_{hi} \left(\frac{C_{hi}}{\overline{C}_{hi}} \right)^{\frac{\varrho_h - 1}{\varrho_h}} \right)^{\frac{\varrho_h}{\varrho_h - 1}}, \qquad E_h = p_h^c C_h = \sum_{i \in \mathcal{N}} p_i C_{hi} \leq w_h L_h + \sum_{i \in \mathcal{N}} \kappa_{ih} \, \pi_i + T_h.$$

The first-order conditions relative to C_{hi}, C_h, and L_h are respectively given by

$$c_h^{-\sigma} \left(1 - E_h^{-\gamma_h} \widetilde{L}_h\right)^{\varphi_h(1-\sigma)} \left(1 + \varphi_h \gamma_h \frac{E_h^{-\gamma_h} \widetilde{L}_h}{1 - E_h^{-\gamma_h} \widetilde{L}_h}\right) \frac{\partial C_h}{\partial C_{hi}} = \varkappa_h n_h p_i, \tag{76}$$

$$c_h^{-\sigma} \left(1 - E_h^{-\gamma_h} \widetilde{L}_h\right)^{\varphi_h(1-\sigma)} \left(1 + \varphi_h \gamma_h \frac{E_h^{-\gamma_h} \widetilde{L}_h}{1 - E_h^{-\gamma_h} \widetilde{L}_h}\right) = \varkappa_h n_h p_h^c, \tag{77}$$

$$\varphi_h c_h^{1-\sigma} \left(1 - E_h^{-\gamma_h} \widetilde{L}_h\right)^{\varphi_h(1-\sigma)-1} E_h^{-\gamma_h} = \varkappa_h n_h w_h, \tag{78}$$

where $\frac{\partial C_h}{\partial C_{hi}} = \beta_{hi} \left(\frac{\overline{C}_h}{\overline{C}_{hi}} \right)^{\frac{\varrho_h - 1}{\varrho_h}} \left(\frac{C_h}{C_{hi}} \right)^{\frac{1}{\varrho_h}}$, and $\varkappa_h$ denotes the Lagrange multiplier associated with household h's budget constraint.

Combining equations (76) and (77), the household's optimal demand for good i satisfies

$$p_i \, C_{hi} = \beta_{hi}^{\varrho_h} \left(\frac{p_h^c \, \overline{C}_h}{p_i \, \overline{C}_{hi}} \right)^{\varrho_h - 1} E_h, \tag{79}$$

or $\overline{p}_i \, \overline{C}_{hi} = \beta_{hi} \, \overline{C}_h$ at the point of normalization.

Combining the first-order conditions (77) and (78), household factor income satisfies:

$$w_h L_h = \frac{n_h w_h E_h^{\gamma_h} - \varphi_h E_h}{1 - \varphi_h \gamma_h}. \tag{80}$$

This expression characterizes factor supply as a nonlinear function of consumption expenditure, disutility curvature γ_h, and the parameter φ_h.

Finally, the price index for household h's composite consumption bundle is given by

$$p_h^c = \frac{1}{\overline{C}_h} \left(\sum_{i \in \mathcal{N}} \beta_{hi}^{\varrho_h} \left(p_i \, \overline{C}_{hi} \right)^{1-\varrho_h} \right)^{\frac{1}{1-\varrho_h}}. \tag{81}$$

2.2 Proof for Proposition 2

From equation (80), the-first order approximation for the factor supply schedule

$$L_h \widehat{L}_h = n_h \frac{\partial L_h}{\partial n_h} \widehat{n}_h + w_h \frac{\partial L_h}{\partial w_h} \widehat{w}_h + E_h \frac{\partial L_h}{\partial E_h} \widehat{E}_h.$$

$$\frac{\partial L_h}{\partial n_h} = \frac{E_h^{\gamma_h}}{1 - \varphi_h \gamma_h}, \quad \frac{\partial L_h}{\partial w_h} = \frac{\varphi_h E_h}{(1 - \varphi_h \gamma_h) \, w_h^2}, \quad \frac{\partial L_h}{\partial E_h} = -\frac{1}{1 - \varphi_h \gamma_h} \left(\frac{\varphi_h}{w_h} - \gamma_h n_h E_h^{\gamma_h - 1} \right),$$

$$\widehat{L}_h = \frac{1}{1 - \varphi_h \gamma_h} \left(E_h^{\gamma_h} \frac{n_h}{L_h} \widehat{n}_h + \varphi_h \frac{\chi_h}{\Lambda_h} \widehat{w}_h - \left(\varphi_h \frac{\chi_h}{\Lambda_h} - \gamma_h E_h^{\gamma_h} \frac{n_h}{L_h} \right) \widehat{E}_h \right) = \zeta_h^n \widehat{n}_h + \zeta_h^w \widehat{w}_h - \zeta_h^e \widehat{E}_h \tag{82}$$

$$\zeta_h^n = \frac{E_h^{\gamma_h}}{1 - \varphi_h \gamma_h} \frac{n_h}{L_h}, \qquad \zeta_h^w = \frac{\varphi_h}{1 - \varphi_h \gamma_h} \frac{\chi_h}{\Lambda_h}, \qquad \zeta_h^e = \zeta_h^w - \gamma_h \zeta_h^n.$$

Under KPR preferences ($\zeta_h^w = \zeta_h^e$) we require $\gamma_h = 0$, which leaves us with $\widehat{L}_h = \zeta_h^n \widehat{n}_h + \zeta_h^w \widehat{w}_h - \zeta_h^e \widehat{E}_h$ with $\zeta_h^n = \frac{n_h}{L_h}$ and $\zeta_h^w = \zeta_h^e = \varphi_h \frac{\chi_h}{\Lambda_h}$. Under GHH preferences ($\zeta_h^e = 0$) γ_h and φ_h are jointly characterized by $\varphi_h = \gamma_h \, E_h^{\gamma_h - 1} \, n_h \, w_h$.

2.3 Equilibrium conditions

Goods markets: From equations (72) and (79), the goods produced by firm $i \in \mathcal{N}$ must satisfy

$$S_i = \sum_{j \in \mathcal{N}} \left(\mu_j \omega_j^x\right)^{\theta_j} \omega_{ji}^{\theta_j^x} \left(A_j \frac{p_j \overline{y}_j}{p_j^x \overline{X}_j}\right)^{\theta_j - 1} \left(\frac{p_j^x \overline{X}_j}{p_i \overline{x}_{ji}}\right)^{\theta_j^x - 1} S_j + \sum_{h \in \mathcal{H}} \beta_{hi}^{\varrho_h} \left(\frac{p_h^c \overline{C}_h}{p_i \overline{C}_{hi}}\right)^{\varrho_h - 1} E_h. \tag{83}$$

The first order approximation for equation (83) is given by

$$\begin{aligned}
\lambda_i \widehat{S}_i = &\sum_{h \in \mathcal{H}} \beta_{hi} \chi_h \left(\varrho_h \widehat{\beta}_{hi} + (\varrho_h - 1)\left(\widehat{p}_h^c - \widehat{p}_i\right) + \widehat{E}_h\right) \\
&+ \sum_{j \in \mathcal{N}} \Omega_{ji}^x \lambda_j \left[\theta_j \left(\widehat{\mu}_j + \widehat{\omega}_j^x\right) + \theta_j^x \, \widehat{\omega}_{ji} + (\theta_i - 1)\left(\widehat{A}_j + \widehat{p}_j\right) + \left(\theta_j^x - \theta_j\right) \widehat{p}_j^x - \left(\theta_j^x - 1\right) \widehat{p}_i + \widehat{S}_j\right].
\end{aligned}$$

In matrix form this equation is given by

$$\begin{aligned}
diag(\lambda)\, \widehat{S} = \Psi_x' \Big\{ &\left(\beta \circ \widehat{\beta}\right)' diag(\varrho)\, \chi + \beta' diag(\chi) \left(diag(\varrho - \mathbb{1}_H)\, \widehat{p}_c + \widehat{E}\right) \\
&- diag\left(\beta' diag(\varrho - \mathbb{1}_H)\, \chi\right) \widehat{p} + \left(\Omega_x \circ \widehat{\mathscr{W}}\right)' diag(\theta_x)\, \lambda - diag\left(\Omega_x' diag(\theta_x - \mathbb{1}_N)\, \lambda\right) \widehat{p} \\
&+ \Omega_x' diag(\lambda) \left(diag(\theta)\left(\widehat{\omega}_x + \widehat{\mu}\right) + diag(\theta - \mathbb{1}_N)\left(\widehat{A} + \widehat{p}\right) + diag(\theta_x - \theta)\, \widehat{p}_x\right) \Big\}.
\end{aligned} \tag{84}$$

Households' budget constraints: The first order approximation for the households' budget constraints are given by

$$\chi_h \widehat{E}_h = \Lambda_h \widehat{J}_h + \sum_{i \in \mathcal{N}} \kappa_{ih} \lambda_i \left((1 - \mu_i)\left(\widehat{\kappa}_{ih} + \widehat{S}_i\right) - \mu_i \widehat{\mu}_i\right) + \tau_h \, \widehat{T}_h.$$

In matrix form this equation is given by

$$\begin{aligned}
diag(\chi)\, \widehat{E} = &\; diag(\Lambda)\, \widehat{J} + \kappa' \, diag(\lambda) \left(diag(\mathbb{1}_N - \mu)\, \widehat{S} - diag(\mu)\, \widehat{\mu}\right) \\
&+ (\kappa \circ \widehat{\kappa})' \, diag(\mathbb{1}_N - \mu)\, \lambda + diag(\tau)\, \widehat{T}.
\end{aligned} \tag{85}$$

Factor Markets: From equation (71), equilibrium in the factor market for household h must satisfy

$$w_h L_h = \sum_{i \in \mathcal{N}} w_h \ell_{ih} = \sum_{i \in \mathcal{N}} \left(\mu_i \omega_i^\ell\right)^{\theta_i} \alpha_{ih}^{\theta_i^\ell} \left(A_i \frac{p_i \overline{y}_i}{p_i^\ell \overline{L}_i}\right)^{\theta_i - 1} \left(\frac{p_i^\ell \overline{L}_i}{w_h \overline{\ell}_{ih}}\right)^{\theta_i^\ell - 1} S_i. \tag{86}$$

The first order approximation for equation (86) is given by

$$\Lambda_h \widehat{J}_h = \sum_{i \in \mathcal{N}} \Omega_{ih}^\ell \lambda_i \left[\theta_i \left(\widehat{\mu}_i + \widehat{\omega}_i^\ell\right) + \theta_i^\ell \, \widehat{\alpha}_{ih} + (\theta_i - 1)\left(\widehat{A}_i + \widehat{p}_i\right) + \left(\theta_i^\ell - \theta_i\right) \widehat{p}_i^\ell - \left(\theta_i^\ell - 1\right) \widehat{w}_h + \widehat{S}_i\right].$$

In matrix form this equation is given by

$$\begin{aligned} diag(\Lambda)\,\widehat{J} &= (\Omega_\ell \circ \widehat{\alpha})'\, diag(\theta_\ell)\,\lambda - diag\big(\Omega_\ell' diag(\theta_\ell - \mathbb{1}_N)\,\lambda\big)\,\widehat{w} \\ &+ \Omega_\ell' diag(\lambda)\left(diag(\theta)\ (\widehat{\omega}_\ell + \widehat{\mu}) + diag(\theta - \mathbb{1}_N)\left(\widehat{A} + \widehat{p}\right) + diag(\theta_\ell - \theta)\,\widehat{p}_\ell + \widehat{S}\right). \end{aligned} \tag{87}$$

Prices: The first-order approximation for equations (73), (74), (75), and (81) are given by $\widehat{p}_i^\ell = \sum_{h\in\mathscr{H}} \alpha_{ih}\,\widehat{w}_h$, $\widehat{p}_i^x = \sum_{j\in\mathscr{N}} \omega_{ij}\,\widehat{p}_j$, $\widehat{p}_i = \omega_i^\ell\,\widehat{p}_i^\ell + \omega_i^x\,\widehat{p}_i^x - \widehat{A}_i - \widehat{\mu}_i$, and $\widehat{p}_h^c = \sum_{i\in\mathscr{N}} \beta_{hi}\,\widehat{p}_i$.

In matrix form these equation are given by

$$\widehat{p}_\ell = \alpha\,\widehat{w}, \tag{88}$$

$$\widehat{p} = \widetilde{\Psi}_\ell\,\widehat{w} - \widetilde{\Psi}_x\left(\widehat{A} + \widehat{\mu}\right), \tag{89}$$

$$\widehat{p}_x = \mathscr{W}\,\widetilde{\Psi}_x\left(\widetilde{\Omega}_\ell\,\widehat{w} - \widehat{A} - \widehat{\mu}\right), \tag{90}$$

$$\widehat{p}_c = \widetilde{\mathscr{B}}\left(\widetilde{\Omega}_\ell\,\widehat{w} - \widehat{A} - \widehat{\mu}\right). \tag{91}$$

2.4 Sufficient Equations: Theorem 7

Theorem 7 characterizes a linear system of dimension $2H + N$ that determines the endogenous first-order response of consumption expenditure, factor prices, and firm sales. This system incorporates both partial equilibrium (PE) and general equilibrium (GE) equilibrium effects.

Theorem 7. In a CES economy, the first-order response of consumption expenditure, factor prices, and firm sales, in response to shocks in productivity, markups, population size, preferences and equity composition are, to a first-order,

$$\begin{aligned} d\log E_h = &\overbrace{\frac{\zeta_h^n \Lambda_h}{\chi_h + \zeta_h^e \Lambda_h}}^{\substack{\text{Demographic Effect} \\ \text{on Expenditure (PE)}}} d\log n_h + \overbrace{\frac{(1+\zeta_h^w)\,\Lambda_h}{\chi_h + \zeta_h^e \Lambda_h}}^{\substack{\text{Factor Price Effect} \\ \text{on Expenditure (GE)}}} d\log w_h + \overbrace{\frac{\tau_h}{\chi_h + \zeta_h^e \Lambda_h}}^{\substack{\text{Lump-sum Effect} \\ \text{on Expenditure (PE)}}} d\log T_h \\ &+ \overbrace{\sum_{i\in\mathscr{N}} \frac{\lambda_i}{\chi_h + \zeta_h^e \Lambda_h} \left((1-\mu_i)\,(d\,\kappa_{ih} + \kappa_{ih}\,d\log S_i) - \kappa_{ih}\,\mu_i\,d\log \mu_i\right)}^{\substack{\text{Corporate Income Effect} \\ \text{on Expenditure (PE + GE)}}}; \end{aligned}$$

$$
\begin{aligned}
d\log S_i = {} & \overbrace{\sum_{h\in\mathscr{H}} \frac{\beta_{hi}\chi_h}{\lambda_i}\, d\log E_h}^{\substack{\textit{Expenditure Effect}\\ \textit{on Sales (GE)}}} + \overbrace{\sum_{j\in\mathscr{N}} \frac{\Omega^x_{ji}\lambda_j}{\lambda_i}\, d\log S_j}^{\substack{\textit{Sales Effect}\\ \textit{on Sales (GE)}}} + \overbrace{\sum_{j\in\mathscr{N}} \frac{\Omega^x_{ji}\lambda_j}{\lambda_i}\left(\left(\theta_j-1\right) d\log A_j + \theta_j\, d\log \mu_j\right)}^{\substack{\textit{Direct Effect}\\ \textit{on Sales (PE)}}} \\
& \overbrace{+\sum_{j\in\mathscr{N}}\left(\sum_{h\in\mathscr{H}} \frac{\beta_{hi}\chi_h}{\lambda_i}\left(\varrho_h-1\right)\left(\widetilde{\psi}^x_{ij}-\widetilde{\mathscr{B}}_{hj}\right)+\sum_{q\in\mathscr{N}} \frac{\Omega^x_{qi}\lambda_q}{\lambda_i}\left(\theta_q-1\right)\left(\widetilde{\psi}^x_{ij}-\widetilde{\psi}^x_{qj}\right)\right)\left(d\log A_j + d\log\mu_j\right)}^{\substack{\textit{Supplier Effect}\\ \textit{on Sales (PE)}}} \\
& \overbrace{+\sum_{j\in\mathscr{N}}\left(\sum_{q\in\mathscr{N}} \frac{\Omega^x_{qi}\lambda_q}{\lambda_i}\left(\theta^x_q-\theta_q\right)\left(\widetilde{\psi}^x_{ij}-\sum_{m\in\mathscr{N}}\omega_{qm}\,\widetilde{\psi}^x_{mj}\right)\right)\left(d\log A_j + d\log\mu_j\right)}^{\substack{\textit{Indirect Supplier Effect}\\ \textit{on Sales (PE)}}} \\
& + \overbrace{\sum_{h\in\mathscr{H}} \frac{\chi_h}{\lambda_i}\varrho_h\, d\beta_{hi}}^{\substack{\textit{Household Preference}\\ \textit{Effect on Sales (PE)}}} + \overbrace{\sum_{j\in\mathscr{N}} \frac{\mu_j\lambda_j}{\lambda_i}\left(\theta_j\,\omega_{ji}\, d\omega^x_j + \theta^x_j\,\omega^x_j\, d\omega_{ji}\right)}^{\substack{\textit{Firm Technology Effect}\\ \textit{on Sales (PE)}}} \\
& \overbrace{+\sum_{h\in\mathscr{H}}\left(\sum_{b\in\mathscr{H}} \frac{\beta_{bi}\chi_b}{\lambda_i}\left(\varrho_b-1\right)\left(\widetilde{\mathscr{C}}_{bh}-\widetilde{\psi}^\ell_{ih}\right)+\sum_{j\in\mathscr{N}} \frac{\Omega^x_{ji}\lambda_j}{\lambda_i}\left(\theta_j-1\right)\left(\widetilde{\psi}^\ell_{jh}-\widetilde{\psi}^\ell_{ih}\right)\right) d\log w_h}^{\substack{\textit{Supplier Substitution Effect}\\ \textit{on Sales (GE)}}} \\
& \overbrace{+\sum_{h\in\mathscr{H}}\left(\sum_{j\in\mathscr{N}} \frac{\Omega^x_{ji}\lambda_j}{\lambda_i}\left(\theta^x_j-\theta_j\right)\left(\sum_{q\in\mathscr{N}}\omega_{jq}\widetilde{\psi}^\ell_{qh}-\widetilde{\psi}^\ell_{ih}\right)\right) d\log w_h}^{\substack{\textit{Indirect Supplier Substitution Effect}\\ \textit{on Sales (GE)}}};
\end{aligned}
$$

$$
\begin{aligned}
d\log w_h = {} & \overbrace{\frac{\zeta^e_h}{1+\zeta^w_h}\, d\log E_h}^{\substack{\textit{Expenditure Effect}\\ \textit{on Factor Prices (GE)}}} - \overbrace{\frac{\zeta^n_h}{1+\zeta^w_h} d\log n_h}^{\substack{\textit{Demographic Effect}\\ \textit{on Factor Prices (PE)}}} + \overbrace{\sum_{i\in\mathscr{N}} \frac{\mu_i\,\lambda_i}{\left(1+\zeta^w_h\right)\Lambda_h}\left(\theta_i\,\alpha_{ih}\, d\omega^\ell_i + \theta^\ell_i\,\omega^\ell_i\, d\alpha_{ih}\right)}^{\substack{\textit{Firm Technology Effect}\\ \textit{on Factor Prices (PE)}}} \\
& \overbrace{+\sum_{i\in\mathscr{N}} \frac{\Omega^\ell_{ih}\lambda_i}{\left(1+\zeta^w_h\right)\Lambda_h}\left(\left(\theta_i-1\right) d\log A_i + \theta_i\, d\log\mu_i\right)}^{\substack{\textit{Direct Effect}\\ \textit{on Factor Prices (PE)}}} - \overbrace{\left(\sum_{i\in\mathscr{N}} \frac{\Omega^\ell_{ih}\lambda_i}{\left(1+\zeta^w_h\right)\Lambda_h}\left(\theta^\ell_i-1\right)\right) d\log w_h}^{\substack{\textit{Direct Substitution Effect}\\ \textit{on Factor Prices (GE)}}} \\
& \overbrace{-\sum_{j\in\mathscr{N}}\left(\sum_{i\in\mathscr{N}} \frac{\Omega^\ell_{ih}\lambda_i}{\left(1+\zeta^w_h\right)\Lambda_h}\left(\theta_i-1\right)\widetilde{\psi}^x_{ij}\right)\left(d\log A_j + d\log\mu_j\right)}^{\substack{\textit{Supplier Effect}\\ \textit{on Factor Prices (PE)}}} + \overbrace{\sum_{i\in\mathscr{N}} \frac{\Omega^\ell_{ih}\lambda_i}{\left(1+\zeta^w_h\right)\Lambda_h}\, d\log S_i}^{\substack{\textit{Sales Effect}\\ \textit{on Factor Prices (GE)}}} \\
& \overbrace{+\sum_{b\in\mathscr{H}}\left(\sum_{i\in\mathscr{N}} \frac{\Omega^\ell_{ih}\lambda_i}{\left(1+\zeta^w_h\right)\Lambda_h}\left(\theta_i-1\right)\left(\widetilde{\psi}^\ell_{ib}-\alpha_{ib}\right)+\left(\theta^\ell_i-1\right)\alpha_{ib}\right) d\log w_b}^{\substack{\textit{Supplier Substitution Effect}\\ \textit{on Factor Prices (GE)}}}.
\end{aligned}
$$

For a household of type h, the first-order variation in consumption expenditure operates through four channels. First, the *demographic effect*: an increase in population size n_h raises factor supply by ζ^n_h, and its impact on consumption expenditure is scaled by the household's income share Λ_h. Second, the *factor price effect*: an increase in factor prices raises income directly, but also induces a substitution effect on factor supply governed by ζ^w_h. The overall impact on expenditure is again scaled by Λ_h. Third, the *lump-sum effect*: an increase in T_h raises consumption expenditure, and its impact is scaled by τ_h. Finally, the *corporate income effect*: household h's dividends from firm i depend on their equity share κ_{ih}, the firm's revenue S_i, and its cost-share μ_i: (i) an increase in equity share increases dividends proportionally to the rent-extraction rate $1-\mu_i$; (ii) higher firm sales increase dividends by $\kappa_{ih}\left(1-\mu_i\right)$; and (iii) a higher markup reduces dividens by $\kappa_{ih}\ \mu_i$. These three channels raise consumption expenditure and induce an income effect that lowers factor supply, thereby attenuating the overall response by the term $\chi_h + \zeta_h^e \Lambda_h$.

For firm i, the first-order variation in sales arises through nine distinct channels. Channels related to household demand for final goods are weighted by expenditure shares β_{hi} and the Domar weight ratio χ_h/λ_i, while channels involving intermediate demand from firm j are scaled by revenue exposure Ω^x_{ji} and the ratio λ_j/λ_i. First, the *expenditure effect* reflects how rising household spending increases demand for firm i's output. Second, the *sales effect* captures how higher sales by downstream firms raise demand for firm i's intermediates. Third, the *direct effect* captures the increase in intermediate demand from firm j in response to shocks to its productivity and markups. These responses depend on firm j's elasticity of substitution: productivity increases raise demand if inputs are substitutable, and lower markups do so unless the production function is Leontief. Fourth, the *supplier effect* describes how households and firms shift demand across suppliers in response to productivity or markup shocks further upstream. Under substitutability, if household h or firm q are less exposed to supplier j than firm i is — i.e., if $\widetilde{\psi}^x_{ij} > \widetilde{\mathscr{B}}_{hj}$ or $\widetilde{\psi}^x_{ij} > \widetilde{\psi}^x_{qj}$ — then they reallocate demand toward good i. Fifth, the *indirect supplier effect* compares firm i's exposure to a shock in supplier j against the average exposure of firm q's suppliers. If firm q has greater substitutability across intermediates than across factors ($\theta^x_q > \theta_q$), it reallocates demand toward firm i when $\widetilde{\psi}^x_{ij} > \sum_{m \in \mathcal{N}} \omega_{qm}\, \widetilde{\psi}^x_{mj}$. Sixth, the *household preference effect* reflects changes in β_{hi} scaled by the substitution elasticity ϱ_h. Seventh, the *firm technology effect* reflects changes in input weights ω^x_j and ω_{ji}, weighted by the elasticities θ_j and θ^x_j. Eight, the *supplier substitution effect* captures how households and firms reallocate demand across goods in response to changes in factor prices. Under substitutability, if household b or firm j are more exposed to factor h than firm i is — i.e., if $\widetilde{\mathscr{C}}_{bh} > \widetilde{\psi}^\ell_{ih}$ and $\widetilde{\psi}^\ell_{jh} > \widetilde{\psi}^\ell_{ih}$ — then they reallocate demand towards firm i. Finally, the *indirect supplier substitution effect* compares firm i's exposure to higher w_h against the average exposure of firm j's suppliers. If firm j has greater substitutability across intermediates than across factors ($\theta^x_j > \theta_h$), it reallocates demand toward firm i when $\sum_{m \in \mathcal{N}} \omega_{jq}\, \widetilde{\psi}^\ell_{qh} > \widetilde{\psi}^\ell_{ih}$.

For household type h, the first-order variation in their factor prices w_h arises through eight distinct channels. Each channel induces a substitution effect that raises factor supply, attenuating the response of factor prices by the term $1 + \zeta^w_h$. The effect of firm-level shocks on w_h is proportional to the revenue-based centrality Ω^ℓ_{ih} and the sales-to-factor-income ratio λ_i/Λ_h. First, the *expenditure effect* reflects that an increase in household income reduces factor supply by ζ^e_h, thereby raising wages. Second, the *demographic effect* captures the factor-price depressing impact of population increase, which raises supply by ζ^n_h. Third, the *firm technology effect* reflects changes in factor weights ω^ℓ_j and α_{ih}, weighted by the elasticities θ_j and θ^ℓ_j. Fourth, the *direct effect* captures increased demand for factor h from firms receiving productivity or markup shocks. Firm i increases demand for type-h factor in response to a productivity shock if inputs are substitutable ($\theta_i > 1$), and in response to lower markups as long as $\theta_i > 0$. Fifth, the *supplier effect* reflects how firm i's demand for factor h decreases when upstream suppliers j experience productivity gains or markup reductions — provided production is substitutable. The strength of this effect scales with firm-to-firm cost-based centrality $\widetilde{\psi}^x_{ij}$. Sixth, the *sales effect* captures how rising firm sales increase scale and thus factor demand. Seventh, the *direct substitution effect* reflects how firm i reduces demand for factor h when w_h increases as long as $\theta^\ell_i > 1$. Finally, the *supplier substitution effect* captures how firms reallocate demand towards factor h when the price of another factor b rises. Demand from firm i for factor h rises when w_b increases when its cost-based factor-to-firm exposure dominates its direct exposure ($\widetilde{\psi}^\ell_{ib} > \alpha_{ib}$) as long as $\theta_i > 0$, and when $\theta^\ell_i > 1$.

The solution in Theorem 7 offer an alternative to Baqaee & Farhi (2024) along five key dimensions: (1) it opens up and decomposes the propagation channels that are summarized by the production network covariance operator introduced in Baqaee & Farhi (2019); (2) it relies on bilateral centrality measures introduced in this paper; (3) it explicitly incorporates demographic, substitution, and income elasticities of factor supply; (4) it decomposes the channels for shocks into direct effects and propagation through intermediate input linkages; and (5) variations are expressed in nominal terms and not in Domar weights.

Proof.

Factor Income. Introducing equations (82), (88), and (89) in equation (87)

$$\begin{aligned}
& diag(\Lambda)\left(I_H + diag(\zeta_w)\right)\widehat{w} = \left(\Omega_\ell \circ \widehat{\alpha}\right)' diag(\theta_\ell)\,\lambda - diag\left(\Omega_\ell' diag(\theta_\ell - \mathbb{1}_N)\,\lambda\right)\widehat{w} \\
& + diag(\zeta_e)\, diag(\Lambda)\,\widehat{E} - diag(\zeta_n)\, diag(\Lambda)\,\widehat{n} \\
& + \Omega_\ell' diag(\lambda)\left(diag(\theta)\,\left(\widehat{\omega}_\ell + \widehat{\mu}\right) + diag(\theta - \mathbb{1}_N)\left(\widetilde{\Psi}_\ell \widehat{w} + \left(I_N - \widetilde{\Psi}_x\right)\widehat{A} - \widetilde{\Psi}_x \widehat{\mu}\right) + diag(\theta_\ell - \theta)\,\alpha \widehat{w} + \widehat{S}\right) \\
& = \Omega_\ell' diag(\lambda)\, diag(\theta - \mathbb{1}_N)\left(I_N - \widetilde{\Psi}_x\right)\widehat{A} + \Omega_\ell' diag(\lambda)\left(diag(\theta) - diag(\theta - \mathbb{1}_N)\,\widetilde{\Psi}_x\right)\widehat{\mu} \\
& + \Omega_\ell' \, diag(\lambda)\, diag(\theta)\,\widehat{\omega}_\ell + \left(\Omega_\ell \circ \widehat{\alpha}\right)' diag(\theta_\ell)\,\lambda - diag(\zeta_n)\, diag(\Lambda)\,\widehat{n} \\
& + \Omega_\ell' diag(\lambda)\,\widehat{S} + diag(\zeta_e)\, diag(\Lambda)\,\widehat{E} + diag\left(\Omega_\ell' diag(\theta_\ell - \mathbb{1}_N)\,\lambda\right)\left(\widetilde{\mathscr{C}} - I_H\right)\widehat{w} \\
& + \left(\Omega_\ell' diag(\lambda)\, diag(\theta - \mathbb{1}_N)\,\widetilde{\Psi}_\ell - diag\left(\Omega_\ell' diag(\theta - \mathbb{1}_N)\,\lambda\right)\widetilde{\mathscr{C}}\right)\widehat{w} \\
& + \left(\Omega_\ell' diag(\lambda)\, diag(\theta_\ell - \theta)\,\alpha - diag\left(\Omega_\ell' diag(\theta_\ell - \theta)\,\lambda\right)\widetilde{\mathscr{C}}\right)\widehat{w}
\end{aligned}$$

This implies that

$$\begin{aligned}
\Lambda_h\left(1+\zeta_h^w\right)\widehat{w}_h = & \sum_{i\in\mathscr{N}}\left(\Omega_{ih}^\ell \lambda_i\left(\theta_i - 1\right) - \sum_{j\in\mathscr{N}} \Omega_{jh}^\ell \lambda_j\left(\theta_j - 1\right)\widetilde{\psi}_{ji}^x\right)\widehat{A}_i + \sum_{i\in\mathscr{N}} \Omega_{ih}^\ell\,\lambda_i\left(\theta_i\,\widehat{\omega}_i^\ell + \theta_i^\ell\,\widehat{\alpha}_{ih}\right) \\
& + \sum_{i\in\mathscr{N}}\left(\Omega_{ih}^\ell \lambda_i \theta_i - \sum_{j\in\mathscr{N}} \Omega_{jh}^\ell \lambda_j\left(\theta_j - 1\right)\widetilde{\psi}_{ji}^x\right)\widehat{\mu}_i + \sum_{i\in\mathscr{N}} \Omega_{ih}^\ell \lambda_i \widehat{S}_i + \Lambda_h\left(\zeta_h^e \widehat{E}_h - \zeta_h^n \widehat{n}_h\right) \\
& - \left(\sum_{i\in\mathscr{N}} \Omega_{ih}^\ell \lambda_i\left(\theta_i^\ell - 1\right)\right)\widehat{w}_h + \sum_{b\in\mathscr{H}}\left(\sum_{i\in\mathscr{N}} \Omega_{ih}^\ell \lambda_i\left(\left(\theta_i^\ell - 1\right)\alpha_{ib} + \left(\theta_i - 1\right)\left(\widetilde{\psi}_{ib}^\ell - \alpha_{ib}\right)\right)\right)\widehat{w}_b .
\end{aligned} \tag{92}$$

Final Expenditure. Introducing equations (82) in equation (85)

$$\begin{aligned}
& \left(diag(\chi) + diag(\Lambda)\, diag(\zeta_e)\right)\widehat{E} = diag(\Lambda)\left(\left(I_H + diag(\zeta_w)\right)\widehat{w} + diag(\zeta_n)\,\widehat{n}\right) \\
& \quad + \left(\kappa \circ \widehat{\kappa}\right)' diag(\mathbb{1}_N - \mu)\,\lambda - \kappa'\, diag(\mu)\, diag(\lambda)\,\widehat{\mu} + \Omega_\pi' diag(\lambda)\,\widehat{S} + diag(\tau)\,\widehat{T}.
\end{aligned}$$

This implies that

$$\widehat{E}_h = \frac{1}{\chi_h + \zeta_h^e \Lambda_h}\left(\Lambda_h\left(\left(1+\zeta_h^w\right)\widehat{w}_h + \zeta_h^n \widehat{n}_h\right) + \sum_{i\in\mathscr{N}} \kappa_{ih}\left(\left(1-\mu_i\right)\lambda_i\left(\widehat{\kappa}_{ih} + \widehat{S}_i\right) - \mu_i \lambda_i \widehat{\mu}_i\right) + \tau_h\,\widehat{T}_h\right). \tag{93}$$

Sales. Now, introducing equations (89), (90), and (91) in equation (84)

$$\begin{aligned}
diag(\lambda)\,\widehat{S} = & \,\mathscr{B}' diag(\chi)\,\widehat{E} + \mathscr{B}' diag(\chi)\, diag(\varrho - \mathbb{1}_H)\,\widehat{p}_c + \Psi_x' \Omega_x' diag(\lambda)\, diag(\theta_x - \theta)\,\widehat{p}_x \\
& + \Psi_x'\left(\Omega_x' diag(\lambda)\, diag(\theta - \mathbb{1}_N) - diag\left(\beta' diag(\varrho - \mathbb{1}_H)\,\chi\right) - diag\left(\Omega_x' diag(\theta_x - \mathbb{1}_N)\,\lambda\right)\right)\widehat{p} \\
& + \Psi_x'\left(\Omega_x' diag(\lambda)\left(diag(\theta)\left(\widehat{\omega}_x + \widehat{\mu}\right) + diag(\theta - \mathbb{1}_N)\,\widehat{A}\right)\right) \\
& + \Psi_x'\left(\left(\beta \circ \widehat{\beta}\right)' diag(\varrho)\,\chi + \left(\Omega_x \circ \widehat{\mathscr{W}}\right)' diag(\theta_x)\,\lambda\right)
\end{aligned}$$

$$
\begin{aligned}
&= \mathscr{B}' diag(\chi)\, \widehat{E} + \Psi_x' \Omega_x' diag(\lambda) \left(diag(\theta - \mathbb{1}_N)\, \widehat{A} + diag(\theta)\, \widehat{\mu} \right) \\
&- \Psi_x' \left(\beta' diag(\chi)\, diag(\varrho - \mathbb{1}_H)\, \widetilde{\mathscr{B}} - diag(\beta' diag(\varrho - \mathbb{1}_H)\, \chi)\, \widetilde{\Psi}_x \right) \left(\widehat{A} + \widehat{\mu} \right) \\
&- \Psi_x' \left(\Omega_x' diag(\lambda)\, diag(\theta - \mathbb{1}_N) - diag(\Omega_x' diag(\theta - \mathbb{1}_N)\, \lambda) \right) \widetilde{\Psi}_x \left(\widehat{A} + \widehat{\mu} \right) \\
&- \Psi_x' \left(\Omega_x' diag(\lambda)\, diag(\theta_x - \theta)\, \mathscr{W} - diag(\Omega_x' diag(\theta_x - \theta)\, \lambda) \right) \widetilde{\Psi}_x \left(\widehat{A} + \widehat{\mu} \right) \\
&+ \Psi_x' \left(\left(\beta \circ \widehat{\beta} \right)' diag(\varrho)\, \chi + \Omega_x'\, diag(\lambda)\, diag(\theta)\, \widehat{\omega}_x + \left(\Omega_x \circ \widehat{\mathscr{W}} \right)' diag(\theta_x)\, \lambda \right) \\
&+ \Psi_x' \left(\beta' diag(\chi)\, diag(\varrho - \mathbb{1}_H)\, \widetilde{\mathscr{C}} - diag(\beta' diag(\varrho - \mathbb{1}_H)\, \chi)\, \widetilde{\Psi}_\ell \right) \widehat{w} \\
&+ \Psi_x' \left(\Omega_x' diag(\lambda)\, diag(\theta - \mathbb{1}_N) - diag(\Omega_x' diag(\theta - \mathbb{1}_N)\, \lambda) \right) \widetilde{\Psi}_\ell \widehat{w} \\
&+ \Psi_x' \left(\Omega_x' diag(\lambda)\, diag(\theta_x - \theta)\, \mathscr{W} - diag(\Omega_x' diag(\theta_x - \theta)\, \lambda) \right) \widetilde{\Psi}_\ell \widehat{w}.
\end{aligned}
\tag{94}
$$

This implies that

$$
\begin{aligned}
\lambda_i \widehat{S}_i &= \sum_{h \in \mathscr{H}} \mathscr{B}_{hi} \chi_h \widehat{E}_h + \sum_{j \in \mathscr{N}} \psi_{ji}^x \sum_{m \in \mathscr{N}} \Omega_{mj}^x \lambda_m \left((\theta_m - 1)\, \widehat{A}_m + \theta_m \widehat{\mu}_m \right) \\
&- \sum_{j \in \mathscr{N}} \left(\sum_{m \in \mathscr{N}} \psi_{mi}^x \sum_{h \in \mathscr{H}} \beta_{hm} \chi_h (\varrho_h - 1) \left(\widetilde{\mathscr{B}}_{hj} - \widetilde{\psi}_{mj}^x \right) \right) \left(\widehat{A}_j + \widehat{\mu}_j \right) \\
&- \sum_{j \in \mathscr{N}} \left(\sum_{m \in \mathscr{N}} \psi_{mi}^x \sum_{n \in \mathscr{N}} \Omega_{nm}^x \lambda_n (\theta_n - 1) \left(\widetilde{\psi}_{nj}^x - \widetilde{\psi}_{mj}^x \right) \right) \left(\widehat{A}_j + \widehat{\mu}_j \right) \\
&- \sum_{j \in \mathscr{N}} \left(\sum_{m \in \mathscr{N}} \psi_{mi}^x \sum_{n \in \mathscr{N}} \Omega_{nm}^x \lambda_n (\theta_n^x - \theta_n) \left(\sum_{q \in \mathscr{N}} \omega_{nq} \widetilde{\psi}_{qj}^x - \widetilde{\psi}_{mj}^x \right) \right) \left(\widehat{A}_j + \widehat{\mu}_j \right) \\
&+ \sum_{j \in \mathscr{N}} \psi_{ji}^x \left(\sum_{h \in \mathscr{H}} \beta_{hj}\, \chi_h\, \varrho_h\, \widehat{\beta}_{hj} + \sum_{q \in \mathscr{N}} \Omega_{qj}^x\, \lambda_q \left(\theta_q\, \widehat{\omega}_q^x + \theta_q^x\, \widehat{\omega}_{qj} \right) \right) \\
&+ \sum_{h \in \mathscr{H}} \left(\sum_{j \in \mathscr{N}} \psi_{ji}^x \sum_{b \in \mathscr{H}} \beta_{bj} \chi_b (\varrho_b - 1) \left(\widetilde{\mathscr{C}}_{bh} - \widetilde{\psi}_{jh}^\ell \right) \right) \widehat{w}_h \\
&+ \sum_{h \in \mathscr{H}} \left(\sum_{j \in \mathscr{N}} \psi_{ji}^x \sum_{m \in \mathscr{N}} \Omega_{mj}^x \lambda_m (\theta_m - 1) \left(\widetilde{\psi}_{mh}^\ell - \widetilde{\psi}_{jh}^\ell \right) \right) \widehat{w}_h \\
&+ \sum_{h \in \mathscr{H}} \left(\sum_{j \in \mathscr{N}} \psi_{ji}^x \sum_{m \in \mathscr{N}} \Omega_{mj}^x \lambda_m (\theta_m^x - \theta_m) \left(\sum_{q \in \mathscr{N}} \omega_{mq} \widetilde{\psi}_{qh}^\ell - \widetilde{\psi}_{jh}^\ell \right) \right) \widehat{w}_h.
\end{aligned}
$$

Summary of Sufficient Equations. Equations (92), (93), and (94) represent a system of $2H + N$ equations on $2H + N$ unknowns that captures the elasticities of factor rates, consumption expenditure and sales in response to exogenous productivity, markup, demographicy, preferences, and equity allocation shocks. This solution can be used to capture the variation of prices from equations (88), (90), (89), and (91). From here using equations (87) it is possible to obtain the variations of factor income.

2.5 Proof for Proposition 3

To illustrate how the numéraire choice is non-neutral in an environment with elastic labor supply, let me assume as in section (2) that labor supply has a substitution and an income effect, i.e

$$
\widehat{L}_h = \zeta_h^n \widehat{n}_h + \zeta_h^w \widehat{w}_h - \zeta_h^e \widehat{E}_h.
$$

After adding and subtracting $\zeta_h^w \widehat{L}_H$ and $\frac{\zeta_h^w - \zeta_h^e}{1+\zeta_h^w} \widehat{GDP}$ this can be represented by

$$
\widehat{L}_h = \frac{\zeta_h^n}{1+\zeta_h^w} \widehat{n}_h + \frac{\zeta_h^w}{1+\zeta_h^w} \widehat{\Lambda}_h - \frac{\zeta_h^e}{1+\zeta_h^w} \widehat{\chi}_h + \frac{\zeta_h^w - \zeta_h^e}{1+\zeta_h^w} \widehat{GDP}.
$$

Introducting this expression in equations (54) and (61):

1. With nominal GDP as the numéraire

$$\widehat{L}_h = \frac{\zeta_h^n}{1+\zeta_h^w}\widehat{n}_h + \frac{\zeta_h^w}{1+\zeta_h^w}\widehat{\Lambda}_h - \frac{\zeta_h^e}{1+\zeta_h^w}\widehat{\chi}_h.$$

This implies that

$$\bullet\ \widehat{Y}\mid_{\overline{GDP}} = \sum_{i\in\mathcal{N}}\widetilde{\lambda}_i\left(\widehat{\mathcal{A}}_i+\widehat{\mu}_i\right) + \sum_{h\in\mathcal{H}}\widetilde{\Lambda}_h\frac{\zeta_h^n\widehat{n}_h - \widehat{\Lambda}_h - \zeta_h^e\widehat{\chi}_h}{1+\zeta_h^w};$$

$$\bullet\ \widehat{C}_h\mid_{\overline{GDP}} = \sum_{i\in\mathcal{N}}\widetilde{\mathscr{B}}_{hi}\left(\widehat{\mathcal{A}}_i+\widehat{\mu}_i\right) + \widehat{\chi}_h + \sum_{b\in\mathcal{H}}\widetilde{\mathscr{C}}_{hb}\frac{\zeta_b^n\widehat{n}_b - \widehat{\Lambda}_b - \zeta_b^e\widehat{\chi}_b}{1+\zeta_b^w}.$$

2. With real GDP as the numéraire

$$\widehat{L}_h = \frac{\zeta_h^n}{1+\zeta_h^w}\widehat{n}_h + \frac{\zeta_h^w}{1+\zeta_h^w}\widehat{\Lambda}_h - \frac{\zeta_h^e}{1+\zeta_h^w}\widehat{\chi}_h + \frac{\zeta_h^w-\zeta_h^e}{1+\zeta_h^w}\widehat{Y}.$$

This implies that

$$\bullet\ \widehat{Y}\mid_{\overline{p}_Y} = \left(\sum_{h\in\mathcal{H}}\widetilde{\Lambda}_h\frac{1+\zeta_h^e}{1+\zeta_h^w}\right)^{-1}\left(\sum_{i\in\mathcal{N}}\widetilde{\lambda}_i\left(\widehat{\mathcal{A}}_i+\widehat{\mu}_i\right) + \sum_{h\in\mathcal{H}}\widetilde{\Lambda}_h\frac{\zeta_h^n\widehat{n}_h - \widehat{\Lambda}_h - \zeta_h^e\widehat{\chi}_h}{1+\zeta_h^w}\right);$$

$$\bullet\ \widehat{C}_h\mid_{\overline{p}_Y} = \sum_{i\in\mathcal{N}}\left(\widetilde{\mathscr{B}}_{hi} + \widetilde{\lambda}_i\frac{\sum_{b\in\mathcal{H}}\widetilde{\mathscr{C}}_{hb}\frac{\zeta_b^w-\zeta_b^e}{1+\zeta_b^w}}{\sum_{b\in\mathcal{H}}\widetilde{\Lambda}_b\frac{1+\zeta_b^e}{1+\zeta_b^w}}\right)\left(\widehat{\mathcal{A}}_i+\widehat{\mu}_i\right) + \widehat{\chi}_h$$
$$+ \sum_{b\in\mathcal{H}}\left(\widetilde{\mathscr{C}}_{hb} + \widetilde{\Lambda}_b\frac{\sum_{i\in\mathcal{H}}\widetilde{\mathscr{C}}_{hi}\frac{\zeta_i^w-\zeta_i^e}{1+\zeta_i^w}}{\sum_{i\in\mathcal{H}}\widetilde{\Lambda}_i\frac{1+\zeta_i^e}{1+\zeta_i^w}}\right)\frac{\zeta_b^n\widehat{n}_b - \widehat{\Lambda}_b - \zeta_b^e\widehat{\chi}_b}{1+\zeta_b^w}.$$

2.6 Proof for Theorem 5

Cobb Douglas Economy. Under the assumptions that $\theta_i = \theta_i^\ell = \theta_i^x = 1$ for all $i \in \mathcal{N}$, and $\varrho_h = 1$ for all $h \in \mathcal{H}$, the system of equations (92), (93), and (94) simplifies in response to productivity and demographic shocks

$$diag(\Lambda)\,(I_H + diag(\zeta_w))\,\widehat{w} = -diag(\zeta_n)\,diag(\Lambda)\,\widehat{n} + \Omega_\ell' diag(\lambda)\,\widehat{S} + diag(\zeta_e)\,diag(\Lambda)\,\widehat{E},$$

$$(diag(\chi) + diag(\Lambda)\,diag(\zeta_e))\,\widehat{E} = diag(\Lambda)\,((I_H + diag(\zeta_w))\,\widehat{w} + diag(\zeta_n)\,\widehat{n}) + \Omega_\pi' diag(\lambda)\,\widehat{S} + diag(\tau)\,\widehat{T},$$

$$diag(\lambda)\,\widehat{S} = \mathscr{B}' diag(\chi)\,\widehat{E}.$$

Using Proposition 2, the log deviation in factor supply is given by $\widehat{L} = diag(\zeta_n)\,\widehat{n} + diag(\zeta_w)\,\widehat{w} - diag(\zeta_e)\,\widehat{e}$. Using GDP ($Y$) as the numéraire, and adding and subtracting $\widehat{GDP}$

$$diag(\Lambda)\,\widehat{\Lambda} = \Omega_\ell' diag(\lambda)\,\widehat{\lambda} + \underbrace{(\Omega_\ell'\lambda - \Lambda)}_{=0}\widehat{Y},$$

$$diag(\chi)\,\widehat{\chi} = diag(\Lambda)\,\widehat{\Lambda} + \Omega_\pi' diag(\lambda)\,\widehat{\lambda} + \underbrace{(\Omega_\ell'\lambda + \Omega_\pi'\lambda - \chi)}_{=0}\widehat{Y},$$

$$diag(\lambda)\,\widehat{\lambda} = \mathscr{B}' diag(\chi)\,\widehat{\chi} + \underbrace{(\mathscr{B}'\chi - \lambda)}_{=0}\widehat{Y}.$$

It follows that $\widehat{\Lambda} = \widehat{\chi} = 0_H$ and $\widehat{\lambda} = 0_N$, so all allocative terms vanish. Therefore, the TFP decomposition reduces to

$$d \log TFP = \sum_{i \in \mathcal{N}} \widetilde{\lambda}_i \, d \log \mathcal{A}_i.$$

Leontief Economy. Under the assumptions that $\theta_i = \theta_i^{\ell} = \theta_i^x = 0$ for all $i \in \mathcal{N}$, $\varrho_h = 0$, and $\zeta_h^w = \zeta_h^e = 0$ for all $h \in \mathcal{H}$, the system of equations (92), (93), and (94) simplifies in response to markup, preferences, and technology shocks

$$\Omega_\ell' diag(\lambda)\, \widetilde{\Psi}_\ell \widehat{w} = \Omega_\ell' diag(\lambda)\, \widetilde{\Psi}_x \widehat{\mu} + \Omega_\ell' diag(\lambda)\, \widehat{S},$$

$$diag(\chi)\widehat{E} = diag(\Lambda)\, \widehat{w} - \kappa'\, diag(\mu)\, diag(\lambda)\, \widehat{\mu} + \Omega_\pi' diag(\lambda)\, \widehat{S},$$

$$\begin{aligned} diag(\lambda)\, \widehat{S} = \mathscr{B}' diag(\chi)\, \widehat{E} &+ \Psi_x' \left(\beta' diag(\chi)\, \beta - diag(\beta'\chi) + \Omega_x' diag(\lambda) - diag(\Omega_x'\lambda)\right) \widetilde{\Psi}_x \widehat{\mu} \\ &- \Psi_x' \left(\beta' diag(\chi)\, \beta - diag(\beta'\chi) + \Omega_x' diag(\lambda) - diag(\Omega_x'\lambda)\right) \widetilde{\Psi}_\ell \widehat{w}. \end{aligned}$$

Taking GDP (Y) as the numéraire and using the identity $\widetilde{\Psi}_\ell \mathbb{1}_H = \mathbb{1}_N$, the system of equations in terms of log deviations is given by

$$\Omega_\ell' diag(\lambda)\, \widetilde{\Psi}_\ell \widehat{\Lambda} = \Omega_\ell' diag(\lambda)\, \widetilde{\Psi}_x \widehat{\mu} + \Omega_\ell' diag(\lambda)\, \widehat{\lambda} + \Omega_\ell' \underbrace{(\lambda - \lambda)}_{=0_N} \widehat{Y},$$

$$diag(\chi)\widehat{\chi} = diag(\Lambda)\, \widehat{\Lambda} - \kappa'\, diag(\mu)\, diag(\lambda)\, \widehat{\mu} + \Omega_\pi' diag(\lambda)\, \widehat{\lambda} + \underbrace{(\Lambda + \Omega_\pi'\lambda - \chi)}_{=0_N} \widehat{Y},$$

$$\begin{aligned} diag(\lambda)\, \widehat{\lambda} = \mathscr{B}' diag(\chi)\, \widehat{\chi} &+ \Psi_x' \left(\beta' diag(\chi)\, \beta - diag(\beta'\chi) + \Omega_x' diag(\lambda) - diag(\Omega_x'\lambda)\right) \widetilde{\Psi}_x \widehat{\mu} \\ &- \Psi_x' \left(\beta' diag(\chi)\, \beta - diag(\beta'\chi) + \Omega_x' diag(\lambda) - diag(\Omega_x'\lambda)\right) \widetilde{\Psi}_\ell \widehat{\Lambda} \\ &+ \Psi_x' \left(\underbrace{diag(\beta'\chi + \Omega_x'\lambda)\, \mathbb{1}_N}_{=\lambda} - \underbrace{(\beta'\chi + \Omega_x'\lambda)}_{=\lambda} \right) Y. \end{aligned}$$

Therefore, the system is represented by

$$\Omega_\ell' diag(\lambda)\, \widetilde{\Psi}_\ell \widehat{\Lambda} = \Omega_\ell' diag(\lambda)\, \widetilde{\Psi}_x \widehat{\mu} + \Omega_\ell' diag(\lambda)\, \widehat{\lambda}, \tag{95}$$

$$diag(\chi)\widehat{\chi} = diag(\Lambda)\, \widehat{\Lambda} - \kappa'\, diag(\mu)\, diag(\lambda)\, \widehat{\mu} + \Omega_\pi' diag(\lambda)\, \widehat{\lambda}, \tag{96}$$

$$\begin{aligned} diag(\lambda)\, \widehat{\lambda} &= \mathscr{B}' diag(\chi)\, \widehat{\chi} \\ &+ \Psi_x' \left(\beta' diag(\chi)\, \beta - diag(\beta'\chi) + \Omega_x' diag(\lambda) - diag(\Omega_x'\lambda)\right) \widetilde{\Psi}_x \left(\widehat{\mu} - \widetilde{\Omega}_\ell \widehat{\Lambda}\right). \end{aligned} \tag{97}$$

It follows that $\widehat{\Lambda} = \widehat{\chi} = 0_H$ and $\widehat{\lambda} = 0_N$ in response to preference and technology shocks. Therefore allowing for *allocative neutrality*. However, this is not the case for markup shocks, and the rest of this proofs provides sufficient conditions for *allocative neutrality* under markup variations.

To illustrate the logic of the proof, we begin with a representative household environment. In this setting, we know that $\chi = 1$, $\widetilde{\Psi}_\ell = \mathbb{1}_N$, $\kappa = \mathbb{1}_N$ $\Omega_\pi = \mathbb{1}_N - \mu$, $\widetilde{\mathscr{C}} = 1$, $\alpha = \mathbb{1}_N$, $\Omega_\ell = diag(\mu)\, \omega_\ell$, $\lambda = \mathscr{B}'$, $\widetilde{\lambda} = \widetilde{\mathscr{B}}'$, and $\widehat{\chi} = 0$. Under these assumptions, the general equilibrium system simplifies to

$$\Lambda\widehat{\Lambda} = \Omega_\ell' diag(\lambda)\, \widetilde{\Psi}_x \widehat{\mu} + \Omega_\ell' diag(\lambda)\, \widehat{\lambda},$$

$$0 = \Lambda\widehat{\Lambda} - \mu' diag(\lambda)\,\widehat{\mu} + (\mathbb{1}_N - \mu)'\, diag(\lambda)\,\widehat{\lambda},$$

$$diag(\lambda)\,\widehat{\lambda} = \Psi_x'\left(\beta'\beta - diag(\beta) + \Omega_x' diag(\lambda) - diag(\Omega_x'\lambda)\right)\widetilde{\Psi}_x\widehat{\mu} - \Psi_x'\left(\underbrace{\beta' + \Omega_x'\lambda}_{=\lambda} - diag\left(\underbrace{\beta + \Omega_x'\lambda}_{=\lambda}\right)\mathbb{1}_N\right)\widehat{\Lambda}.$$

Substituting the sales share identity into the factor income equation yields

$$\Lambda\widehat{\Lambda} = \Omega_\ell'\left(\Psi_x'\beta'\beta + \Psi_x'\left(diag\left(\underbrace{\lambda - \beta - \Omega_x'\lambda}_{=0_N}\right)\right)\right)\widetilde{\Psi}_x\widehat{\mu},$$

$$\Lambda\widehat{\Lambda} = \mathscr{C}\widetilde{\lambda}'\widehat{\mu},$$

$$\widehat{\Lambda} = \widetilde{\lambda}'\widehat{\mu},$$

where the last line comes from $\mathscr{C} = \Lambda$. Substituting into equation (61) implies $\widehat{Y} = 0$.

In the heterogenous household case, we can no longer guarantee that $\widehat{\chi} = 0_H$. Substituting the sales share condition from equation (97) into the household expenditure equation (96), we obtain

$$\begin{aligned}(I_H - \Omega_\pi'\mathscr{B}')\, diag(\chi)\,\widehat{\chi} &= diag(\Lambda)\,\widehat{\Lambda} - \kappa'\, diag(\mu)\, diag(\lambda)\,\widehat{\mu} \\ &+ \Omega_\pi'\Psi_x'\left(\beta' diag(\chi)\,\beta - diag(\beta'\chi) + \Omega_x' diag(\lambda) - diag(\Omega_x'\lambda)\right)\widetilde{\Psi}_x\left(\widehat{\mu} - \widetilde{\Omega}_\ell\widehat{\Lambda}\right).\end{aligned}$$

Substituting this expression for $\widehat{\chi}$ into equation (97), and assuming that all eigenvalues of $\mathscr{B}\Omega_\pi$ lie strictly within the unit circle, we obtain

$$\begin{aligned}diag(\lambda)\,\widehat{\lambda} &= \mathscr{B}'\left(I_H - \Omega_\pi'\mathscr{B}'\right)^{-1} diag(\Lambda)\,\widehat{\Lambda} - \mathscr{B}'\left(I_H - \Omega_\pi'\mathscr{B}'\right)^{-1}\kappa'\, diag(\mu)\, diag(\lambda)\,\widehat{\mu} \\ &+ \mathscr{B}'\left(I_H - \Omega_\pi'\mathscr{B}'\right)^{-1}\Omega_\pi'\Psi_x'\left(\beta' diag(\chi)\,\beta - diag(\beta'\chi) + \Omega_x' diag(\lambda) - diag(\Omega_x'\lambda)\right)\widetilde{\Psi}_x\left(\widehat{\mu} - \widetilde{\Omega}_\ell\widehat{\Lambda}\right) \\ &+ \Psi_x'\left(\beta' diag(\chi)\,\beta - diag(\beta'\chi) + \Omega_x' diag(\lambda) - diag(\Omega_x'\lambda)\right)\widetilde{\Psi}_x\left(\widehat{\mu} - \widetilde{\Omega}_\ell\widehat{\Lambda}\right).\end{aligned}$$

Substituting the expression for $\widehat{\lambda}$ into the factor income equation (95) we obtain

$$\begin{aligned}0_H &= \mathscr{C}'\left(I_H - \Omega_\pi'\mathscr{B}'\right)^{-1} diag(\Lambda)\,\widehat{\Lambda} - \mathscr{C}'\left(I_H - \Omega_\pi'\mathscr{B}'\right)^{-1}\kappa'\, diag(\mu)\, diag(\lambda)\,\widehat{\mu} \\ &+ \mathscr{C}'\left(I_H - \Omega_\pi'\mathscr{B}'\right)^{-1}\Omega_\pi'\Psi_x'\left(\beta' diag(\chi)\,\beta - diag(\beta'\chi) + \Omega_x' diag(\lambda) - diag(\Omega_x'\lambda)\right)\widetilde{\Psi}_x\left(\widehat{\mu} - \widetilde{\Omega}_\ell\widehat{\Lambda}\right) \\ &+ \Omega_\ell'\left(\Psi_x'\beta' diag(\chi)\,\beta - \Psi_x' diag(\beta'\chi) + (I_N + \Psi_x'\Omega_x')\, diag(\lambda) - \Psi_x' diag(\Omega_x'\lambda)\right)\widetilde{\Psi}_x\left(\widehat{\mu} - \widetilde{\Omega}_\ell\widehat{\Lambda}\right).\end{aligned}$$

Now add and subtract the term $\mathscr{C}'\left(I_H - \Omega_\pi'\mathscr{B}'\right)^{-1}\Omega_\pi' diag(\lambda)\,\widetilde{\Psi}_x\left(\widehat{\mu} - \widetilde{\Omega}_\ell\widehat{\Lambda}\right)$, and use the identities $I_N + \Psi_x\Omega_x = I_N + \left(\sum_{q=0}^{\infty}\Omega_x^q\right)\Omega_x = \Psi_x$ and $\lambda = \beta'\chi + \Omega_x'\lambda$, yielding

$$\begin{aligned}0_H &= \mathscr{C}'\left(I_H - \Omega_\pi'\mathscr{B}'\right)^{-1} diag(\Lambda)\,\widehat{\Lambda} - \mathscr{C}'\left(I_H - \Omega_\pi'\mathscr{B}'\right)^{-1}\kappa'\, diag(\mu)\, diag(\lambda)\,\widehat{\mu} \\ &- \mathscr{C}'\left(I_H - \Omega_\pi'\mathscr{B}'\right)^{-1}\Omega_\pi' diag(\lambda)\,\widetilde{\Psi}_x\left(\widehat{\mu} - \widetilde{\Omega}_\ell\widehat{\Lambda}\right) \\ &+ \mathscr{C}'\left(I_H - \Omega_\pi'\mathscr{B}'\right)^{-1}\Omega_\pi'\Psi_x'\left(\beta' diag(\chi)\,\beta + diag(\lambda - \beta'\chi - \Omega_x'\lambda)\right)\widetilde{\Psi}_x\left(\widehat{\mu} - \widetilde{\Omega}_\ell\widehat{\Lambda}\right) \\ &+ \Omega_\ell'\Psi_x'\left(\beta' diag(\chi)\,\beta + diag(\lambda - \beta'\chi - \Omega_x'\lambda)\right)\widetilde{\Psi}_x\left(\widehat{\mu} - \widetilde{\Omega}_\ell\widehat{\Lambda}\right),\end{aligned}$$

$$
\begin{aligned}
0_H = &\mathscr{C}' \left(I_H - \Omega_\pi'\mathscr{B}'\right)^{-1} diag(\Lambda)\,\widehat{\Lambda} - \mathscr{C}' \left(I_H - \Omega_\pi'\mathscr{B}'\right)^{-1} \kappa'\, diag(\mu)\, diag(\lambda)\,\widehat{\mu} \\
&- \mathscr{C}' \left(I_H - \Omega_\pi'\mathscr{B}'\right)^{-1} \Omega_\pi' diag(\lambda)\,\widetilde{\Psi}_x \left(\widehat{\mu} - \widetilde{\Omega}_\ell \widehat{\Lambda}\right) \\
&+ \mathscr{C}' \left(I_H + \left(I_H - \Omega_\pi'\mathscr{B}'\right)^{-1} \Omega_\pi'\mathscr{B}'\right) diag(\chi)\,\beta \widetilde{\Psi}_x \left(\widehat{\mu} - \widetilde{\Omega}_\ell \widehat{\Lambda}\right).
\end{aligned}
$$

Using the identity $I_H + \left(I_H - \mathscr{B}\Omega_\pi\right)^{-1} \mathscr{B}\Omega_\pi = I_N + \left(\sum_{q=0}^{\infty} \left(\mathscr{B}\Omega_\pi\right)^q\right) \mathscr{B}\Omega_\pi = I_N + \sum_{q=1}^{\infty} \left(\mathscr{B}\Omega_\pi\right)^q = \left(I_H - \mathscr{B}\Omega_\pi\right)^{-1}$. It follows that the factor income equilibrium condition simplifies to

$$
0_H = \mathscr{C}' \left(I_H - \Omega_\pi'\mathscr{B}'\right)^{-1} \left(diag(\Lambda)\,\widehat{\Lambda} - \kappa'\, diag(\mu)\, diag(\lambda)\,\widehat{\mu} + \left(diag(\chi)\,\beta - \Omega_\pi' diag(\lambda)\right) \widetilde{\Psi}_x \left(\widehat{\mu} - \widetilde{\Omega}_\ell \widehat{\Lambda}\right)\right).
$$

Assuming $\mathscr{C}$ is invertible and summing across households yields

$$
\begin{aligned}
0 &= \widetilde{\lambda}'\widehat{\mu} - \widetilde{\Lambda}'\widehat{\Lambda} + \lambda' \left(diag(\mu) + diag(\mathbb{1}_N - \mu)\,\widetilde{\Psi}_x\right) \left(\widetilde{\Omega}_\ell \widehat{\Lambda} - \widehat{\mu}\right) \\
0 &= \widetilde{\lambda}'\widehat{\mu} - \widetilde{\Lambda}'\widehat{\Lambda} + \lambda' \left(\widetilde{\Psi}_x - diag(\mu) \left(\widetilde{\Psi}_x - I_N\right)\right) \left(\widetilde{\Omega}_\ell \widehat{\Lambda} - \widehat{\mu}\right) \\
0 &= \widetilde{\lambda}'\widehat{\mu} - \widetilde{\Lambda}'\widehat{\Lambda} + \lambda' \left(\widetilde{\Psi}_x - diag(\mu)\,\widetilde{\Omega}_x \widetilde{\Psi}_x\right) \left(\widetilde{\Omega}_\ell \widehat{\Lambda} - \widehat{\mu}\right) \\
0 &= \widetilde{\lambda}'\widehat{\mu} - \widetilde{\Lambda}'\widehat{\Lambda} + \lambda' \left(I_N - \Omega_x\right) \left(\widetilde{\Psi}_\ell \widehat{\Lambda} - \widetilde{\Psi}_x \widehat{\mu}\right).
\end{aligned}
$$

The first two terms in this expression correspond to the aggregate *reallocation* wedge in the TFP decomposition. When the last term equals zero, there are no first-order aggregate gains from reallocation. Suppose $\widetilde{\Lambda}'\widehat{\Lambda} = \widetilde{\lambda}'\widehat{\mu}$. Equation (32) implies

$$
\lambda' \left(I_N - \Omega_x\right) = \chi'\mathscr{B} \left(I_N - \Omega_x\right) = \chi'\beta\Psi_x \left(I_N - \Omega_x\right) = \chi'\beta.
$$

This confirms the assumption that $\widetilde{\Lambda}'\widehat{\Lambda} = \widetilde{\lambda}'\widehat{\mu}$ is internally consistent and yields a valid solution.

In summary, two additional conditions must hold in the heterogeneous household environment for the neutrality result to apply. First, the matrix $\mathscr{B}\Omega_\pi$ must have all eigenvalues strictly within the unit circle. A sufficient condition is that the sum of each row of $\mathscr{B}\Omega_\pi$ is strictly less than one. By the Gershgorin circle theorem, this implies the spectral radius is less than one, which guarantees the convergence of the matrix of geometric series. Second, the consumer-to-factor centrality matrix $\mathscr{C}$ has to be nonsingular.

Outside the class of economies satisfying the two structural conditions above, we can characterize the structural conditions that are necessary for allocative-driven growth to arise. Substituting equation (97) into equation (95), and using the identity from equation (32), we obtain obtain

$$
0_H = \mathscr{C}' diag(\chi) \left(\widehat{\chi} + \widetilde{\mathscr{B}} \left(\widehat{\mu} - \widetilde{\Omega}_\ell \widehat{\Lambda}\right)\right).
$$

Adding and subtracting $\widehat{GDP}$, and using the substitutions $\widehat{E} = \widehat{p}_c + \widehat{C}$, $\widehat{p}_c = \widetilde{\mathscr{B}} \left(\widetilde{\Omega}_\ell \widehat{w} - \widehat{\mu}\right)$, using Y as the numéraire and $\widetilde{C}\,\mathbb{1}_H$, the previous condition can be expressed as

$$
0_H = \mathscr{C}' diag(\chi) \left(\widehat{C} - \left(I_H - \widetilde{\mathscr{C}}\right) \mathbb{1}_H \widehat{Y}\right) = \mathscr{C}' diag(\chi)\,\widehat{C}.
$$

Finally, summing over households and applying the GDP identity in (57) yields

$$
d \log Y = \sum_{h \in \mathscr{H}} \chi_h \left(1 - \sum_{b \in \mathscr{H}} \mathscr{C}_{hb}\right) d \log C_h. \tag{98}
$$

From equation (98), we can derive an additional sufficient condition for neutrality. Suppose all consumption bundles satisfy $\sum_{b\in\mathscr{H}}\mathscr{C}_{hb}=\tau$ for all $h\in\mathscr{H}$. This condition is equivalent to $\mathscr{C}\mathbb{1}_H=\mathbb{1}_H\tau$, which implies $\chi'\widehat{C}=0$ and, by equation (57), $\widehat{Y}=0$.

In sum, two sufficient conditions guarantee allocative neutrality in response to distortion shocks under Leontief production with heterogenous households: 1) $\mathscr{B}\,\Omega_\pi$ has spectral radius strictly less than one and $\mathscr{C}$ is nonsingular; or 2) all consumers have identical added consumer-to-vector centralities, i.e., $\sum_{b\in\mathscr{H}}\mathscr{C}_{hb}=\tau$ for all $h\in\mathscr{H}$.

The second sufficient condition is not implied by the first. For example, consider a two-household economy with $\mathscr{C}=\begin{pmatrix}0.2 & 0.4\\ 0.2 & 0.4\end{pmatrix}$. Moreover, equal row-sum in the consumer-to-factor centrality matrices does not require households to have identical consumption bundles. For instance, the matrix $\mathscr{C}=\begin{pmatrix}0.5 & 0.3\\ 0.2 & 0.6\end{pmatrix}$.

Horizontal Economy. In the horizontal economy with N firms, we have that $\omega_x=0_N$, $\omega_\ell=\mathbb{1}_N$,

$$\alpha=\widetilde{\Omega}_\ell=\widetilde{\Psi}_\ell=\begin{pmatrix}\alpha_{11} & \cdots & \alpha_{1H}\\ \vdots & \ddots & \vdots\\ \alpha_{N1} & \cdots & \alpha_{NH}\end{pmatrix},\qquad \Omega_\ell=\Psi_\ell=\begin{pmatrix}\mu_1\alpha_{11} & \cdots & \mu_1\alpha_{1H}\\ \vdots & \ddots & \vdots\\ \mu_N\alpha_{N1} & \cdots & \mu_N\alpha_{NH}\end{pmatrix},$$

$$\widetilde{\Omega}_x=\Omega_x=0_N0_N',\qquad \widetilde{\Psi}_x=\Psi_x=I_N,$$

$$\widetilde{\mathscr{B}}=\mathscr{B}=\beta=\begin{pmatrix}\beta_{11} & \cdots & \beta_{1N}\\ \vdots & \ddots & \vdots\\ \beta_{H1} & \cdots & \beta_{HN}\end{pmatrix},\qquad \widetilde{\lambda}=\lambda=\begin{pmatrix}\sum_{h\in\mathscr{H}}\chi_h\beta_{h1}\\ \vdots\\ \sum_{h\in\mathscr{H}}\chi_h\beta_{hN}\end{pmatrix},$$

$$\widetilde{\Lambda}=\begin{pmatrix}\sum_{i\in\mathscr{N}}\alpha_{i1}\lambda_i\\ \vdots\\ \sum_{i\in\mathscr{N}}\alpha_{iH}\lambda_i\end{pmatrix},\qquad \Lambda=\begin{pmatrix}\sum_{i\in\mathscr{N}}\alpha_{i1}\mu_i\lambda_i\\ \vdots\\ \sum_{i\in\mathscr{N}}\alpha_{iH}\mu_i\lambda_i\end{pmatrix},$$

$$\widetilde{\mathscr{C}}=\begin{pmatrix}\sum_{i\in\mathscr{N}}\beta_{1i}\alpha_{i1} & \cdots & \sum_{i\in\mathscr{N}}\beta_{1i}\alpha_{iH}\\ \vdots & \ddots & \vdots\\ \sum_{i\in\mathscr{N}}\beta_{Hi}\alpha_{i1} & \cdots & \sum_{i\in\mathscr{N}}\beta_{Hi}\alpha_{iH}\end{pmatrix},\ \mathscr{C}=\begin{pmatrix}\sum_{i\in\mathscr{N}}\beta_{1i}\mu_i\alpha_{i1} & \cdots & \sum_{i\in\mathscr{N}}\beta_{1i}\mu_i\alpha_{iH}\\ \vdots & \ddots & \vdots\\ \sum_{i\in\mathscr{N}}\beta_{Hi}\mu_i\alpha_{i1} & \cdots & \sum_{i\in\mathscr{N}}\beta_{Hi}\mu_i\alpha_{iH}\end{pmatrix}.$$

In this economy, $GDP=\sum_{i\in\mathscr{N}}S_i=\sum_{h\in\mathscr{H}}J_h+\sum_{i\in\mathscr{N}}\pi_i$. The corresponding expressions for factor prices, consumption expenditure, and firm sales from equations (92), (93), and (94) are respectively given by

$$\begin{aligned}\Lambda_h\left(1+\zeta_h^w\right)\widehat{w}_h=&\sum_{i\in\mathscr{N}}\mu_i\,\lambda_i\,\alpha_{ih}\left(\widehat{\mu}_i+\widehat{S}_i\right)+\Lambda_h\left(\zeta_h^e\widehat{E}_h-\zeta_h^n\widehat{n}_h\right)+\sum_{i\in\mathscr{N}}\mu_i\,\lambda_i\,\theta_i\,d\,\alpha_{ih}\\ &+\sum_{b\in\mathscr{H}}\left(\sum_{i\in\mathscr{N}}\mu_i\,\lambda_i\,\alpha_{ih}\,\alpha_{ib}\left(\theta_i-1\right)\right)\widehat{w}_b-\left(\sum_{i\in\mathscr{N}}\mu_i\,\lambda_i\,\alpha_{ih}\left(\theta_i-1\right)\right)\widehat{w}_h,\end{aligned}$$

$$\widehat{E}_h=\frac{1}{\chi_h+\zeta_h^e\Lambda_h}\left(\Lambda_h\left(\left(1+\zeta_h^w\right)\widehat{w}_h+\zeta_h^n\widehat{n}_h\right)+\sum_{i\in\mathscr{N}}\kappa_{ih}\left(\left(1-\mu_i\right)\lambda_i\left(\widehat{\kappa}_{ih}+\widehat{S}_i\right)-\mu_i\lambda_i\widehat{\mu}_i\right)+\tau_h\,\widehat{T}_h\right),$$

$$\lambda_i \widehat{S}_i = \sum_{h \in \mathscr{H}} \beta_{hi}\, \chi_h\, (\varrho_h - 1) \left(\widehat{A}_i + \widehat{\mu}_i - \sum_{j \in \mathscr{N}} \beta_{hj} \left(\widehat{A}_j + \widehat{\mu}_j \right) \right) + \sum_{h \in \mathscr{H}} \chi_h\, \varrho_h\, d\, \beta_{hi}$$
$$+ \sum_{h \in \mathscr{H}} \beta_{hi}\, \chi_h\, \widehat{E}_h + \sum_{h \in \mathscr{H}} \left(\sum_{b \in \mathscr{H}} \beta_{bi}\, \chi_b\, (\varrho_b - 1) \left(\widetilde{\mathscr{C}}_{bh} - \alpha_{ih} \right) \right) \widehat{w}_h.$$

Take $2H + N - 1$ of these equations, and normalize relative p_Y to 1, which implies that

$$\sum_{h \in \mathscr{H}} \widetilde{\Lambda}_h \widehat{w}_h = \sum_{i \in \mathscr{N}} \widetilde{\lambda}_i \left(\widehat{A}_i + \widehat{\mu}_i \right).$$

Suppose all firms face the same distortion, $\mu_i = \mu$ for all $i \in \mathscr{N}$. Then equations (92), and (94) are given by

$$diag\left(\widetilde{\Lambda}\right) (I_H + diag(\zeta_w))\, \widehat{w} = \alpha' diag(\lambda)\, \widehat{\mu} + \alpha' diag(\lambda)\, \widehat{S} + diag\left(\widetilde{\Lambda}\right) diag(\zeta_e)\, \widehat{E} - diag\left(\widetilde{\Lambda}\right) diag(\zeta_n)\, \widehat{n}$$
$$+ (d\, \alpha)'\ diag(\theta)\ \lambda + \alpha' diag(\lambda)\, diag(\theta - \mathbb{1}_N)\, \alpha \widehat{w} - diag(\alpha' diag(\theta - \mathbb{1}_N)\, \lambda)\, \widehat{w},$$

$$diag(\lambda)\, \widehat{S} = (\beta' diag(\chi)\, diag(\varrho - \mathbb{1}_H)\, \beta - diag(\beta' diag(\varrho - \mathbb{1}_H)\, \chi)) \left(\alpha\, \widehat{w} - \widehat{A} - \widehat{\mu} \right)$$
$$+ \beta' diag(\chi)\, \widehat{E} + (d\, \beta)'\ diag(\rho)\ \chi$$

We now express the system in terms of factor income elasticities by substituting $\widehat{L} = diag(\zeta_n)\, \widehat{n} + diag(\zeta_w)\, \widehat{w} - diag(\zeta_e)\, \widehat{E}$ into the equilibrium conditions

$$diag\left(\widetilde{\Lambda}\right) \widehat{J} = (\alpha' diag(\lambda)\, diag(\theta - \mathbb{1}_N)\, \alpha - diag(\alpha' diag(\theta - \mathbb{1}_N)\, \lambda)) \left(\widehat{J} - \widehat{L} \right)$$
$$+ \alpha' diag(\lambda)\, \widehat{\mu} + \alpha' diag(\lambda)\, \widehat{S} + (d\, \alpha)'\ diag(\theta)\ \lambda$$

$$diag(\lambda)\, \widehat{S} = (\beta' diag(\chi)\, diag(\varrho - \mathbb{1}_H)\, \beta - diag(\beta' diag(\varrho - \mathbb{1}_H)\, \chi)) \left(\alpha \left(\widehat{J} - \widehat{L} \right) - \widehat{A} - \widehat{\mu} \right)$$
$$+ \beta' diag(\chi)\, \widehat{E} + (d\, \beta)'\ diag(\rho)\ \chi$$

We express the system of GDP ratios by adding and subtracting $\widehat{GDP}$, and normalizing with aggregate output Y as the numéraire, such that $\widehat{GDP} = \widehat{Y}$

$$diag\left(\widetilde{\Lambda}\right) \widehat{\Lambda} = \alpha' diag(\lambda)\, \widehat{\mu} + \alpha' diag(\lambda)\, \widehat{\lambda} + (\alpha' diag(\lambda)\, diag(\theta - \mathbb{1}_N)\, \alpha - diag(\alpha' diag(\theta - \mathbb{1}_N)\, \lambda)) \left(\widehat{\Lambda} - \widehat{L} \right)$$
$$+ \underbrace{\left(\alpha' \lambda - \widetilde{\Lambda} \right)}_{= 0_N} \widehat{Y} + (d\, \alpha)'\ diag(\theta)\ \lambda + (\alpha' diag(\lambda)\, diag(\theta - \mathbb{1}_N)\, \alpha - diag(\alpha' diag(\theta - \mathbb{1}_N)\, \lambda))\, \mathbb{1}_H \widehat{Y},$$

$$diag(\lambda)\, \widehat{\lambda} = \beta' diag(\chi)\, \widehat{\chi} + (\beta' diag(\chi)\, diag(\varrho - \mathbb{1}_H)\, \beta - diag(\beta' diag(\varrho - \mathbb{1}_H)\, \chi)) \left(\alpha \left(\widehat{\Lambda} - \widehat{L} \right) - \widehat{A} - \widehat{\mu} \right)$$
$$+ \underbrace{(\beta' \chi - \lambda)}_{= 0_N} \widehat{Y} + (d\, \beta)'\ diag(\rho)\ \chi + (\beta' diag(\chi)\, diag(\varrho - \mathbb{1}_H)\, \beta - diag(\beta' diag(\varrho - \mathbb{1}_H)\, \chi))\, \alpha \mathbb{1}_H \widehat{Y}.$$

Summing over all elements of each vector, observe that $\mathbb{1}' diag(x) = x'$. In addition, the equilibrium constraints imply $\sum_{i \in \mathscr{N}} d\, \beta_{hi} = 0$, $\sum_{h \in \mathscr{H}} d\, \alpha_{ih} = 0$

$$\widetilde{\Lambda}' \widehat{\Lambda} = \lambda' \widehat{\mu} + \lambda' \widehat{\lambda} + (\lambda' diag(\theta - \mathbb{1}_N)\, \alpha - \lambda' diag(\theta - \mathbb{1}_N)\, \alpha)\, \widehat{\Lambda} - (\lambda' diag(\theta - \mathbb{1}_N)\, \alpha - \lambda' diag(\theta - \mathbb{1}_N)\, \alpha)\, \widehat{L},$$

$$\lambda' \widehat{\lambda} = \chi' \widehat{\chi} + (\chi' diag(\varrho - \mathbb{1}_H)\, \beta - \chi' diag(\varrho - \mathbb{1}_H)\, \beta) \left(\alpha \left(\widehat{\Lambda} - \widehat{L} \right) - \widehat{A} - \widehat{\mu} \right).$$

Using the fact that $\chi' \widehat{\chi} = 0$, we have that $\widetilde{\Lambda}' \widehat{\Lambda} = \lambda' \widehat{\mu} + \lambda' \widehat{\lambda}$ and $\lambda' \widehat{\lambda} = 0$. Finally, given that $\lambda = \widetilde{\lambda}$, then from equation (61) $\widehat{Y} = \widetilde{\lambda}'\widehat{A} + \widetilde{\Lambda}'\widehat{L}$.

Vertical Economy. In a vertical economy with N firms, the primary factor is used exclusively by the final firm $\omega_\ell' = \begin{pmatrix} 0 & \cdots & 0 & 1 \end{pmatrix}$, while all intermediate inputs are supplied by upstream firms $\omega_x' = \begin{pmatrix} 1 & \cdots & 1 & 0 \end{pmatrix}$. Factor weights for the final firm satisfy $a' = \begin{pmatrix} \alpha_{N1} & \cdots & \alpha_{NH} \end{pmatrix} = \begin{pmatrix} \alpha_1 & \cdots & \alpha_H \end{pmatrix}$.

$$\mathscr{W} = \widetilde{\Omega}_x = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ 0 & 0 & 0 & \cdots & 0 \end{pmatrix}, \qquad \alpha = \widetilde{\Omega}_\ell = \begin{pmatrix} 0 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ \alpha_1 & \cdots & \alpha_H \end{pmatrix} = o_N(N)\, a',$$

$$\Omega_x = \begin{pmatrix} 0 & \mu_1 & 0 & \cdots & 0 \\ 0 & 0 & \mu_2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \mu_{N-1} \\ 0 & 0 & 0 & \cdots & 0 \end{pmatrix}, \qquad \Omega_\ell = \begin{pmatrix} 0 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ \alpha_1\mu_N & \cdots & \alpha_H\mu_N \end{pmatrix} = \mu_N\, o_N(N)\, a',$$

$$\widetilde{\Psi}_x = \begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ 0 & 1 & 1 & \cdots & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{pmatrix}, \qquad \Psi_x = \begin{pmatrix} 1 & \mu_1 & \mu_1\mu_2 & \cdots & \prod_{i=1}^{N-2}\mu_i & \prod_{i=1}^{N-1}\mu_i \\ 0 & 1 & \mu_2 & \cdots & \prod_{i=2}^{N-2}\mu_i & \prod_{i=2}^{N-1}\mu_i \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & \mu_{N-1} \\ 0 & 0 & 0 & \cdots & 0 & 1 \end{pmatrix},$$

$$\beta = \mathbb{1}_H o_N(1)', \qquad \widetilde{\mathscr{B}} = \mathbb{1}_H \mathbb{1}_N', \qquad \mathscr{B} = \begin{pmatrix} 1 & \mu_1 & \cdots & \prod_{i=1}^{N-2}\mu_i & \prod_{i=1}^{N-1}\mu_i \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 1 & \mu_1 & \cdots & \prod_{i=1}^{N-2}\mu_i & \prod_{i=1}^{N-1}\mu_i \end{pmatrix},$$

$$\widetilde{\mathscr{C}} = \begin{pmatrix} \alpha_1 & \cdots & \alpha_H \\ \vdots & \ddots & \vdots \\ \alpha_1 & \cdots & \alpha_H \end{pmatrix} = \mathbb{1}_H\, a', \qquad \mathscr{C} = \begin{pmatrix} \alpha_1\prod_{i\in\mathscr{N}}\mu_i & \cdots & \alpha_H\prod_{i\in\mathscr{N}}\mu_i \\ \vdots & \ddots & \vdots \\ \alpha_1\prod_{i\in\mathscr{N}}\mu_i & \cdots & \alpha_H\prod_{i\in\mathscr{N}}\mu_i \end{pmatrix} = \prod_{i\in\mathscr{N}}\mu_i \mathbb{1}_H\, a',$$

$$\widetilde{\Psi}_\ell = \begin{pmatrix} \alpha_1 & \cdots & \alpha_H \\ \alpha_1 & \cdots & \alpha_H \\ \vdots & \ddots & \vdots \\ \alpha_1 & \cdots & \alpha_H \\ \alpha_1 & \cdots & \alpha_H \end{pmatrix} = \mathbb{1}_N\, a', \quad \Psi_\ell = \begin{pmatrix} \alpha_1\prod_{i=1}^{N}\mu_i & \cdots & \alpha_H\prod_{i=1}^{N}\mu_i \\ \alpha_1\prod_{i=2}^{N}\mu_i & \cdots & \alpha_H\prod_{i=2}^{N}\mu_i \\ \vdots & \ddots & \vdots \\ \alpha_1\prod_{i=N-1}^{N}\mu_i & \cdots & \alpha_H\prod_{i=N-1}^{N}\mu_i \\ \alpha_1\mu_N & \cdots & \alpha_H\mu_N \end{pmatrix} = \begin{pmatrix} \prod_{i\in\mathscr{N}}\mu_i \\ \prod_{i=2}^{N}\mu_i \\ \vdots \\ \mu_{N-1}\mu_N \\ \mu_N \end{pmatrix} a',$$

$$\lambda = \begin{pmatrix} 1 \\ \vdots \\ \prod_{i=1}^{k-1}\mu_i \\ \vdots \\ \prod_{i=1}^{N-1}\mu_i \end{pmatrix}, \quad \widetilde{\lambda} = \begin{pmatrix} 1 \\ \vdots \\ 1 \\ \vdots \\ 1 \end{pmatrix}, \quad \Lambda = \Omega_\ell'\lambda = \begin{pmatrix} \alpha_1\mu_N\lambda_N \\ \vdots \\ \alpha_H\mu_N\lambda_N \end{pmatrix} = \mu_N\lambda_N a, \quad \widetilde{\Lambda} = \widetilde{\Omega}_\ell'\widetilde{\lambda} = \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_H \end{pmatrix} = a.$$

Observe that in the vertical economy $\mu_N\lambda_N = \sum_{h\in\mathscr{H}}\Lambda_h$, $\lambda_k = \left(\prod_{j=i}^{k-1}\mu_j\right)\lambda_i$, and $\delta_h = \mu_N\lambda_N =$

$\prod_{i \in \mathcal{N}} \mu_i$ for all $h \in \mathcal{H}$. Moreover nominal GDP is given by

$$GDP = \mathbb{1}_N' \left(I_N - diag(\mu)\, diag(\omega_x)\right) S = \sum_{i=1}^{N-1} \left(1 - \mu_i\right) S_i + S_N = S_1 = \sum_{h \in \mathcal{H}} E_h.$$

Distortion centralities are symmetric across households: $\delta_h = \sum_{h \in \mathcal{H}} \Lambda_h = \Lambda$. From equation (61), it follows that

$$d \log TFP = \sum_{i \in \mathcal{H}} d \log A_i + \sum_{i \in \mathcal{H}} d \log \mu_i - d \log \Lambda.$$

Since $\Lambda = \prod_{i \in \mathcal{N}} \mu_i$, $d \log \Lambda = \sum_{i \in \mathcal{H}} d \log \mu_i$, and

$$d \log TFP = \sum_{i \in \mathcal{H}} d \log A_i.$$

These conditions imply allocative neutrality with respect to all shocks.

2.7 Example 1

Assume inelastic labor supply, $\beta_h \neq \beta_l$, $\mu_m \neq \mu_a$, $\alpha_{mh} = \alpha_{ah} = \frac{1}{2}$, and $\kappa_{mh} = \kappa_{ah} = \frac{1}{2}$. Then $\widetilde{\lambda}_m = \lambda_m = \beta_l + (\beta_h - \beta_l)\,\chi_h$, $\widetilde{\lambda}_a = \lambda_a = 1 - \lambda_m$, $\widetilde{\Lambda}_h = \widetilde{\Lambda}_l = \frac{1}{2}$, $\Lambda_h = \Lambda_l = \frac{\mu_a}{2} + (\mu_m - \mu_a)\,\frac{\lambda_m}{2}$, $\chi_h = \Lambda_h + \frac{1}{2}\left(1 - \mu_a + (\mu_a - \mu_m)\,\lambda_m\right) + \tau_h$, and $\chi_l = 1 - \chi_h$. The solution for this system of equations is given by $\chi_h = \frac{1}{2} + \tau_h$, $\chi_l = \frac{1}{2} - \tau_h$, $\lambda_m = \frac{1}{2}(\beta_h + \beta_l) + (\beta_h - \beta_l)\,\tau_h$, $\lambda_a = 1 - \frac{1}{2}(\beta_h + \beta_l) - (\beta_h - \beta_l)\,\tau_h$, $\Lambda_h = \Lambda_l = \frac{\mu_a}{2} + \frac{1}{4}(\beta_h + \beta_l)(\mu_m - \mu_a) + \frac{1}{2}(\beta_h - \beta_l)(\mu_m - \mu_a)\,\tau_h$.

From equations (92), (93), and (94):

$$\Lambda_h\, d \log w_h = \Lambda_l\, d \log w_l = \frac{1}{2}\left(\mu_m\, \lambda_m\, d \log S_m + \mu_a\, \lambda_a\, d \log S_a\right),$$

$$\chi_h\, d \log E_h = \Lambda_h\, d \log w_h + \frac{1}{2}\left((1 - \mu_m)\,\lambda_m\, d \log S_m + (1 - \mu_a)\,\lambda_a\, d \log S_a\right) + \tau_h\, \widehat{T}_h,$$

$$\chi_l\, d \log E_l = \Lambda_l\, d \log w_l + \frac{1}{2}\left((1 - \mu_m)\,\lambda_m\, d \log S_m + (1 - \mu_a)\,\lambda_a\, d \log S_a\right) + \tau_l\, \widehat{T}_l,$$

$$\lambda_m\, d \log S_m = \beta_h\, \chi_h\, d \log E_h + \beta_l\, \chi_l\, d \log E_l,$$

$$\lambda_a\, d \log S_a = (1 - \beta_h)\,\chi_h\, d \log E_h + (1 - \beta_l)\,\chi_l\, d \log E_l.$$

Add and subtract $d \log GDP$ to obtain:

$$d\,\Lambda_h = d\,\Lambda_l = \frac{1}{2}\left(\mu_m\, d\,\lambda_m + \mu_a\, d\,\lambda_a\right),$$

$$d\,\chi_h = d\,\Lambda_h + \frac{1}{2}\left((1 - \mu_m)\, d\,\lambda_m + (1 - \mu_a)\, d\,\lambda_a\right) + d\,\tau_h,$$

$$d\,\chi_l = d\,\Lambda_l + \frac{1}{2}\left((1 - \mu_m)\, d\,\lambda_m + (1 - \mu_a)\, d\,\lambda_a\right) - d\,\tau_h,$$

$$d\,\lambda_m = \beta_h\, d\,\chi_h + \beta_l\, d\,\chi_l,$$

$$d\,\lambda_a = (1 - \beta_h)\, d\,\chi_h + (1 - \beta_l)\, d\,\chi_l.$$

Using the property from horizontal economies $d\,\lambda_m + d\,\lambda_l = 0$ and $d\,\chi_h = d\,\chi_l$, the previous system of equations can be simplified into

$$d\,\Lambda_h = d\,\Lambda_l = \frac{1}{2}\left(\mu_m - \mu_a\right) d\,\lambda_m, \qquad d\,\chi_h = d\,\Lambda_h + \frac{1}{2}\left(\mu_a - \mu_m\right) d\,\lambda_m + d\,\tau_h, \qquad d\,\lambda_m = \left(\beta_h - \beta_l\right) d\,\chi_h.$$

Thus

$$d\,\Lambda_h = d\,\Lambda_l = \frac{1}{2}\left(\beta_h - \beta_l\right)\left(\mu_m - \mu_a\right) d\,\tau_h, \qquad d\,\lambda_m = \left(\beta_h - \beta_l\right) d\,\tau_h, \qquad d\,\chi_h = d\,\tau_h.$$

Using the *entropic terms of trade* from Theorem 2

$$d\,\log\,Y = -\frac{1}{2\,\Lambda_h}\left(d\,\Lambda_h + d\,\Lambda_l\right) = \frac{\left(\beta_h - \beta_l\right)\left(\mu_a - \mu_m\right)}{\mu_a + \left(\mu_m - \mu_a\right)\left(\left(\beta_h - \beta_l\right)\tau_h + \frac{1}{2}\left(\beta_h + \beta_l\right)\right)} d\,\tau_h.$$

Additionally, we have

$$\delta_h = \delta_l = \frac{1}{2\,\Lambda_h}, \qquad \mathscr{C}_{hh} = \mathscr{C}_{hl} = \frac{\mu_a}{2} + \frac{1}{2}\left(\mu_m - \mu_a\right)\beta_h, \qquad \mathscr{C}_{lh} = \mathscr{C}_{ll} = \frac{\mu_a}{2} + \frac{1}{2}\left(\mu_m - \mu_a\right)\beta_l,$$

$$M_h = \frac{\mu_a + \left(\mu_m - \mu_a\right)\beta_h}{\mu_a + \left(\beta_h - \beta_l\right)\left(\mu_m - \mu_a\right)\tau_h + \frac{1}{2}\left(\beta_h + \beta_l\right)\left(\mu_m - \mu_a\right)},$$

$$M_l = \frac{\mu_a + \left(\mu_m - \mu_a\right)\beta_l}{\mu_a + \left(\beta_h - \beta_l\right)\left(\mu_m - \mu_a\right)\tau_h + \frac{1}{2}\left(\beta_h + \beta_l\right)\left(\mu_m - \mu_a\right)}.$$

Using the *distributive terms of trade* from Theorem 2

$$d\,\log\,Y = -\left(M_h - M_l\right) d\,\chi_h = \frac{\left(\beta_h - \beta_l\right)\left(\mu_a - \mu_m\right)}{\mu_a + \left(\mu_m - \mu_a\right)\left(\left(\beta_h - \beta_l\right)\tau_h + \frac{1}{2}\left(\beta_h + \beta_l\right)\right)} d\,\tau_h,$$

which is the same expression as before.

The household-specific distortions centralities are given by $\delta_{h|b} = \frac{1}{2\,\Lambda_h}$ and $\delta_{l|h} = \frac{1}{2\,\Lambda_l}$ for $b \in \{h, l\}$, and their *expenditure centralities* are given by

$$M_{h|h} = M_{h|l} = M_h = \frac{\mu_a + \left(\mu_m - \mu_a\right)\beta_h}{\mu_a + \left(\beta_h - \beta_l\right)\left(\mu_m - \mu_a\right)\tau_h + \frac{1}{2}\left(\beta_h + \beta_l\right)\left(\mu_m - \mu_a\right)},$$

$$M_{l|h} = M_{l|l} = M_l = \frac{\mu_a + \left(\mu_m - \mu_a\right)\beta_l}{\mu_a + \left(\beta_h - \beta_l\right)\left(\mu_m - \mu_a\right)\tau_h + \frac{1}{2}\left(\beta_h + \beta_l\right)\left(\mu_m - \mu_a\right)}.$$

Then from Theorem 4

$$d\,\log\,C_h = d\,log\,\chi_h - \sum_{b\in\{h,l\}} M_{b|h}\,d\,\chi_b = \left(\frac{2}{1+2\,\tau_h} + \frac{\left(\beta_h - \beta_l\right)\left(\mu_a - \mu_m\right)}{\mu_a + \left(\mu_m - \mu_a\right)\left(\left(\beta_h - \beta_l\right)\tau_h + \frac{1}{2}\left(\beta_h + \beta_l\right)\right)}\right) d\,\tau_h,$$

$$d\,\log\,C_l = d\,log\,\chi_l - \sum_{b\in\{h,l\}} M_{b|l}\,d\,\chi_b = \left(-\frac{2}{1-2\,\tau_h} + \frac{\left(\beta_h - \beta_l\right)\left(\mu_a - \mu_m\right)}{\mu_a + \left(\mu_m - \mu_a\right)\left(\left(\beta_h - \beta_l\right)\tau_h + \frac{1}{2}\left(\beta_h + \beta_l\right)\right)}\right) d\,\tau_h.$$

With $d\,\tau_h > 0$ we have that $d\,\log\,C_l > 0$ as long as

$$\left(\beta_h - \beta_l\right)\left(\mu_a - \mu_m\right)\left(1 - 2\tau_h\right) > 2\mu_a + \left(\beta_h + \beta_l\right)\left(\mu_m - \mu_a\right) + 2\left(\beta_h - \beta_l\right)\left(\mu_m - \mu_a\right)\tau_h$$

$$\beta_h\left(\mu_a - \mu_m\right) > \mu_a.$$

Case 1: $\mu_a > \mu_m$. In this case, for $d \log C_l > 0$ we require $\beta_h > \frac{\mu_a}{\mu_a - \mu_m}$, which is never satisfied because $\frac{\mu_a}{\mu_a - \mu_m} > 1$ and $\beta_h \leq 1$.

Case 2: $\mu_m > \mu_a$. In this case, for $d \log C_l > 0$ we require $-\frac{\mu_a}{\mu_m - \mu_a} > \beta_h$, which is never satisfied because $\beta_h \geq 0$.

2.8 Example 2

Consider the same economy as in Section 4.3, but now centralize household decisions under the constrained social planner in Theorem 3.

The planner chooses $\{C_{hm}, C_{ha}, C_{lm}, C_{la}\}$ to equalize households' marginal rates of substitution with the social MRS between each good $i \in \{m, a\}$ and aggregate output:

$$\frac{p_m}{P_Y Y} = \chi_h \frac{\beta_h}{C_{hm}} = (1 - \chi_h) \frac{\beta_l}{C_{lm}}, \qquad \frac{p_a}{P_Y Y} = \chi_h \frac{1 - \beta_h}{C_{ha}} = (1 - \chi_h) \frac{1 - \beta_l}{C_{la}}.$$

Here we use the property of the aggregator Q_Y that $\frac{d \log Y}{d \log C_h} = \chi_h$, so $\partial Y / \partial C_h = \chi_h Y / C_h$; with Cobb-Douglas C_h, $\partial C_h / \partial C_{hi} = \beta_{hi} C_h / C_{hi}$.

Labor demand for firms m and a is given by (FOCs under Cobb-Douglas with equal factor shares):

$$\ell_{mh} = \frac{1}{2} \frac{\mu_m \, p_m \, y_m}{w_h}, \qquad \ell_{ml} = \frac{1}{2} \frac{\mu_m \, p_m \, y_m}{w_l}, \qquad \ell_{ah} = \frac{1}{2} \frac{\mu_a \, p_a \, y_a}{w_h}, \qquad \ell_{al} = \frac{1}{2} \frac{\mu_a \, p_a \, y_a}{w_l}.$$

Hence, prices and wages for the firms are related by $y_i = \frac{1}{2} \mu_i \, p_i \, y_i \, (w_h \, w_l)^{-\frac{1}{2}}$. Equivalently $p_i = \frac{2}{\mu_i} \sqrt{w_h \, w_l}$ for $i \in \{m, a\}$.

By definition, $GDP = p_m \, y_m + p_a \, y_a$, with $\lambda_i = \frac{p_i y_i}{GDP}$, in a horizontal economy $\tilde{\lambda}_i = \lambda_i$, and by symmetry $\tilde{\Lambda}_h = \tilde{\Lambda}_l = \frac{1}{2} \left(\tilde{\lambda}_m + \tilde{\lambda}_a \right) = \frac{1}{2}$.

Using the labor market clearing conditions (normalizing each type's endowment to one, $L_h = L_l = 1$)

$$1 = \ell_{mh} + \ell_{ah} = \frac{1}{2 \, w_h} \left(\mu_m \, p_m \, y_m + \mu_a \, p_a \, y_a \right), \qquad 1 = \ell_{ml} + \ell_{al} = \frac{1}{2 \, w_l} \left(\mu_m \, p_m \, y_m + \mu_a \, p_a \, y_a \right).$$

Since the bracketed term is common across both equations, it follows that $w_h = w_l$. Define the common wage:

$$w = w_h = w_l = \frac{1}{2} \left(\mu_m \, p_m \, y_m + \mu_a \, p_a \, y_a \right).$$

Consequently, $\Lambda_h = \Lambda_l$, and total labor income equals $2 \, w = \mu_m \, p_m \, y_m + \mu_a \, p_a \, y_a$. Wage symmetry implies homogeneous labor demand, so that per-type demands are equal $\ell_m = \ell_{mh} = \ell_{ml}$ and $\ell_a = \ell_{ah} = \ell_{al}$. Hence, output and prices satisfy

$$y_m = \ell_m, \qquad y_a = \ell_a = 1 - \ell_m, \qquad p_m = \frac{2 \, w}{\mu_m}, \qquad p_a = \frac{2 \, w}{\mu_a}.$$

Using nominal GDP and defining $\Gamma = \frac{2 \, w}{P_Y \, Y}$

$$P_Y \, Y = p_m \, y_m + p_a \, y_a = 2w \left(\frac{\ell_m}{\mu_m} + \frac{1 - \ell_m}{\mu_a} \right)$$

$$1 = \Gamma \left(\frac{\ell_m}{\mu_m} + \frac{1 - \ell_m}{\mu_a} \right).$$

Note $\lambda_m = \chi_h \beta_h + (1-\chi_h) \beta_l$ and $\Gamma = \mu_m \lambda_m + \mu_a (1-\lambda_m) = \mu_a + \lambda_m (\mu_m - \mu_a)$. Therefore

$$y_m = \ell_m = \frac{\mu_m}{\Gamma} \lambda_m.$$

Normalize by setting $w = 1$.

Case 1: No distortions ($\mu_m = \mu_a = 1$) and no transfers ($\tau_h = 0$)

In the frictionless benchmark, prices equal marginal costs, and the decentralized allocation coincides with the planner's allocation. The expenditure distribution is given by $\chi_h = \chi_l = \frac{1}{2}$ by symmetry and the absence of transfers. Then, $\lambda_m = \frac{\beta_h+\beta_l}{2}$, $\Gamma = 1$, $y_m = \frac{\beta_h+\beta_l}{2}$, $w = 1$, and $p_m = p_a = 2$. Because $y_m + y_a = 1$, it follows that $P_Y Y = 2$, and hence $\Gamma = 1$. Hence, with Cobb-Douglas preferences, the consumption allocations are $C_{hm} = \frac{\beta_h}{2}$, $C_{ha} = \frac{1-\beta_h}{2}$, $C_{lm} = \frac{\beta_l}{2}$, and $C_{la} = \frac{1-\beta_l}{2}$. This implies that

$$C_h = \frac{1}{2} \beta_h^{\beta_h} (1-\beta_h)^{1-\beta_h}, \qquad C_l = \frac{1}{2} \beta_l^{\beta_l} (1-\beta_l)^{1-\beta_l}.$$

Define the aggregator Q_Y as $Y_{\text{eff}} \equiv (C_h C_l)^{1/2}$, which is consistent with the assumption $\frac{d \log Y}{d \log C_h} = \chi_h$. Then, real GDP is given by

$$Y_{\textit{eff}} = \frac{1}{2} \beta_h^{\frac{\beta_h}{2}} (1-\beta_h)^{\frac{1-\beta_h}{2}} \beta_l^{\frac{\beta_l}{2}} (1-\beta_l)^{\frac{1-\beta_l}{2}}.$$

Case 2: Distortions and no transfers ($\tau_h = 0$)

The expenditure distribution is given by $\chi_h = \chi_l = \frac{1}{2}$ by symmetry and zero transfers. Let $\overline{\beta} = \frac{\beta_h+\beta_l}{2} \in [0,1]$. Then $\lambda_m = \overline{\beta}$, $p_m = \frac{2}{\mu_m}$, $p_a = \frac{2}{\mu_a}$, $w = 1$, $\Gamma = \overline{\beta}\mu_m + (1-\overline{\beta}) \mu_a$, $y_m = \ell_m = \frac{\overline{\beta}\mu_m}{\Gamma}$, $y_a = 1 - \ell_m = \frac{(1-\overline{\beta})\mu_a}{\Gamma}$, and $P_Y Y = \frac{2}{\Gamma}$. These imply the following real consumption allocations

$$C_{hm} = \frac{\mu_m \beta_h}{2\Gamma}, \qquad C_{lm} = \frac{\mu_m \beta_l}{2\Gamma}, \qquad C_{ha} = \frac{\mu_a (1-\beta_h)}{2\Gamma}, \qquad C_{la} = \frac{\mu_a (1-\beta_l)}{2\Gamma}.$$

$$C_h = \frac{(\mu_m \beta_h)^{\beta_h} (\mu_a (1-\beta_h))^{1-\beta_h}}{2\Gamma}, \qquad C_l = \frac{(\mu_m \beta_l)^{\beta_l} (\mu_a (1-\beta_l))^{1-\beta_l}}{2\Gamma}.$$

Using the same real GDP aggregator provides

$$Y = \frac{\mu_m^{\overline{\beta}} \mu_a^{1-\overline{\beta}} \beta_h^{\frac{\beta_h}{2}} (1-\beta_h)^{\frac{1-\beta_h}{2}} \beta_l^{\frac{\beta_l}{2}} (1-\beta_l)^{\frac{1-\beta_l}{2}}}{2\Gamma}$$

The TFP wedge relative to the efficient economy is given by

$$\frac{Y}{Y_{\textit{eff}}} = \frac{\mu_m^{\overline{\beta}} \mu_a^{1-\overline{\beta}}}{\overline{\beta}\mu_m + (1-\overline{\beta}) \mu_a}$$

It remains to show that $Y_{\mu,\tau=0} \leq Y_{\mu,\tau=0}$. When $\mu_m = \mu_a$, then $Y_{\mu,\tau=0} = Y_{\mu,\tau=0}$, consistent with Theorem 5, where the allocative neutrality for horizontal economies with symmetric markups is proved. For $\mu_m \neq \mu_a$, we need

$$\mu_m^{\overline{\beta}} \mu_a^{1-\overline{\beta}} \leq \overline{\beta}\mu_m + (1-\overline{\beta}) \mu_a.$$

This is satisfied because the arithmetic average weakly dominates the geometric average.

Case 3: Efficient labor allocation via lump-sum transfers

The expenditure distribution is $\chi_h = \frac{1}{2} + \tau_h$ and $\chi_l = \frac{1}{2} - \tau_h$. Hence, $\lambda_m = \overline{\beta} + (\beta_h - \beta_l)\tau_h$, $p_m = \frac{2}{\mu_m}$,

$p_a = \frac{2}{\mu_a}$, $w = 1$, and $\Gamma = \overline{\beta}\mu_m + \left(1-\overline{\beta}\right)\mu_a + \left(\beta_h - \beta_l\right)\left(\mu_m - \mu_a\right)\tau_h$. Firm-level output

$$y_m = \ell_m = \frac{\mu_m\left(\overline{\beta} + \left(\beta_h - \beta_l\right)\tau_h\right)}{\overline{\beta}\mu_m + \left(1-\overline{\beta}\right)\mu_a + \left(\beta_h - \beta_l\right)\left(\mu_m - \mu_a\right)\tau_h}.$$

The transfer τ_h that restores the undistorted labor allocation ($y_m = \overline{\beta}$) is given by

$$\frac{\mu_m\left(\overline{\beta} + \left(\beta_h - \beta_l\right)\tau_h\right)}{\overline{\beta}\mu_m + \left(1-\overline{\beta}\right)\mu_a + \left(\beta_h - \beta_l\right)\left(\mu_m - \mu_a\right)\tau_h} = \overline{\beta}$$

$$\tau_h = \frac{\left(\mu_a - \mu_m\right)\overline{\beta}\left(1-\overline{\beta}\right)}{\left(\beta_h - \beta_l\right)\left(\overline{\beta}\mu_a + \left(1-\overline{\beta}\right)\mu_m\right)}.$$

Therefore, $\tau_h > 0$ when $\frac{\mu_a - \mu_m}{\beta_h - \beta_l} > 0$. Feasibility requires $|\tau_h| < \frac{1}{2}$.

Additionally $\Gamma = \frac{\mu_m\mu_a}{\overline{\beta}\,\mu_a + \left(1-\overline{\beta}\right)\mu_m}$, $\lambda_m = \frac{\overline{\beta}\mu_a}{\overline{\beta}\,\mu_a + \left(1-\overline{\beta}\right)\mu_m}$, $GDP = 2\frac{\overline{\beta}\mu_a + \left(1-\overline{\beta}\right)\mu_m}{\mu_m\mu_a}$,

$$C_{hm} = \frac{\beta_h}{\mu_a}\left(\frac{1}{2} + \frac{\left(\mu_a - \mu_m\right)\overline{\beta}\left(1-\overline{\beta}\right)}{\left(\beta_h - \beta_l\right)\left(\overline{\beta}\mu_a + \left(1-\overline{\beta}\right)\mu_m\right)}\right)\left(\overline{\beta}\mu_a + \left(1-\overline{\beta}\right)\mu_m\right),$$

$$C_{lm} = \frac{\beta_l}{\mu_a}\left(\frac{1}{2} - \frac{\left(\mu_a - \mu_m\right)\overline{\beta}\left(1-\overline{\beta}\right)}{\left(\beta_h - \beta_l\right)\left(\overline{\beta}\mu_a + \left(1-\overline{\beta}\right)\mu_m\right)}\right)\left(\overline{\beta}\mu_a + \left(1-\overline{\beta}\right)\mu_m\right),$$

$$C_{am} = \frac{1-\beta_h}{\mu_m}\left(\frac{1}{2} + \frac{\left(\mu_a - \mu_m\right)\overline{\beta}\left(1-\overline{\beta}\right)}{\left(\beta_h - \beta_l\right)\left(\overline{\beta}\mu_a + \left(1-\overline{\beta}\right)\mu_m\right)}\right)\left(\overline{\beta}\mu_a + \left(1-\overline{\beta}\right)\mu_m\right),$$

$$C_{al} = \frac{1-\beta_l}{\mu_m}\left(\frac{1}{2} - \frac{\left(\mu_a - \mu_m\right)\overline{\beta}\left(1-\overline{\beta}\right)}{\left(\beta_h - \beta_l\right)\left(\overline{\beta}\mu_a + \left(1-\overline{\beta}\right)\mu_m\right)}\right)\left(\overline{\beta}\mu_a + \left(1-\overline{\beta}\right)\mu_m\right),$$

$$C_h = \left(\frac{\beta_h}{\mu_a}\right)^{\beta_h}\left(\frac{1-\beta_h}{\mu_m}\right)^{1-\beta_h}\left(\frac{1}{2} + \frac{\left(\mu_a - \mu_m\right)\overline{\beta}\left(1-\overline{\beta}\right)}{\left(\beta_h - \beta_l\right)\left(\overline{\beta}\mu_a + \left(1-\overline{\beta}\right)\mu_m\right)}\right)\left(\overline{\beta}\mu_a + \left(1-\overline{\beta}\right)\mu_m\right),$$

$$C_l = \left(\frac{\beta_l}{\mu_a}\right)^{\beta_l}\left(\frac{1-\beta_l}{\mu_m}\right)^{1-\beta_l}\left(\frac{1}{2} - \frac{\left(\mu_a - \mu_m\right)\overline{\beta}\left(1-\overline{\beta}\right)}{\left(\beta_h - \beta_l\right)\left(\overline{\beta}\mu_a + \left(1-\overline{\beta}\right)\mu_m\right)}\right)\left(\overline{\beta}\mu_a + \left(1-\overline{\beta}\right)\mu_m\right).$$

With $Y = \left(C_h C_l\right)^{1/2}$

$$Y = \frac{\beta_h^{\frac{\beta_h}{2}}\left(1-\beta_h\right)^{\frac{1-\beta_h}{2}}\beta_l^{\frac{\beta_l}{2}}\left(1-\beta_l\right)^{\frac{1-\beta_l}{2}}}{\mu_a^{\overline{\beta}}\,\mu_m^{1-\overline{\beta}}}\sqrt{\frac{1}{4} - \left(\frac{\left(\mu_a - \mu_m\right)\overline{\beta}\left(1-\overline{\beta}\right)}{\left(\beta_h - \beta_l\right)\left(\overline{\beta}\mu_a + \left(1-\overline{\beta}\right)\mu_m\right)}\right)^2}\left(\overline{\beta}\mu_a + \left(1-\overline{\beta}\right)\mu_m\right).$$

Case 4: Efficient lump-sum transfers

Just as in the previous case

$$y_m = \frac{\mu_m\left(\overline{\beta} + \left(\beta_h - \beta_l\right)\tau_h\right)}{\overline{\beta}\mu_m + \left(1-\overline{\beta}\right)\mu_a - \left(\beta_h - \beta_l\right)\left(\mu_a - \mu_m\right)\tau_h}, \qquad y_a = \frac{\mu_a\left(1 - \overline{\beta} - \left(\beta_h - \beta_l\right)\tau_h\right)}{\overline{\beta}\mu_m + \left(1-\overline{\beta}\right)\mu_a - \left(\beta_h - \beta_l\right)\left(\mu_a - \mu_m\right)\tau_h}.$$

$$GDP = \frac{2}{\overline{\beta}\mu_m + \left(1-\overline{\beta}\right)\mu_a - \left(\beta_h - \beta_l\right)\left(\mu_a - \mu_m\right)\tau_h}.$$

Hence, consumption allocations are given by

$$C_{hm} = \frac{\beta_h \, \mu_m \left(\frac{1}{2} + \tau_h\right)}{\overline{\beta}\mu_m + \left(1 - \overline{\beta}\right)\mu_a - \left(\beta_h - \beta_l\right)\left(\mu_a - \mu_m\right)\tau_h}, \quad C_{lm} = \frac{\beta_l \, \mu_m \left(\frac{1}{2} - \tau_h\right)}{\overline{\beta}\mu_m + \left(1 - \overline{\beta}\right)\mu_a - \left(\beta_h - \beta_l\right)\left(\mu_a - \mu_m\right)\tau_h},$$

$$C_{ha} = \frac{\left(1 - \beta_h\right)\mu_m \left(\frac{1}{2} + \tau_h\right)}{\overline{\beta}\mu_m + \left(1 - \overline{\beta}\right)\mu_a - \left(\beta_h - \beta_l\right)\left(\mu_a - \mu_m\right)\tau_h}, \quad C_{la} = \frac{\left(1 - \beta_l\right)\mu_m \left(\frac{1}{2} - \tau_h\right)}{\overline{\beta}\mu_m + \left(1 - \overline{\beta}\right)\mu_a - \left(\beta_h - \beta_l\right)\left(\mu_a - \mu_m\right)\tau_h},$$

$$C_h = \frac{\left(\beta_h \mu_m\right)^{\beta_h}\left(\left(1 - \beta_h\right)\mu_a\right)^{1-\beta_h}\left(\frac{1}{2} + \tau_h\right)}{\overline{\beta}\mu_m + \left(1 - \overline{\beta}\right)\mu_a - \left(\beta_h - \beta_l\right)\left(\mu_a - \mu_m\right)\tau_h}, \quad C_l = \frac{\left(\beta_l \mu_m\right)^{\beta_l}\left(\left(1 - \beta_l\right)\mu_a\right)^{1-\beta_l}\left(\frac{1}{2} - \tau_h\right)}{\overline{\beta}\mu_m + \left(1 - \overline{\beta}\right)\mu_a - \left(\beta_h - \beta_l\right)\left(\mu_a - \mu_m\right)\tau_h}.$$

Hence

$$Y = \frac{\beta_h^{\frac{\beta_h}{2}}\left(1 - \beta_h\right)^{\frac{1-\beta_h}{2}}\beta_l^{\frac{\beta_l}{2}}\left(1 - \beta_l\right)^{\frac{1-\beta_l}{2}}\mu_m^{\overline{\beta}}\mu_a^{1-\overline{\beta}}\sqrt{\frac{1}{4} - \tau_h^2}}{\overline{\beta}\mu_m + \left(1 - \overline{\beta}\right)\mu_a - \left(\beta_h - \beta_l\right)\left(\mu_a - \mu_m\right)\tau_h}.$$

Then $\tau_h^* = \frac{1}{4}\frac{(\beta_h - \beta_l)(\mu_a - \mu_m)}{\overline{\beta}\mu_m + \left(1 - \overline{\beta}\right)\mu_a}$.

3 Nonlinear Effects using Differentiated Hat Algebra

Nonlinear general equilibrium effects are incorporated using the differentiated hat algebra method. The exact hat-algebra solves a nonlinear fixed-point mapping of equilibrium conditions for finite shocks. Differentiated hat-algebra instead integrates local first-order responses along a path from the baseline to the shock economy. Practically, the algorithm applies the shock in small increments, and at each increment:

1. recomputes local Jacobians at the current (updated) structure;
2. solves a linear system for first-order changes in endogenous variables; and
3. updates the structure by these small changes (renormalizing shares).

This iterative implementation captures nonlinear general equilibrium feedback while maintaining a linear algebraic approach at each step.

Notation:

- The algorithm allows for distinct index sets for firms/goods $i = 1, \cdots, N$; households $h = 1, \cdots, H$; and factors $f = 1, \cdots, F$.
- Structural objects (updated across iterations):
  - Final demand shares $\beta \in [0,1]^{H \times N}$.
  - Intermediate and factor cost shares: $\widetilde{\Omega}_\ell \in [0,1]^{N \times F}$ and $\widetilde{\Omega}_x \in [0,1]^{N \times N}$.
  - Their normalized components: $\alpha \in [0,1]^{N \times F}$, $W \in [0,1]^{N \times N}$, $\omega_\ell \in [0,1]^N$, and $\omega_x = \mathbb{1}_N - \omega_\ell$.
  - Inverse markups $\mu \in \mathbb{R}_+^N$, equity ownership $\kappa \in [0,1]^{N \times H}$, and lump-sum transfers $\tau \in [-1,1]^H$.
- Elasticities and parameters (held fixed when computing Jacobians): θ, θ_ℓ, θ_x, ϱ, ζ_w, ζ_e, and ζ_n.
- Endogenous objects: sector multipliers $\lambda \in \mathbb{R}^N$, $\widetilde{\lambda} \in \mathbb{R}^N$, $\Lambda \in \mathbb{R}^F$, $\widetilde{\Lambda} \in \mathbb{R}^F$, and $\chi \in \mathbb{R}^H$.

- Shocks: $d\log A$ (productivity), $d\log \mu$ (markup variation), dT (transfers), $d\beta$ (preferences), $d\kappa$ (equity ownership), $d\omega_\ell$, $d\alpha$, dW (technology).

Numerical Steps:

- **Shock:** To trace a finite shock, move along a path in small increments with maximum step size *unit* (e.g., 1%). Let the remaining shock vector at iteration k be *shock stock*$_k$. The code computes:
  - *Max Shock* equals the max over components of the absolute value of *shock stock*$_k$.
  - If $Max\ Shock > unit$, take the incremental shock $shock_k = unit \times \frac{shock\ stock_k}{Max\ Shock}$, else take $shock_k = shock\ stock_k$. Then reduce the remaining shocks $shock\ stock_{k+1} = shock\ stock_k - shock_k$.

  These steps are captured by

$$shock_k = Min\left\{1, \frac{unit}{||shock\ stock_{k+1}||_\infty}\right\} shock\ stock_k, \qquad shock\ stock_{k+1} = shock\ stock_k - shock_k.$$

- **Equilibrium:** Solve for $\lambda \in \mathbb{R}^N$, $\widetilde{\lambda} \in \mathbb{R}^N$, $\Lambda \in \mathbb{R}^F$, $\widetilde{\Lambda} \in \mathbb{R}^F$, and $\chi \in \mathbb{R}^H$ using $\lambda = \mathscr{B}'\chi$, $\Lambda = \mathscr{C}'\chi$, $\widetilde{\lambda} = \widetilde{\mathscr{B}}'\chi$, $\widetilde{\Lambda} = \widetilde{\mathscr{C}}'\chi$, and $\chi = (\Omega_\ell + \Omega_\pi)'\lambda + \tau$. Replace one equation in the last system with the normalization $\mathbb{1}_H'\chi = 1$. If needed, solve linear systems with Tikhonov (ridge) regularization rather than explicitly investing $I_N - \Omega_x$ or $I_N - \widetilde{\Omega}_x$.
- **First-order approximation:** For the corresponding structure and incremental shock $shock_k$, assemble the Jacobian linear system consistent with Theorem 7

$$\begin{pmatrix} S_S & S_E & S_w \\ E_S & E_E & E_w \\ w_S & w_E & w_w \end{pmatrix} \begin{pmatrix} d\log S_k \\ d\log E_k \\ d\log w_k \end{pmatrix} = \begin{pmatrix} S_{0,k} \\ E_{0,k} \\ w_{0,k} \end{pmatrix}$$

  with

$$S_S = diag(\lambda), \qquad S_E = -\mathscr{B}'diag(\chi),$$

$$\begin{aligned} S_w = &-\Psi_x'\left(\beta' diag(\chi)\, diag(\rho - \mathbb{1}_H)\,\mathscr{C} - diag\left(\beta' diag(\rho - \mathbb{1}_H)\,\chi\right)\widetilde{\Psi}_\ell\right) \\ &-\Psi_x'\left(\Omega_x' diag(\lambda)\, diag(\theta - \mathbb{1}_N) - diag\left(\Omega_x' diag(\theta - \mathbb{1}_N)\,\lambda\right)\right)\widetilde{\Psi}_\ell \\ &-\Psi_x'\left(\Omega_x' diag(\lambda)\, diag(\theta_x - \theta)\, W - diag\left(\Omega_x' diag(\theta_x - \theta)\,\lambda\right)\right)\widetilde{\Psi}_\ell, \end{aligned}$$

$$\begin{aligned} S_{0,k} = &\Psi_x'\,\Omega_x'\, diag(\lambda)\left(diag(\theta - \mathbb{1}_N)\, d\log A_k + diag(\theta)\, d\log \mu_k\right) \\ &-\Psi_x'\left(\beta' diag(\chi)\, diag(\rho - \mathbb{1}_H)\,\mathscr{B} - diag\left(\beta' diag(\rho - \mathbb{1}_H)\,\chi\right)\widetilde{\Psi}_x\right)\left(d\log A_k + d\log \mu_k\right) \\ &-\Psi_x'\left(\Omega_x' diag(\lambda)\, diag(\theta - \mathbb{1}_N) - diag\left(\Omega_x' diag(\theta - \mathbb{1}_N)\,\lambda\right)\right)\widetilde{\Psi}_x\left(d\log A_k + d\log \mu_k\right) \\ &-\Psi_x'\left(\Omega_x' diag(\lambda)\, diag(\theta_x - \theta)\, W - diag\left(\Omega_x' diag(\theta_x - \theta)\,\lambda\right)\right)\widetilde{\Psi}_x\left(d\log A + d\log \mu\right) \\ &+\Psi_x'\left(d\beta_k'\, diag(\rho)\,\chi - W' diag(\mu)\, diag(\lambda)\, diag(\theta)\, d\omega_{k,\ell} + \left(diag(\omega_x)\, diag(\mu)\ \ dW_k\right)' diag(\theta_x)\,\lambda\right), \end{aligned}$$

$$E_S = -\Omega_\pi'\, diag(\lambda), \qquad E_E = \left(diag(\chi) + diag\begin{pmatrix}\Lambda \\ 0_{H-F}\end{pmatrix} diag\begin{pmatrix}\zeta_e \\ 0_{H-F}\end{pmatrix}\right),$$

$$E_w = -diag\begin{pmatrix}\Lambda \\ 0_{H-F}\end{pmatrix}\left(I_H + diag\begin{pmatrix}\zeta_e \\ 0_{H-F}\end{pmatrix}\right),$$

$$E_{0,k} = diag\begin{pmatrix}\Lambda \\ 0_{H-F}\end{pmatrix} diag\begin{pmatrix}\zeta_n \\ 0_{H-F}\end{pmatrix} d\log n_k + d\kappa_k' diag(\mathbb{1}_N - \mu)\,\lambda - \kappa' diag(\mu)\, diag(\lambda)\, d\log \mu_k + d\tau_k,$$

$$w_S = -\Omega_\ell' diag(\lambda)\,, \qquad\qquad w_E = -diag\begin{pmatrix}\Lambda \\ 0_{H-F}\end{pmatrix} diag\begin{pmatrix}\zeta_e \\ 0_{H-F}\end{pmatrix},$$

$$\begin{aligned} w_w =& diag(\Lambda)\,(I_N + diag(\zeta_e)) + diag\left(\Omega_\ell' diag(\theta_\ell - \mathbb{1}_N)\,\lambda\right) \\ & - \Omega_\ell' diag(\Lambda)\, diag(\theta - \mathbb{1}_N)\,\widetilde{\Psi}_\ell - \Omega_\ell' diag(\Lambda)\, diag(\theta_\ell - \theta)\,\alpha, \end{aligned}$$

$$\begin{aligned} w_{0,k} =& \Omega_\ell' diag(\lambda)\, diag(\theta - \mathbb{1}_N)\left(I_N - \widetilde{\Psi}_x\right) d\log A_k + \Omega_\ell' diag(\lambda)\left(diag(\theta) - diag(\theta - \mathbb{1}_N)\,\widetilde{\Psi}_\ell\right) d\log \mu_k \\ & + (diag(\mu)\,\alpha)'\, diag(\lambda)\, diag(\theta)\, d\,\omega_{k,\ell} + (diag(\mu)\, diag(\omega_\ell)\, d\,\alpha_k)'\, diag(\Omega_\ell)\,\lambda. \end{aligned}$$

Replace the last row of this system of equations with $\widetilde{\Lambda}' d\log w_k = \widetilde{\lambda}'\,(d\log A + d\log \mu)$, which imposes the normalization $d\log p_Y = 0$. Solve this system in step k for $d\log S_k$, $d\log E_k$, and $d\log w_k$.

- **Factor supply variation:**

$$d\log L_k = diag(\zeta_w)\, d\log w_k - diag(\zeta_e)\, d\log E_{k;1:F} + .diag(\zeta_n)\, d\log n_{k\,1:F},$$

$$d\log J_k = d\log w_k + d\log L_k.$$

- **Real GDP in step k:** Using the second representation of $d\log TFP$ in Theorem 2

$$\begin{aligned} d\log Y_k =& \widetilde{\lambda}'\, d\log A_k + \left(\widetilde{\lambda} - diag(\mu)\,\lambda\right)'\, d\log \mu_k - ((\delta - \mathbb{1}_F)\circ\Lambda)'\, d\log J_k \\ & + (\lambda\circ(\mathbb{1}_N - \mu))'\, d\log S_k + \widetilde{\Lambda}'\, d\log L_k. \end{aligned}$$

- **Distributional updates:**

$$\begin{aligned} d\log \lambda_k &= d\log S_k - d\log Y_k, \\ d\log \Lambda_k &= d\log J_k - d\log Y_k, \\ d\log \chi_k &= d\log E_k - d\log Y_k. \end{aligned}$$

- **Price variation in step k:**

$$d\log p_k = \widetilde{\Psi}_\ell\, d\log w_k - \widetilde{\Psi}_x\,(d\log A_k + d\log \mu_k)\,,$$

$$d\log p_k^\ell = \alpha\, d\log w_k, \qquad\qquad d\log p_k^x = W\, d\log p_k,$$

$$d\log p_k^c = \widetilde{\mathscr{C}}\, d\log w_k - \widetilde{\mathscr{B}}\,(d\log A_k + d\log \mu_k)\,.$$

- **Structural updating after the local step:**

$$\begin{aligned} d\log \widetilde{\Omega}_{k,\ell} =& diag(\theta - \mathbb{1}_N)\left(d\log A_k + d\log \mu_k + d\log p_k - d\log p_k^\ell\right)\mathbb{1}_F' \\ & + diag(\theta_\ell - \mathbb{1}_N)\left(d\log p_k^\ell \mathbb{1}_F' - \mathbb{1}_N\, d\log w_k'\right) \\ & + \widetilde{\Omega}_\ell^{\circ -1}\ (diag(\omega_\ell)\ d\,\alpha_k + diag(d\,\omega_{k,\ell})\ \alpha)\,, \end{aligned}$$

$$\begin{aligned} d\log \widetilde{\Omega}_{k,x} =& diag(\theta - \mathbb{1}_N)\,(d\log A_k + d\log \mu_k + d\log p_k - d\log p_k^x)\,\mathbb{1}_N' \\ & + diag(\theta_x - \mathbb{1}_N)\,(d\log p_k^x \mathbb{1}_N' - \mathbb{1}_N\, d\log p_k') \\ & + \widetilde{\Omega}_x^{\circ -1}\,(diag(\mathbb{1}_N - \omega_\ell)\ d\,W_k - diag(d\,\omega_{k,\ell})\ W)\,, \end{aligned}$$

$$d\log\beta_k = diag(\rho - \mathbb{1}_H)\left(d\log p_k^c\,\mathbb{1}_N' - \mathbb{1}_H\,d\log p_k'\right) + \beta^{\circ -1}\,d\,\beta_k.$$

Using these variations, update shares and levels via the first-order multiplicative rule and normalize to keep shares in $[0,1]$ and summing to 1 (function $\mathrm{Renorm}_1(.)$); updates apply only to active links.

$$\chi^* = \mathrm{Renorm}_1(\chi + \chi \circ d\log\chi_k), \qquad d\,\chi_k = \chi^* - \chi,$$

$$\beta^* = \mathrm{Renorm}_1(\beta + \beta \circ d\log\beta_k), \qquad d\,\beta_k = \beta^* - \beta,$$

$$\widetilde{\Omega}_\ell^* = \widetilde{\Omega}_\ell + \widetilde{\Omega}_\ell \circ d\log\widetilde{\Omega}_{k,\ell}, \qquad \widetilde{\Omega}_x^* = \widetilde{\Omega}_x + \widetilde{\Omega}_x \circ d\log\widetilde{\Omega}_{k,x}.$$

Normalize $\widetilde{\Omega}_\ell$ and $\widetilde{\Omega}_x$ by $\aleph = \widetilde{\Omega}_\ell^*\,\mathbb{1}_F + \widetilde{\Omega}_x^*\,\mathbb{1}_N$. Set $\widetilde{\Omega}_\ell^* \leftarrow diag(\aleph^{\circ -1})\,\widetilde{\Omega}_\ell^*$ and $\widetilde{\Omega}_x^* \leftarrow diag(\aleph^{\circ -1})\,\widetilde{\Omega}_x^*$, which guarantess CRS in cost shares. Then

$$\omega_\ell^* = \mathrm{Clamp}_{[0,1]}\left(\widetilde{\Omega}_\ell^*\,\mathbb{1}_F\right), \qquad \omega_x^* = \mathbb{1}_N - \omega_\ell^*,$$

$$\alpha_i^* = \begin{cases}\widetilde{\Omega}_{\ell,i}^*/\omega_{\ell,i}^* & \omega_{\ell,i}^* > 0\\ 0 & \text{otherwise}\end{cases}, \qquad W_i^* = \begin{cases}\widetilde{\Omega}_{x,i}^*/\omega_{x,i}^* & \omega_{x,i}^* > 0\\ 0 & \text{otherwise.}\end{cases}$$

Finally, row-renormalize α and W to sum to one. Use the updated objects, compute

$$d\,\widetilde{\Omega}_{k,\ell} = \widetilde{\Omega}_\ell^* - \widetilde{\Omega}_\ell, \qquad d\,\widetilde{\Omega}_{k,x} = \widetilde{\Omega}_x^* - \widetilde{\Omega}_x.$$

- **Aggregate terms of trade in step k:** Now define

$$\delta = diag\left(\widetilde{\Lambda}^{\circ -1}\right)\Lambda, \qquad M = \mathscr{C}\,\delta, \qquad F = \Psi_\ell\,\delta,$$

$$\textit{Distributive } TT_k = M'\,d\,\chi_k, \qquad \textit{Final-Demand } TT_k = \chi'\,(d\,\beta_k\,F),$$

$$\textit{Intermediate-Demand } TT_k = (\mu \circ \lambda)'\left(d\,\widetilde{\Omega}_{k,x}\,F\right), \qquad \textit{Factor-Demand } TT_k = (\mu \circ \lambda)'\left(d\,\widetilde{\Omega}_{k,\ell}\,\delta\right),$$

$$\textit{Competitive } TT_k = (F \circ \lambda)'\,d\log\mu_k.$$

$$\begin{aligned}\textit{Distributional Reallocation}_k = &- \textit{Distributive } TT_k - \textit{Final-Demand } TT_k - \textit{Factor-Demand } TT_k\\ &- \textit{Intermediate-Demand } TT_k - \textit{Competitive } TT_k,\end{aligned}$$

$$\textit{Technology}_k = \widetilde{\lambda}'\,d\log A_k, \qquad \textit{Competitiveness}_k = \widetilde{\lambda}'\,d\log\mu_k,$$

$$d\log TFP_k = \textit{Technology}_k + \textit{Competitiveness}_k + \textit{Distributional Reallocation}_k.$$

- **Idiosyncratic terms of trade in step k:**

$$\delta_{Dist} = diag\left(\widetilde{\Lambda}^{\circ -1}\right)\widetilde{\mathscr{C}}, \qquad M_{Dist} = \delta_{Dist}\,\mathscr{C}', \qquad F_{Dist} = \delta_{Dist}\,\Psi_\ell',$$

$$\textit{Distributive } TT_{Dist,k} = M_{Dist}\,d\,\chi_k, \qquad \textit{Final-Demand } TT_{Dist,k} = F_{Dist}\,d\,\beta_k'\,\chi,$$

$$\textit{Intermediate-Demand } TT_{Dist,k} = F_{Dist}\,d\,\widetilde{\Omega}_{k,x}'\,(\mu \circ \lambda),$$

$$\textit{Factor-Demand } TT_{Dist,k} = \delta_{Dist}\,d\,\widetilde{\Omega}_{k,\ell}'\,(\mu \circ \lambda),$$

$$\textit{Competitive } TT_{Dist,k} = F_{Dist}\left(\lambda \circ d \log \mu_k\right).$$

$$\textit{Distributional Reallocation}_{Dist,k} = d \log \chi_k - \textit{Distributive } TT_{Dist,k} - \textit{Final-Demand } TT_{Dist,k} - \textit{Factor-Demand } TT_{Dist,k} - \textit{Intermediate-Demand } TT_{Dist,k} - \textit{Competitive } TT_{Dist,k},$$

$$\textit{Technology}_{Dist,k} = \widetilde{\mathscr{B}}' \, d \log A_k, \qquad \textit{Competitiveness}_{Dist,k} = \widetilde{\mathscr{B}}' \, d \log \mu_k,$$

$$d \log PPT_k = \textit{Technology}_{Dist,k} + \textit{Competitiveness}_{Dist,k} + \textit{Distributional Reallocation}_{Dist,k},$$

$$d \log C_k = d \log PPT_k + \widetilde{\mathscr{C}} \, d \log L_k.$$

- **Accumulate variations from step k:** For $X \in$ { Y, TFP, C, PTT, *Technology*, *Competitiveness*, *Distributional Reallocation*, *Distributive TT*, *Final-Demand TT*, *Factor-Demand TT*, *Intermediate-Demand TT*, *Competitive TT*, *Technology*$_{Dist}$, *Competitiveness*$_{Dist}$, *Distributive TT*$_{Dist}$, *Final-Demand TT*$_{Dist}$, *Intermediate-Demand TT*$_{D}$ *Factor-Demand TT*$_{Dist}$, *Competitive TT*$_{Dist}$}, the cummulative variation up to step k is

$$d \log X = \sum_{s=1}^{k} d \log X_s.$$

- **Update markups and equity distribution:** update markups, equity shares, and lump-sum transfers

$$\mu \leftarrow \text{clamp}_{[0,1]}\left(\mu + \mu \circ d \log \mu_k\right), \qquad \kappa_i \leftarrow \text{Renorm}_1\left(\kappa_i + d\,\kappa_i\right), \qquad T \leftarrow T + d\,T.$$

- **Convergence:** Check convergence on the norms of the distributional updates:

$$\|d \log \lambda_k\| \leq tol, \qquad \|d \log \Lambda_k\| \leq tol, \qquad \|d \log \chi_k\| \leq tol,$$

  where tol is a preset tolerance, and also impose $\textit{Max Shock} = 0$. Equivalently, iterate until distributional changes are within tolerance and the residual shock stock is exhausted.

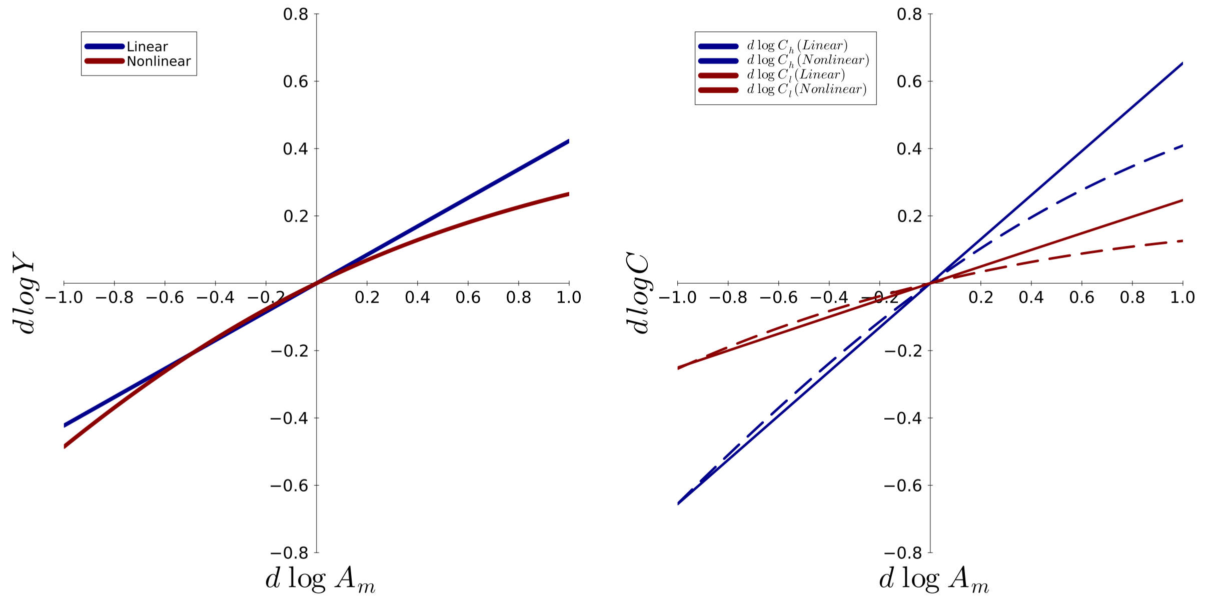
Figure 15 compares baseline first-order ("linear") predictions with the path-integrated via differentiated hat algebra for the horizontal economy in Figure 2 in response to productivity shocks to manufacturing. The curves coincide for small shocks — reflecting first-order equivalence — but diverge as shock magnitudes increase. Along the nonlinear path, prices, Domar weight, and both cost and expenditure shares are re-updated each step, which introduces general equilibrium feedback that systematically modify the impact on aggregate output $d \log Y$ (left panel) relative to a single baseline linearization. The same effect is observed in household consumption, as shown by $d \log C_h$ and $d \log C_l$ (right panel), with differences across households arising from preference and income heterogeneity. Overall, the figure shows that nonlinear estimations bend responses away from linear prefictions, so large shocks require the differentiated hat-algebra treatment to capture nonlinear general equilibrium effects.

4 Data

4.1 Input Output Tables

These steps are implemented in the Julia script `Data Clean IO.jl`

Figure 15: Linear vs. Nonlinear responses to a productivity shock



Note: Left panel plots aggregate output changes $d \log Y$ for the horizontal economy in Figure 2 in response to a manufacturing productivity shock $d \log A_m \in [-1, 1]$ under $(\beta_h, \beta_l) = (0.8, 0.3)$, $(\mu_m, \mu_a) = (0.2, 0.7)$, $\alpha_{mh} = \alpha_{al} = 0.8$, $\alpha_{ml} = \alpha_{ah} = 0.2$, $\omega_m^\ell = \omega_a^\ell = 1$, $\tau_h = \tau_l = 0$, $\kappa_{ih} = 0.5\ \forall i \in \mathcal{N}, h \in \mathcal{H}$, $\theta_m = \theta_a = 0.5$, and $\rho_h = \rho_l = 0.5$. Right panel plots idiosyncratic consumption variation for households h and l in the same economy. *Linear* curve applies a single first-order approximation at the baseline; *nonlinear* curves aggregate incremental responses via differentiated hat algebra, updating cost and expenditure shares each step.

4.1.1 Capital Compensation

Use the `Industry Level Production Accounts (ILPA)` and the associated `Extended Capital Detail` datasets from the BEA-BLS integrated accounts.

1. **Capital asset typology.** We harmonize nine capital asset classes (i) communications equipment, (ii) computer hardware, (iii) R&D, (iv) software capital, (v) entertainment originals, (vi) instruments and other office equipment, (vii) structures, land, and inventories, (viii) transportation equipment, and (ix) other equipment.

2. **Capital series harmonization.** For each asset class, we import the quantity and capital compensation series and build a balanced asset–industry–year panel for 1997–2021. Each observation is mapped to $(year, NAICS\ code)$, linking quantities and payments across assets.

4.1.2 Use and Make Input Output Tables

Use the BEA `USE` and `MAKE` tables for 1997-2021.

1. **USE/MAKE Matrix cleaning and normalization.** Drop rows and columns for: (i) federal general government (defense), (ii) federal general government (nondefense), (iii) federal government enterprises, (iv) state and local general government, (v) state and local government enterprises, and (vi) scrap, used and secondhand goods. This leaves 66 private industries. `USE` is expressed as *industry* $\times$ *commodity* and `MAKE` as *commodity* $\times$ *industry*.

2. **Weighted `USE` and `MAKE` matrices.** Row-normalize to obtain `WUSE` and `WMAKE` with rows that sum to 1. Define the input-output matrix $\mathtt{W} = \mathtt{WUSE} \times \mathtt{WMAKE}$.

3. **Industry accounts and capital allocation.** Construct 66-element vectors for (i) gross `Sales` (row `T018` in `USE`); (ii) `Value Added` (row `VABAS` in `USE`); (iii) `Labor Cost` (row `V001` in `USE`); (iv) `Intermediate Input Cost` (column sums of `USE`); (v) `Capital Cost` for each of the nine asset classes; and (vi) `Capital Quantity` for each of the nine asset classes from the `Industry Level Production Accounts`.

   For three sector groups, additional assumptions are required to allocate `Capital Cost` and `Capital Quantity`. (1) *Retail trade* (`44RT` in `Industry Level Production Accounts`) is split in `USE/MAKE` into: (i) *motor vehicle and part dealers* (`441`); (ii) *food and beverage stores* (`445`); (iii) *general merchandise stores* (`452`); and (iv) *other retail* (`4A0`). (2) *Real estate* (`531` in `Industry Level Production Accounts`) is split in `USE/MAKE` into: (i) *housing* (`HR`); and (ii) *other real estate* (`ORE`). (3) *Hospitals and nursing and residential care facilities* (`622HO` in `Industry Level Production Accounts`) is split in `USE/MAKE` into: (i) *hospitals* (`622`), and (ii) *nursing and residential care facilities* (`623`).

   For these sectors, `Capital Cost` and `Capital Quantity` are allocated in proportion to value added net of labor costs. For example, for *housing* (`HR`) and *other real estate* (`ORE`)

$$\begin{matrix}\texttt{Capital}\\\texttt{Cost}\end{matrix}(\texttt{HR}) = \frac{\texttt{Value Added}(\texttt{HR}) - \texttt{Labor Costs}(\texttt{HR})}{\sum_{g\in\{\texttt{HR},\texttt{ORE}\}}\texttt{Value Added}(g) - \texttt{Labor Costs}(g)} \times \begin{matrix}\texttt{Capital}\\\texttt{Cost}\end{matrix}(\textit{real estate}),$$

$$\begin{matrix}\texttt{Capital}\\\texttt{Quantity}\end{matrix}(\texttt{HR}) = \frac{\texttt{Value Added}(\texttt{HR}) - \texttt{Labor Costs}(\texttt{HR})}{\sum_{g\in\{\texttt{HR},\texttt{ORE}\}}\texttt{Value Added}(g) - \texttt{Labor Costs}(g)} \times \begin{matrix}\texttt{Capital}\\\texttt{Quantity}\end{matrix}(\textit{real estate}).$$

   Total production `Cost` is the sum of `Labor Cost`, `Capital Cost` for the nine types of capital, and `Intermediate Input Cost`.

$$\texttt{Cost} = \texttt{Labor Cost} + \sum_{\textit{Capital types}} \texttt{Capital Cost} + \texttt{Intermediate Input Cost}.$$

   `Intermediate Sales` are computed via a two-step procedure: (1) sum across rows of `USE` matrix to obtain commodity-level sales; and (ii) weight these commodity totals by industry production share from `WMAKE`, i.e.,

$$\texttt{Intermediate Sales} = \min\left\{\texttt{WMAKE}'\ \left(\texttt{USE}'\ \mathbb{1}\right), \texttt{Sales}\right\}.$$

   Then define `Final Sales` as the residual

$$\texttt{Final Sales} = \texttt{Sales} - \texttt{Intermediate Sales}.$$

   Across the sample, the following seven sectors record zero `Final Sales` (all years unless noted): (i) *forestry, fishing, and related activities* (`113FF`), (ii) *oil and gas extraction* (sector `211`), (iii) *wood products* (`321`), (iv) *plastics and rubber products* (`326`), (v) *nonmetallic mineral products* (`327`), (6) *primary metals* (`331`), and (7) *fabricated metal products* (`332`). *Mining, except oil and gas* (`212`) has zero `Final Sales` through 2007 and positive values from 2008 onward. Additionally, *textile mills and textile product mills* (`313TT`) show zero `Final Sales` from 2008 onward, and *electrical equipment, appliances, and components* (`335`) from 2013 onward. Finally, *plastics and rubber products* (`326`) show positive `Final Sales` in 2000–2001, and *apparel and leather and allied products* (`315AL`) report zero `Final Sales` in 2014–2015.

   Inverse markups are computed as $\mu = $ `Cost./Sales`. This procedure does not impose $\mu \leq 1$; values $\mu > 1$ indicate operating losses for the sector. The following is a list of the combinations of sectors, years, and inverse markups for which operational losses are estimated. **1999:** *Data processing, internet publishing, and other information services* (514) $\mu = 1.041$. **2000:** (i) *Data processing, internet publishing, and other information services* (514) $\mu = 1.256$; (ii) *securities, commodity contracts, and investment* (523) $\mu = 1.009$; and (iii) *computer systems design and related activities* (5415) $\mu = 1.032$. **2001:** (i) *Data processing, internet publishing, and other information services* (514) $\mu = 1.035$; and (ii) *computer systems design and related activities* (5415) $\mu = 1.003$. **2008:** *Securities, commodity contracts, and investment* (523) $\mu = 1.222$. **2009:** *Motor vehicles, bodies and trailers, and parts* (3361MV) $\mu = 1.027$. **2011:** *Securities, commodity contracts, and investment* (523) $\mu = 1.025$. **2013:** (i) *Support activities for mining* (213) $\mu = 1.029$; and (ii) *rail transportation* (482) $\mu = 1.005$. **2014:** (i) *Support activities for mining* (213) $\mu = 1.069$; (ii) *motor vehicle and part dealers* (441) $\mu = 1.001$, (iii) *rail transportation* (482) $\mu = 1.012$; and (iv) *publishing industries, except internet (includes software)* (511) $\mu = 1.004$. **2015:** (i) *Support activities for mining* (213) $\mu = 1.057$; (ii) *motor vehicle and part dealers* (441) $\mu = 1.01$; (iii) *food and beverage stores* (445) $\mu = 1.007$; (iv) *general mechandise stores* (452) $\mu = 1.001$; (v) *other retail* (4A0) $\mu = 1.008$; (vi) *rail transportation* (482) $\mu = 1.019$; (vii) *publishing industries, except internet (includes software)* (511) $\mu = 1.005$; (viii) *hospitals* (622) $\mu = 1.001$; and (ix) *accommodation* (721) $\mu = 1.006$. **2016:** (i) *Support activities for mining* (213) $\mu = 1.028$; (ii) *food and beverage and tobacco products* (311FT) $\mu = 1.001$; (iii) *motor vehicle and part dealers* (441) $\mu = 1.022$; (iv) *food and beverage stores* (445) $\mu = 1.019$; (v) *general mechandise stores* (452) $\mu = 1.007$; (vi) *other retail* (4A0) $\mu = 1.022$; (vii) *rail transportation* (482) $\mu = 1.026$; (viii) *publishing industries, except internet (includes software)* (511) $\mu = 1.017$; (ix) *broadcasting and telecommunications* (513) $\mu = 1.003$; (x) *hospitals* (622) $\mu = 1.003$; and (xi) *accommodation* (721) $\mu = 1.012$. **2017:** (i) *Support activities for mining* (213) $\mu = 1.083$; (ii) *food and beverage and tobacco products* (311FT) $\mu = 1.002$; (iii) *motor vehicle and part dealers* (441) $\mu = 1.034$; (iv) *food and beverage stores* (445) $\mu = 1.028$; (v) *general mechandise stores* (452) $\mu = 1.012$; (vi) *other retail* (4A0) $\mu = 1.034$; (vii) *rail transportation* (482) $\mu = 1.034$; (viii) *publishing industries, except internet (includes software)* (511) $\mu = 1.026$; (ix) *broadcasting and telecommunications* (513) $\mu = 1.006$; (x) *insurance carriers and related activities* (524) $\mu = 1.002$; (xi) *hospitals* (622) $\mu = 1.005$; (xii) *nursing and residential care facilities* (623) $\mu = 1.001$; and (xiii) *accommodation* (721) $\mu = 1.019$. **2018:** (i) *Support activities for mining* (213) $\mu = 1.138$; (ii) *motor vehicles, bodies and trailers, and parts* (3361MV) $\mu = 1.001$; (iii) *chemical products* (325) $\mu = 1.018$; (iv) *motor vehicle and part dealers* (441) $\mu = 1.034$; (v) *food and beverage stores* (445) $\mu = 1.028$; (vi) *general mechandise stores* (452) $\mu = 1.012$; (vii) *other retail* (4A0) $\mu = 1.035$; (viii) *rail transportation* (482) $\mu = 1.031$; (ix) *publishing industries, except internet (includes software)* (511) $\mu = 1.007$; (x) *broadcasting and telecommunications* (513) $\mu = 1.004$; (xi) *housing* (*HS*) $\mu = 1.01$; (xii) *hospitals* (622) $\mu = 1.004$; (xii) *nursing and residential care facilities* (623) $\mu = 1.001$; and (xiii) *accommodation* (721) $\mu = 1.026$. **2019:** (i) *Support activities for mining* (213) $\mu = 1.117$; (ii) *motor vehicles, bodies and trailers, and parts* (3361MV) $\mu = 1.002$; (iii) *plastics and rubber products* (326) $\mu = 1.001$; (iv) *motor vehicle and part dealers* (441) $\mu = 1.037$; (v) *food and beverage stores* (445) $\mu = 1.028$; (vi) *general mechandise stores* (452) $\mu = 1.014$; (vii) *other retail* (4A0) $\mu = 1.037$; (viii) *rail transportation* (482) $\mu = 1.03$; (ix) *publishing industries, except internet (includes software)* (511) $\mu = 1.036$; (x) *broadcasting and telecommunications* (513) $\mu = 1.017$; (xi) *housing* (*HS*) $\mu = 1.006$; (xii) *hospitals* (622) $\mu = 1.005$; (xii) *nursing and residential care facilities* (623) $\mu = 1.001$; and (xiii) *accommodation* (721) $\mu = 1.026$. **2020:** (i) *Mining, except oil and gas* (212) $\mu = 1.011$; (ii) *support activities for mining* (213) $\mu = 1.117$; (iii) *apparel and leather and allied products* (315AL) $\mu = 1.094$; (iv) *printing and related support activities* (323) $\mu = 1.006$; (v) *plastics and rubber products* (326) $\mu = 1.002$; (vi) *wholesale trade* (42) $\mu = 1.009$; (vii) *motor vehicle and part dealers* (441) $\mu = 1.064$; (viii) *food and beverage stores* (445) $\mu = 1.05$; (ix) *general mechandise stores* (452) $\mu = 1.011$; (x) *other retail* (4A0) $\mu = 1.062$; (xi) *air transportation* (481) $\mu = 1.042$; (xii) *rail transportation* (482) $\mu = 1.023$;

(xiii) *transit and ground and passenger transportation* (484) $\mu = 1.005$; (xiv) *other transportation and support activities* (487OS) $\mu = 1.029$; (xv) *publishing industries, except internet (includes software)* (511) $\mu = 1.006$; (xvi) *broadcasting and telecommunications* (513) $\mu = 1.01$; (xvii) *insurance carriers and related activities* (524) $\mu = 1.002$; (xviii) *housing* (*HS*) $\mu = 1.023$; (xix) *other real estate* (*ORE*) $\mu = 1.005$; (xx) *rental and leasing services and lessors of intengible assets* (532RL) $\mu = 1.028$; (xxi) *hospitals* (622) $\mu = 1.015$; (xxii) *nursing and residential care facilities* (623) $\mu = 1.001$; (xxiii) *accommodation* (721) $\mu = 1.084$; and (xxiv) *food services and drinking places* (722) $\mu = 1.003$. **2021:** (i) *Mining, except oil and gas* (212) $\mu = 1.054$; (ii) *support activities for mining* (213) $\mu = 1.108$; (iii) *apparel and leather and allied products* (315AL) $\mu = 1.090$; (iv) *printing and related support activities* (323) $\mu = 1.015$; (v) *petroleum and coal products* (324) $\mu = 1.018$; (vi) *chemical products* (325) $\mu = 1.026$; (vii) *motor vehicle and part dealers* (441) $\mu = 1.067$; (viii) *food and beverage stores* (445) $\mu = 1.047$; (ix) *general mechandise stores* (452) $\mu = 1.021$; (x) *other retail* (4A0) $\mu = 1.059$; (xi) *air transportation* (481) $\mu = 1.068$; (xii) *rail transportation* (482) $\mu = 1.035$; (xii) *water transportation* (483) $\mu = 1.073$; (xiv) *transit and ground and passenger transportation* (484) $\mu = 1.012$; (xv) *other transportation and support activities* (487OS) $\mu = 1.031$; (xvi) *warehousing and storage* (493) $\mu = 1.003$; (xvii) *publishing industries, except internet (includes software)* (511) $\mu = 1.024$; (xviiii) *broadcasting and telecommunications* (513) $\mu = 1.0006$; (xix) *housing* (*HS*) $\mu = 1.049$; (xx) *other real estate* (*ORE*) $\mu = 1.014$; (xxi) *rental and leasing services and lessors of intengible assets* (532RL) $\mu = 1.059$; (xxii) *waste management and remediation services* (562) $\mu = 1.004$; (xxiii) *hospitals* (622) $\mu = 1.012$; (xxiv) *nursing and residential care facilities* (623) $\mu = 1.003$; (xxv) *amusements, gambling, and recreation industries* (713) $\mu = 1.006$; (xxvi) *accommodation* (721) $\mu = 1.053$; and (xxvii) *food services and drinking places* (722) $\mu = 1.016$.

4.2 Sectoral Productivities

The Julia script *Data Clean TFP.jl* uses the Industry Level Production Accounts (ILPA) and proceeds in two steps.

1. **Map NAICS codes to Production Account industries.** Harmonize the BEA–BLS KLEMS sectors with the production-account taxonomy by reassigning each NAICS code from KLEMS to its target code in Production Accounts. Agriculture and resource activities map as 111-112 $\to$ 111CA and 113-115 $\to$ 113FF. Manufacturing codes are aggregated to 311-312 $\to$ 311FT, 313-314 $\to$ 313TT, 315-316 $\to$ 315AL, 3361-3363 $\to$ 3361MV, 3364-3369 $\to$ 3364OT. Retail codes maps one-to-many 44,45 $\to$ { 441, 445, 452, 4A0}. Other transportation maps 487,488,492 $\to$ 487OS. Broadcasting and telecommunications maps 515,517 $\to$ 513. Data processing and other information maps 518-519 $\to$ 514. Credit intermediation maps 521-522 $\to$ 521CI. Housing maps one-to-many 531 $\to$ { HS, ORE}. Rental and leasing maps 532-533 $\to$ 532RL. Other professional services maps 5412-5414,5416-5419 $\to$ 5412OP. Hospitals and nursing care maps one-to-many 622-623 $\to$ { 622, 623}. Arts and spectator sports maps 711-712 $\to$ 711AS. All the other matches are one-to-one and share the same codes.

2. **Compute yearly productivity growth (1997-2022).** For sector i in year $t \in \{1998, \cdots, 2022\}$, use $d \log A_{i,t} = \log\left(TFP_{i,t} \,/\, TFP_{i,t-1}\right)$ using ILPA sectoral TFP levels.

4.3 State Level Personal Consumer Expenditure Bundles

The Julia script *Data Clean PCE.jl* uses for 1997-2021 the BEA state-level Personal Consumption Expenditure (PCE) tables with the PCE Bridge that links National Income and product accounts (NIPA) to commodity codes that match target codes in Production Accounts. The estimation of the state-level consumption vectors proceeds in the following steps.

1. **Define weighted crosswalk from NIPA to commodity codes.** For each year $t \in \{1997, \ldots, 2023\}$ and each NIPA PCE line (category) n, let PV_{cnt} denote the *purchasers' value* in the `PCE Bridge` attributed to IO commodity c by that NIPA line. Define the crosswalk weights.

$$weight_{c|n,t} = \frac{PV_{cnt}}{\sum_{c'} PV_{c'nt}}, \quad \text{so that} \quad \sum_{c} weight_{c|n,t} = 1 \quad \text{for each} \quad (n, t).$$

   For example, in 1999, for *jewelry and watches* (NIPA code `20`), the crosswalk vector is:

$$\left(w_{334|20,1999},\ w_{339|20,1999},\ w_{\mathtt{USED}|20,1999}\right) = \left(0.1119,\ 0.8637,\ 0.0244\right),$$

   i.e., 11.19% to *computer and electronic products* (`334`), 86.37% to *miscellaneous manufacturing* (`339`), and 2.44% to *used*.

2. **Define expenditure grid.** Let be the 52 geographies (50 states + DC + national), the 42 non redundant NIPA PCE lines, and the 70 commodity codes (66 private + `used` + `other` + `state and local government enterprises`, and `federal government enterprises`).

3. **Distribute spending across commodities.** For year t, let E_{gnt} denote purchasers' value in geography g on NIPA n. For each triple (t, g, c), accumulate NIPA dollars using the crosswalk

$$X_{gct} \;=\; \sum_{n \in NIPA} weight_{c|n,t}\, E_{gnt}.$$

4. **Compute expenditure shares.** For the strict 66 private-commodity list define $\bar{X}_{gt} = \sum_c X_{gct}$. The share for (t, g, c) is $s_{gct} = \frac{X_{gct}}{\bar{X}_{gt}}$, so that $\sum_c s_{gct} = 1$.

4.4 Quarterly Census of Employment and Wages

The Julia script `Data Clean QCEW.jl` uses for 1997-2021 the BLS `Quarterly Census of Employment and Wages (QCEW)`. The estimation of the industry-level labor compensation by state proceeds in the following steps.

1. **Identify NAICS industries and crosswalk to product accounts.** Identify the level of NAICS disaggregation in the `QCEW` that matches the industry codes from the BEA product accounts. This leaves 105 NAICS codes, which are later mapped to the industry classification used in product accounts. For example, values for *other retail* (`4A0`) are estimated using the following 3-digit NAICS industries: *furniture and home furnishing stores* (`442`), *electronics and appliance stores* (`443`), *building materal and garden equipment and supplies dealers* (`444`), *health and personal care stores* (`446`), *gasoline stations* (`447`), *clothing and clothing accessories stores* (`448`), *sporting goods, hobby, book, and music stores* (`451`), *miscellaneous store retailers* (`453`), and *nonstore retailers* (`454`).

2. **Raw file filtering and aggregation.** Computationally, a key challenge with `QCEW` is that each NAICS classification has independent files that need to be independently uploaded and matched. For each of the 105 preselected NAICS industries, I ingest the required records and merge them into a single annual file to facilitate subsequent calculations.

3. **Employment and labor compensation estimate.** For each year, geographic entity (states or US aggregate), and Production Accounts industry, compute the sum of annual average employment and labor compensation. For each `( year, state, industry)` triple, store the tuple `( employment, compensation)` in a master dictionary.

4.5 Consumer Expenditure Survey

The Julia script *`Data Clean CEX.jl`* processes Census Bureau and BLS `Consumer Expenditure Survey (CEX)` household expenditure data and maps it to the Production Accounts commodity classification using the following steps for years 1997 to 2021. The script uses CEX data as filtered in Markevych (2024).

1. **Mapping from NIPA to commodity codes.** Analogous to the state-level PCE tables, construct year-specific mappings from NIPA lines to Production Accounts commodity codes using the BEA's PCE Bridge. This yields a comprehensive mapping to 66 commodity codes for translating household expenditure from NIPA lines.

2. **Annual Income Calculation.** For each household-year combination, create a standardized income measure, which is used as the basis for identifying income groups: (i) extract all monthly salary observations, (ii) compute the mean monthly household salary, (iii) annualize to obtain household annual income, and (iv) assign this value to all records for that household-year.

3. **Income-based household grouping.** Group households by income similarity within each year to ensure comparability of consumption profiles. (i) For each household-year pair, identify the target household's annual income. (ii) Within the same year, compute income distances between the target household and all other households. (iii) Select the 100 nearest neighbors (by income distance) as "similar peers"; including the target, yields groups of size 101.

4. **Expenditure aggregation by NIPA classification.** For each income-similar group, construct group-level NIPA expenditures. (i) Aggregate expenditure data across all households in an income-similar group. (ii) Distribute NIPA-level expenditures to Production Accounts commodity codes using the bridge mapping shares specified before. (iii) Sum across NIPA lines to obtain commodity-level totals for the 66 codes.

5. **Expenditure share normalization.** For each household-year, use the group-level commodity totals to form shares. (i) Compute total expenditure across the 66 commodity codes. (ii) Divide each commodity's expenditure by the total to obtain normalized shares. (iii) Return a 66-element share vector summing to 1. (iv) Store the expenditure share vector with household metadata (annual income, year, Census region (1 = Northeast, 2 = Midwest, 3 = South, 4 = West), and occupation codes.

4.6 Occupational Employment Wage Statistics

The Julia script *`Data Clean OEWS.jl`* processes BLS `Occupational Employment Wage Statistics (OEWS)` and performs data extraction, transformation, and aggregation spanning for 1997—2024. It proceeds as follows.

1. **Data loading.** Load 127 Excel files containing `OEWS` data across: (i) industry-level data (by 2-3 digit SIC or 4-digit NAICS); (ii) state-level data; (iii) national-level aggregates; and (iv) metropolitan statistical area (MSA) data. The data span from 1997 to 2024, with separate files for each year and disaggregation available for industry, state, national, and MSA. Some years are split into multiple files due to size constraints (e.g., MSA data splits across 2—4 files).

2. **Industry classification crosswalks.** Create a hierarchical system of mapping dictionaries to connect industry classifications across time. This is crucial because `OEWS` data comes in different industry coding schemes depending on the year: Standard Industrial Classification (SIC) for 1997—2001; and NAICS for 2002—2024. This mapping uses an official U.S. Census Bureau crosswalk that contains (i) 2022 NAICS codes (current standard), (ii) 2017 NAICS codes (prior standard), and SIC

codes. This provides a mapping between industrial classification names, codes, and a crosswalk between 2017 NAICS codes and SIC. From here, the script creates a chain of mappings that translate between all NAICS versions 2002 → 2007 → 2012 → 2017 → 2022. This allows for backwards and forward conversion. Finally, the code then creates a mapping from 73 Production Accounts sectors to 4-digits 2017 NAICS codes (e.g., *motor vehicles and parts* (`3361MV`) to `3361`, `3362` and `3363`).

3. **Occupational classification crosswalks.** Harmonize occupational codes across legacy OES and three SOC vintages: OES (1997–1999), SOC 2000 (2000–2010), SOC 2010 (2011–2018), and SOC 2018 (2019–2024).

   SOC 2018 ↔ SOC 2010. Handle three mapping types. **(a) One-to-one mapping**, e.g., SOC 2018 *Chief Executive Officers* (`11-1011`) maps into SOC 2010 *Chief Executive Officers* (`11-1011`). **(b) Many-to-one mapping** in which multiple 2018 codes collapse into one 2010 code; e.g., SOC 2018 *Agents and Business Mgrs of Artists* (`13-1011`) and SOC 2018 *Agents and Business Mgrs of Athletes* (`13-1012`) map into a single SOC 2010 *Agents and Business Managers of Artists, Performers, and Athletes* (`13-1011`). For each source code, 2018 `OEWS` employment weights are used to distribute across 2010 codes, e.g., if in 2018 `13-1011` has 45,000 workers and `13-1012` has 35,000 workers, their corresponding weights are 0.5625 and 0.4375. **(c) Many-to-many mapping** (complex 1-to-multiple or multiple-to-multiple), e.g., SOC 2018 *Administrative Services Managers* (`11-3011`) can map to one of three SOC 2010: *Administrative Services Managers* (`11-3011`), *Facilities Managers* (`11-3012`), and *Transportation, Storage, Distribution Managers* (`11-3013`). For (c), apply a **residual weighting algorithm** with the following steps: (i) identify "*exclusive*" mappings as those SOC 2018 codes that map to only one SOC 2010 code; (ii) identify "*ambiguous*" mappings as those SOC 2018 codes that map to multiple SOC 2010 codes; (iii) calculate the residual employment for "*ambiguous*" codes as the total employment excluding "*exclusive*" code workers; (iv) allocate residual proportionally to the employment for the SOC 2010 codes observed in 2018, i.e., the last year in which SOC 2010 codes were used. The mapping **SOC 2010** ↔ **SOC 2000** and **SOC 2000** ↔ **OES** follow the same logic and steps. These mappings provide a crosswalk chaing for **SOC 2018** ↔ **SOC 2010** ↔ **SOC 2000** ↔ **OES**.

4. **Employment and labor compensation measurement.** For each year (1997—2024) and for each of the four disaggregations (industry, state, national, and MSA): (i) map SIC or NAICS codes to BEA Production Account industries; (ii) for each SOC 2018 occupation, use the crosswalk mappings to construct an equivalent occupation mix, distributing workers and wages by the derived employment weights for one-to-many or many-to-many cases; and (ii) sum total employment and compensation for each (year, industry/state/msa/national, occupation) combination.

4.7 Representative Household Specification

The Julia script *`Representative Household.jl`* computes aggregate TFP growth for 1997-2021 and compares eight model specifications, evaluating their predictive performance against the official BLS TFP growth rates. The script integrates three preprocessed datasets: (i) the BEA `USE` and `MAKE` tables; (ii) BEA Industry-Level Production Accounts (ILPA) sectoral productivity; and (iii) the BEA `PCE` tables.

1. **GDP calculation and factor shares.** The script computes aggregate GDP using the income approach by summing all factor payments (labor, 9 capital types, and economic profits) across industries. This GDP measure serves as the denominator for computing factor shares (λ for sectoral shares, Λ for factor shares, and Π for sectoral profit shares), which will be used as inputs for the model.

2. **Markup variation.** $d \log \mu$ are computed year-over-year and capture changes in sectoral competitiveness across time.

3. **Consumption bundle estimation.** Construct two alternative household consumption specifications. (a) **IO-based bundles (β_{IO}):** derived from BEA final demand patterns, capturing aggregate economy-wide spending. (b) **PCE-based bundles (β_{PCE}):** derived from consumer expenditure surveys. The PCE vectors capture shares at the commodity levels, and by multiplying them by the corresponding `WMAKE` matrix, the vector is expressed in terms of industrial consumption shares.

4. **Equilibrium estimation.** The function `equilibrium()` solves for sectoral market shares (λ), factor shares (Λ), and household consumption (χ), by enforcing three equilibrium conditions: (i) goods market clearing, (ii) household budget constraints, and (iii) factor market equilibrium. Inputs are the vector β showing household allocation of spending across sectors, the labor cost share vector ω_ℓ, the factor cost share matrix α, the input output intermediate network W, the inverse-markup vector μ, the profit distribution matrix k (for the representative household model this is a vector of ones), and the transfer vector T (for the representative household model this is a zero scalar). The function internalizes the network multiplier effects via the Leontief inverse matrices Ψ_x and Ψ_ℓ.

5. **Eight model specifications.** Evaluate the $2 \times 2 \times 2$ model specifications that arise from combining (*Labor* & *Capital*, *Only Labor*) $\times$ (*Production Network with Intermediate Input Markets*, *Horizontal Economy*) $\times$ (β_{IO}, β_{PCE}).

6. **TFP decomposition.** For each model specification and year, TFP growth is decomposed into *technology*, *competitiveness*, and *distributional reallocation*. The sum of these components captures the predicted TFP growth from the model, which is then compared with the observed BLS data.

7. **Model validation and performance metrics.** The script evaluates each of the 8 model specifications using the following metrics:

   (a) **Mean absolute error (MAE):** $\frac{1}{T}\sum_{t=1}^{T}|d \log TFP_t - d\, \widehat{\log TFP}_t|$. Captures the mean percentage point deviation from observed TFP. Lower is better.

   (b) **Root mean squared error (RMSE):** $\sqrt{\frac{1}{T}\sum_{t=1}^{T}\left(d \log TFP_t - d\, \widehat{\log TFP}_t\right)^2}$. Penalizes large errors more heavily than MAE. Lower is better.

   (c) **Bias:** $\frac{1}{T}\sum_{t=1}^{T}\left(d \log TFP_t - d\, \widehat{\log TFP}_t\right)$. Systemic over/under-prediction. If positive (negative), the model overestimates (underestimates) TFP growth. Closer to zero is better.

   (d) **Correlation (r):** $\frac{\sum_{t=1}^{T}\left(d \log TFP_t - \overline{d \log TFP}\right)\left(d\, \widehat{\log TFP}_t - \overline{d\, \widehat{\log TFP}}\right)}{\sqrt{\sum_{t=1}^{T}\left(d \log TFP_t - \overline{d \log TFP}\right)^2}\sqrt{\sum_{t=1}^{T}\left(d\, \widehat{\log TFP}_t - \overline{d\, \widehat{\log TFP}}\right)^2}}$ where $\overline{TFP}$ captures mean of observed values and $\overline{d\, \widehat{\log TFP}}$ the mean of predicted values. Ranges between $[-1, 1]$. Higher is better.

   (e) **Nash-Sutcliffe Efficiency (NSE):** $1 - \frac{\sum_{t=1}^{T}\left(d \log TFP_t - d\, \widehat{\log TFP}_t\right)}{\sum_{t=1}^{T}\left(d \log TFP_t - \overline{d \log TFP}\right)}$. Captures the fraction of variance explained by the model, and it's equivalent to R^2 for residual analysis. It ranges between $[-\infty, 1]$, $NSE = 1$ perfect fit, $NSE = 0$ no better than mean.

   (f) **Index of Agreement (IA):** $1 - \frac{\sum_t^T\left(d \log TFP_t - d\, \widehat{\log TFP}_t\right)^2}{\sum_t^T\left(|d \log TFP_t - \overline{d \log TFP}| + |d\, \widehat{\log TFP}_t - \overline{d \log TFP}|\right)^2}$. Normalized agreement measure accounting for magnitude. It ranges between $[0, 1]$ with $1 =$ perfect agreement and $0 =$ no agreement. Higher is better.

(g) **Theil's U Statistics (U):** $\frac{\sqrt{\frac{1}{T}\sum_{t=1}^{T} d \log TFP_t - d \widehat{\log TFP_t})^2}}{\sqrt{\frac{1}{T}\sum_{t=1}^{T} d \log TFP_t^2} + \sqrt{\frac{1}{T}\sum_{t=1}^{T} d \widehat{\log TFP_t}^2}}$. Compares forecast to naive forecast $d\,log\,TFP_{t-1}$. If $U = 0$ forecast is perfect, if $U < 1$ better than naive, and if $U > 1$ worse than forecast.

(h) **Kling-Gupta Efficiency (KGE):** $1 - \sqrt{(r-1)^2 + (\beta-1)^2 + (\gamma-1)^2}$, $\beta = \frac{\overline{d \widehat{\log TFP}}}{\overline{d\,log\,TFP}}$ captures the ratio of mean values, and $\gamma = \frac{\sigma_{d \widehat{\log TFP}}}{\sigma_{d \log TFP}}$ the ratio of model to observed standard deviations. It ranges between $[-\infty, 1]$, and the higher, the better.

8. **Variance Decomposition.** For each element in the decomposition, estimate its corresponding covariance share

4.8 State-Level Specification

The Julia script *State Level.jl* extends the representative household framework to incorporate state-level heterogeneity across the 50 US states and the District of Columbia. Whereas the representative household model evaluates multiple specifications, the state-level model fixes the best-performing specification: multi-factor production with intermediate inputs and state-specific consumption patterns. This geographic disaggregation enables analysis of how state-level heterogeneity in labor income and consumption patterns drives aggregate variation.

1. **State-level labor market integration via QCEW data.** Rather than treating labor as homogeneous, the script incorporates geographic labor heterogeneity by using employment and labor compensation from the `QCEW`. For each industry-year pair, state-level labor cost shares are computed for state s as $\alpha_{is}^{QCEW} = \frac{Labor\ Income_{is}}{\sum_{s'} Labor\ Income_{is'}}$, capturing each state's share of national labor income within that sector. This yields a labor cost share matrix of dimension 66×52 (industries $\times$ types of labor). Labor types comprise the 50 states, the District of Columbia, and one US aggregate worker that reconciles differences between `USE` and `MAKE` tables and the `QCEW`. This matrix is then integrated with capital-cost data to construct the full factor cost share matrix α.

2. **State-specific consumption patterns.** The script constructs state-level PCE consumption shares $\beta_{s,i}^{PCE}$ for each state-industry pair using household-level consumption survey data. For the residual national worker, the national PCE bundle is assigned. Additionally, a non-labor consumption vector is constructed from the gap between Production Accounts aggregate consumption and state-level PCE worker consumption, scaled by the non-labor share of GDP, i.e., $\beta^{non\text{-}labor} = (1 - \sum_s \Lambda_s)^{-1} \left(\beta^{IO} - \sum_s \Lambda_s\, \beta_s^{PCE}\right)$. This captures consumption patterns of capital and equity owners, enabling distributional decomposition beyond labor markets.

3. **Equilibrium solution with geographic structure.** The extended `equilibrium()` function solves the system with the same three equilibrium conditions as in the representative household model, but over the higher-dimensional state–industry–factor space. The sectoral market shares λ remain 66×1, but factor shares Λ are now 61×1 (capturing state-specific labor, US labor residual, and 9 types of capital), and household consumption χ is now 62×1 (adding one additional household that centralizes all profits). The profit distribution matrix κ is 66×62 and has zeros everywhere except the last column of ones, capturing that the last household receives all profit payments.

4. **TFP and PTT decompositions with geographic specificity.** TFP growth is estimated. Additionally, the script also estimates state-level variations in PTT. Distortion centralities δ and expenditure centralities M are now disaggregated by state and household type. These disaggregated statistics identify which states and households exert greater network leverage on aggregate production and incidence.

4.9 Occupation-Level Specification

The Julia script *`Occupation Level.jl`* integrates OEWS industry-by-occupation employment and wages (with SOC crosswalks to 2018 SOC and mappings to Production Accounts sectors), builds income-conditioned occupation-specific CE consumption bundles, and solves the network equilibrium with an occupation-augmented factor block (833 occupations + 9 capital types) to produce sectoral market shares, factor shares, household expenditure, and occupation-level TFP/PTT decompositions.

1. **Occupation-level labor market integration via OEWS data.** Instead of state- or aggregate-level labor, the script uses OEWS industry-by-occupation employment and wage data by year. For each industry–occupation–year triplet, compute occupation-specific labor cost shares as $\alpha_{o,i}^{OEWS} = \frac{\textit{Labor Income}_{o,i}}{\sum_{o'} \textit{Labor Income}_{o',i}}$, the occupation o share of sector i's labor income. This creates a labor cost share matrix of dimension 66×833 (industries $\times$ occupations), which is integrated with capital cost data to construct the comprehensive factor cost share matrix α.

2. **Cccupational data cleaning and mapping.** To handle OEWS coverage and breaks, (i) exclusion of 34 occupations in public administration, military, and private services (which fall outside the private business sector); and (ii) forward-filling and backward-filling strategies for occupations with missing data in specific years (e.g., using 2004 data for years 1997—2003 when earlier data is unavailable).

3. **Income-specific consumption patterns from CE.** For each occupation–year (o, t), compute mean earnings from OEWS, then map to a CEX expenditure-share vector via nearest-income bin (or linear interpolation across adjacent bins) to obtain $\beta_{o,t}^{CE}$. This yields occupation-specific consumption bundles aligned with empirically observed income-conditioned CE shares.

4. **Equilibrium with occupational structure.** The extended `equilibrium()` imposes (i) goods-market clearing, (ii) household budget balance, and (iii) factor-market clearing in the occupation-augmented space. Sectoral market shares λ remain 66×1, but factor shares Λ are now 842×1 (capturing 833 occupations and 9 types of capital), and household consumption χ is now 843×1 (adding one additional household that centralizes all profits). The profit distribution matrix κ is 66×843 matrix of zeros where the last column equals 1, capturing the idea that the last household centralizes all profit payments.

5. **TFP and PTT decompositions with occupational specificity.** Estimate TFP growth and occupation-level PTTs, and compute distortion centralities δ and expenditure centralities M by occupation. These statistics quantify incidence and network leverage of shocks across occupations over time.

4.10 Generalized Method of Moments (GMM) Estimation: Global and Local Stages

The Julia scripts *`Representative Household Parameterization Inelastic 4 parameters.jl`*, *`State Level Parameterization Inelastic 4 parameters.jl`*, and *`Occupation Level Parameterization Inelastic 4 parameters.jl`* a nested CES general-equilibrium model via a two-stage GMM procedure that jointly estimates four elasticities of substitution. This section documents bootstrapping and the global and local optimization stages implemented in the Julia code in sufficient technical detail for replication.

1. **Model parameters and moment conditions.** The model estimates four elasticities: (i) the economy-wide elasticity of substitution between primary factor and intermediate input aggregators θ; (ii) the elasticity of substitution among primary factors θ_ℓ; (iii) the elasticity of substitution among intermediate inputs θ_x; and (iv) the household elasticity of substitution ρ. These elasticities are collected in $\boldsymbol{\theta} = [\theta, \theta_\ell, \theta_x, \rho]$.

2. **Parameter bounds.** Parameters are constrained to $\boldsymbol{\theta} \in [0, R_{\max}]^4$ with $R_{\max} = 3.0$.

3. **Moment conditions.** The GMM estimation targets TFP growth observed in official BLS TFP data. The moment conditions are defined as

$$g_t(\boldsymbol{\theta}) = d \log TFP_t^{model}(\boldsymbol{\theta}) - d \log TFP_t^{BLS}.$$

where $d \log TFP_t^{model}(\boldsymbol{\theta})$ captures the model's predicted TFP growth with parameters $\boldsymbol{\theta}$, $d \log TFP_t^{BLS}$ stands for the official BLS TFP growth rate, and $t \in [1997, 2000]$.

The moment vector is

$$g(\boldsymbol{\theta}) = [g_{1997}(\boldsymbol{\theta}), g_{1998}(\boldsymbol{\theta}), \ldots, g_{2019}(\boldsymbol{\theta}), g_{2020}(\boldsymbol{\theta})]' \in \mathbb{R}^{24}.$$

4. **GMM Objective Function.** The two-stage GMM objective function is

$$Q(\boldsymbol{\theta}) = g(\boldsymbol{\theta})' \ \mathbf{W}\, g(\boldsymbol{\theta}).$$

Where $Q(\boldsymbol{\theta})$ is the weighteing matrix. Stage 1 captures global optimization, and stage 2 captures local optimization. For both stages $\mathbf{W} = I$.

5. **Stage 1 - Global optimization with BlackBoxOptim.** The GMM objective function has the potential to be highly nonlinear and nonconvex due to: (i) matrix inversion through the Leontief matrices Ψ_x and Ψ_ℓ; (ii) even under linearized equilibrium conditions, parameters interact nonlinearly; (iii) network effects have the potential to create multiple equilibria and saddle points; and (iv) different regions in the parameter space have the potential to produce locally optimal but globally suboptimal fits.

The code implements BlackBoxOptim's **adaptive differential evolution (DE)** algorithm, a population-based, derivative-free optimizer well-suited to solve the potentially non-convex and non-smooth problem $\min_{\boldsymbol{\theta}} \ Q(\boldsymbol{\theta})$ subject to $\boldsymbol{\theta} \in [0, R_{\max}]^4$. The algorithm is configured as follows:

   (a) **Input.** GMM objective $Q(\boldsymbol{\theta})$, population size $N_p = 5$, and max steps $S_{\max} = 100$.

   (b) **Initialize population.** $\mathcal{P} = \{\boldsymbol{\theta}^{\boldsymbol{i}}\}_{i=1}^{N_p}$ uniformly in $[0, R_{\max}]^4$.

   (c) **Evaluate fitness.** $f_i = Q(\boldsymbol{\theta}^{\boldsymbol{i}})$ for all i.

   (d) **Track best.** $\boldsymbol{\theta}_{best} = \arg\min_i \ f_i$.

   (e) For step $s = 1$ to $S_{\max}$:

      i. For individual $i = 1$ to N_p: (1) randomly select $r_1, r_2, r_3 \in \{1, \cdots, N_p\} \setminus \{i\}$.

      ii. Define daptative scaling factor $F(s)$ for mutations.

      iii. Create mutant $\boldsymbol{u}^{\boldsymbol{i}} = \boldsymbol{\theta}^{\boldsymbol{r_1}} + F(s)(\boldsymbol{\theta}^{\boldsymbol{r_2}} - \boldsymbol{\theta}^{\boldsymbol{r_3}})$.

      iv. Constraint to bounds $\boldsymbol{u}^{\boldsymbol{i}} \leftarrow \max\{0, \min\{\boldsymbol{u}^{\boldsymbol{i}}, R_{\max}\}\}$.

      v. Crossover with probability $CR(s,i)$ by defining $\boldsymbol{v}^{\boldsymbol{i}} = \begin{cases} \boldsymbol{u}^{\boldsymbol{i}} & \text{with probability } CR(s,i) \\ \boldsymbol{\theta}^{\boldsymbol{i}} & \text{with probability } 1 - CR(s,i) \end{cases}$.

      vi. Evaluate $f_v = Q(\boldsymbol{v}^{\boldsymbol{i}})$. If $f_v < f_i$, then accept mutation and $\boldsymbol{\theta}^{\boldsymbol{i}} \leftarrow \boldsymbol{v}^{\boldsymbol{i}}$ and $f_i \leftarrow f_v$. Additionally, if $f_v < f_{best}$ update $\boldsymbol{\theta}_{best} \leftarrow \boldsymbol{v}^{\boldsymbol{i}}$ and $f_{best} \leftarrow f_v$.

      vii. Together, $CR(s,i)$ and $F(s)$ create a self-tuning algorithm that naturally transitions from global exploration (early) to local exploration (late). **1. Early exploration (scattered individuals):** Most individuals start with $CR \approx 0.5 - 0.7$ and radius-limited allows for large step sizes in $F(s) \sim 0.8 - 1.2$. This allows mutations to jump far across the parameter space, enabling fast exploration of unexplored regions, creating diversity in trial vectors, and producing some radically new individuals. **2. Adaptive divergence (clustering around promising regions):** individuals with a successful improvement keep high CR, while individuals with failures drop CR, creating a population split into two types, (i) *exploiters* with low CR, and (ii) *explorers* with high CR. Within emerging clusters, population spread decreases, shrinking $F(s)$. The *explorers* are making bold jumps into promising regions, and the *exploiters* are making incremental refinements around the best found. **3. Convergence phase (strong consensus, most individuals near the same region):** the successful strategies dominate, unsuccessful strategies (low CR + large F, or high CR but in a poor region) die out. At this stage, mutation steps become tiny, and it is all about fine-tuning the parameters.

(f) **Output:** $\boldsymbol{\theta}_{best}$ with objective value f_{best}.

6. **Stage 2 - Local Refinement with LBFGS.** While global adaptive DE search explores the parameter space efficiently, it uses only function values (no gradient information). LBFGS (Limited-memory BFGS) combines: (i) the Quasi-Newton method using gradient approximation, (ii) the limited-memory Hessian approximation, and (iii) a box constraint enforcement algorithm. This is as follows:

(a) **Input:** GMM objective $Q(\boldsymbol{\theta})$, starting point $\boldsymbol{\theta}_0$ from Stage 1, and box $[0, R_{\max}]^4$.

(b) **Compute gradient function:** $\nabla Q(\boldsymbol{\theta})$ via automatic differentiation, using the package `ForwardDiff.jl`. The gradient with respect to parameter j is

$$\frac{\partial Q}{\partial \theta_j} = 2 \sum_{t=1998}^{2020} \left[d \log TFP_t^{\text{model}}(\boldsymbol{\theta}) - d \log TFP_t^{\text{BLS}} \right] \cdot \frac{\partial d \log TFP_t^{\text{model}}}{\partial \theta_j}.$$

(c) **Initialize LBFGS:** memory $M = 10$ (stores 10 most recent steps). For iteration $k = 0$ to $K_{\max} = 1000$

i. Evaluate $Q(\boldsymbol{\theta}_k)$ and $\nabla Q(\boldsymbol{\theta}_k)$.

ii. Check convergence $|Q(\boldsymbol{\theta}_k)|_\infty < \varepsilon$, then stop.

iii. Compute LBFGS search direction $\boldsymbol{d}_k$ approximation Newton direction. **Gradient descent** is a simple algorithm for search direction $\boldsymbol{d}_k^{GD} = -\nabla Q(\boldsymbol{\theta}_k)$. However, it converges slowly if the objective function is elongated or curved. For this reason, **Newton's Method** implements $\boldsymbol{d}_k^{Newton} = -H_k^{-1} \nabla Q(\boldsymbol{\theta}_k)$ where $H_k = \nabla^2 Q(\boldsymbol{\theta}_k)$ is the Hessian matrix. LBFGS approximates H_k with $\widehat{H}_k$ using recent information. With $M = 10$, take parameter updates $\{\boldsymbol{\theta}_{k-1} - \boldsymbol{\theta}_{k-2}, \cdots, \boldsymbol{\theta}_{k-10} - \boldsymbol{\theta}_{k-11}\}$ and gradient changes $\{\nabla Q(\boldsymbol{\theta}_{k-1}) - \nabla Q(\boldsymbol{\theta}_{k-2}), \cdots, \nabla Q(\boldsymbol{\theta}_{k-10}) - \nabla Q(\boldsymbol{\theta}_{k-11})\}$, and estimate $\widehat{H}_k = \widehat{H}_{k-1} + \frac{y_{k-1} y_{k-1}'}{y_{k-1}' s_{k-1}} - \frac{\widehat{H}_{k-1} s_{k-1} s_{k-1}' \widehat{H}_{k-1}}{s_{k-1}' \widehat{H}_{k-1} s_{k-1}}$ with $s_i = \boldsymbol{\theta}_i - \boldsymbol{\theta}_i$ and $y_i = \nabla Q(\boldsymbol{\theta}_i) - \nabla Q(\boldsymbol{\theta}_{i-1})$.

iv. Line search to find step α_k maintaining descent. Choose α_k to modify parameters through $\boldsymbol{\theta}_{k+1} = \boldsymbol{\theta}_k + \alpha_k \boldsymbol{d}_k$. The tradeoff is that if α_k is to small, then convergence is slow and iterations are wasted. If α_k is too large, may overshoot or diverge. **Exact line search** is a computationally expensive algorithm that solves $\alpha_k^* = \arg\min_{\alpha > 0} Q(\boldsymbol{\theta}_k + \alpha_k \boldsymbol{d}_k)$. The problem with this search algorithm is that it requires many objective evaluations and may be expensive. For this reason, LBFGS uses the **Hager-Zhang line search** algorithm, which combines two conditions that account for a sufficient decrease and account for curvature in the objective function. (i) The **Armijo condition** chosses an α that satisfies $Q\left(\boldsymbol{\theta}_k + \alpha \boldsymbol{d}_k\right) \leq Q\left(\boldsymbol{\theta}_k\right) + 10^{-4}\alpha\, \nabla\, Q_k\left(\boldsymbol{\theta}_k\right)'\ \boldsymbol{d}_k$ and guarantees a sufficient decrease in the objective, while aboiding overshooting. (ii) The **Wolfe curvature condition** $|\nabla\, Q\left(\boldsymbol{\theta}_k + \alpha \boldsymbol{d}_k\right)'\ \boldsymbol{d}_k| \leq 0.1 |\nabla\, Q_k\left(\boldsymbol{\theta}_k\right)'\ \boldsymbol{d}_k|$, which ensures that the gradient projection on the search direction is reduced. While Armijo allows for small α, Wolfe guarantees that the α is not small enough.

v. Update with box projection $\boldsymbol{\theta}_{k+1} = \max\left(0, \min\left(\boldsymbol{\theta}_k + \alpha_k \boldsymbol{d}_k, R_{\max}\right)\right)$.

(d) **Output:** $\boldsymbol{\theta}_{refined}$ with objective value $f_{refined}$.

7. **Bootstrap GMM.** To assess parameter uncertainty, the code implements a **nonparametric bootstrap** that resembles the time series of moment conditions. This generates a distribution of parameter estimates reflecting sampling variability. For this, the code sets a bootstrap limit B, which is set to 100 or 1000 depending on the estimation. For each bootstrap sample $b = 1$ to B, after running Stage 1 and 2, store parameters $\hat{\boldsymbol{\theta}}_b$, objective function $Q_b = Q_b(\hat{\boldsymbol{\theta}}_b)$, and metrics (MAE, RMSE, etc.)

5 Tables and Figures

Table 6: Sector-level Inverse Markups Statistics (1997-2021)

Sector	Mean	Std	Min	Max	Sector	Mean	Std	Min	Max
Arts, Sports & Museums	0.76	0.03	0.71	0.84	Insurance & Related	0.98	0.01	0.96	1.01
Farms	0.82	0.04	0.77	0.95	Data & Internet Services	0.98	0.07	0.87	1.26
Primary Metals	0.87	0.02	0.81	0.90	Wood Products	0.98	0.01	0.96	0.99
Legal Services	0.87	0.06	0.75	0.98	Motor vehicle parts dealers	0.98	0.04	0.93	1.07
Water Transport	0.90	0.04	0.85	1.07	Waste & Remediation	0.98	0.01	0.97	1.00
Transit & Ground Pass	0.91	0.03	0.80	0.94	Textiles & Textile Mills	0.98	0.01	0.97	0.99
Other Prof & Tech Svcs	0.92	0.02	0.88	0.95	Nonmetallic Minerals	0.98	0.01	0.96	0.99
Securities & Investments	0.92	0.08	0.83	1.22	Rental & Leasing	0.98	0.02	0.94	1.06
Construction	0.93	0.02	0.89	0.99	Gnrl merchandise stores	0.98	0.02	0.96	1.02
Other Services	0.93	0.02	0.91	0.98	Oil & Gas Extraction	0.98	0.02	0.95	1.00
Air Transport	0.93	0.04	0.90	1.07	Utilities	0.98	0.01	0.97	0.99
Social Assistance	0.93	0.03	0.89	0.98	Electrical Equip & Comp	0.99	0.01	0.95	0.99
Ambulatory Health Care	0.93	0.03	0.87	0.98	Credit & Banking	0.99	0.01	0.95	1.00
Funds & Trusts	0.93	0.05	0.83	1.00	Fabricated Metals	0.99	0.01	0.96	1.00
Forestry & Fishing	0.94	0.02	0.91	0.99	Broadcast & Telecom	0.99	0.01	0.97	1.02
Administrative Support	0.94	0.01	0.92	0.97	Machinery	0.99	0.01	0.96	1.00
Truck Transport	0.94	0.04	0.91	1.01	Petroleum & Coal Products	0.99	0.01	0.95	1.02
Film & Sound Recording	0.95	0.01	0.93	0.98	Publish Ind exc Internet	0.99	0.02	0.97	1.04
IT Systems Design	0.96	0.03	0.91	1.03	Warehousing & Storage	0.99	0.00	0.98	1.00
Computers & Electronics	0.96	0.04	0.88	1.00	Chemical Products	0.99	0.01	0.97	1.03
Amusements & Recreation	0.96	0.02	0.93	1.01	Housing	0.99	0.02	0.98	1.05
Education Services	0.96	0.01	0.95	0.98	Mining, except oil and gas	0.99	0.02	0.94	1.05
Misc Manufacturing	0.96	0.01	0.94	0.98	Accommodation	0.99	0.03	0.97	1.08
Other Transport & Support	0.97	0.02	0.94	1.03	Food, Beverage & Tobacco	0.99	0.01	0.98	1.00
Wholesale Trade	0.97	0.01	0.95	1.01	Company Management	0.99	0.00	0.99	1.00
Furniture & Related	0.97	0.01	0.96	0.98	Other real estate	0.99	0.01	0.99	1.01
Food Services & Drinking	0.97	0.01	0.96	1.02	Other Transport Equip	0.99	0.00	0.98	1.00
Pipeline Transport	0.97	0.03	0.91	1.00	Nursing & Residential Care	1.00	0.00	0.99	1.00
Other retail	0.97	0.04	0.93	1.06	Hospitals	1.00	0.01	0.99	1.02
Paper Products	0.98	0.01	0.96	0.98	Plastics & Rubber	1.00	0.00	0.99	1.00
Apparel & Leather	0.98	0.04	0.93	1.09	Motor Vehicles & Parts	1.00	0.01	0.98	1.03
Food and beverage stores	0.98	0.04	0.93	1.05	Mining Support	1.01	0.06	0.93	1.14
Printing & Support	0.98	0.01	0.97	1.01	Rail Transport	1.01	0.01	1.00	1.03

Notes: Sector-level inverse markups ($\mu_i = \text{Cost}_i/\text{Sales}_i$), 1997–2021 summary statistics Mean and Std (standard deviation), Min/Max (lowest/highest sectoral growth) for the 66 private industries. Values are constructed from BEA USE/MAKE and ILPA Extended Capital Detail; $\mu_i > 1$ denotes operating losses.

Table 7: Inverse Markups Statistics by Year

Year	Mean	Std	Min	P10	Q1	Median	Q3	P90	Max
1997	0.950	$5.559e^{-2}$	0.753	0.882	0.919	0.969	0.991	0.996	0.999
1998	0.953	$5.300e^{-2}$	0.757	0.892	0.932	0.969	0.991	0.996	0.999
1999	0.956	$5.222e^{-2}$	0.752	0.900	0.936	0.973	0.991	0.996	1.041
2000	0.962	$6.315e^{-2}$	0.748	0.901	0.939	0.976	0.993	0.997	1.256
2001	0.957	$5.132e^{-2}$	0.752	0.900	0.939	0.974	0.992	0.997	1.035
2002	0.956	$4.885e^{-2}$	0.742	0.904	0.943	0.973	0.991	0.995	0.999
2003	0.956	$4.889e^{-2}$	0.721	0.906	0.936	0.971	0.991	0.995	0.999
2004	0.957	$5.080e^{-2}$	0.707	0.909	0.943	0.972	0.991	0.995	0.999
2005	0.958	$4.935e^{-2}$	0.724	0.912	0.939	0.973	0.992	0.996	0.999
2006	0.959	$4.953e^{-2}$	0.735	0.907	0.942	0.978	0.991	0.995	0.999
2007	0.961	$4.643e^{-2}$	0.754	0.915	0.941	0.979	0.992	0.996	1.000
2008	0.964	$5.787e^{-2}$	0.744	0.914	0.946	0.980	0.993	0.996	1.222
2009	0.958	$4.905e^{-2}$	0.771	0.904	0.936	0.976	0.991	0.995	1.027
2010	0.961	$4.365e^{-2}$	0.780	0.912	0.938	0.977	0.992	0.995	1.000
2011	0.964	$4.180e^{-2}$	0.773	0.916	0.946	0.979	0.991	0.996	1.025
2012	0.962	$4.264e^{-2}$	0.776	0.915	0.946	0.978	0.991	0.996	1.000
2013	0.965	$4.359e^{-2}$	0.759	0.921	0.948	0.980	0.991	0.996	1.029
2014	0.966	$4.272e^{-2}$	0.779	0.926	0.950	0.977	0.993	0.997	1.069
2015	0.966	$4.267e^{-2}$	0.769	0.923	0.946	0.977	0.994	1.003	1.057
2016	0.964	$4.699e^{-2}$	0.768	0.917	0.946	0.973	0.995	1.009	1.028
2017	0.964	$5.117e^{-2}$	0.785	0.903	0.943	0.969	0.998	1.016	1.083
2018	0.964	$5.590e^{-2}$	0.778	0.872	0.946	0.974	0.999	1.015	1.138
2019	0.965	$5.577e^{-2}$	0.795	0.895	0.945	0.974	0.994	1.021	1.117
2020	0.983	$5.612e^{-2}$	0.814	0.913	0.974	0.986	1.006	1.035	1.117
2021	0.989	$5.634e^{-2}$	0.780	0.946	0.966	0.983	1.018	1.056	1.108

Notes: Inverse markups by year (1997–2021). Cross-sector distributional statistics for $\mu_{i,t} = \text{Cost}_{i,t}/\text{Sales}_{i,t}$ across the 66 private industries. Columns report mean, standard deviation, minimum, P10, Q1, median, Q3, P90, and maximum. Costs equal labor + capital (nine asset classes) + intermediate inputs, constructed from BEA USE/MAKE and ILPA. Values $\mu > 1$ indicate operating losses.

Table 8: Sector-level Productivities Statistics (1997—2023)

Sector	Mean	Std	Min	Max
Education Services	-0.76%	3.08%	-7.18%	4.45%
Rental & Leasing	-0.74%	3.57%	-9.09%	5.92%
Mining, except oil and gas	-0.64%	4.73%	-10.5%	9.42%
Construction	-0.64%	1.40%	-4.44%	1.37%
Funds & Trusts	-0.59%	5.11%	-11.7%	10.62%
Credit & Banking	-0.59%	5.77%	-8.81%	15.2%
Other Services	-0.53%	1.95%	-7.81%	1.87%
Amusements & Recreation	-0.47%	3.65%	-9.83%	8.44%
Chemical Products	-0.38%	3.27%	-6.33%	4.28%
Other Transport & Support	-0.36%	3.18%	-9.38%	3.95%
Securities & Investments	-0.30%	11.3%	-25.7%	32.9%
Hospitals	-0.21%	1.06%	-2.56%	1.95%
Nursing & Residential Care	-0.21%	1.06%	-2.56%	1.95%
Truck Transport	-0.10%	1.90%	-6.14%	3.44%
Fabricated Metals	-0.09%	2.75%	-8.19%	4.84%
Furniture & Related	-0.08%	2.59%	-7.07%	4.30%
Wood Products	-0.05%	2.14%	-5.33%	4.07%
Legal Services	-0.01%	3.10%	-9.62%	5.40%
Food, Beverage & Tobacco	0.09%	1.54%	-3.15%	3.99%
Utilities	0.10%	5.81%	-14.3%	17.2%
Primary Metals	0.13%	3.91%	-15.5%	4.10%
Other Transport Equip	0.14%	3.13%	-7.08%	6.24%
Nonmetallic Minerals	0.14%	2.18%	-4.95%	3.26%
Plastics & Rubber	0.21%	1.84%	-3.63%	3.46%
Social Assistance	0.22%	1.50%	-2.33%	3.81%
Water Transport	0.24%	5.62%	-8.79%	12.4%
Petroleum & Coal Products	0.26%	6.07%	-8.23%	18.4%
Waste & Remediation	0.26%	3.70%	-9.69%	11.0%
Wholesale Trade	0.29%	2.14%	-4.34%	4.15%
Food Services & Drinking	0.32%	2.15%	-5.65%	7.33%
Machinery	0.38%	3.18%	-5.75%	6.10%
Electrical Equip & Comp	0.38%	3.54%	-7.71%	6.02%
Other Prof & Tech Svcs	0.42%	2.15%	-2.32%	5.51%
Paper Products	0.46%	2.44%	-4.80%	4.72%
Textiles & Textile Mills	0.47%	3.06%	-6.49%	8.87%
Warehousing & Storage	0.51%	4.65%	-8.25%	10.5%
Housing	0.56%	2.38%	-4.07%	5.40%
Other real estate	0.56%	2.38%	-4.07%	5.40%
Ambulatory Health Care	0.56%	1.32%	-2.41%	2.89%
Company Management	0.68%	3.15%	-6.53%	5.76%
Transit & Ground Pass	0.69%	3.37%	-6.06%	6.04%
Food and beverage stores	0.72%	2.05%	-3.44%	5.40%
Gnrl merchandise stores	0.72%	2.05%	-3.44%	5.40%
Motor vehicle parts dealers	0.72%	2.05%	-3.44%	5.40%
Other retail	0.72%	2.05%	-3.44%	5.40%
Apparel & Leather	0.73%	4.70%	-7.10%	12.5%
Insurance & Related	0.73%	5.90%	-11.37%	14.6%
Arts, Sports & Museums	0.76%	4.28%	-14.9%	5.6%
Misc Manufacturing	0.77%	2.68%	-4.68%	6.19%
Farms	0.77%	3.43%	-4.11%	7.73%
Pipeline Transport	0.91%	11.58%	-30.3%	33.1%
Administrative Support	0.94%	1.71%	-1.68%	4.05%
Accommodation	0.97%	5.12%	-8.65%	19.3%
Motor Vehicles & Parts	1.01%	5.89%	-17.9%	16.3%
Rail Transport	1.06%	2.66%	-4.40%	5.55%
Forestry & Fishing	1.09%	3.76%	-7.32%	10.5%
Printing & Support	1.14%	2.01%	-4.09%	4.34%
Publish Ind exc Internet	1.20%	5.86%	-18.8%	9.40%
Oil & Gas Extraction	1.26%	11.73%	-18.9%	24.7%
Broadcast & Telecom	1.36%	3.29%	-4.68%	8.47%
Film & Sound Recording	1.80%	5.59%	-9.78%	9.49%
Data & Internet Services	2.04%	11.08%	-21.4%	29.7%
Air Transport	2.23%	16.29%	-64.9%	37.7%
Mining Support	3.24%	14.89%	-25.6%	31.5%
IT Systems Design	3.37%	2.50%	-1.49%	7.90%
Computers & Electronics	4.22%	4.97%	-4.78%	16.3%

Notes: Annual sectoral productivity growth is computed as $d \log A_{i,t} = \log\left(TFP_{i,t}/TFP_{i,t-1}\right)$ using BEA–BLS Industry-Level Production Accounts (ILPA) TFP levels. Sectors are harmonized to the 66 private Production-Accounts industries (government and "scrap/used" categories excluded). One-to-many mappings follow the conventions described in the data section (e.g., retail split into `441`, `445`, `452`, `4A0`; housing split into `HS`, `ORE`; hospitals/nursing split into `622`, `623`; broadcasting/telecom merged into `513`). For each sector, the table reports the time-series Mean and Std (standard deviation), Min/Max (lowest/highest sectoral growth) of $d \log A_{i,t}$ over 1997–2021; values are reported in percent per year.

Table 9: Productivity Statistics by Year

Year	Mean	Std	Min	P10	Q1	Median	Q3	P90	Max
1998	0.92%	4.62%	-7.95%	-4.64%	-1.75%	0.28%	3.17%	6.11%	17.06%
1999	0.68%	4.39%	-21.39%	-3.23%	-0.69%	0.78%	2.56%	5.35%	9.64%
2000	0.14%	5.16%	-18.95%	-4.52%	-1.00%	0.70%	1.84%	3.89%	16.27%
2001	0.87%	7.04%	-14.30%	-4.29%	-2.28%	-0.01%	2.02%	4.95%	33.07%
2002	0.99%	6.93%	-30.28%	-6.29%	-1.03%	1.17%	2.62%	4.89%	29.73%
2003	1.58%	4.14%	-13.79%	-2.04%	-0.04%	1.85%	3.25%	4.61%	18.36%
2004	1.83%	3.75%	-6.53%	-1.62%	-0.51%	1.42%	3.22%	6.87%	13.58%
2005	0.16%	4.20%	-21.37%	-3.80%	-1.43%	0.48%	2.18%	3.93%	8.61%
2006	0.84%	3.63%	-6.20%	-3.72%	-1.33%	0.10%	2.73%	5.68%	12.43%
2007	-0.10%	5.13%	-14.00%	-6.30%	-3.10%	-0.18%	2.42%	5.96%	18.75%
2008	-0.04%	5.21%	-25.74%	-4.04%	-1.81%	-0.31%	2.50%	5.37%	12.45%
2009	-0.67%	7.14%	-17.94%	-6.78%	-4.34%	-0.84%	1.25%	5.43%	32.90%
2010	2.31%	4.49%	-13.89%	-1.84%	-0.02%	2.48%	4.49%	5.80%	16.28%
2011	-0.13%	3.42%	-11.71%	-3.33%	-1.30%	-0.33%	1.44%	3.42%	12.62%
2012	0.41%	3.92%	-10.75%	-3.64%	-1.21%	-0.25%	1.96%	5.08%	11.53%
2013	0.75%	2.72%	-6.56%	-2.60%	-0.90%	0.44%	2.32%	3.60%	8.27%
2014	0.29%	3.14%	-11.69%	-2.17%	-0.72%	0.34%	0.89%	2.19%	14.58%
2015	0.18%	5.42%	-25.63%	-4.77%	-1.71%	0.14%	1.85%	4.05%	24.70%
2016	-0.30%	3.61%	-17.68%	-4.44%	-1.28%	-0.16%	1.60%	3.51%	6.02%
2017	0.95%	3.43%	-5.57%	-1.80%	-0.98%	0.60%	2.33%	3.53%	20.98%
2018	0.34%	3.11%	-6.83%	-2.72%	-0.96%	0.22%	1.68%	2.49%	15.82%
2019	-0.02%	3.45%	-17.80%	-3.18%	-1.57%	-0.10%	1.98%	2.53%	7.84%
2020	-1.50%	9.05%	-64.95%	-6.06%	-2.49%	-0.67%	2.22%	3.29%	15.92%
2021	3.37%	7.48%	-17.43%	-2.40%	0.11%	2.96%	5.74%	7.88%	37.73%
2022	-0.30%	6.10%	-15.82%	-6.61%	-3.44%	-0.07%	1.88%	3.45%	31.52%
2023	-0.10%	5.19%	-9.89%	-5.52%	-3.02%	-0.11%	2.68%	4.68%	27.10%

Notes: Annual sectoral productivity growth is $d \log A_{i,t} = \log\left(TFP_{i,t}/TFP_{i,t-1}\right)$ from the BEA–BLS Industry-Level Production Accounts (ILPA), mapped to the 66 private Production Accounts industries. For each calendar year, the table reports unweighted cross-industry moments: Mean and Std (standard deviation), Min/Max (lowest/highest sectoral growth), P10/P90 (10th/90th percentiles), and Q1/Median/Q3 (25th/50th/75th percentiles).

Table 10: Sector-level Revenue Centrality Statistics (1997-2021)

Sector	Mean	Std	Min	Max	Sector	Mean	Std	Min	Max
Arts, Sports & Museums	0.77	0.03	0.70	0.84	Computers & Electronics	1.01	0.05	0.91	1.05
Farms	0.80	0.04	0.74	0.94	Other Transport & Support	1.01	0.02	0.99	1.05
Primary Metals	0.84	0.04	0.75	0.90	Education Services	1.01	0.01	1.00	1.02
Funds & Trusts	0.91	0.08	0.75	1.04	Other real estate	1.01	0.00	1.01	1.02
Legal Services	0.92	0.06	0.79	1.00	Wholesale Trade	1.01	0.01	0.98	1.03
Water Transport	0.92	0.03	0.87	1.05	Printing & Support	1.01	0.01	1.01	1.02
Food, Beverage & Tobacco	0.95	0.01	0.93	0.97	Broadcast & Telecom	1.02	0.01	1.00	1.04
Transit & Ground Pass	0.95	0.04	0.83	0.99	Waste & Remediation	1.02	0.01	1.00	1.03
Securities & Investments	0.95	0.10	0.83	1.29	Other retail	1.02	0.04	0.98	1.08
Construction	0.96	0.01	0.94	1.00	Oil & Gas Extraction	1.02	0.03	0.93	1.05
Other Prof & Tech Svcs	0.96	0.01	0.93	0.97	Petroleum & Coal Products	1.02	0.03	0.94	1.05
Air Transport	0.96	0.04	0.94	1.09	Nonmetallic Minerals	1.02	0.02	0.99	1.04
Other Services	0.96	0.01	0.95	0.99	Food and beverage stores	1.02	0.03	0.99	1.07
Forestry & Fishing	0.97	0.03	0.92	1.02	Insurance & Related	1.02	0.02	1.00	1.06
Social Assistance	0.97	0.03	0.92	1.01	Utilities	1.02	0.01	1.00	1.05
Ambulatory Health Care	0.97	0.02	0.92	1.01	Data & Internet Services	1.02	0.08	0.89	1.31
Film & Sound Recording	0.97	0.01	0.95	1.00	Other Transport Equip	1.03	0.01	0.99	1.05
Truck Transport	0.98	0.03	0.95	1.03	Motor vehicle parts dealers	1.03	0.04	0.98	1.09
Fabricated Metals	0.98	0.03	0.90	1.01	Plastics & Rubber	1.03	0.01	1.01	1.05
Administrative Support	0.99	0.01	0.97	1.00	Chemical Products	1.03	0.02	1.00	1.06
Furniture & Related	0.99	0.02	0.95	1.01	Rental & Leasing	1.03	0.02	0.99	1.07
Misc Manufacturing	0.99	0.01	0.97	1.01	Mining, except oil and gas	1.03	0.02	0.96	1.06
Wood Products	0.99	0.01	0.97	1.01	General merchandise stores	1.03	0.02	1.01	1.06
Amusements & Recreation	0.99	0.02	0.97	1.02	Credit & Banking	1.03	0.02	0.97	1.05
Electrical Equip & Comp	1.00	0.03	0.92	1.02	Accommodation	1.03	0.02	1.01	1.11
Paper Products	1.00	0.01	0.98	1.01	Housing	1.03	0.01	1.03	1.04
Textiles & Textile Mills	1.00	0.01	0.98	1.03	Publish Ind exc Internet	1.04	0.02	1.01	1.10
Apparel & Leather	1.00	0.04	0.95	1.11	Company Management	1.04	0.01	1.00	1.05
Food Services & Drinking	1.00	0.01	0.99	1.03	Hospitals	1.04	0.01	1.03	1.05
Pipeline Transport	1.00	0.03	0.93	1.05	Warehousing & Storage	1.04	0.02	1.02	1.08
Machinery	1.00	0.03	0.92	1.03	Nursing & Residential Care	1.04	0.01	1.03	1.05
IT Systems Design	1.00	0.03	0.96	1.08	Rail Transport	1.04	0.01	1.03	1.06
Motor Vehicles & Parts	1.01	0.03	0.92	1.05	Mining Support	1.04	0.06	0.99	1.18

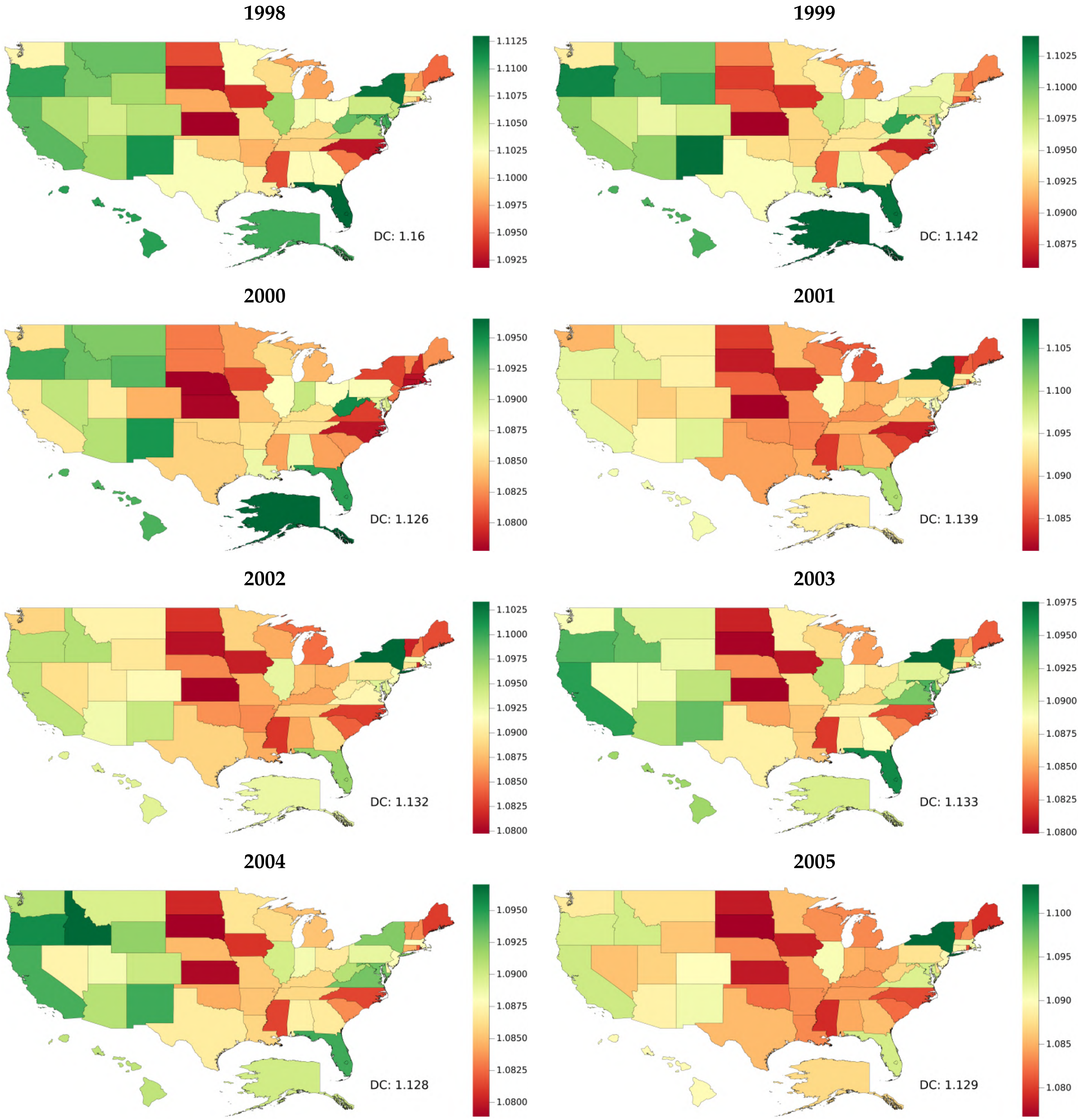
Notes: Sector-level revenue centralities ($F_i = \sum_{h \in \mathscr{H}} \psi^{\ell}_{ih} \delta_h$), 1997–2021 summary statistics Mean and Std (standard deviation), Min/Max (lowest/highest sectoral growth) for the 66 private industries.

Table 11: Revenue Centrality Statistics by Year

Year	Mean	Std	Min	P10	Q1	Median	Q3	P90	Max
1997	0.991	$6.315e^{-2}$	0.772	0.925	0.958	1.011	1.035	1.048	1.079
1998	0.993	$6.093e^{-2}$	0.767	0.939	0.966	1.009	1.036	1.045	1.077
1999	0.994	$5.980e^{-2}$	0.761	0.942	0.980	1.011	1.034	1.043	1.087
2000	0.999	$7.041e^{-2}$	0.755	0.942	0.982	1.010	1.033	1.043	1.309
2001	0.994	$5.860e^{-2}$	0.756	0.944	0.969	1.009	1.032	1.044	1.087
2002	0.993	$5.696e^{-2}$	0.745	0.943	0.968	1.007	1.030	1.041	1.052
2003	0.993	$5.674e^{-2}$	0.719	0.943	0.968	1.007	1.032	1.041	1.048
2004	0.992	$6.024e^{-2}$	0.703	0.940	0.975	1.006	1.033	1.042	1.051
2005	0.992	$5.823e^{-2}$	0.725	0.927	0.973	1.007	1.030	1.040	1.054
2006	0.994	$5.854e^{-2}$	0.740	0.941	0.971	1.012	1.031	1.042	1.051
2007	0.992	$5.482e^{-2}$	0.755	0.942	0.973	1.007	1.027	1.037	1.041
2008	0.991	$6.624e^{-2}$	0.738	0.943	0.974	1.002	1.020	1.030	1.294
2009	0.992	$6.018e^{-2}$	0.739	0.931	0.980	1.012	1.030	1.039	1.048
2010	0.992	$5.340e^{-2}$	0.793	0.944	0.975	1.006	1.027	1.034	1.041
2011	0.991	$4.898e^{-2}$	0.782	0.941	0.976	1.001	1.023	1.030	1.062
2012	0.995	$5.190e^{-2}$	0.796	0.936	0.985	1.009	1.029	1.037	1.045
2013	0.994	$5.306e^{-2}$	0.770	0.938	0.982	1.007	1.025	1.039	1.070
2014	0.993	$5.284e^{-2}$	0.795	0.948	0.979	1.001	1.018	1.039	1.109
2015	0.995	$5.174e^{-2}$	0.776	0.952	0.975	1.004	1.020	1.044	1.090
2016	0.992	$5.835e^{-2}$	0.779	0.936	0.972	0.997	1.027	1.052	1.073
2017	0.991	$6.478e^{-2}$	0.784	0.926	0.970	0.995	1.032	1.059	1.120
2018	0.988	$6.883e^{-2}$	0.755	0.905	0.966	0.995	1.026	1.056	1.175
2019	0.990	$6.971e^{-2}$	0.756	0.925	0.970	0.998	1.029	1.058	1.154
2020	0.992	$7.068e^{-2}$	0.748	0.913	0.972	1.002	1.026	1.064	1.132
2021	0.990	$6.882e^{-2}$	0.764	0.922	0.967	1.001	1.029	1.063	1.111

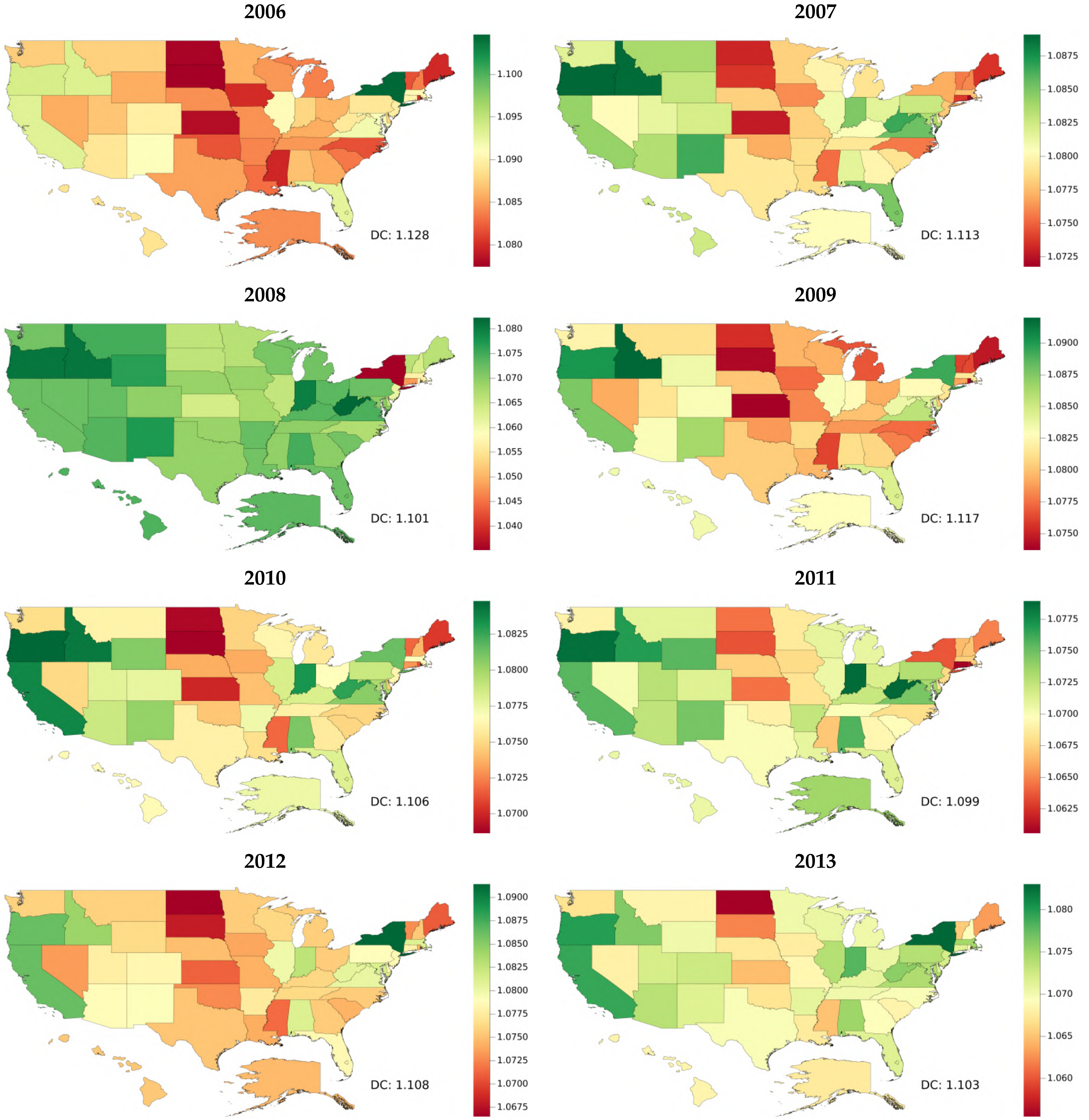
Notes: Revenue centralities by year (1997–2021). Cross-sector distributional statistics for $F_i = \sum_{h \in \mathscr{H}} \psi^{\ell}_{ih} \delta_h$ across the 66 private industries. Columns report mean, standard deviation, minimum, P10, Q1, median, Q3, P90, and maximum.

Figure 16: Distortion Centrality by State (1998-2005)



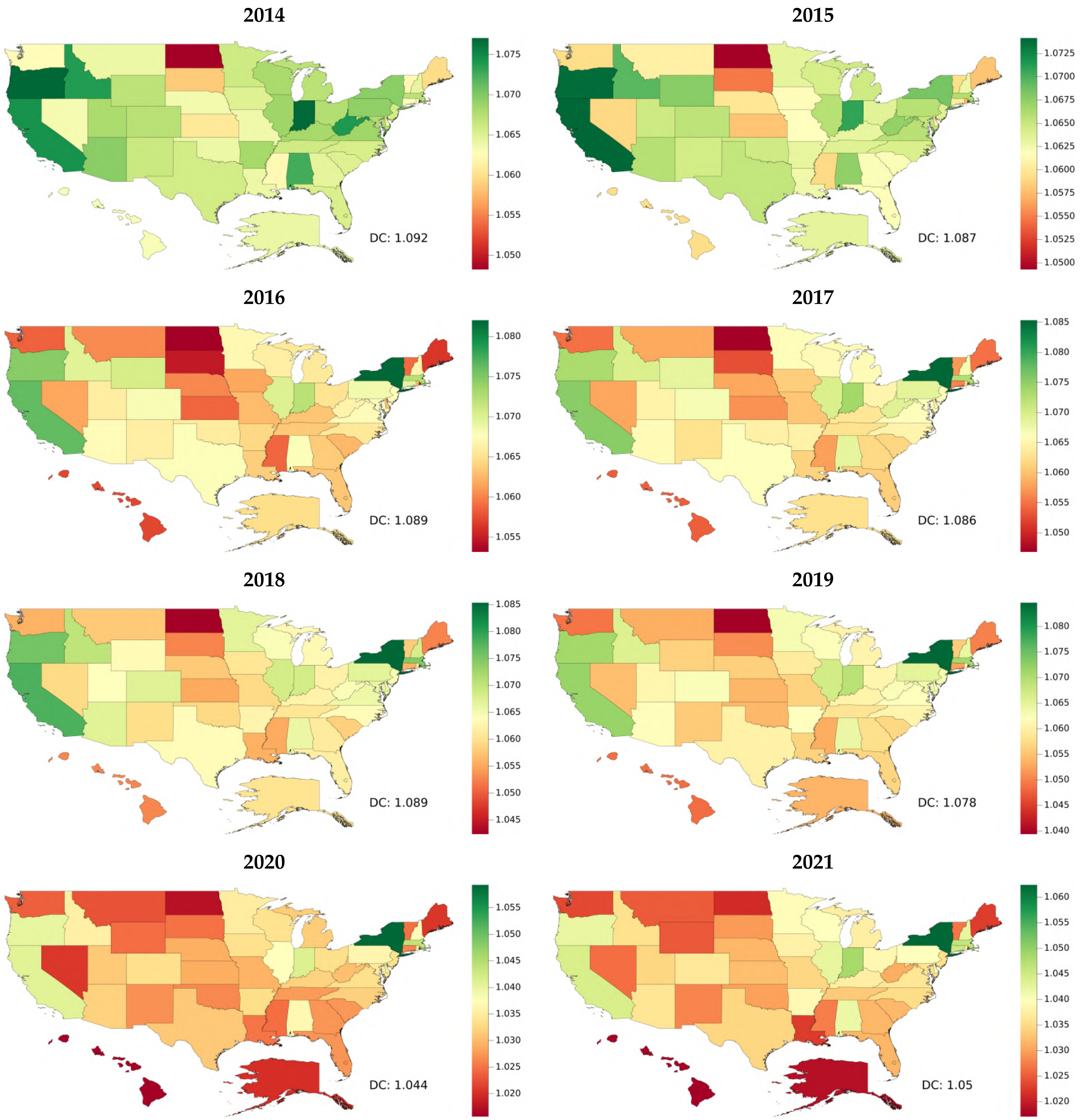
Note: Panels map the distortion centrality index $\delta_{s,t}$ for each state s in year $t \in \{1998, \cdots, 2005\}$. $\delta_{s,t}$ is computed as the value-added to factor income ratios $\widetilde{\Lambda}_{s,t}/\Lambda_{s,t}$, which are estimated taking the parameterization described above. Darker green (red) denotes high-(low-) expenditure centrality. Label next to "DC" gives its value.

Figure 17: Distortion Centrality by State (2006-2013)



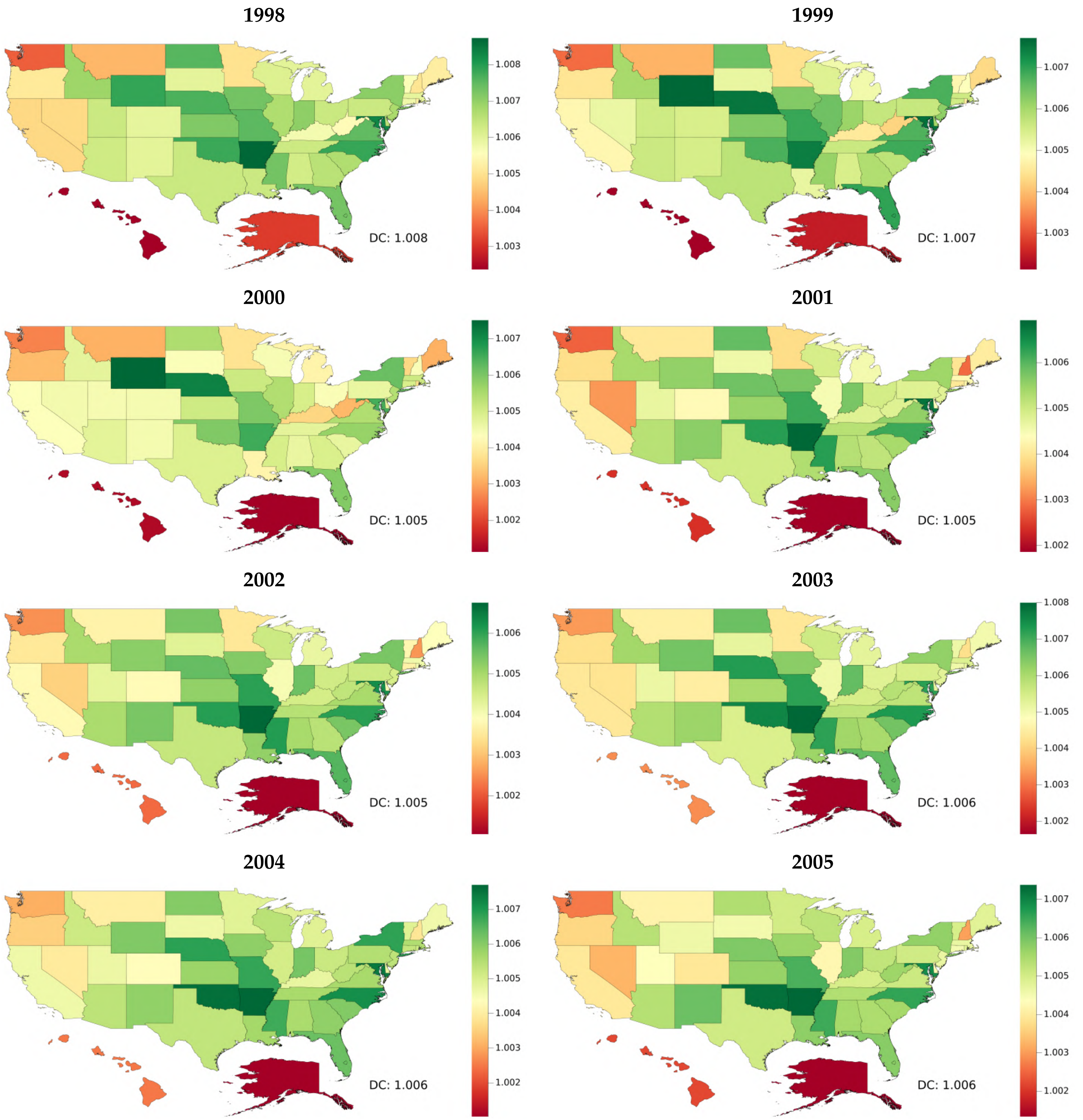
Note: Panels map the distortion centrality index $\delta_{s,t}$ for each state s in year $t \in \{2006, \cdots, 2013\}$. $\delta_{s,t}$ is computed as the value-added to factor income ratios $\widetilde{\Lambda}_{s,t}/\Lambda_{s,t}$, which are estimated taking the parameterization described above. Darker green (red) denotes high-(low-) expenditure centrality. Label next to "DC" gives its value.

Figure 18: Distortion Centrality by State (2014-2021)



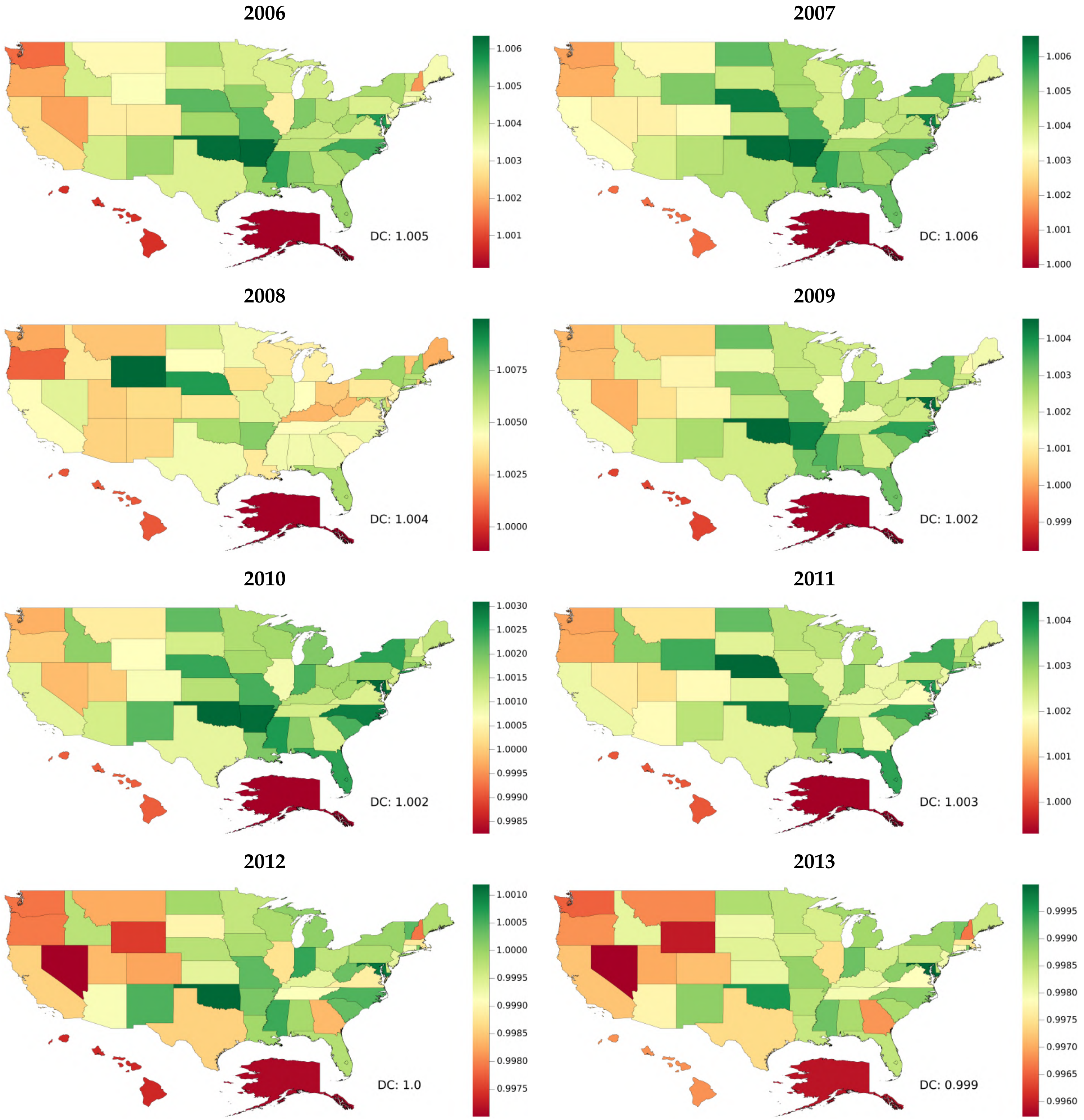
Note: Panels map the distortion centrality index $\delta_{s,t}$ for each state s in year $t \in \{2014, \cdots, 2021\}$. $\delta_{s,t}$ is computed as the value-added to factor income ratios $\widetilde{\Lambda}_{s,t}/\Lambda_{s,t}$, which are estimated taking the parameterization described above. Darker green (red) denotes high-(low-) expenditure centrality. Label next to "DC" gives its value.

Figure 19: Expenditure Centrality by State (1998-2005)



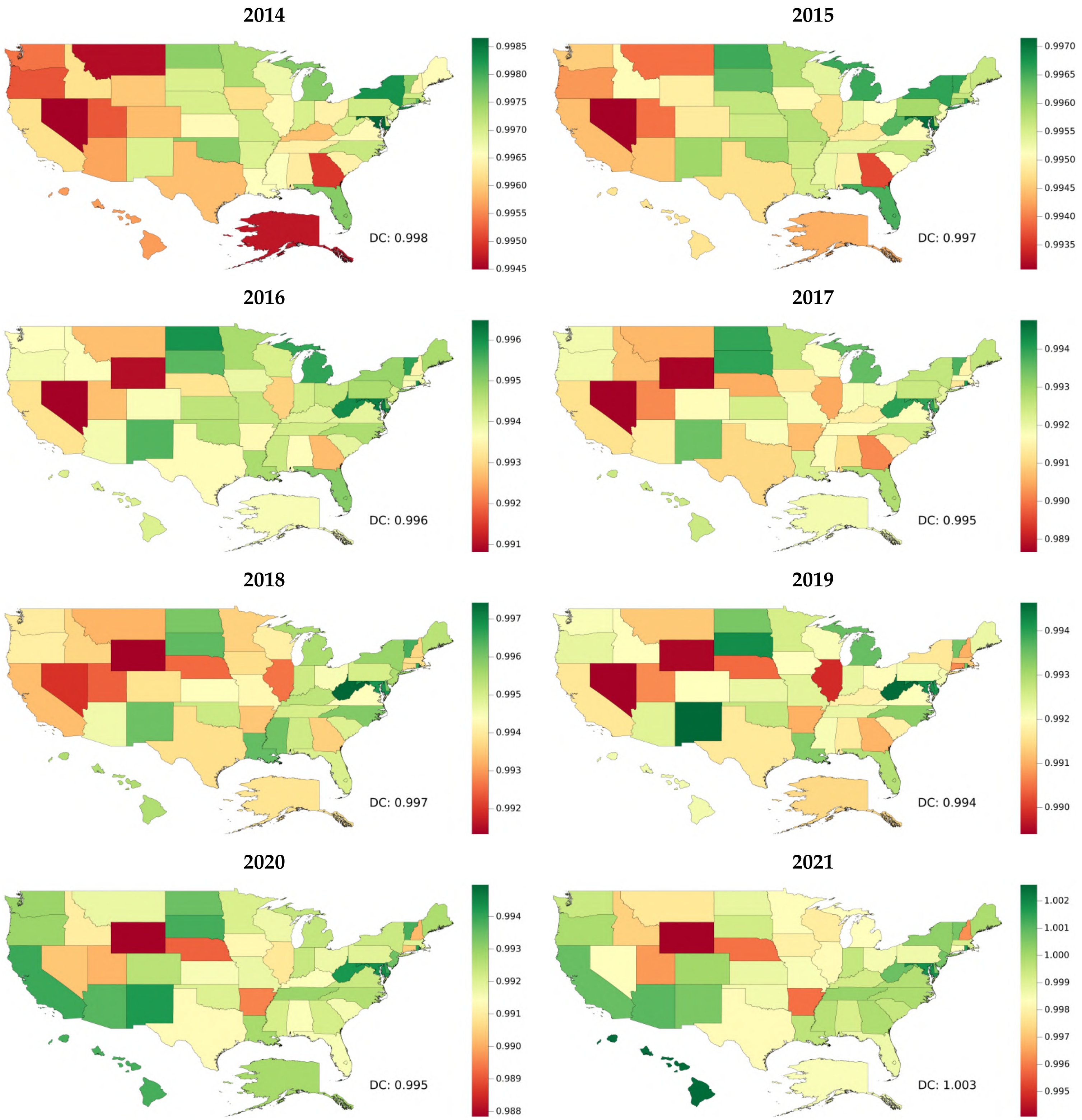
Note: Panels map the expenditure centrality index $M_{s,t}$ for each state s in year $t \in \{1998, \cdots, 2005\}$. $M_{s,t} = \sum_{h \in \mathscr{H}} \mathscr{C}_{sh,t}\ \delta_{h,t}$ stands for the weighted summation of distortion centralities, estimated under the parameterization described above. Darker green (red) denotes high-(low-) expenditure centrality. Label next to "DC" gives its value.

Figure 20: Expenditure Centrality by State (2006-2013)



Note: Panels map the expenditure centrality index $M_{s,t}$ for each state s in year $t \in \{2006, \cdots, 2013\}$. $M_{s,t} = \sum_{h \in \mathscr{H}} \mathscr{C}_{sh,t}\, \delta_{h,t}$ stands for the weighted summation of distortion centralities, estimated under the parameterization described above. Darker green (red) denotes high-(low-) expenditure centrality. Label next to "DC" gives its value.

Figure 21: Expenditure Centrality by State (2014-2021)



Note: Panels map the expenditure centrality index $M_{s,t}$ for each state s in year $t \in \{2014, \cdots, 2021\}$. $M_{s,t} = \sum_{h \in \mathscr{H}} \mathscr{C}_{sh,t}\ \delta_{h,t}$ stands for the weighted summation of distortion centralities, estimated under the parameterization described above. Darker green (red) denotes high-(low-) expenditure centrality. Label next to "DC" gives its value.

Table 12: Distortion Centrality — Occupations

Year	Mean	Std	Min	P10	Q1	Median	Q3	P90	Max	$\text{Corr}_{\text{income}}$
1997	1.117	$4.617e^{-2}$	1.029	1.059	1.089	1.113	1.135	1.160	1.384	-0.016
1998	1.111	$4.493e^{-2}$	1.027	1.057	1.087	1.108	1.128	1.152	1.351	-0.070
1999	1.108	$5.145e^{-2}$	1.029	1.057	1.080	1.101	1.124	1.146	1.361	0.029
2000	1.101	$5.446e^{-2}$	1.014	1.051	1.072	1.093	1.115	1.134	1.370	-0.008
2001	1.104	$4.803e^{-2}$	1.025	1.056	1.077	1.099	1.115	1.141	1.328	-0.001
2002	1.105	$5.982e^{-2}$	1.016	1.056	1.070	1.093	1.116	1.153	1.448	0.024
2003	1.104	$6.134e^{-2}$	1.014	1.053	1.069	1.093	1.115	1.154	1.495	0.012
2004	1.104	$6.440e^{-2}$	1.013	1.054	1.070	1.091	1.114	1.153	1.545	-0.044
2005	1.101	$5.887e^{-2}$	1.012	1.054	1.069	1.091	1.112	1.150	1.494	-0.049
2006	1.101	$5.959e^{-2}$	1.011	1.050	1.068	1.090	1.112	1.150	1.468	-0.034
2007	1.094	$5.607e^{-2}$	1.010	1.046	1.062	1.082	1.107	1.143	1.414	-0.095
2008	1.083	$5.963e^{-2}$	0.828	1.041	1.051	1.071	1.099	1.121	1.411	-0.152
2009	1.096	$5.796e^{-2}$	1.013	1.048	1.060	1.087	1.109	1.139	1.399	-0.115
2010	1.091	$5.127e^{-2}$	1.010	1.046	1.061	1.082	1.110	1.131	1.359	-0.106
2011	1.083	$4.903e^{-2}$	1.001	1.042	1.053	1.074	1.104	1.124	1.361	-0.124
2012	1.091	$5.242e^{-2}$	1.009	1.044	1.055	1.085	1.108	1.137	1.366	-0.114
2013	1.086	$5.231e^{-2}$	1.000	1.038	1.052	1.078	1.101	1.129	1.401	-0.088
2014	1.082	$5.146e^{-2}$	0.993	1.036	1.050	1.075	1.098	1.123	1.374	-0.096
2015	1.080	$5.004e^{-2}$	0.985	1.032	1.052	1.073	1.095	1.123	1.375	-0.073
2016	1.082	$5.377e^{-2}$	0.978	1.031	1.054	1.073	1.098	1.129	1.377	-0.059
2017	1.082	$5.706e^{-2}$	0.970	1.029	1.052	1.072	1.101	1.136	1.401	-0.052
2018	1.081	$5.587e^{-2}$	0.929	1.030	1.052	1.069	1.096	1.138	1.357	-0.030
2019	1.080	$5.760e^{-2}$	0.945	1.029	1.050	1.066	1.094	1.150	1.388	-0.047
2020	1.046	$5.379e^{-2}$	0.912	0.999	1.017	1.034	1.062	1.102	1.417	-0.022
2021	1.048	$5.578e^{-2}$	0.916	0.998	1.019	1.037	1.066	1.091	1.402	-0.032

Notes: Each row reports cross-sectional summary statistics of the *occupation-level distortion centrality* index $\delta_{o,t}$ for the set of occupations in year $t \in \{1997, \cdots, 2021\}$. The index is computed from the occupational model using OEWS-based labor cost shares and the estimated parameterization; formally, $\delta_{o,t} \equiv \widetilde{\Lambda}_{o,t}/\Lambda_{o,t}$. *Mean* and *Std* are unweighted across occupations; *Min*/*Max* are cross-sectional extremes; *P10*, *Q1*, *Median*, *Q3*, and *P90* denote the 10th percentile, 25th percentile, median, 75th percentile, and 90th percentile, respectively. $Corr_{income}$ is the cross-sectional Pearson correlation between $\delta_{o,t}$ and mean OEWS earning $\overline{J}_{o,t}$ in year t (negative values indicate higher-income occupations tend to have lower $\delta_{o,t}$).

Table 13: Distortion Centrality — Capital

Year	Mean	Std	Min	P10	Q1	Median	Q3	P90	Max
1997	1.096	$1.084e^{-2}$	1.076	1.084	1.090	1.100	1.103	1.105	1.113
1998	1.092	$1.031e^{-2}$	1.073	1.082	1.086	1.095	1.098	1.100	1.109
1999	1.085	$8.094e^{-3}$	1.069	1.078	1.081	1.089	1.090	1.093	1.095
2000	1.079	$8.459e^{-3}$	1.066	1.070	1.072	1.080	1.086	1.089	1.091
2001	1.083	$8.888e^{-3}$	1.065	1.072	1.079	1.087	1.090	1.091	1.091
2002	1.080	$8.860e^{-3}$	1.062	1.071	1.076	1.083	1.086	1.089	1.091
2003	1.079	$8.207e^{-3}$	1.064	1.069	1.075	1.080	1.085	1.086	1.090
2004	1.083	$6.952e^{-3}$	1.074	1.076	1.077	1.086	1.088	1.091	1.093
2005	1.082	$7.088e^{-3}$	1.070	1.075	1.076	1.085	1.087	1.089	1.090
2006	1.082	$7.351e^{-3}$	1.068	1.075	1.078	1.085	1.088	1.090	1.090
2007	1.077	$7.140e^{-3}$	1.063	1.069	1.075	1.079	1.082	1.084	1.086
2008	1.066	$9.115e^{-3}$	1.053	1.055	1.061	1.064	1.073	1.076	1.080
2009	1.080	$1.033e^{-2}$	1.056	1.070	1.081	1.082	1.084	1.087	1.094
2010	1.079	$1.053e^{-2}$	1.058	1.066	1.078	1.081	1.083	1.087	1.095
2011	1.073	$1.051e^{-2}$	1.057	1.059	1.070	1.073	1.079	1.084	1.091
2012	1.082	$1.160e^{-2}$	1.064	1.066	1.079	1.082	1.087	1.091	1.102
2013	1.076	$9.040e^{-3}$	1.060	1.061	1.074	1.077	1.082	1.083	1.083
2014	1.072	$1.149e^{-2}$	1.053	1.058	1.066	1.070	1.081	1.084	1.088
2015	1.067	$1.249e^{-2}$	1.047	1.052	1.062	1.068	1.074	1.078	1.087
2016	1.068	$1.480e^{-2}$	1.044	1.049	1.063	1.070	1.076	1.080	1.094
2017	1.067	$1.647e^{-2}$	1.040	1.049	1.057	1.067	1.076	1.081	1.096
2018	1.068	$1.330e^{-2}$	1.044	1.046	1.066	1.073	1.075	1.077	1.080
2019	1.061	$2.023e^{-2}$	1.029	1.040	1.043	1.060	1.072	1.082	1.094
2020	1.036	$1.619e^{-2}$	1.016	1.019	1.023	1.036	1.043	1.054	1.067
2021	1.036	$1.439e^{-2}$	1.011	1.020	1.023	1.037	1.048	1.050	1.052

Notes: Each row reports cross-sectional summary statistics of the *capital asset distortion centrality* index $\delta_{k,t}$ for the nine capital asset classes in year $t \in \{1997, \cdots, 2021\}$. The index is computed from the occupation model with capital using BEA ILPA (Extended Capital Detail) to firm asset-specific capital shares; formally, $\delta_{k,t} \equiv \widetilde{\Lambda}_{k,t}/\Lambda_{k,t}$. *Mean* and *Std* are unweighted across occupations; *Min*/*Max* are cross-sectional extremes; *P10*, *Q1*, *Median*, *Q3*, and *P90* denote the 10th percentile, 25th percentile, median, 75th percentile, and 90th percentile, respectively.

Table 14: Expenditure Centrality — Occupations

Year	Mean	Std	Min	P10	Q1	Median	Q3	P90	Max	$\textbf{Corr}_{\textbf{income}}$
1997	0.998	$2.160e^{-3}$	0.992	0.995	0.996	0.997	0.999	1.000	1.004	0.446
1998	0.997	$2.376e^{-3}$	0.991	0.995	0.996	0.998	0.999	1.000	1.004	0.462
1999	0.999	$2.456e^{-3}$	0.993	0.996	0.998	0.999	1.001	1.002	1.005	0.637
2000	0.999	$2.752e^{-3}$	0.992	0.995	0.997	0.999	1.001	1.002	1.006	0.677
2001	0.997	$1.696e^{-3}$	0.992	0.995	0.996	0.997	0.998	0.999	1.001	0.451
2002	0.997	$1.877e^{-3}$	0.991	0.995	0.996	0.997	0.998	0.999	1.004	0.317
2003	0.999	$1.822e^{-3}$	0.994	0.997	0.998	0.999	1.000	1.002	1.007	0.455
2004	0.995	$3.708e^{-3}$	0.987	0.990	0.993	0.996	0.998	1.000	1.004	0.657
2005	0.994	$2.596e^{-3}$	0.987	0.991	0.993	0.995	0.996	0.998	1.002	0.701
2006	0.994	$1.927e^{-3}$	0.990	0.992	0.993	0.994	0.995	0.997	1.000	0.580
2007	0.997	$2.158e^{-3}$	0.991	0.994	0.995	0.997	0.998	1.000	1.001	0.638
2008	1.002	$4.393e^{-3}$	0.990	0.995	0.999	1.002	1.005	1.008	1.012	0.582
2009	0.994	$2.871e^{-3}$	0.987	0.990	0.991	0.993	0.996	0.997	1.002	0.565
2010	0.993	$1.613e^{-3}$	0.988	0.991	0.992	0.993	0.994	0.995	0.998	0.697
2011	0.996	$1.660e^{-3}$	0.991	0.994	0.995	0.996	0.997	0.998	0.999	0.703
2012	0.989	$1.613e^{-3}$	0.985	0.986	0.988	0.989	0.990	0.991	0.993	0.619
2013	0.987	$1.447e^{-3}$	0.984	0.985	0.987	0.988	0.988	0.989	0.991	0.520
2014	0.988	$1.945e^{-3}$	0.983	0.985	0.986	0.988	0.989	0.990	0.993	0.761
2015	0.989	$1.841e^{-3}$	0.984	0.987	0.988	0.989	0.990	0.991	0.994	0.700
2016	0.988	$1.333e^{-3}$	0.983	0.986	0.987	0.988	0.988	0.989	0.991	0.432
2017	0.985	$1.234e^{-3}$	0.982	0.984	0.984	0.985	0.986	0.987	0.988	0.344
2018	0.986	$1.090e^{-3}$	0.984	0.985	0.985	0.986	0.987	0.987	0.989	0.564
2019	0.985	$1.187e^{-3}$	0.981	0.983	0.984	0.985	0.986	0.987	0.988	0.486
2020	0.983	$1.430e^{-3}$	0.980	0.981	0.982	0.983	0.984	0.985	0.987	0.390
2021	0.984	$1.780e^{-3}$	0.978	0.982	0.983	0.984	0.985	0.986	0.988	-0.028

Notes: Each row reports cross-sectional summary statistics of the *occupation-level expenditure centrality* index $M_{o,t}$ for the set of occupations in year $t \in \{1997, \cdots, 2021\}$. $M_{o,t}$ is computed in the occupational model by mapping CE-based occupation bundles to the 66 industries (via MAKE), propagating through the USE-MAKE production network with sectoral inverse markups, and using OEWS-based factor cost shares where required. *Mean* and *Std* are unweighted across occupations; *Min*/*Max* are cross-sectional extremes; *P10*, *Q1*, *Median*, *Q3*, and *P90* denote the 10th percentile, 25th percentile, median, 75th percentile, and 90th percentile, respectively. $Corr_{income}$ is the cross-sectional Pearson correlation between $M_{o,t}$ and mean OEWS earning $\bar{J}_{o,t}$ in year t (positive values indicate higher-income occupations tend to have higher $\delta_{o,t}$).